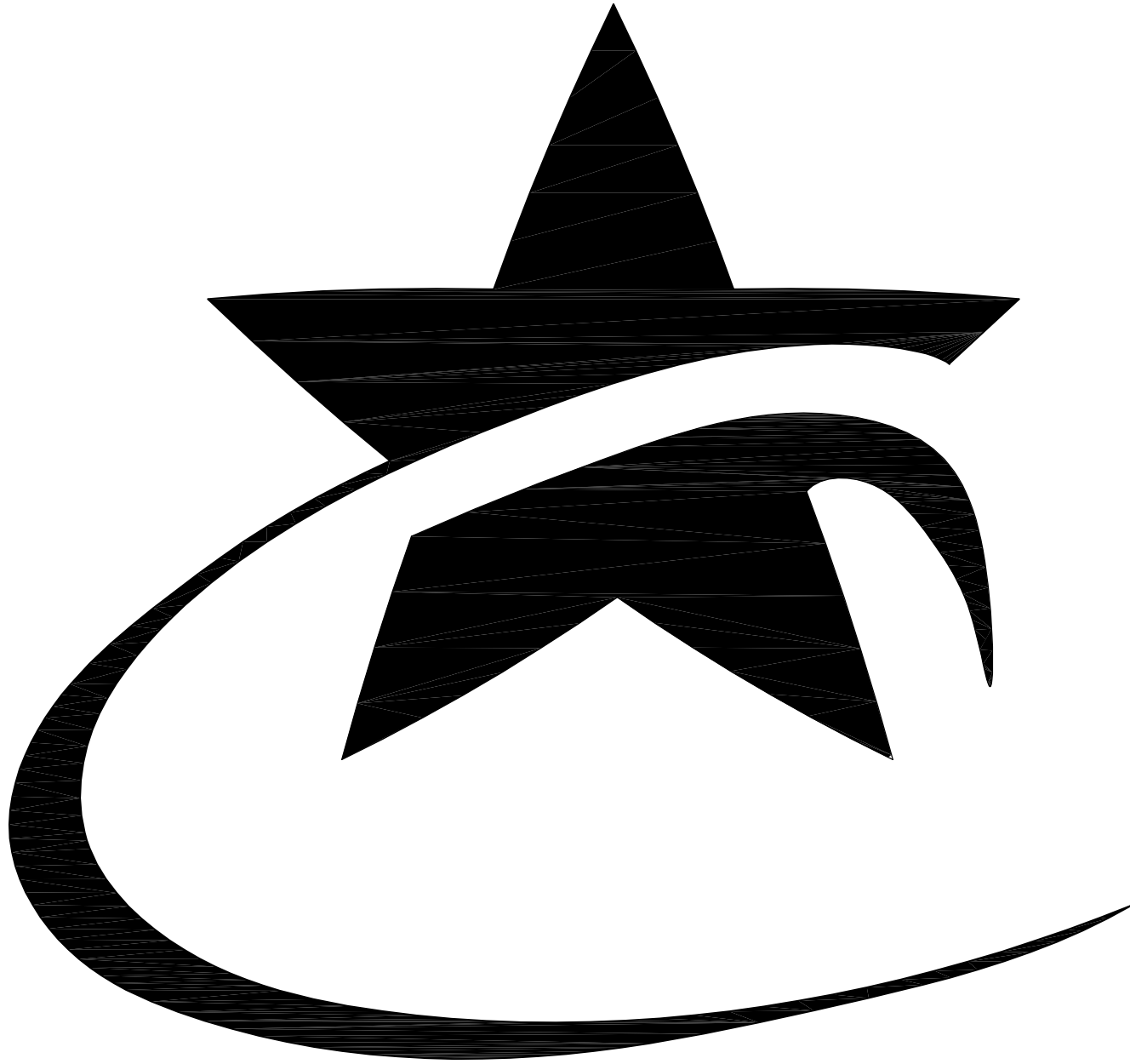


MICHAEL E. DEBAKEY



HOUSTON

VETERANS AFFAIRS
MEDICAL CENTER

2002 HOLCOMBE BLVD.
HOUSTON, TEXAS 77030

BUILDING 104
COOLING TOWER REPLACEMENT

DESIGN BUILDER

York International Corporation
11750 Clay Road
Houston, TX 77043

DRAWING LIST

- | | |
|------|---|
| A0.1 | TITLE SHEET AND DRAWING INDEX |
| A1.0 | BLDG 104–PARTIAL PLAN–ARCHITECTURAL |
| S1.0 | BLDG 104–PARTIAL PLAN–STRUCTURAL |
| M1.0 | BLDG 104–PARTIAL PLAN–MECHANICAL PIPING |
| M3.0 | BLDG 104–MECHANICAL SECTIONS |
| M5.0 | BLDG 104–CONDENSER WATER FLOW DIAGRAM |
| M6.0 | BLDG 104–COOLING TOWER DETAILS |
| E1.0 | BLDG 104–COOLING TOWER–ELECTRICAL |
| E5.0 | BLDG 104–ELECTRICAL SINGLE LINE DIAGRAM |

A0.1

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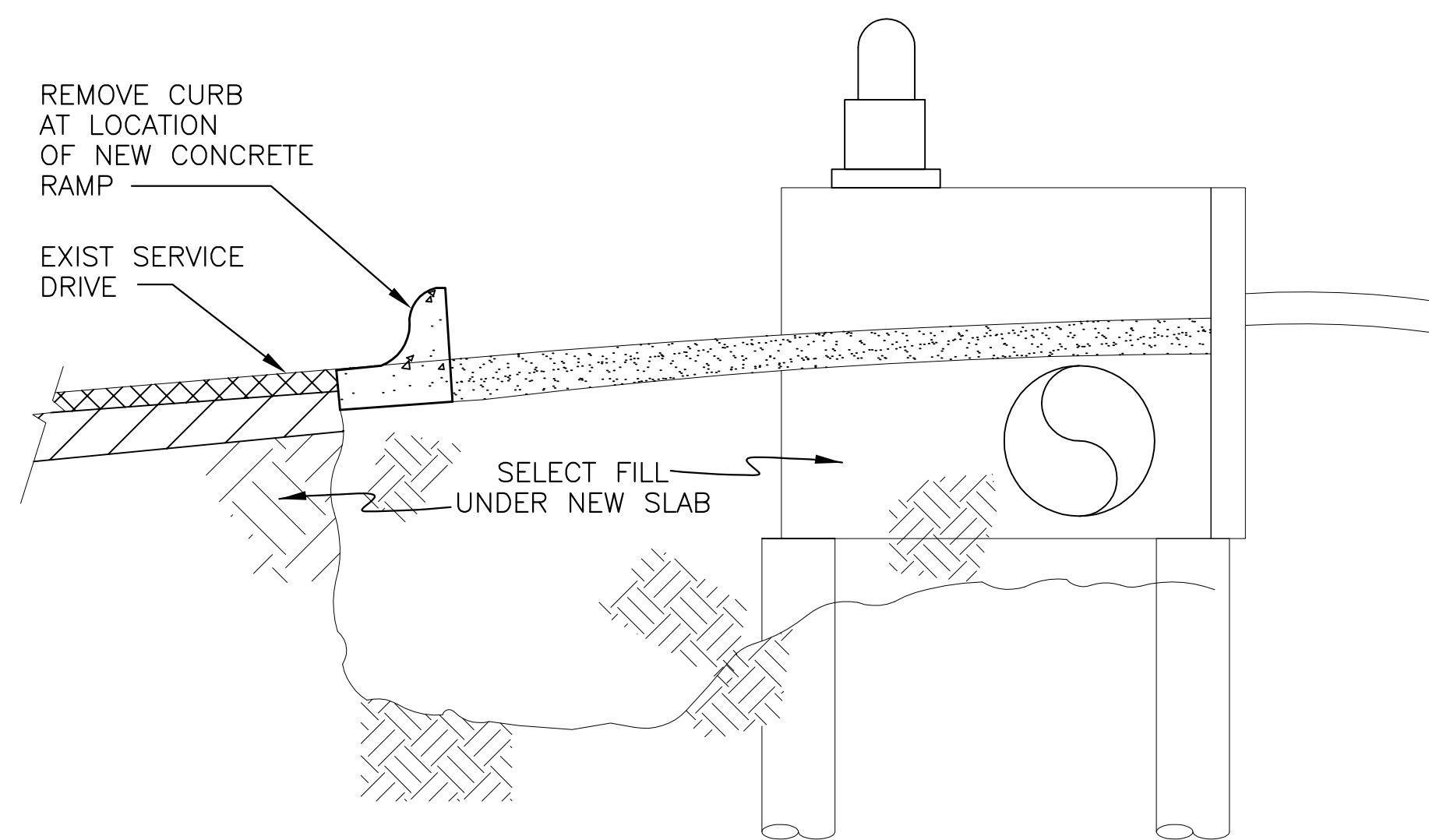
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VAMC–104 COOLING TOWER REPLACEMENT

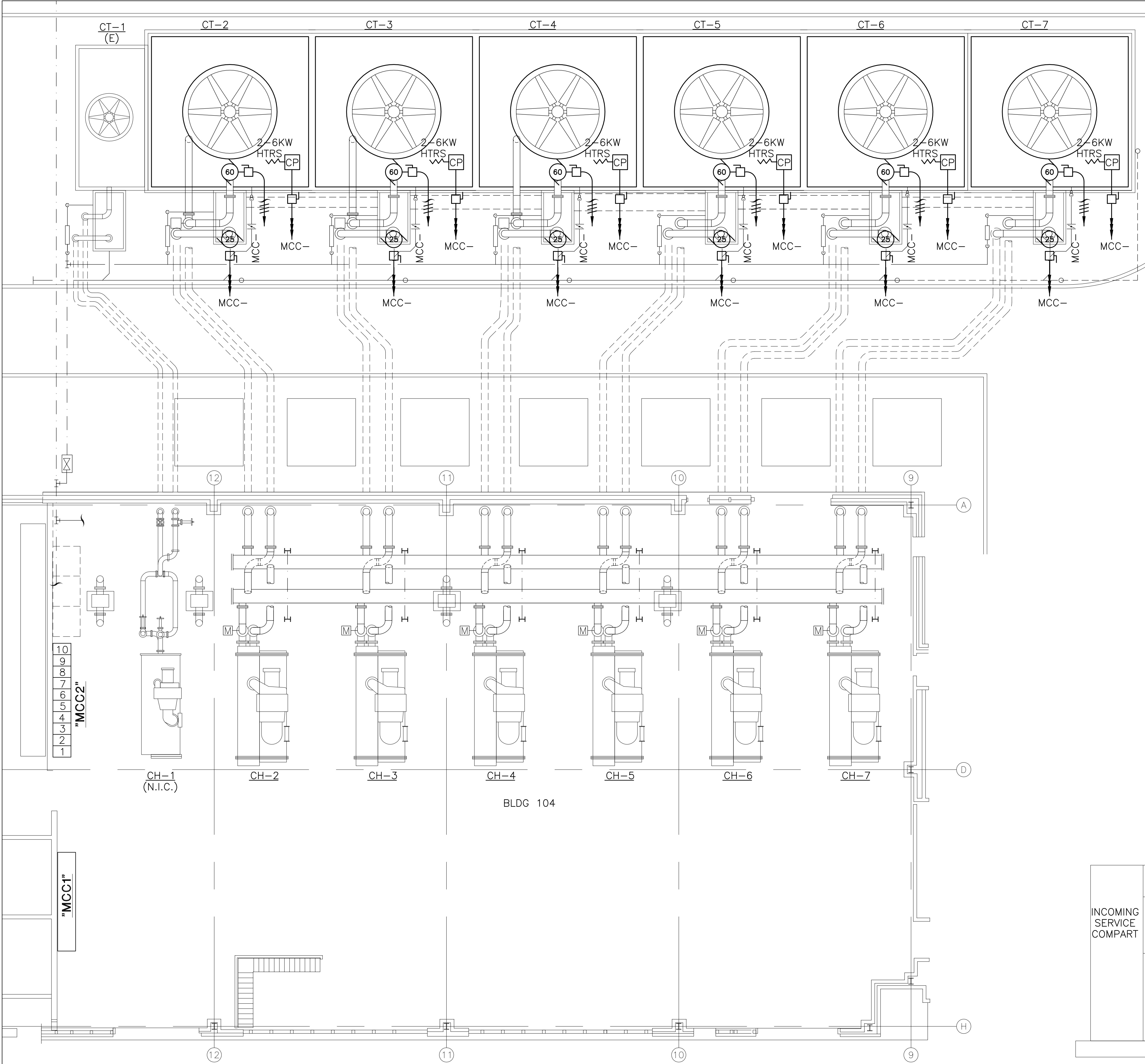
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Email P.Corbett@saturn.rr.com
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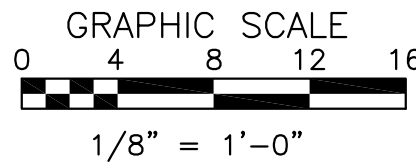


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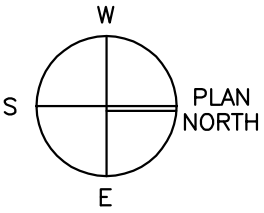


KEYED NOTES:
① .

1/E1.0



1 PARTIAL PLAN - BLDG. 104-CHILLER 2 THRU 7-ELECTRICAL
E1.0 SCALE: 1/8"= 1'-0"



A MCC2- ELEVATION
E1.0 SCALE: NONE

INCOMING SERVICE COMPART	1	4	7	10	13	16	19	25	31
	2	5	8	11	14	17	20	26	32
	3	6	9	12	15	18	22	28	33
							21	27	
							23	29	
							24	30	

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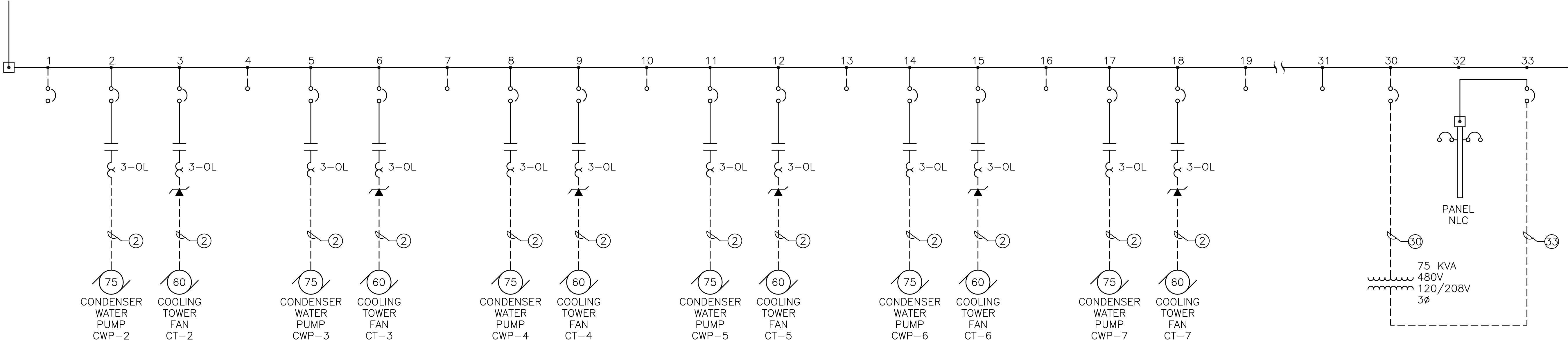
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VA MEDICAL CENTER
VAMC-104 COOLING TOWER REPLACEMENT
COOLING TOWER-ELECTRICAL

E1.0

MCC-2 CIRCUIT AND FEEDER SCHEDULE								
SPACE	CIRCUIT NO.	DESCRIPTION	HP	FLA	CIRCUIT BREAKER	STARTER SIZE	DRIVE	FEEDER CIRCUIT
1		SPACE						
2		CONDENSER PUMP - CWP-2						
3		TOWER FAN - CT-2						
4		SPACE						
5		CONDENSER PUMP - CWP-3						
6		TOWER FAN - CT-3						
7		SPACE						
8		CONDENSER PUMP - CWP-4						
9		TOWER FAN - CT-4						
10		SPACE						
11		CONDENSER PUMP - CWP-5						
12		TOWER FAN - CT-5						
13		SPACE						
14		CONDENSER PUMP - CWP-6						
15		TOWER FAN - CT-6						
16		SPACE						
17		CONDENSER PUMP - CWP-7						
18		TOWER FAN - CT-7						
19		SPACE						
20								
21								
22								
23								
24								
25		SPACE						
26								
27								
28								
29								
30		PRIMARY CIRCUIT BREAKER						
31		SPACE						
32		PANEL NLC						
33		SECONDARY CIRCUIT BREAKER						

PANEL:		NLC		VOLTAGE:		120/208 VOLT, 3PH, 4W, 60 Hz		100% NEUTRAL BUS							
LOCATION:		XXX		MAINS:		XXX AMP MLO		ISOLATED GND BUS							
BUILDING:		XXX				XXX AMP MAIN BREAKER		SHUNT TRIP							
DRAWING:		XXX		MOUNTING:		FLUSH SURFACE		MINIMUM AIC RATING: KA							
DEVICE		BRANCH CIRCUIT					VA LOAD				BRANCH CIRCUIT			DEVICE	
AMP	POLES	WIRE SIZE	DESIGNATION	VOLT-AMPS	NO.	A	B	C	NO.	VOLT-AMPS	DESIGNATION	WIRE SIZE	POLES	AMP	
					1				2						
					3				4						
					5				6						
					7				8						
					9				10						
					11				12						
					13				14						
					15				16						
					17				18						
					19				20						
					21				22						
					23				24						
					25				26						
					27				28						
					29				30						
					31				32						
					33				34						
					35				36						
					37				38						
					39				40						
					41				42						
TOTALS PER PHASE						0	0	0	TOTAL CONNECTED LOAD (KVA)			0.0			
									BALANCED LOAD IN AMPS PER PHASE			0.0			
PANEL:															
BUILDING:															
NOTES:															



A **MCC2, SINGLE LINE DIAGRAM**
E5.0 SCALE: NONE

BY

REVISIONS

NO

DATE

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VA MEDICAL CENTER
VAMC-104 COOLING TOWER REPLACEMENT

ELECTRICAL SINGLE LINE DIAGRAM

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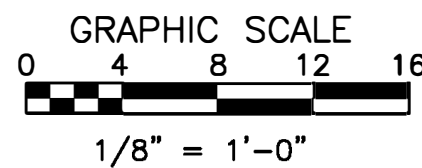
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- 1/M1.0

CCS: COMPOSITE COOLING SOLUTIONS

DATA BASED ON PUMP CURVE: JTS-16HMC-EC0247WP Rev A

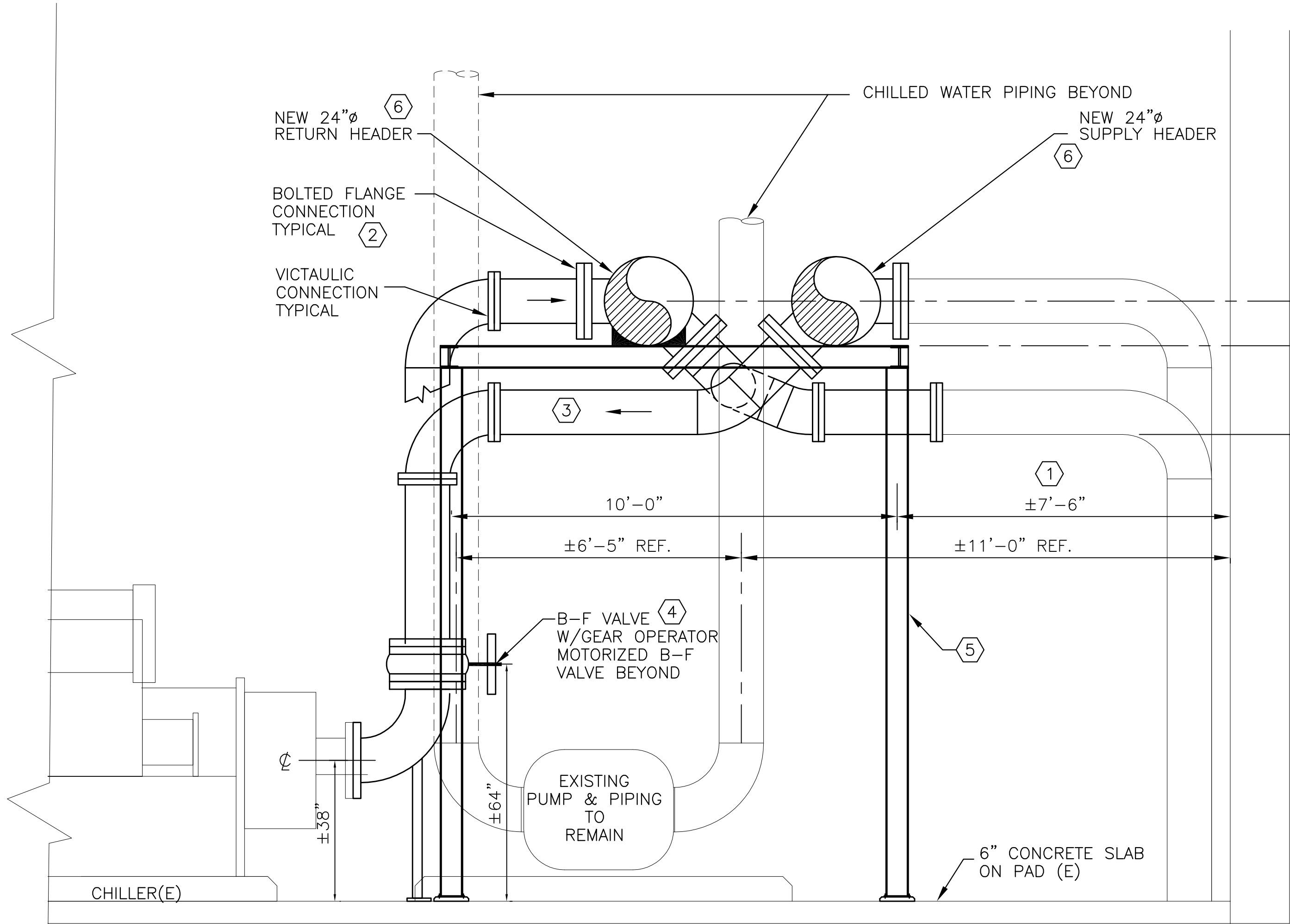
VAMC-104 COOLING TOWER REPLACEMENT

PARTIAL PLAN-MECHANICAL PIPING

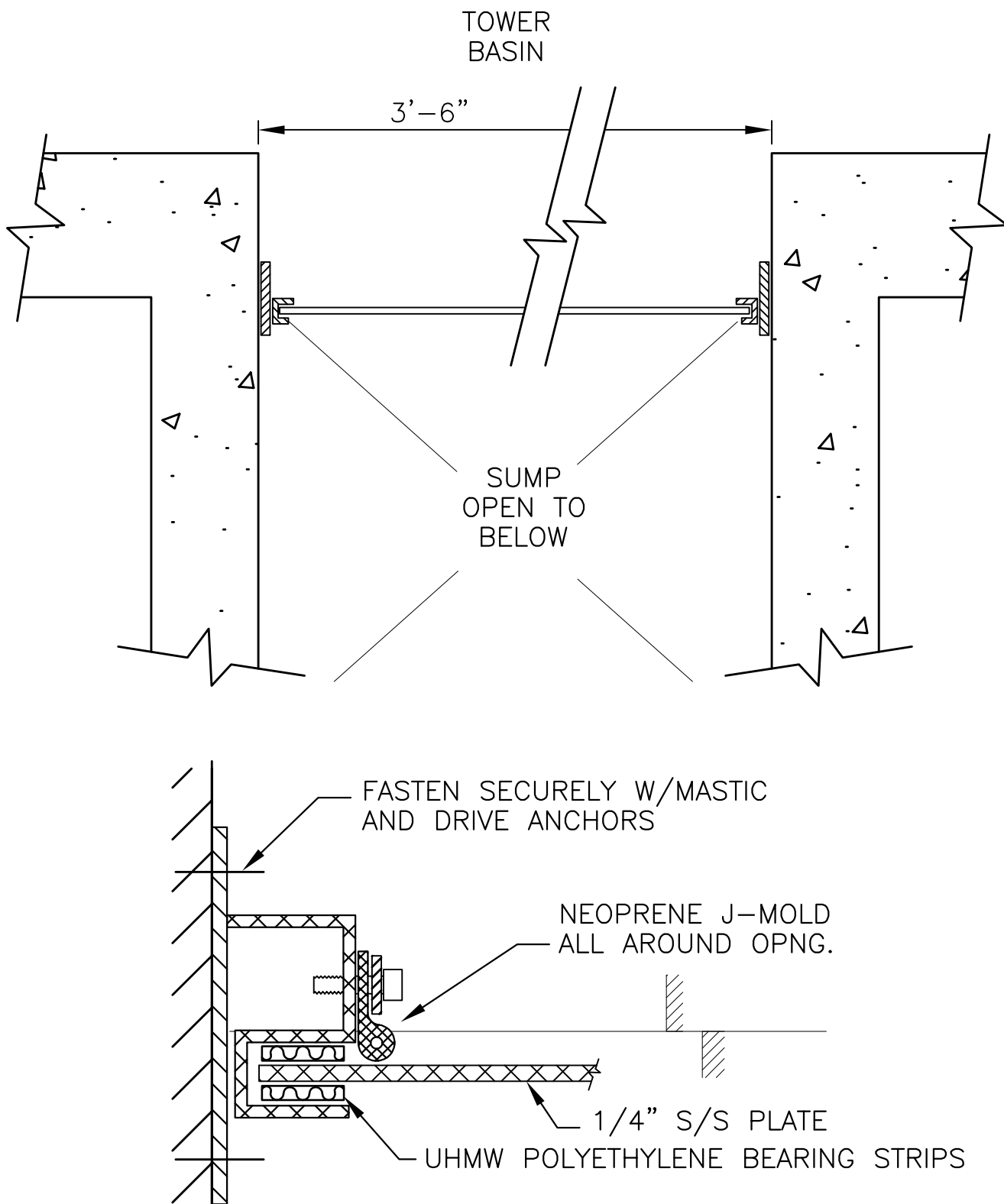
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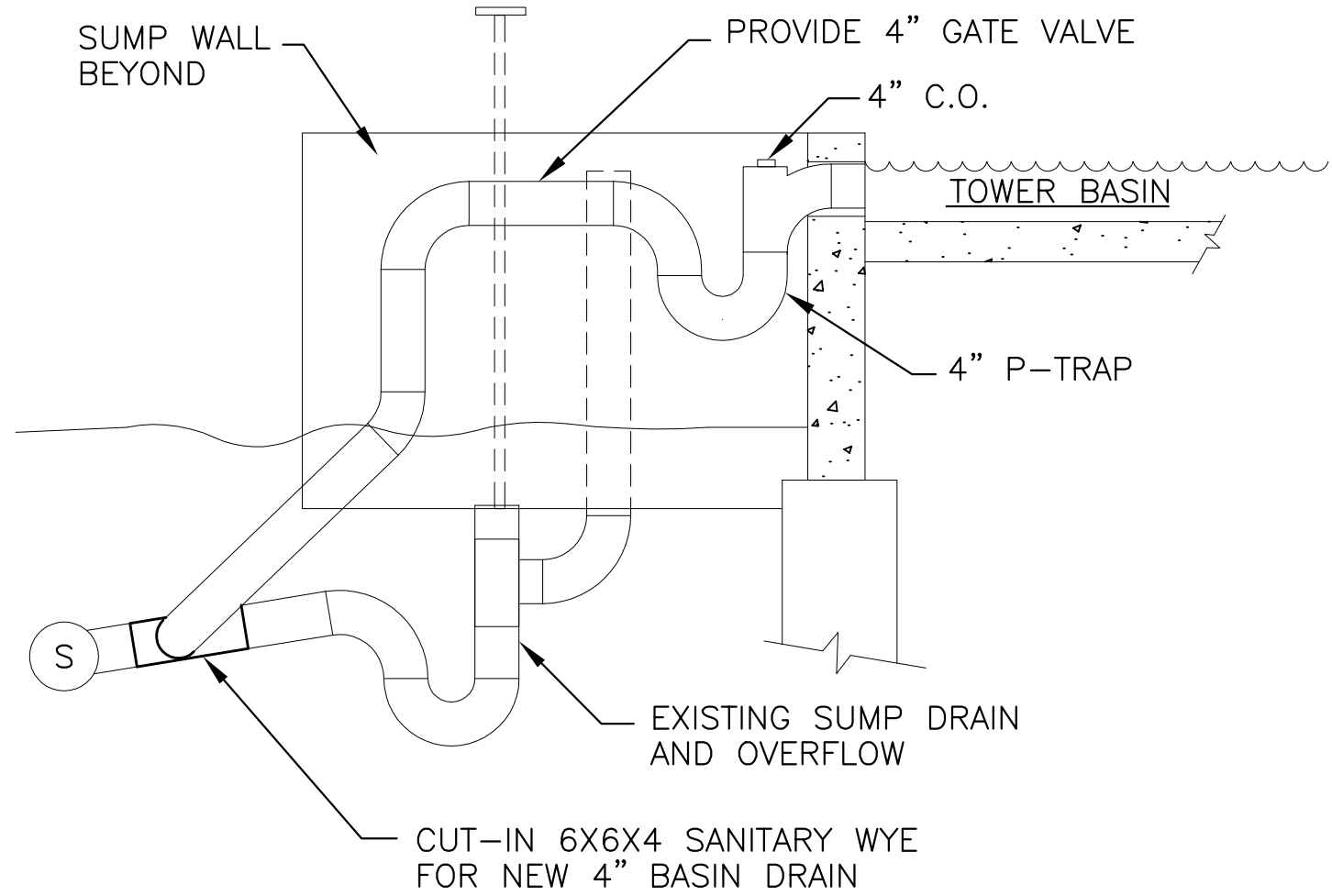
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1 SECTION-VAMC-104 CHILLER ROOM CONDENSER WATER PIPING
M3.0 SCALE: 1/2" = 1'-0"



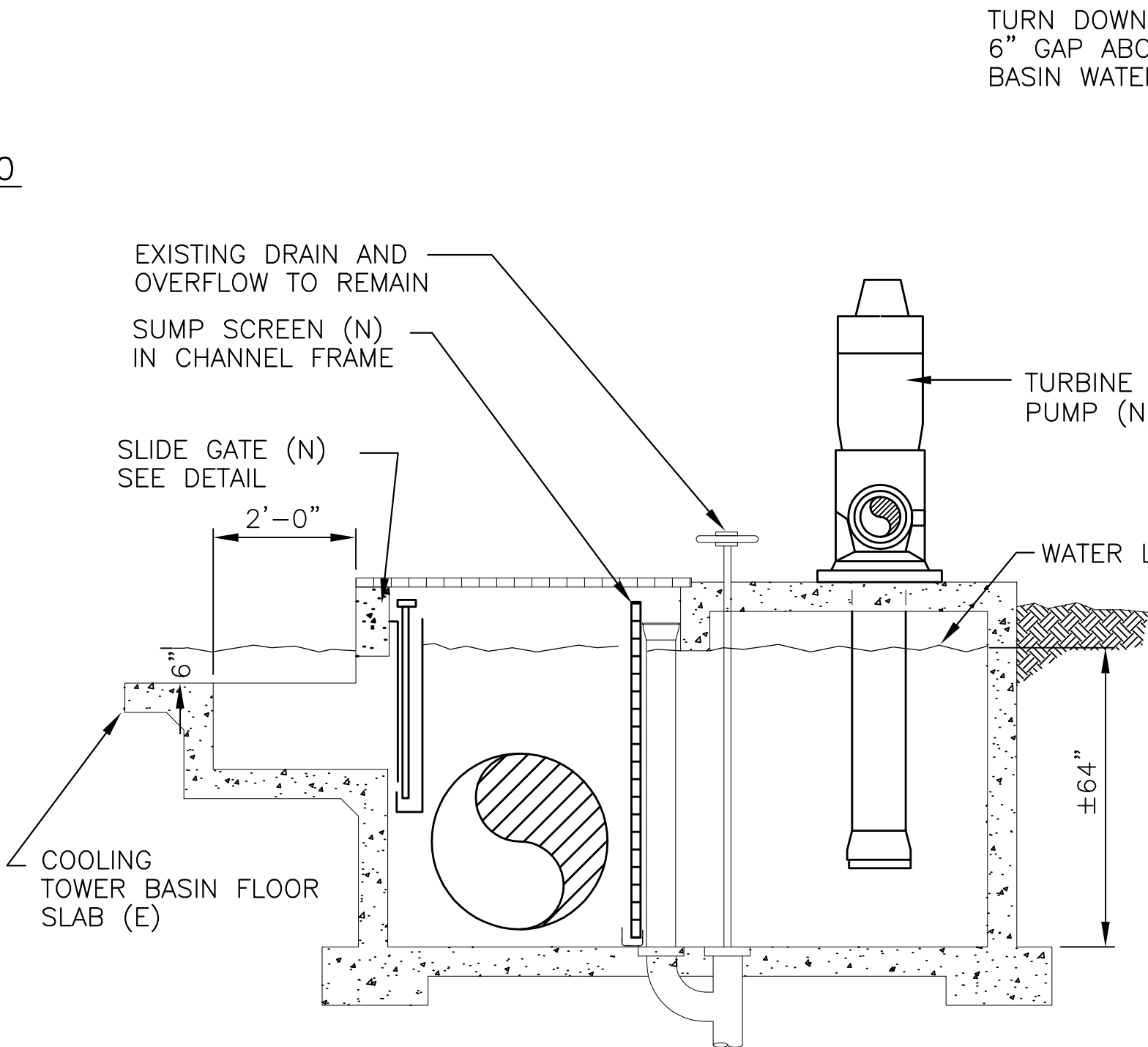
A SECTION - TOWER BASIN
M3.0 SCALE: 1" = 1'-0"



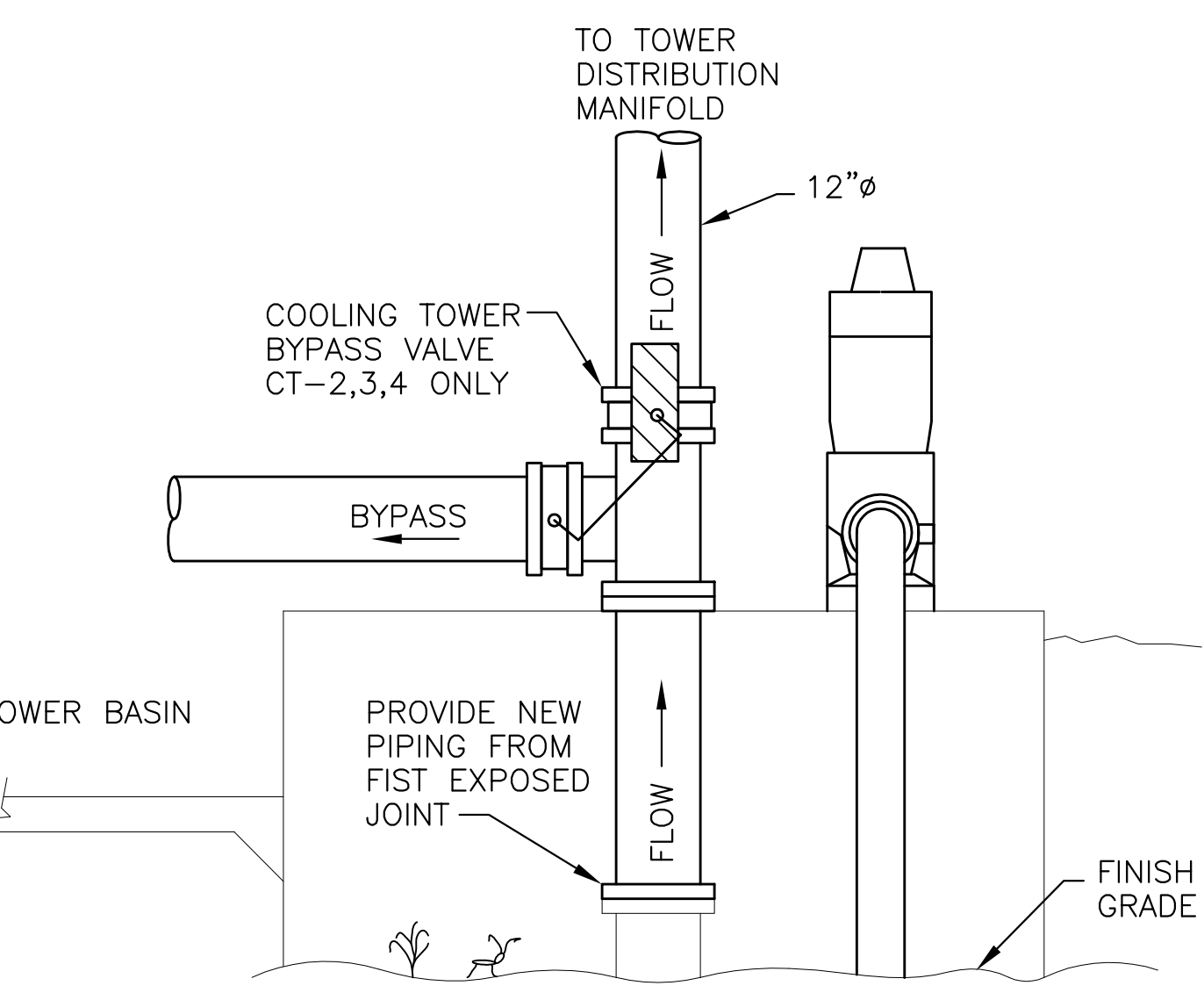
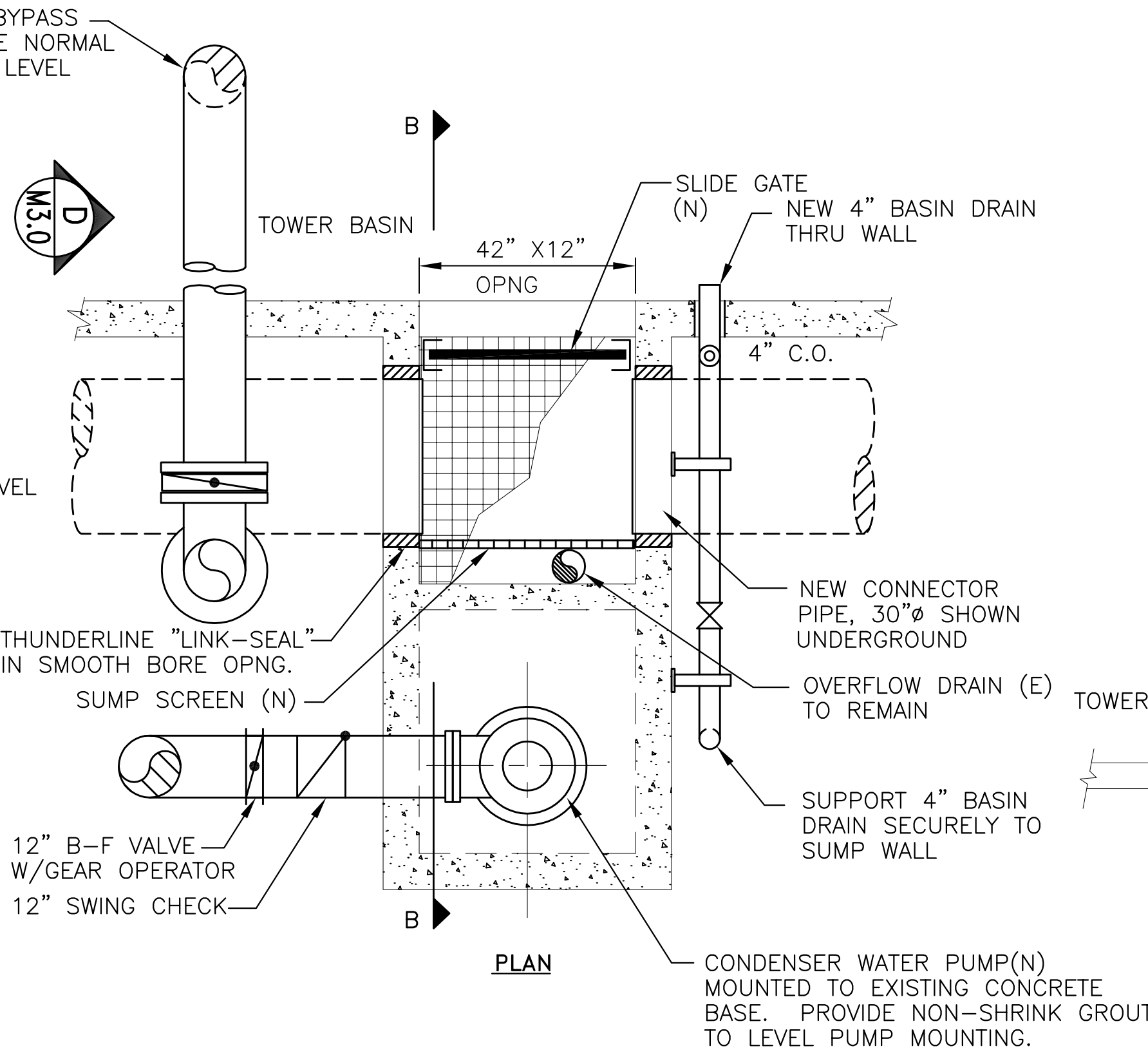
B DETAIL-4" BASIN DRAIN
M3.0 SCALE: NONE

KEYED NOTES: 1-M3.0

- 1 DIMENSIONS SHOWN SHALL BE VERIFIED BY THE CONTRACTOR. ANY DIMENSION REQUIRING ADJUSTMENT OF MORE THAN 6 INCHES, SHOULD BE CLEARED BY THE ENGINEER PRIOR TO CONSTRUCTION.
- 2 PROVIDE SCHEDULE 40 STEEL, ANSI 150# CONNECTIONS AS INDICATED.
- 3 PROVIDE NEW PIPING TO CHILLER USING VICTAULIC COUPLINGS. PROVIDE PIPE STAND AS INDICATED TO RELIEVE STRESS AT CHILLER CONNECTIONS.
- 4 PROVIDE FLANGED CONNECTIONS FOR B-F VALVE.
- 5 STRUCTURAL STEEL FRAME. REFER TO S1.0 FOR DETAILS OF CONSTRUCTION. PAINT COLUMN FIRST 8 FEET SAFETY YELLOW.
- 6 PROVIDE SUITABLE SADDLE TO SUPPORT CONDENSER WATER PIPING ON BEAM.

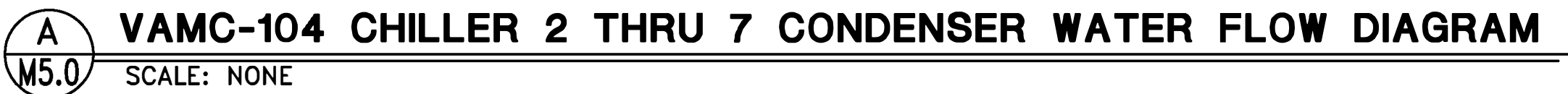


C DETAILS-COOLING TOWER SUMP PIPING
M3.0 SCALE: NONE



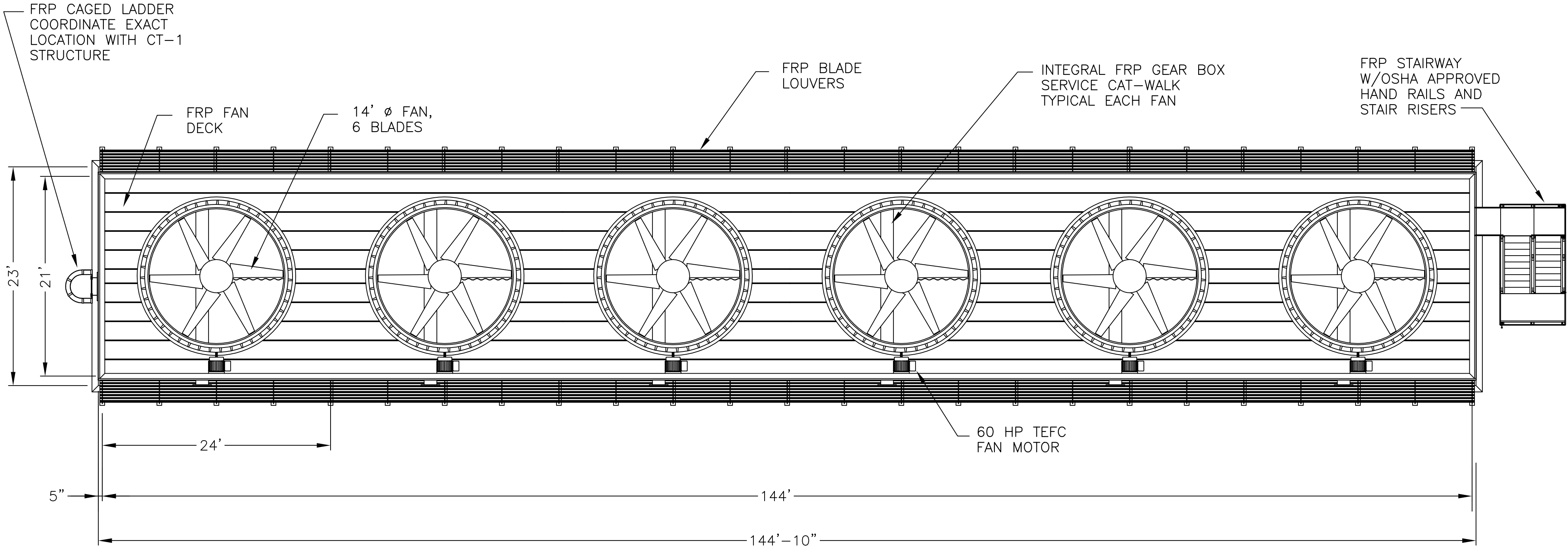
D VIEW-COOLING TOWER SUMP PIPING
M3.0 SCALE: NONE

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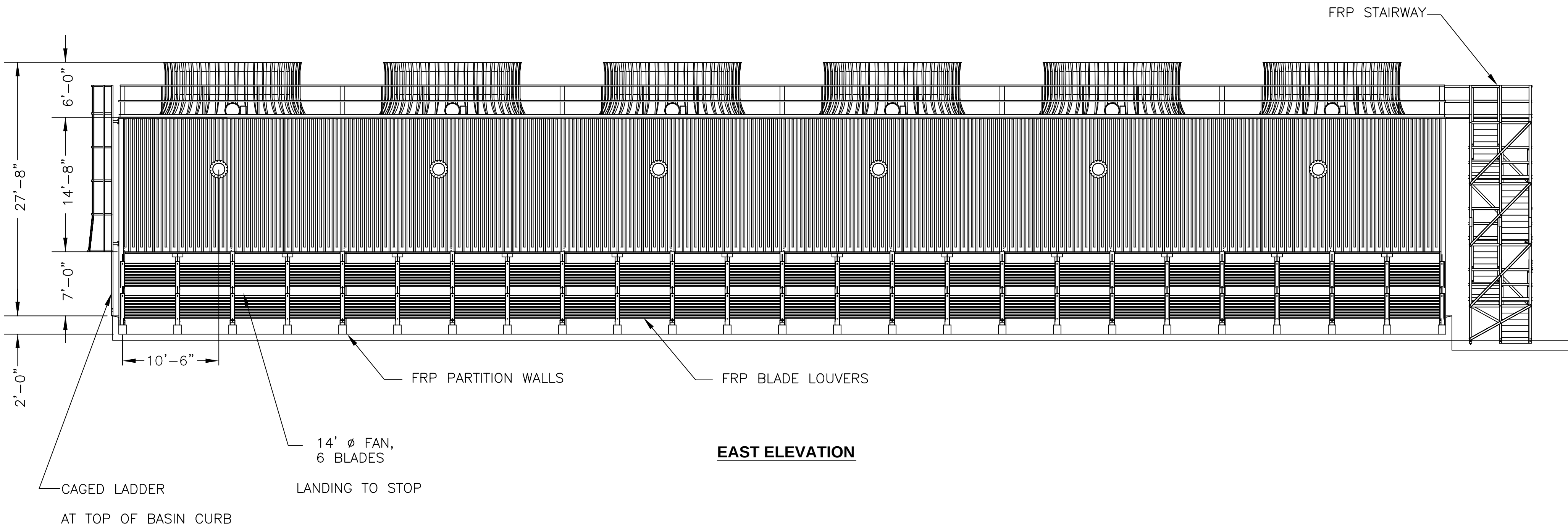


- ① NEW CONDENSER WATER SUPPLY AND RETURN HEADER CONNECTING SIX CHILLERS TO SIX COOLING TOWERS. PROVIDE 24" BUTTERFLY VALVE FOR PHASED CONSTRUCTION.
- ② PROVIDE PNEUMATIC ACTUATOR FOR EXISTING BUTTERFLY ISOLATION VALVE. PROVIDE BRAY SERIES 90, DOUBLE ACTING ACTUATOR WITH SERIES 60 INTEGRAL SOLENOID VALVE AND SERIES 55 SPEED CONTROL.
- ③ PROVIDE BUTTERFLY VALVE W/PNEUMATIC ACTUATOR, BRAY 12" SERIES 30, CAST IRON ASTM A126 CLASS B, BUNA-N SEAT, 304 S/S STEM , COATED DUCTILE IRON DISC. PROVIDE BRAY SERIES 90, DOUBLE ACTING ACTUATOR WITH SERIES 60 INTEGRAL SOLENOID VALVE AND SERIES 55 SPEED CONTROL.
- ④ PROVIDE 3-WAY BUTTERFLY VALVE IM BYPASS PIPING ARRANGEMENT, ____ SIZE, WITH PNEUMATIC ACTUATOR. CAST IRON TEE WITH SERIES 30, SAST IRON ASTM A126 CLASS B, BUNA-N SEAT, 304 S/S STEM , COATED DUCTILE IRON DISC. PROVIDE BRAY SERIES 90, DOUBLE ACTING ACTUATOR WITH SERIES 60 INTEGRAL SOLENOID VALVE AND SERIES 55 SPEED CONTROL AND SERIES 64 PNEUMATIC POSITIONER.
- ⑤ PROVIDE 1" WELDOLET WITH BALL VALVE FOR CHEMICAL FEED LOOP. PIPING, VALVES AND FITTINGS TO BE CPVC CONSTRUCTION. PROVIDE LOCKING QUADRANT ON BALL VALVE FOR FLOW BALANCING.
- ⑥ PROVIDE NEAR CHEMICAL PUMPS, THREE 3/4" FULL PORT BALL VALVES TO BE USED BY V.A. CHEMICAL TREATMENT PERSONNEL FOR CONNECTIONS OF OWNER PROVIDED CHEMICAL TREATMENT EQUIPMENT.
- ⑦ CHEMICAL TREATMENT LOOP CIRCULATING PUMP. GRUNDFOS UPS 32-40, 20 GPM, AT 8 FT-HD, 460V, 3ø.

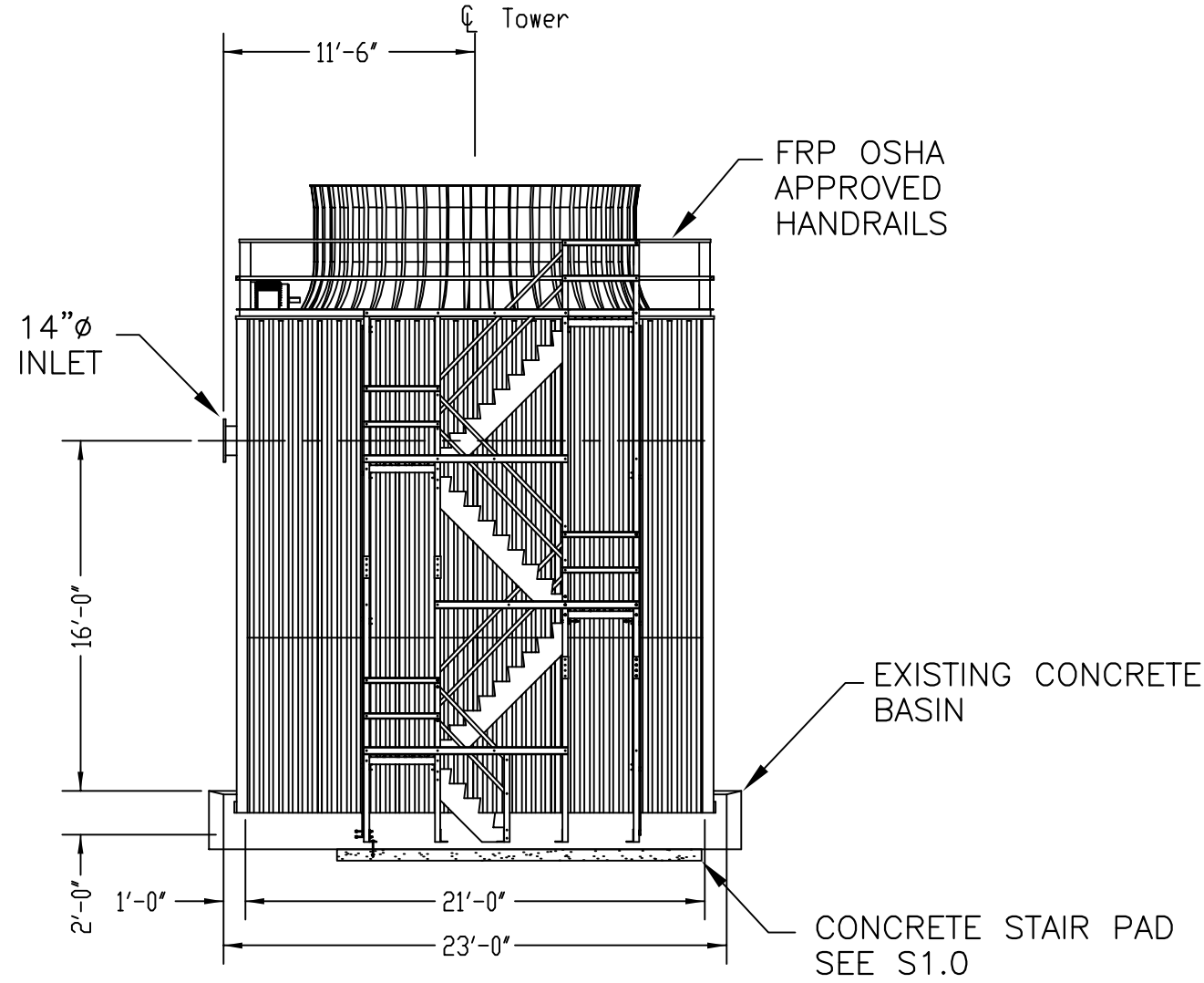
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PLAN VIEW



EAST ELEVATION



END VIEW

A
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VAMC-104 CHILLER 2 THRU 7 COOLING TOWER DETAILS

SCALE: NONE

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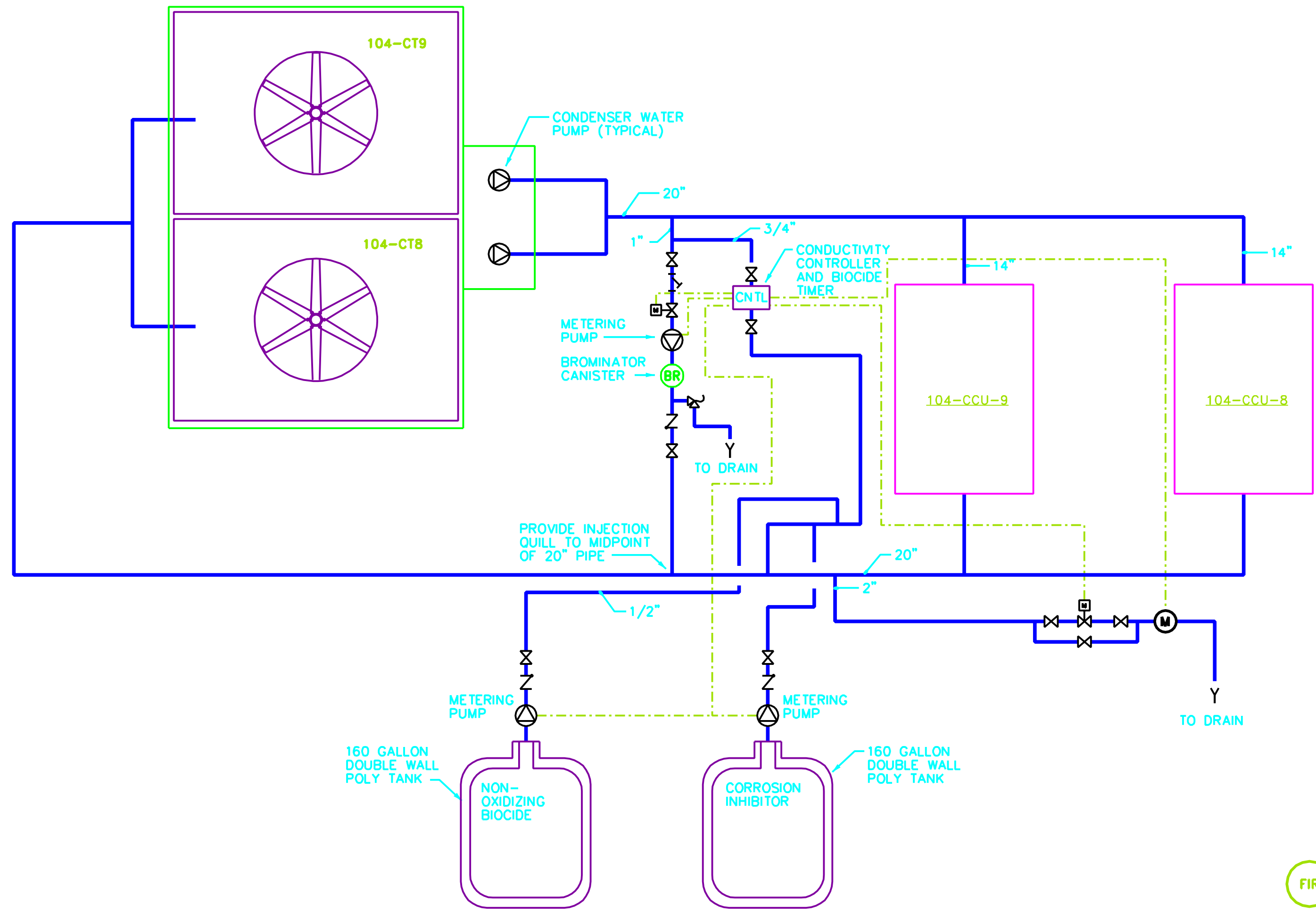
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COOLING TOWER DETAILS

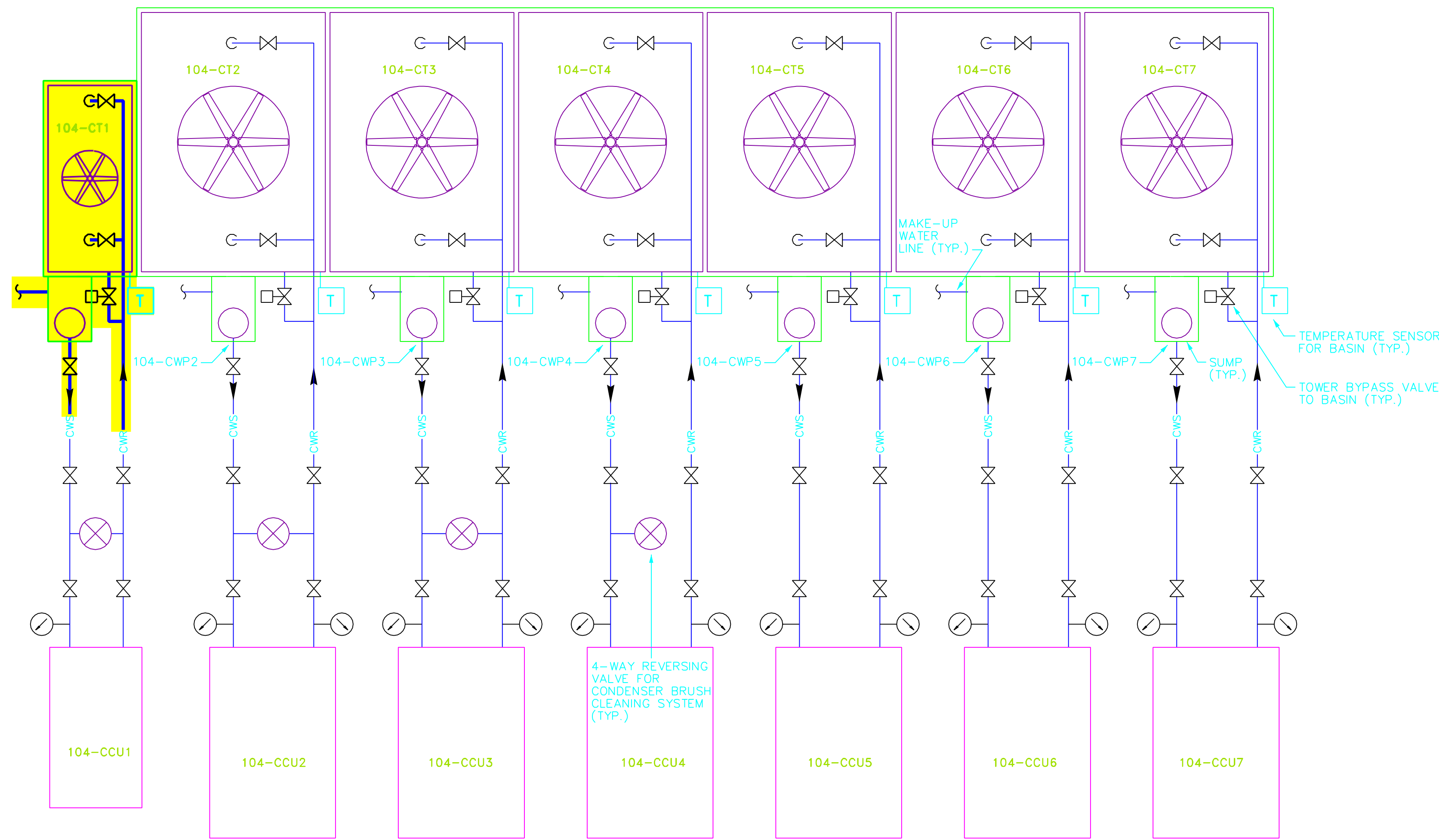
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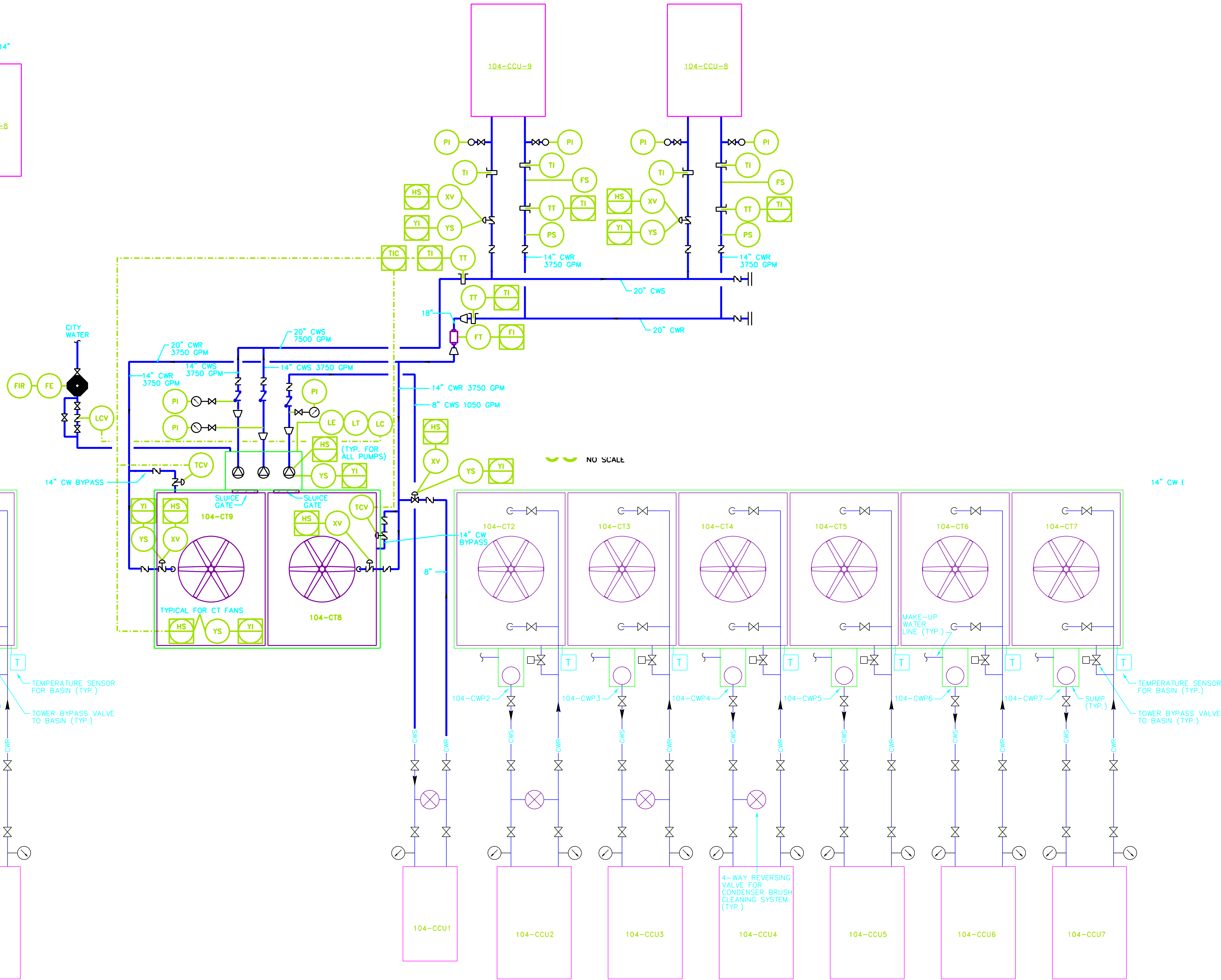
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03 CONDENSER WATER TREATMENT SCHEMATIC
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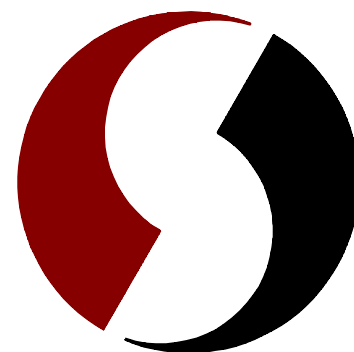
01 CONDENSER WATER FLOW DIAGRAM - DEMOLITION
NO SCALE



02 CONDENSER WATER FLOW DIAGRAM - RENOVATION
NO SCALE

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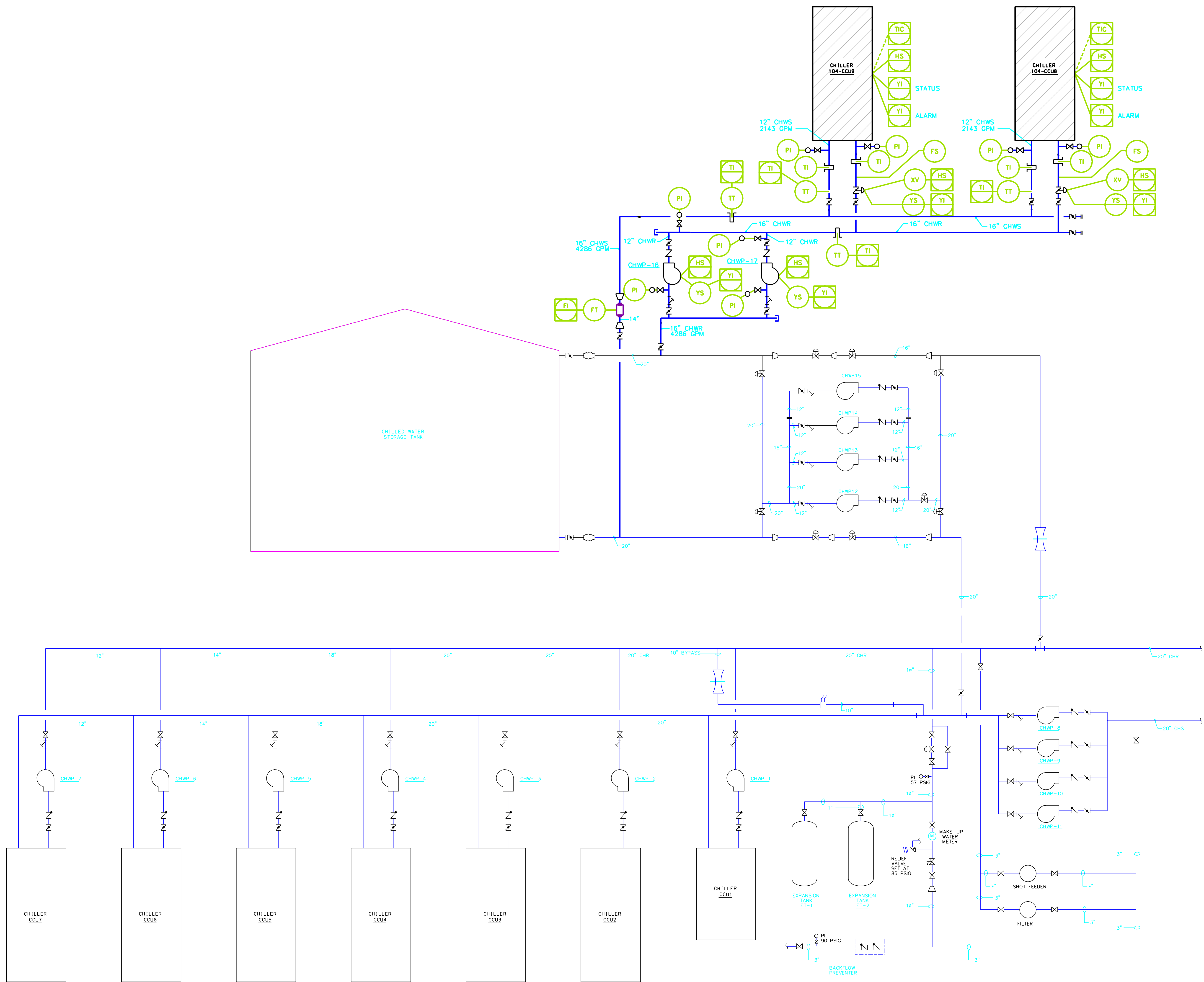
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Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	CONDENSER WATER FLOW DIAGRAM		
Approved: Engineering Svc. Chief			
Approved: Medical Center Director			

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION		
Building Number	104/106	Checked	ASS
Location	VAMC HOUSTON, TX		

Date	02/25/2003
Project No.	VA #560-58-110 SSA #1017-007-1
Drawing No.	104-106-1
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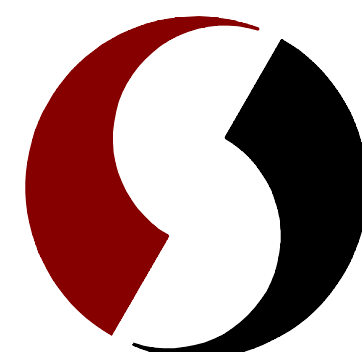
Veterans
Administration



01 CHILLED WATER FLOW DIAGRAM
NOT TO SCALE

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Revisions	Date



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HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
CHILLED WATER FLOW DIAGRAM
Approved: Engineering Svc. Chief
Approved: Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

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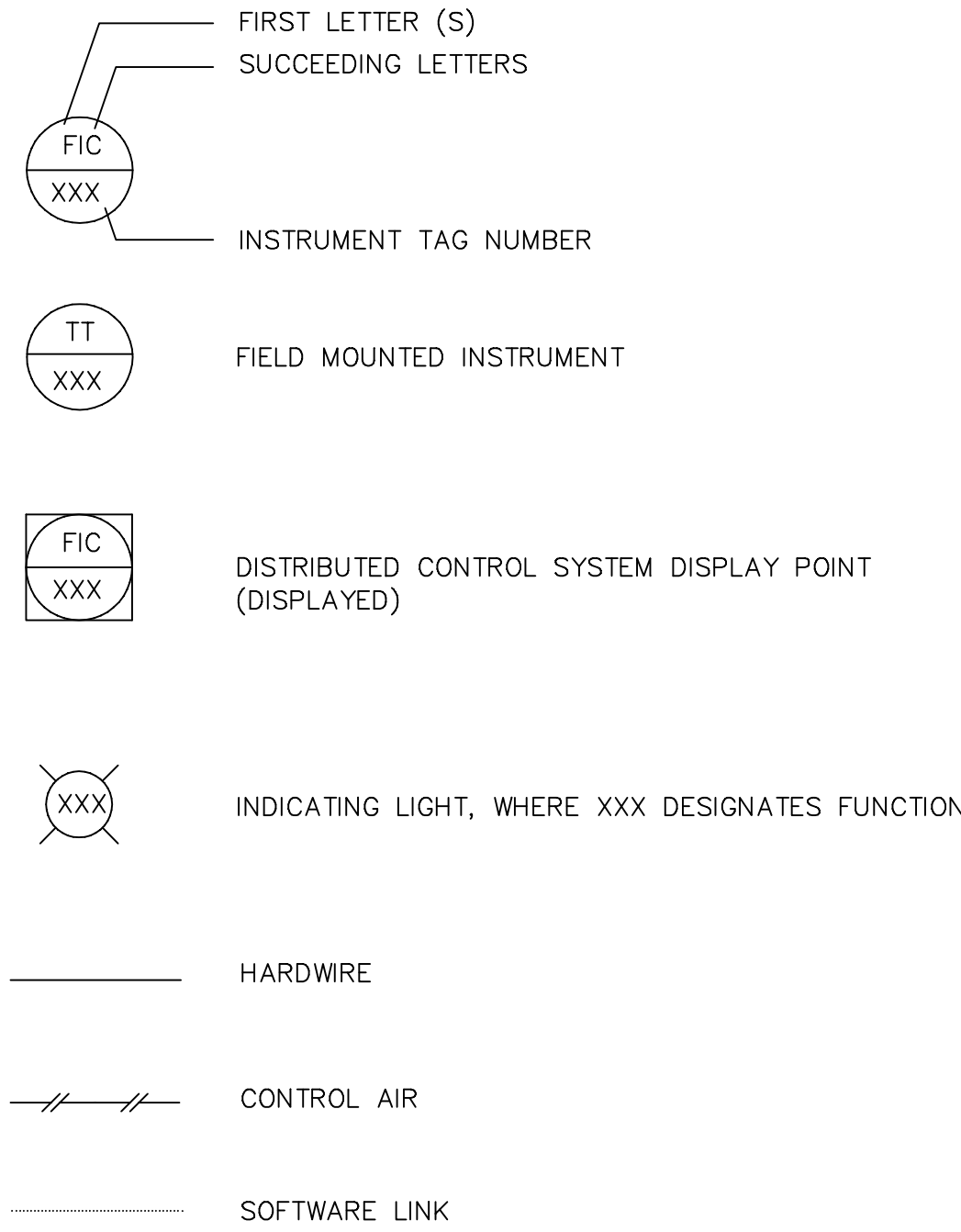
Veterans
Administration

LEGEND								
SYMBOL	ABBREV.	DESCRIPTION	SYMBOL	ABBREV.	DESCRIPTION	SYMBOL	ABBREV.	DESCRIPTION
		SUPPLY DUCT		CHS	CHILLED WATER SUPPLY			SOLENOID VALVE
		O.A. RETURN OR EXH.DUCT		CHR	CHILLED WATER RETURN			BALL VALVE
	VD	VOLUME DAMPER		CWS	CONDENSING WATER SUPPLY			FIRESTAT
	FD or SD	FIRE OR SMOKE DAMPER		CWR	CONDENSING WATER RETURN		HG	HOT GAS
		THERMOSTAT		HWS	HOT WATER SUPPLY		L	REFRIGERANT LIQUID
		HUMIDISTAT		HWR	HOT WATER RETURN		S	REFRIGERANT SUCTION
		SAN.WASTE OR VENT STACK		#S	# OF STEAM SUPPLY			GATE VALVE
		WASTE OR VENT NO.		#R	# OF STEAM RETURN			GLOBE VALVE
	VTR	VENT THRU ROOF		A	COMPRESSED AIR			CHECK VALVE
	FCO	FLOOR CLEANOUT			THERMOMETER WELL			VALVE BOX
	CW	DOM. COLD WATER			THERMOMETER			GAGE COCK
	HW	DOM. HOT WATER SUPPLY			UNION			BUTTERFLY VALVE
		DOM. HOT WATER RETURN			STRAINER			PLUG VALVE
		VENT			REDUCER			TWO-WAY CONTROL VALVE
	SD	STORM DRAIN			GAGE			THREE-WAY CONTROL VALVE
	G	GAS			FLEXIBLE JOINT		PRV	PRESSURE REDUCING VALVE
	F	FIRE			ANCHOR		RV	PRESSURE RELIEF VALVE
	FS	FIRE SPRINKLER			VENTURI FLOW ELEMENT			EXISTING ITEMS TO REMAIN
	ACD	AUXILIARY CONDENSATE DRAIN		MVD	MOTORIZED VOLUME DAMPER			NEW ITEMS
		DIFF; GR. OR REG. MARK			TEMPERATURE SENSOR			EXISTING ITEMS TO BE REMOVED
		CFM			ANNUBAR			

PUMP SCHEDULE									
MARK	SERVICE	TYPE	SIZE		FLOW RATE GPM	TOTAL HEAD FT H ₂ O	HP/VOLTS/PH/HZ	RPM	REMARKS
			SUCT.	DISCH.					
CHWP-12, 13, 14 & 15	CHILLED WTR STOR.	HORIZ. SPLIT	8	10	2840	130	125/460/3/60	1750	AURORA SERIES 410 8X10X15 B OR EQUAL

GENERAL NOTES
THESE GENERAL NOTES APPLY TO ALL MECHANICAL DRAWINGS.
IN ANY CASE WHERE A PIPE SHOWN ON A PLAN SHEET DIFFERS FROM THAT SHOWN IN A SCHEMATIC OR DETAIL, USE THE LARGER OF THE TWO SIZES SHOWN.
ALL ELEVATIONS INDICATED IN THIS WAY (8'-0") ARE THE ELEVATIONS FROM THE FINISHED FLOOR DIRECTLY BELOW TO THE BOTTOM OF THE BARE PIPE.
SEAL OPENINGS AROUND PIPE PENETRATIONS IN EXTERIOR WALLS WITH WEATHER PROOFING MATERIAL.
PIPE HANGER LOCATIONS MAY BE SHOWN ON CERTAIN DRAWINGS, BUT NOT ALL THE HANGER LOCATIONS ARE SHOWN. PROVIDE HANGERS AT MAXIMUM SPANS AS REQUIRED BY SPECIFICATIONS, WHERE NOT INDICATED ON DRAWINGS.
LEGENDS ON THIS SHEET ARE STANDARD LEGENDS AND NOT ALL THE INFORMATION APPLIES TO THIS PROJECT.

INSTRUMENT IDENTIFICATION
EXAMPLE SYMBOLS

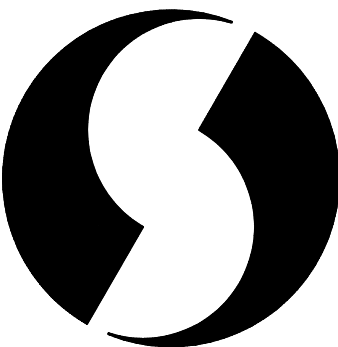


INSTRUMENT IDENTIFICATION LETTERS

FIRST LETTERS		SUCCEEDING LETTERS		
MEASURING OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A ANALYSIS (+)		ALARM		
B BLOCK (ON/OFF)				
C			CONTROL	
D	DIFFERENTIAL			
E VOLTAGE		SENSOR (PRIMARY ELEMENT)		
F FLOW RATE	RATIO (FRACTION)			
G	GAUGE	GLASS, VIEWING DEVICE		
H HAND				HIGH
I CURRENT (ELECTRICAL)		INDICATE		
J POWER	SCAN			
K TIME, TIME SCHEDULE	TIME RATE OF CHANGE		CONTROL STATION	
L LEVEL		LIGHT		LOW
M	MOMENTARY MOTION			MIDDLE, INTERMEDIATE
N SURGE				
O		ORIFICE, RESTRICTION		
P PRESSURE, VACUUM		POINT (TEST) CONNECTION		
Q QUANTITY, EVENT	INTEGRATE, TOTALIZE			
R RADIATION		RECORD		
S SPEED, SOLENOID, FREQUENCY	SAFETY		SWITCH	
T TEMPERATURE			TRANSMIT	
U MULTIVARIABLE		MULTI-FUNCTION	MULTI-FUNCTION	MULTI-FUNCTION
V VIBRATION, MECHANICAL ANALYSIS			VALVE, DAMPER, LOUVER	
W WEIGHT, FORCE		WELL		
X TORQUE	X AXIS			
Y EVENT, STATE OR PRESENCE	Y AXIS		RELAY, COMPUTE, CONVERT	
Z POSITION, DIMENSION	Z AXIS		DRIVER, ACTUATOR, UNCLASSIFIED FINAL CONTROL ELEMENT	

(+) WHEN USED, EXPLANATION IS SHOWN ADJACENT TO INSTRUMENT SYMBOL. SEE ABBREVIATIONS AND LETTER SYMBOL

Revisions	Date



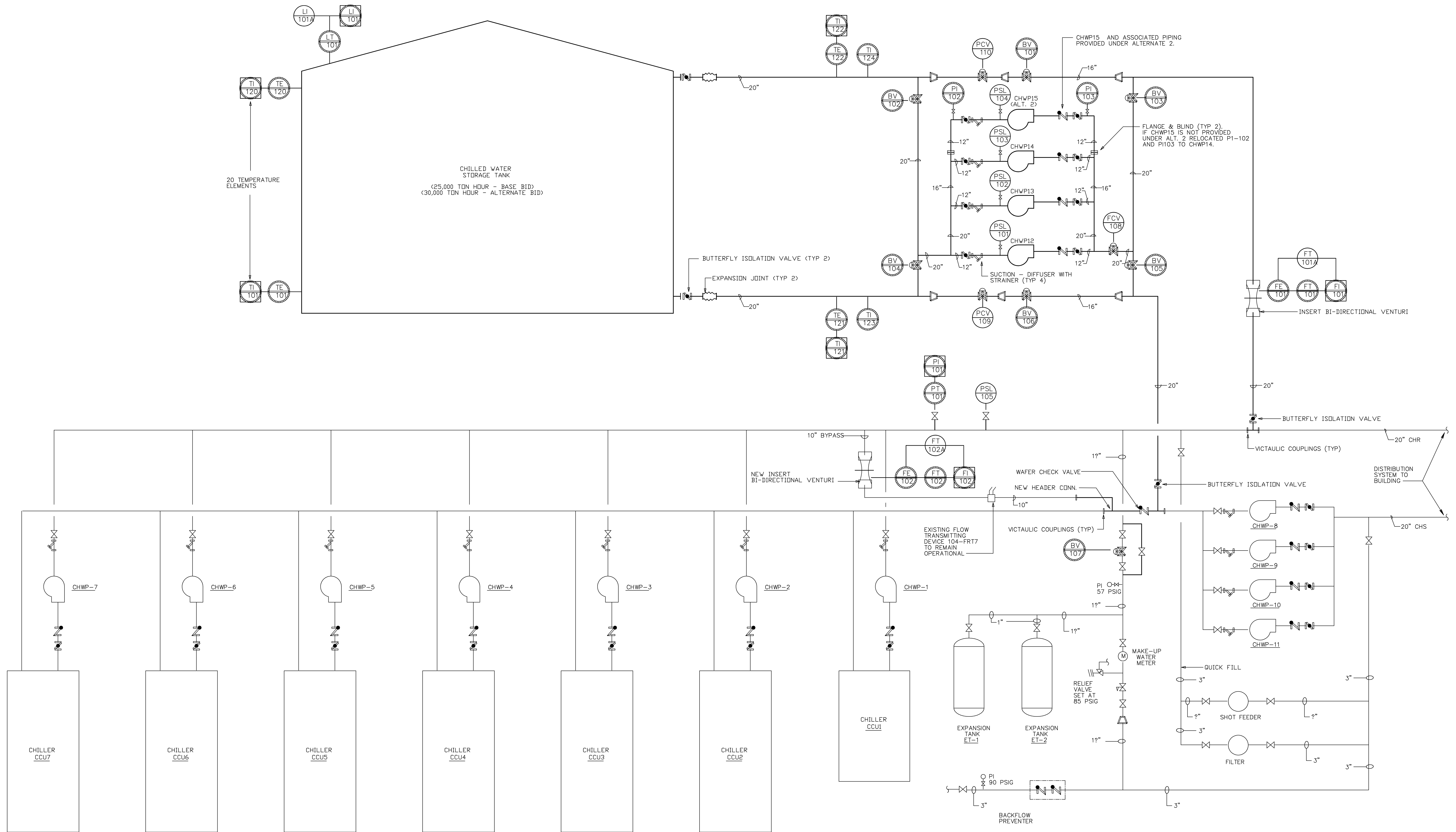
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 420
HOUSTON, TEXAS 77042

Drawing Title
MECHANICAL SCHEDULES AND LEGENDS
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title		
HOUSTON VETERANS AFFAIRS MEDICAL CENTER INSTALL THERMAL STORAGE UNIT		
Building Number	Checked	Drawn
104/105	JDS	JRA
Location		
VAMC HOUSTON, TX		

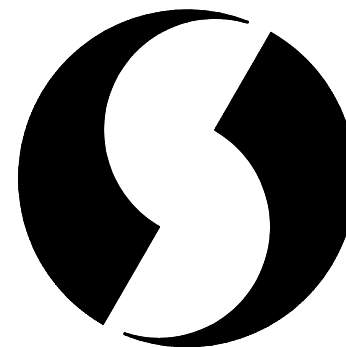
Date
10/95
Project No.
VA #V580P-5840 SSA #1017-001-2
Drawing No.
M1
Dwg. of

Veterans
Administration



1 CHILLED WATER FLOW DIAGRAM
NOT TO SCALE

Revisions	Date



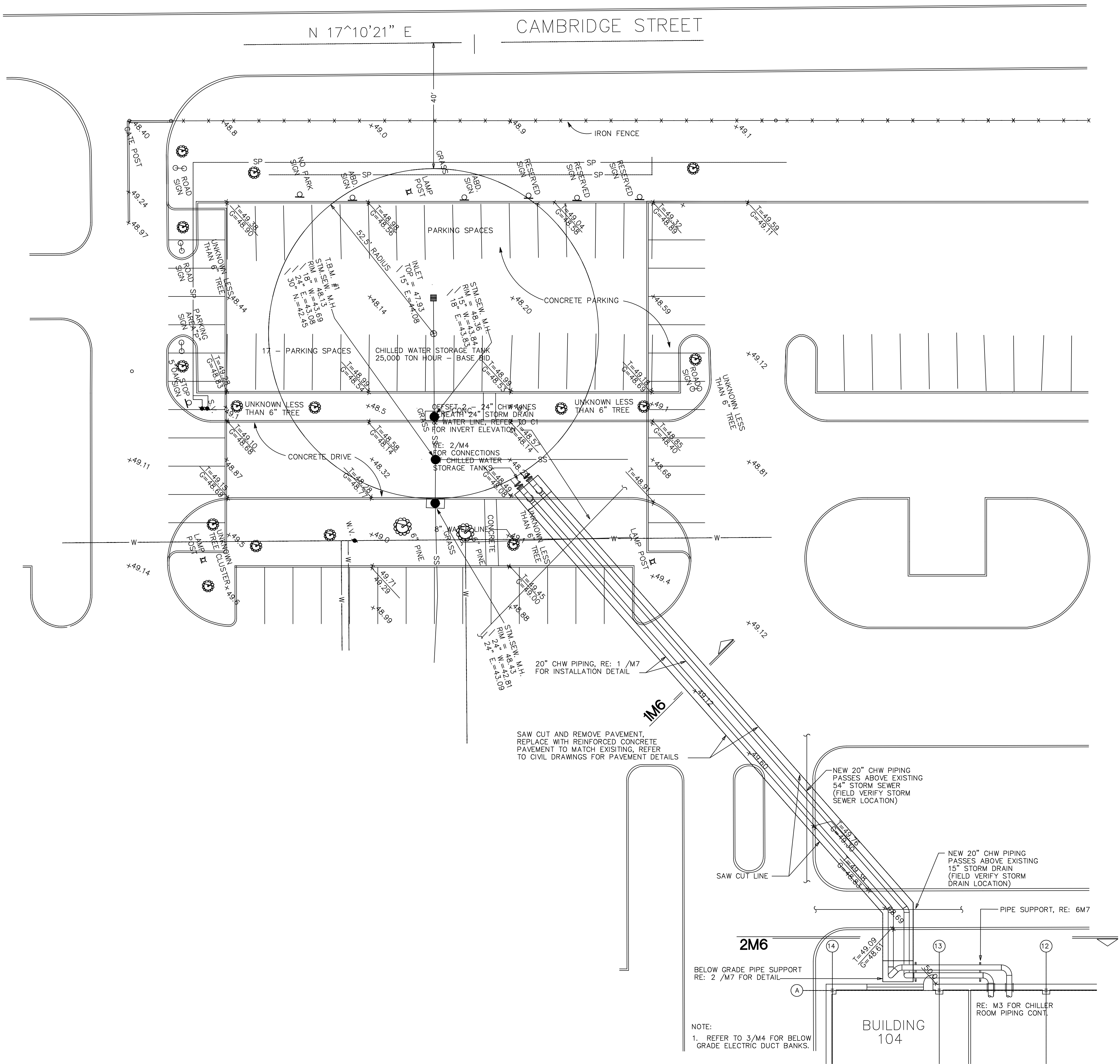
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 420
HOUSTON, TEXAS 77042

Drawing Title
CHILLED WATER FLOW DIAGRAM
Approved: Engineering Svc Chief
Approved: Medical Center Director

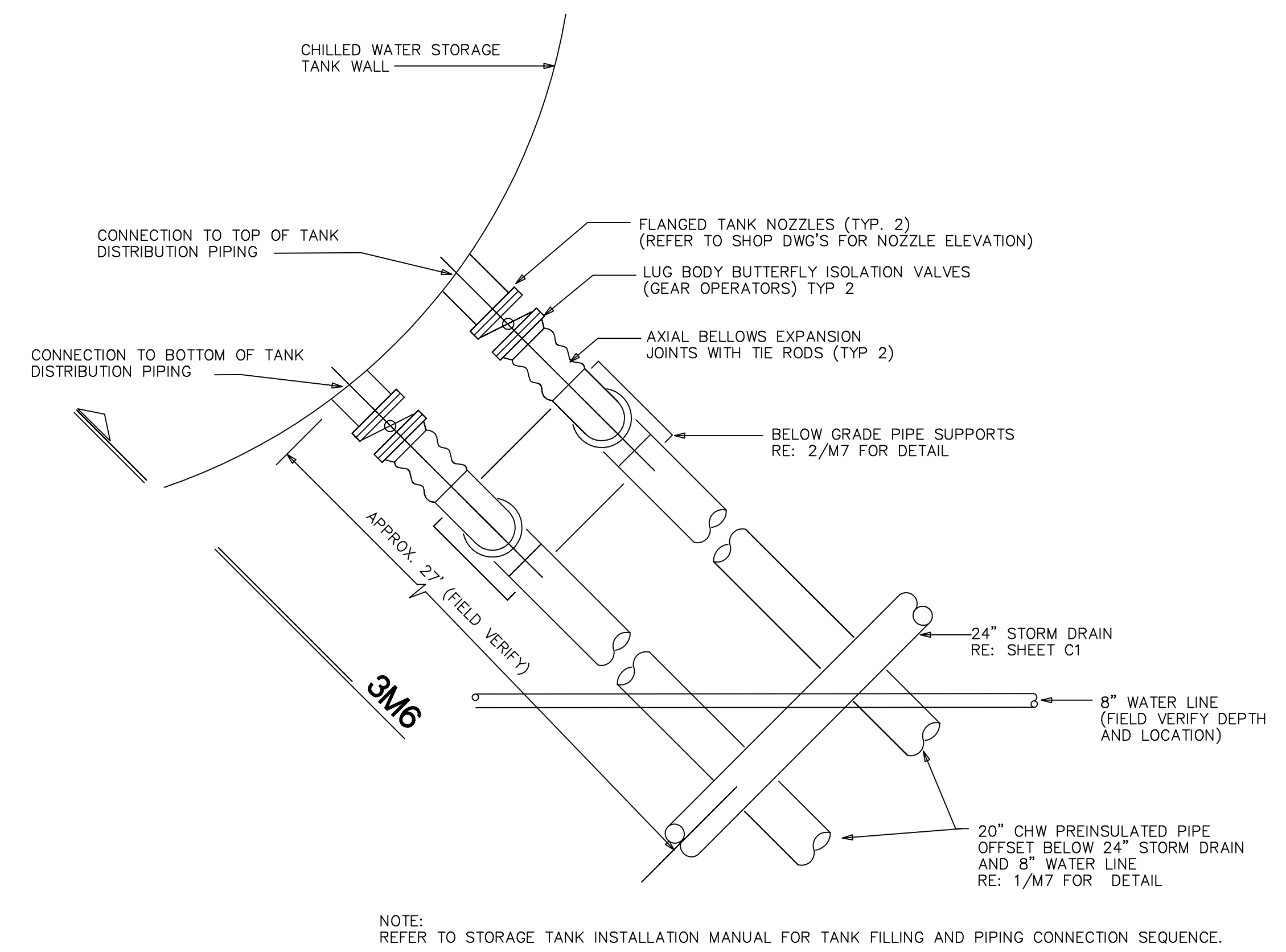
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Building Number
104/105
Location
VAMC HOUSTON, TX

Date
10/95
Project No.
VA #V580P-5840 SSA #1017-001-2
Drawing No.
M2
Dwg. of

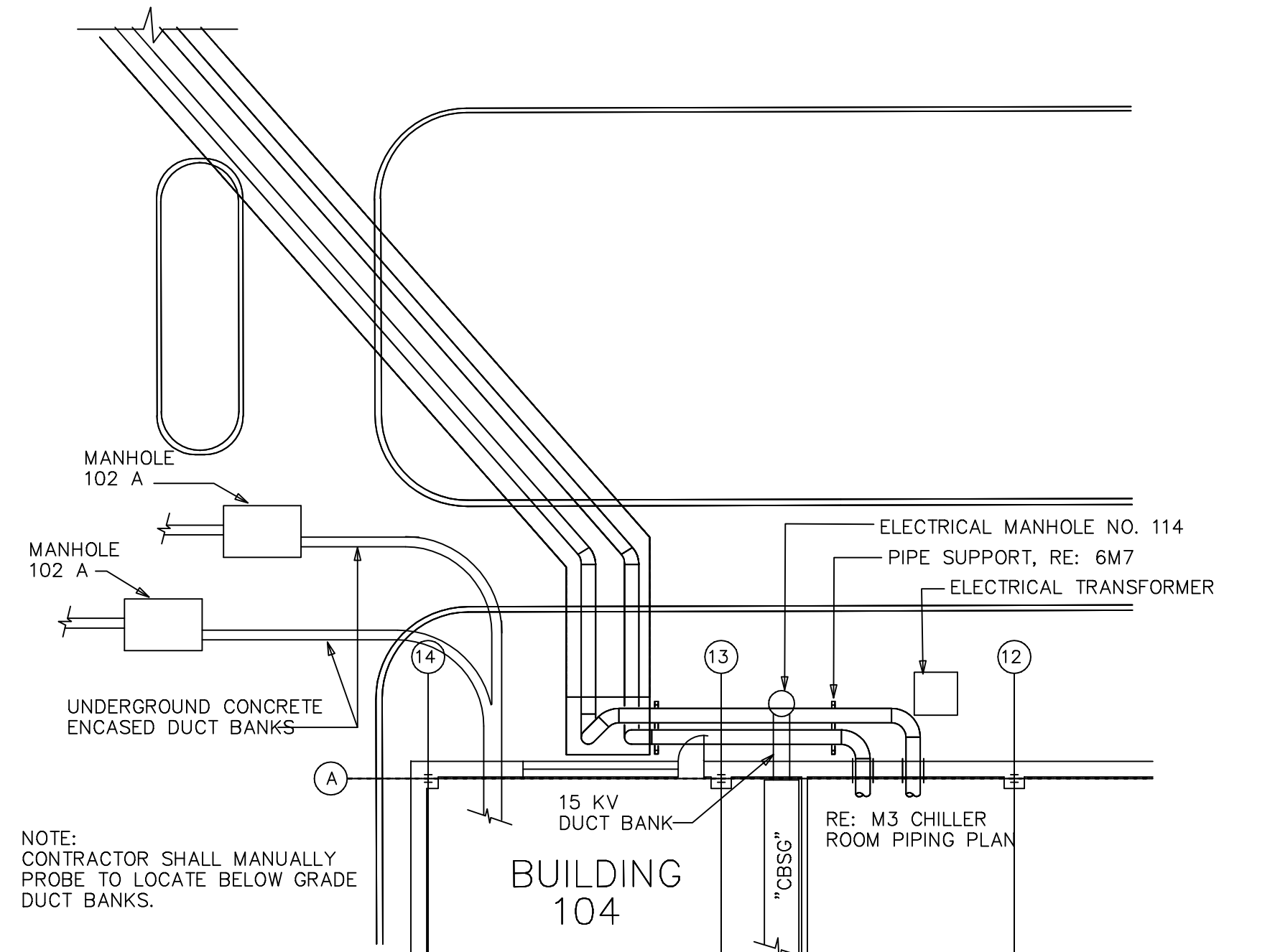
**Veterans
Administration**



1 CHILLED WATER STORAGE TANK SITE PIPING PLAN SCALE : ?" = 1'-0"



2 STORAGE TANK PIPING CONNECTIONS SCALE : ?" = 1'-0"



3 BUILDING EXTERIOR PIPING PLAN SCALE : ?" = 1'-0"

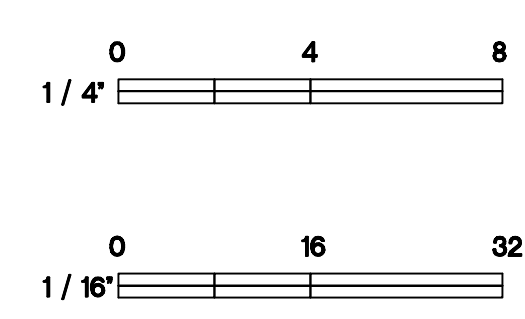
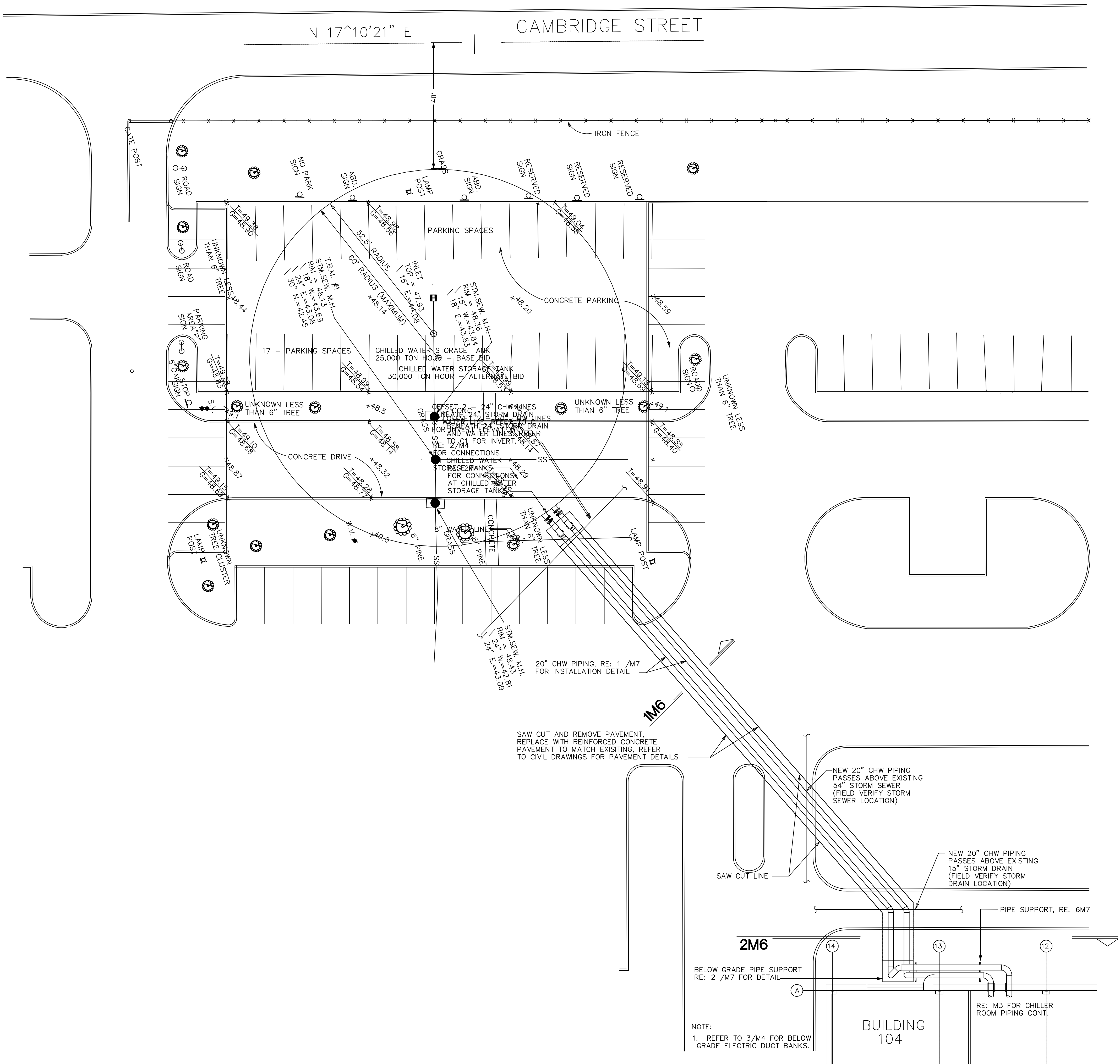


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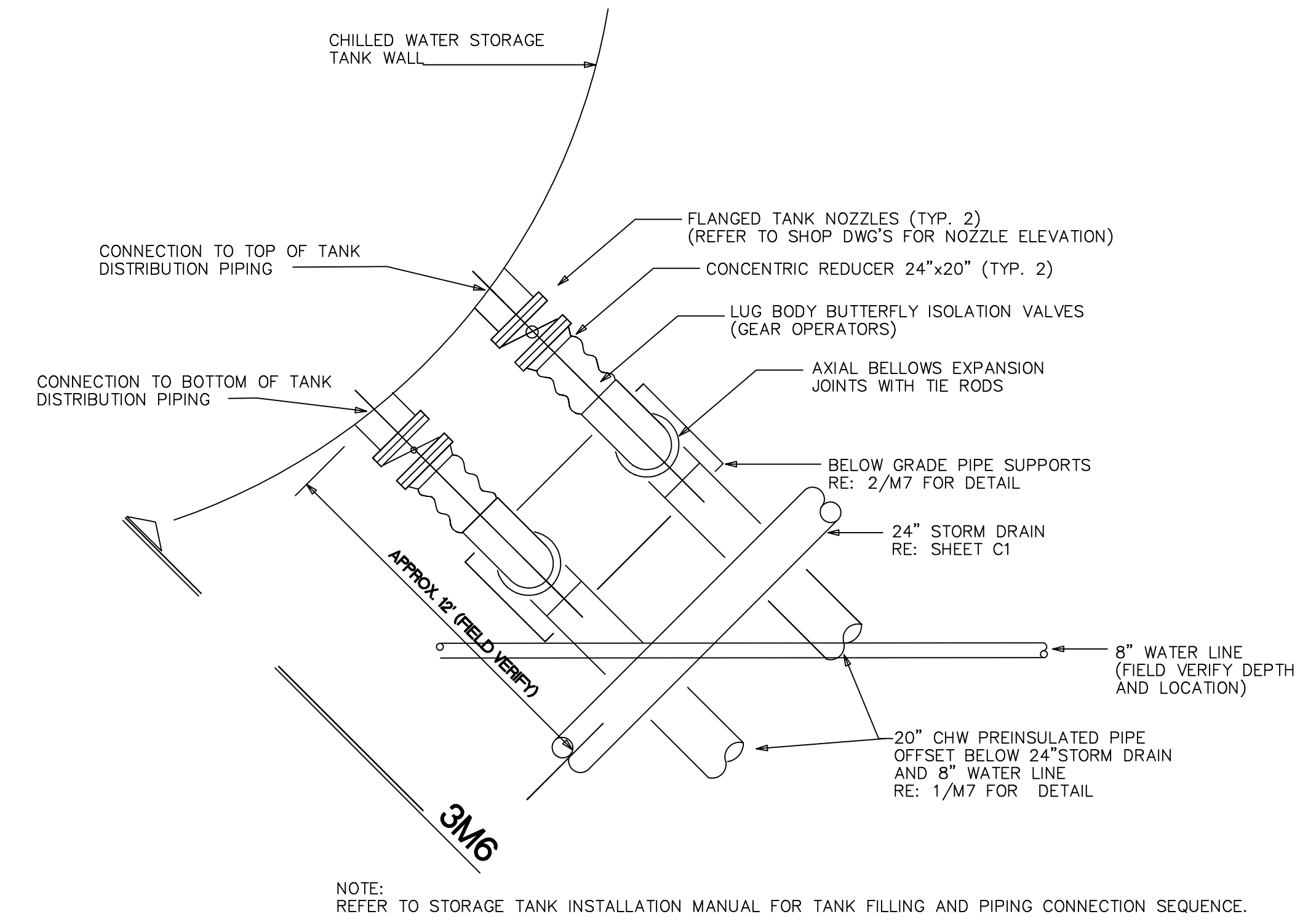
SHAH SMITH & ASSOCIATES, INC. 2825 WILCREST, SUITE 420 HOUSTON, TEXAS 77042

Project Title: BASE BID CHILLED WATER STORAGE TANK SITE PIPING PLANS. Project Title: HOUSTON VETERANS AFFAIRS MEDICAL CENTER INSTALL THERMAL STORAGE UNIT. Includes approval lines for Engineering Svc Chief, Medical Center Director, Building Number, Location, Date, Project No., Drawing No. (M4), and Dwg. of.

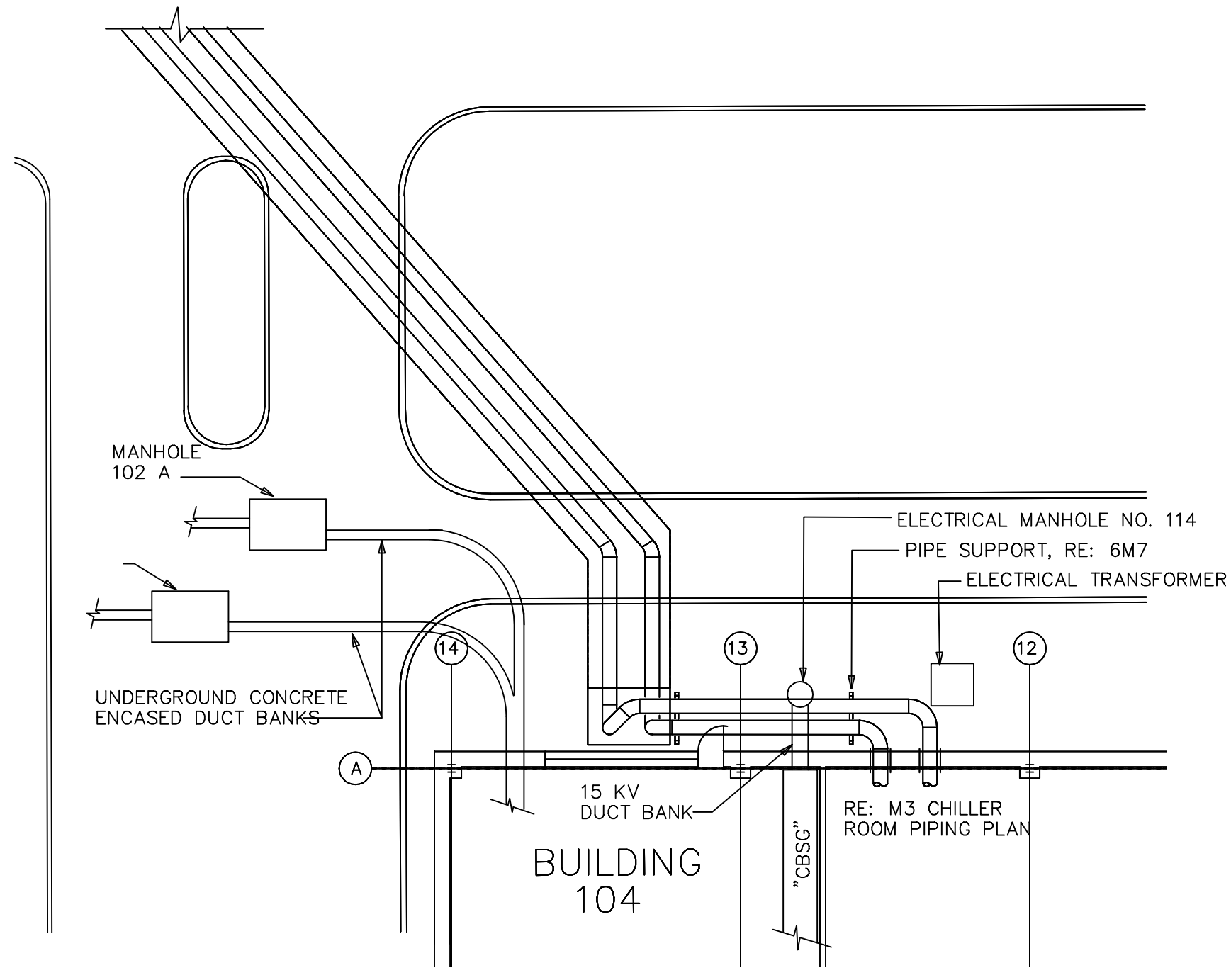
Veterans Administration



1 CHILLED WATER STORAGE TANK SITE PIPING PLAN



2 STORAGE TANK PIPING CONNECTIONS



3 BUILDING EXTERIOR PIPING PLAN

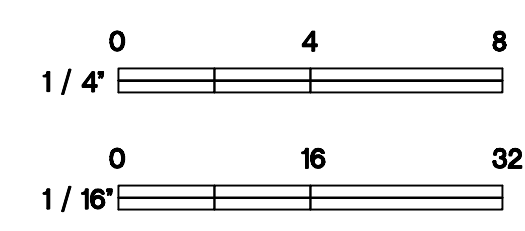
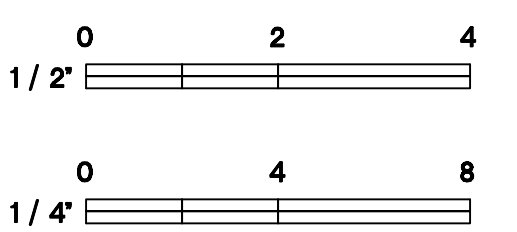
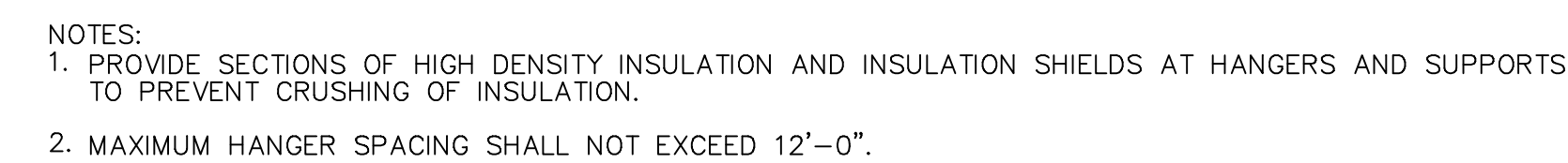


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SHAH SMITH & ASSOCIATES, INC. 2825 WILCREST, SUITE 420 HOUSTON, TEXAS 77042

Project Title: ALTERNATE BID CHILLED WATER STORAGE TANK SITE PIPING PLANS. Includes drawing title, project title, date, project number, building number, location, and drawing number M5.

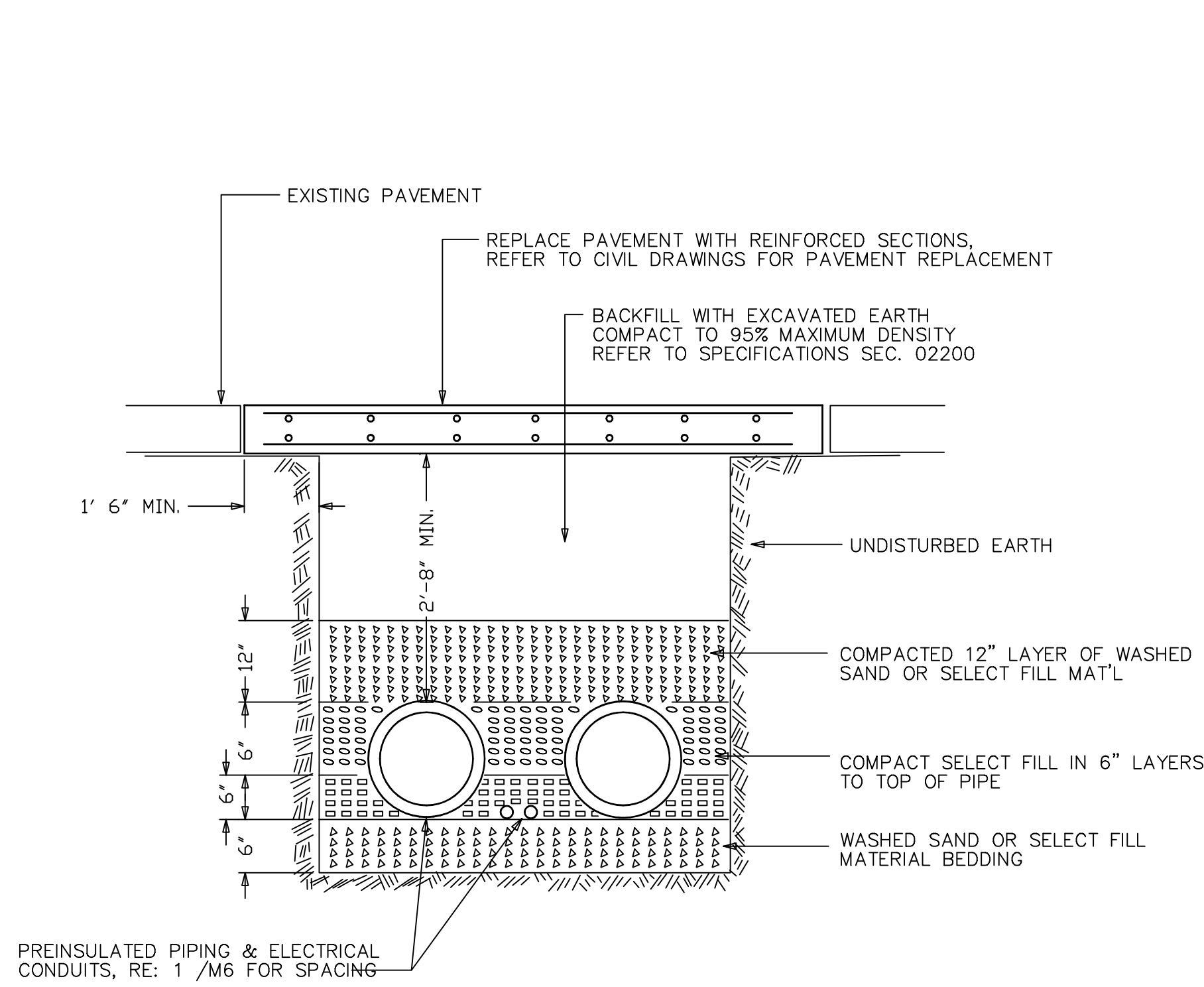
Veterans Administration



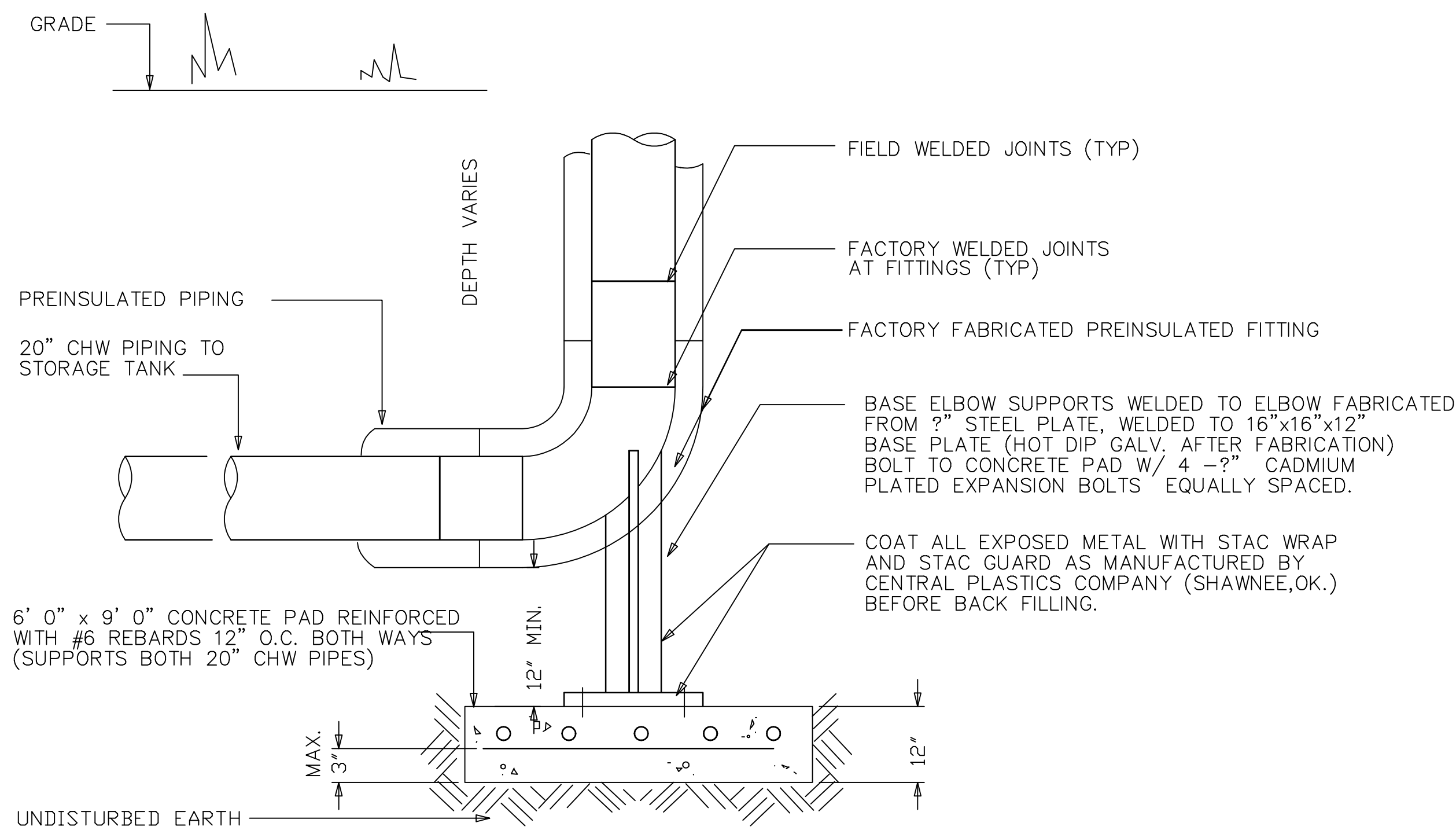
Date	10/95
Project No.	VA #V580P-5840 SSA #1017-001-2
Drawing No.	M6
Dwg. of	

Veterans Administration

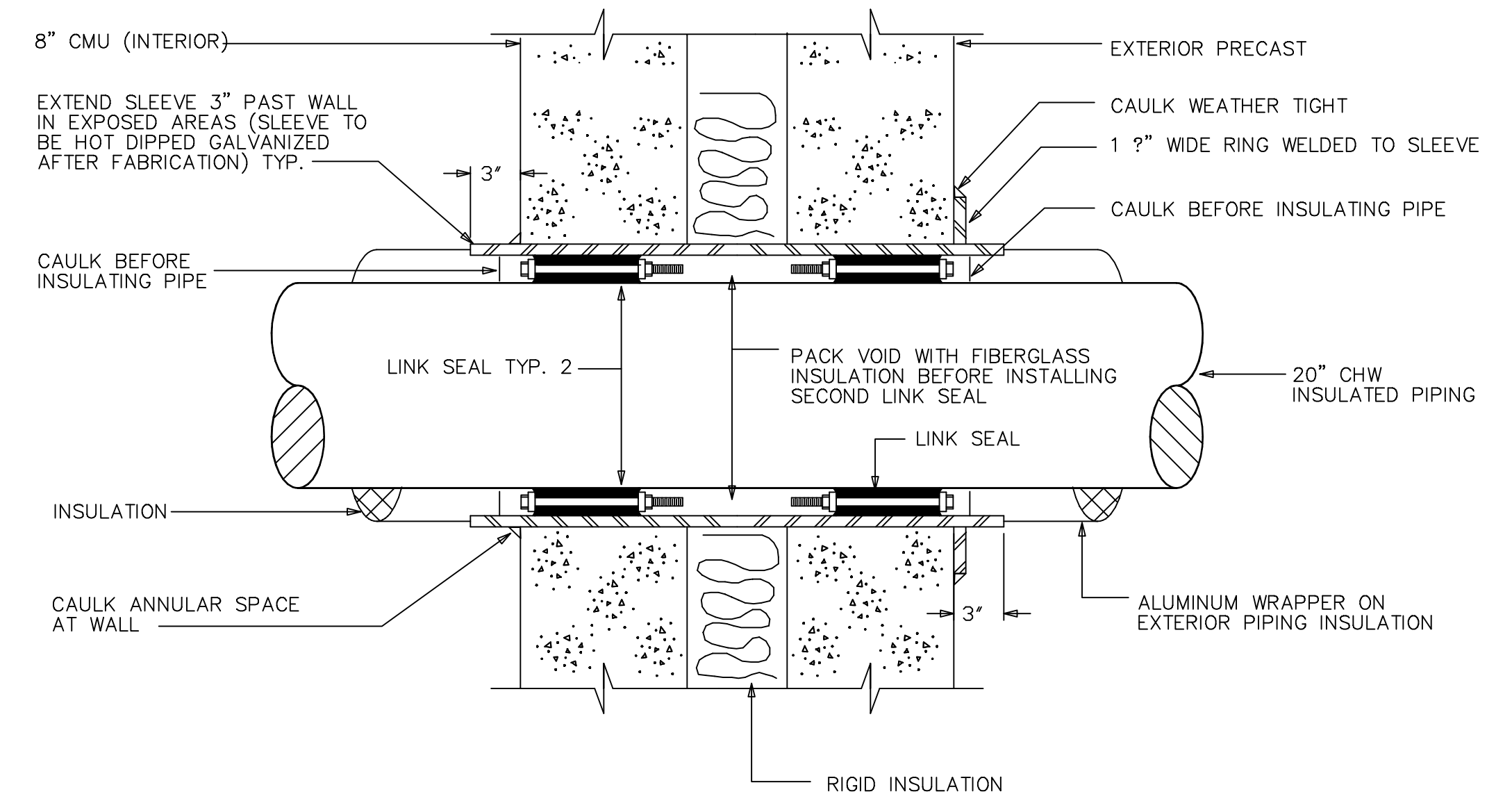
Veterans Administration



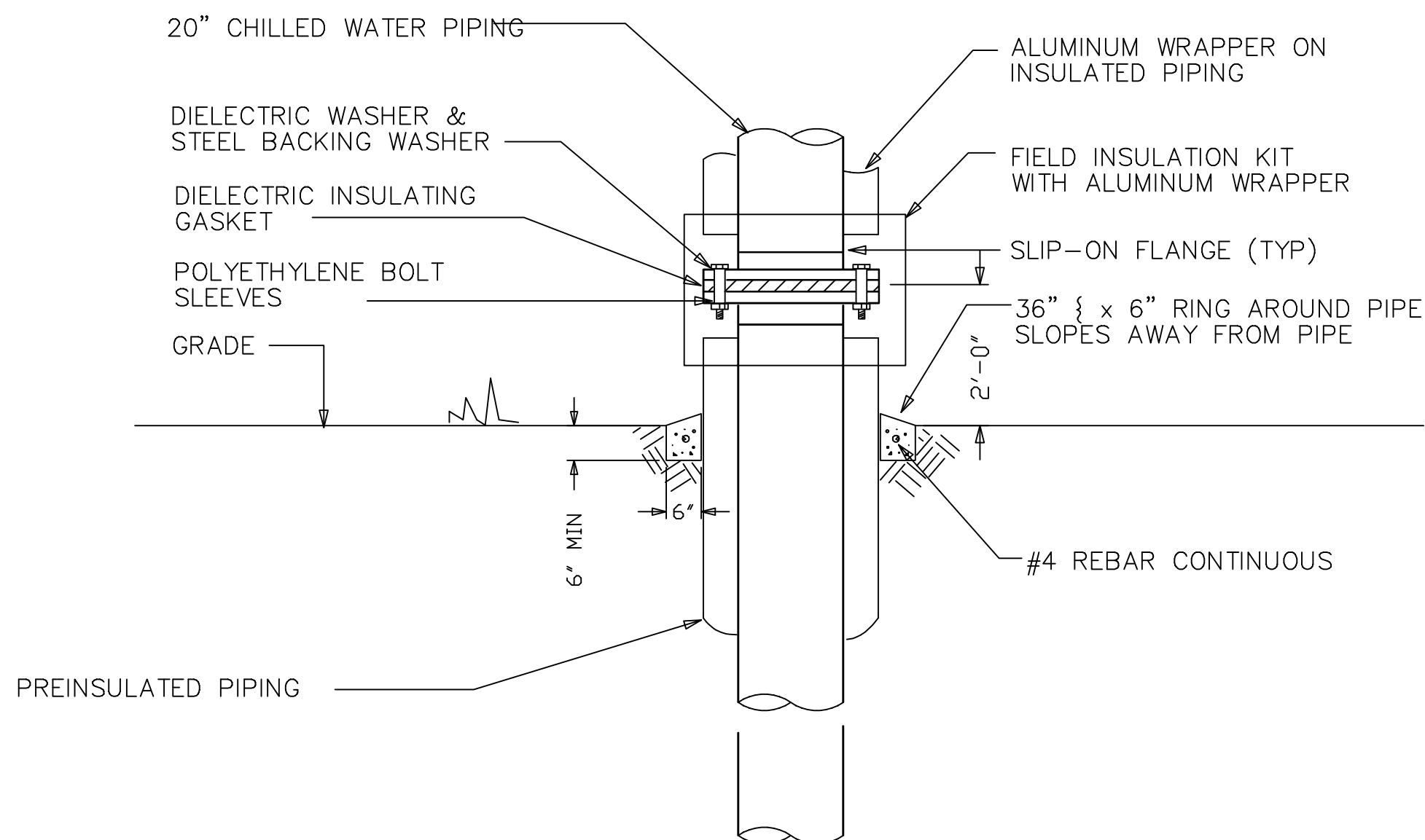
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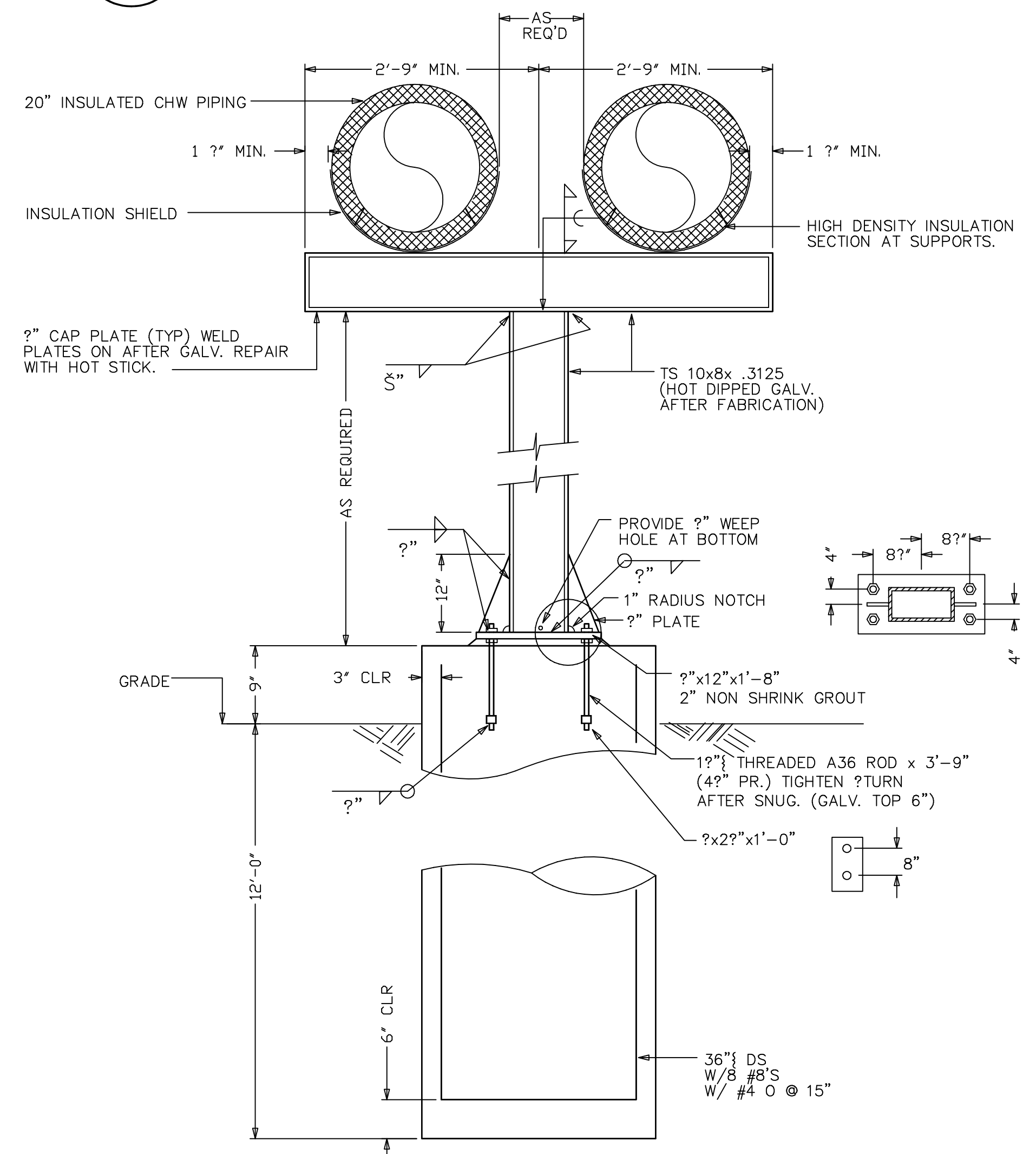
2 BELOW GRADE PIPING SUPPORT DETAIL
NOT TO SCALE



3 PIPE PENETRATION THRU WALL
NOT TO SCALE

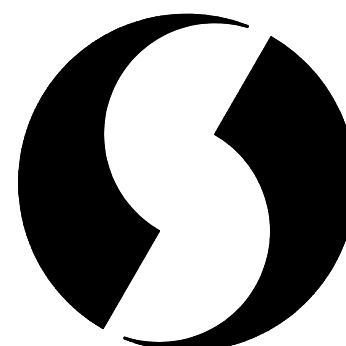


5 DIELECTRIC ISOLATION FLANGE DETAIL
NOT TO SCALE



6 EXTERIOR PIPE SUPPORT DETAIL
NOT TO SCALE

Revisions	Date



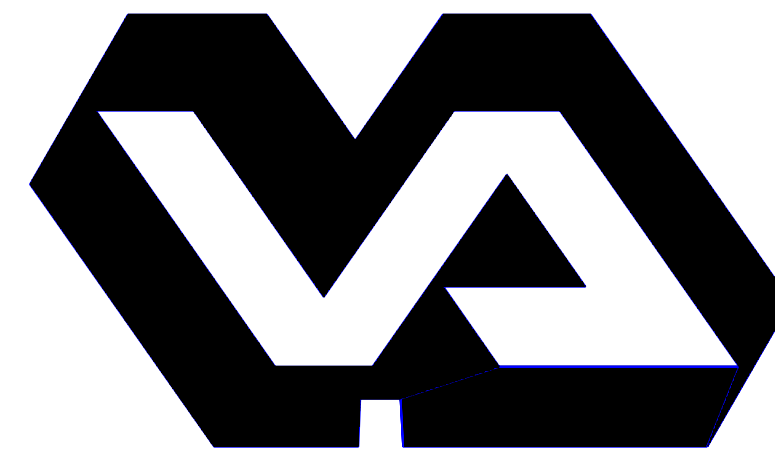
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 420
HOUSTON, TEXAS 77042

Drawing Title
MECHANICAL DETAILS
Approved: Engineering Svc. Chief
Approved: Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER INSTALL THERMAL STORAGE UNIT
Building Number
104/105
Location
VAMC HOUSTON, TX

Date
10/95
Project No.
VA #V580P-5840 SSA #1017-001-2
Drawing No.
M7
Dwg. of

Veterans
Administration



VETERANS AFFAIRS MEDICAL CENTER #580
Houston, Texas

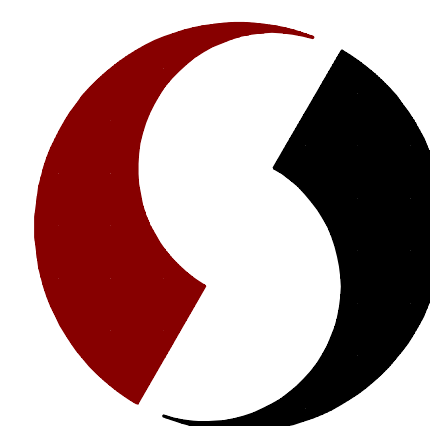
CHILLER PLANT EXPANSION
PROJECT NO. #580-98-110

INDEX OF DRAWINGS

	COVER SHEET	M0.01	MECHANICAL SCHEDULES AND LEGENDS	E0.01	ELECTRICAL LEGEND, NOTES AND SCHEDULE
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G0.01	PROJECT LOCATION PLAN	M2.01	MECHANICAL PIPING PLAN	E2.01	ELECTRICAL SITE PLAN RENOVATION
		M3.01	MECHANICAL PIPING PLAN AND SECTIONS	E2.02	ELECTRICAL SITE PLAN 1247KV DISTRIBUTION
C0.01	SITE PIPING PLAN	M3.02	MECHANICAL PIPING PLAN COOLING TOWER	E3.01	ENLARGED ELECTRICAL POWER, LIGHTING AND COMMUNICATION PLANS
C0.02	SITE PAVING PLAN	M4.01	MECHANICAL DETAILS	E4.01	COMPOSITE ONE LINE DIAGRAM
C0.03	CIVIL DETAILS	M4.02	MECHANICAL DETAILS	E4.02	COMPOSITE ONE LINE DIAGRAM
		M5.01	CONDENSER WATER FLOW DIAGRAM	E4.03	EXISTING MCC2 DEMOLITION & RENOVATION
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S1.2	TYPICAL DETAILS	M5.03	MECHANICAL SCHEMATICS	E4.05	5KV MOTOR CONTROL DIAGRAM
S2.1	NEW CHILLER BUILDING FOUNDATION PLAN			E5.01	ELECTRICAL DETAILS
S2.2	NEW CHILLER BUILDING ROOF FRAMING PLAN	P0.01	PLUMBING LEGEND AND DETAILS	E5.02	ELECTRICAL DETAILS
S2.3	NEW COOLING TOWER FOUNDATION PLAN	P2.01	PLUMBING PIPING PLAN	E5.03	ELECTRICAL DETAILS
S3.1	FOUNDATION DETAILS			E5.04	ELECTRICAL DETAILS
S3.2	COOLING TOWER BASIN FOUNDATION DETAILS			E5.05	ELECTRICAL DETAILS
S4.1	SECTION AND DETAILS				
A1.01	ARCHITECTURAL PLANS AND ELEVATIONS				
A1.02	ARCHITECTURAL DETAILS				

August 2001

PREPARED BY



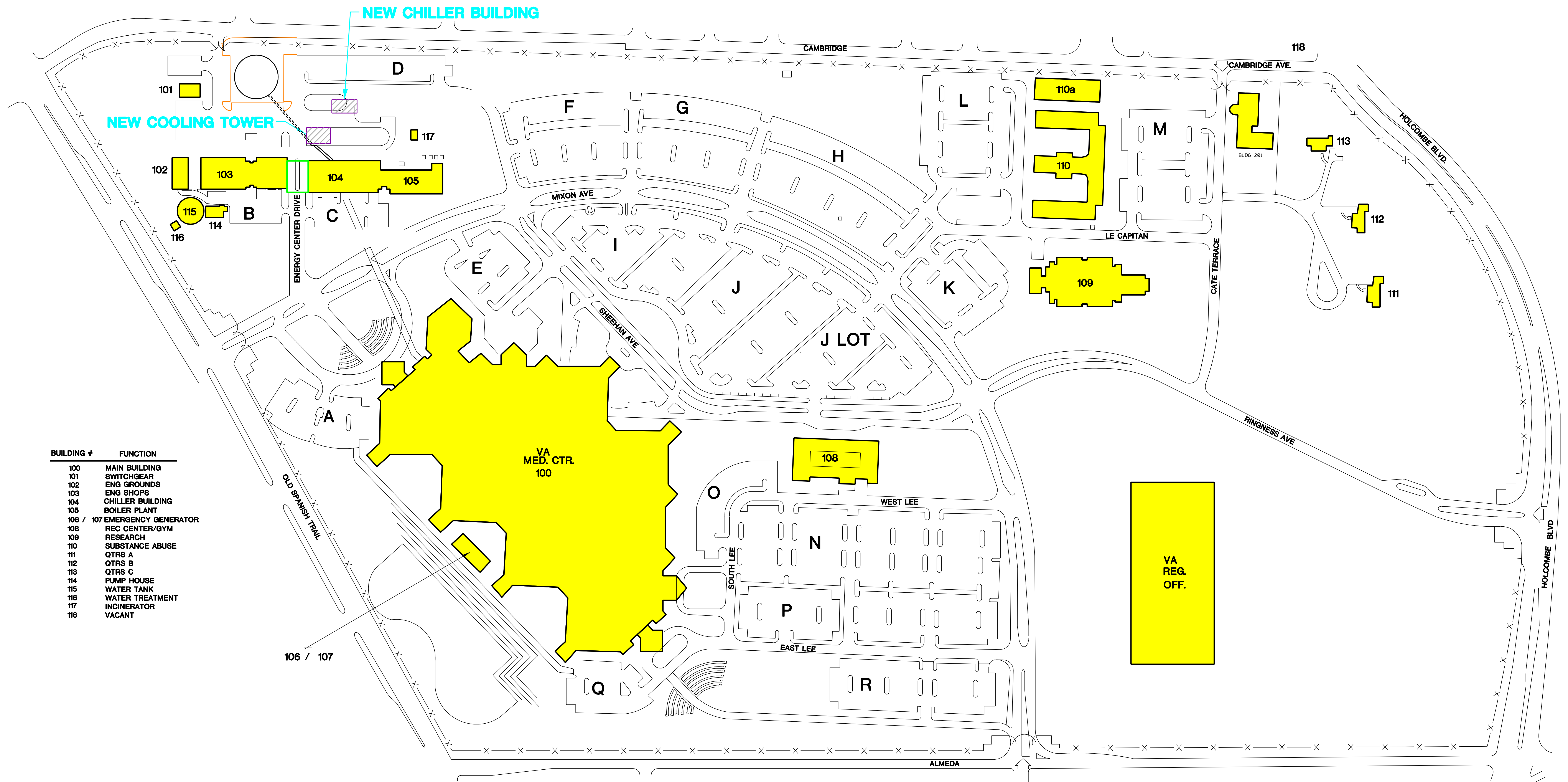
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

RECORD DRAWING

THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

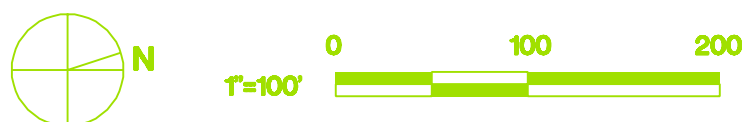
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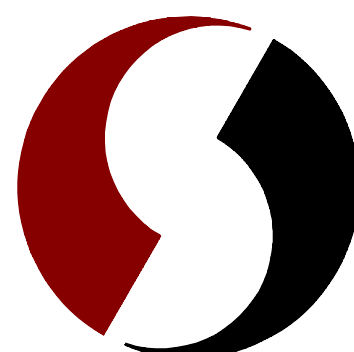


SITE PLAN

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.



Revisions	Date



SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

JERRY D. SMITH
42402

Drawing Title
PROJECT LOCATION PLAN

Approved: Engineering Svc Chief

Approved: Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER

CHILLER PLANT EXPANSION

Building Number

Location
VAMC HOUSTON, TX

Checked
JDS

Drawn
JRA

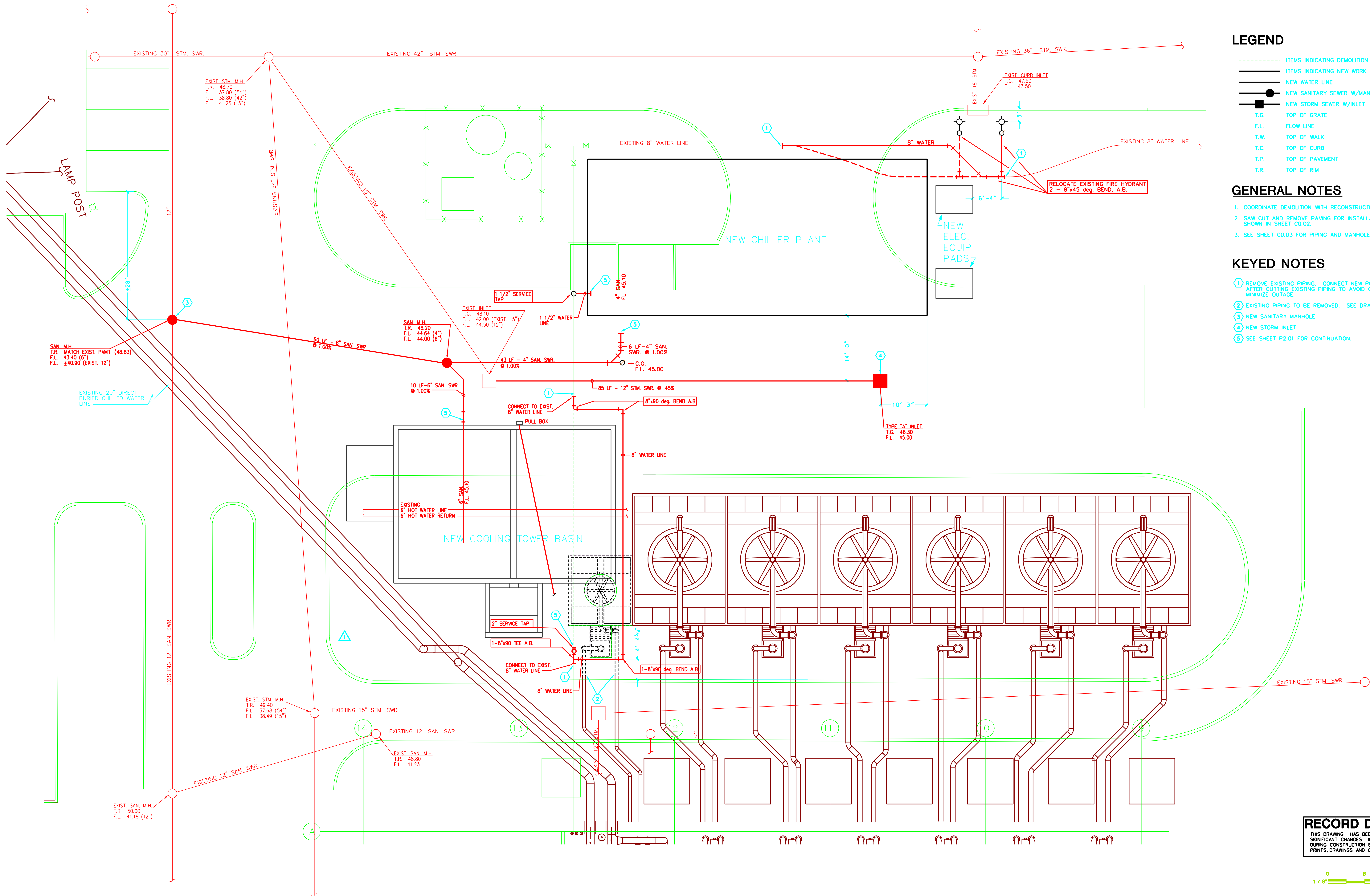
Date
02/25/2003

Project No.
VA #560-58-110
SSA #1017-007-1

Drawing No.
G0.01

Dwg. of

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LEGEND

- ITEMS INDICATING DEMOLITION
- ITEMS INDICATING NEW WORK
- NEW WATER LINE
- NEW SANITARY SEWER W/MANHOLE
- NEW STORM SEWER W/INLET
- T.G. TOP OF GRATE
- F.L. FLOW LINE
- T.W. TOP OF WALK
- T.C. TOP OF CURB
- T.P. TOP OF PAVEMENT
- T.R. TOP OF RIM

GENERAL NOTES

- COORDINATE DEMOLITION WITH RECONSTRUCTION.
- SAW CUT AND REMOVE PAVING FOR INSTALLATION OF NEW PIPING AS SHOWN IN SHEET C0.02.
- SEE SHEET C0.03 FOR PIPING AND MANHOLE DETAILS.

KEYED NOTES

- REMOVE EXISTING PIPING. CONNECT NEW PIPING IMMEDIATELY AFTER CUTTING EXISTING PIPING TO AVOID CONTAMINATION AND MINIMIZE OUTAGE.
- EXISTING PIPING TO BE REMOVED. SEE DRAWING M1.01.
- NEW SANITARY MANHOLE
- NEW STORM INLET
- SEE SHEET P2.01 FOR CONTINUATION.

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.



Revisions	Date



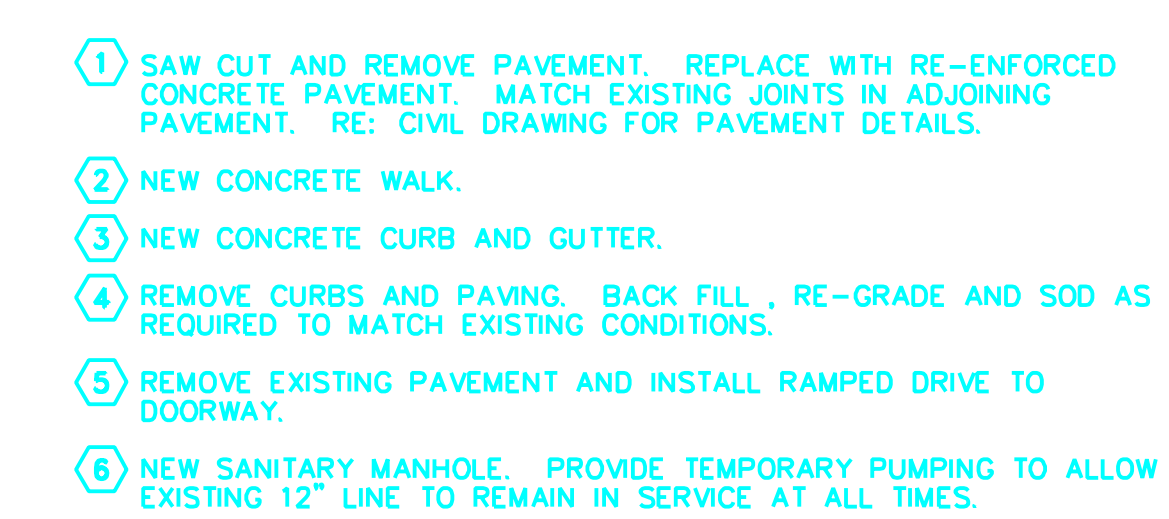
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

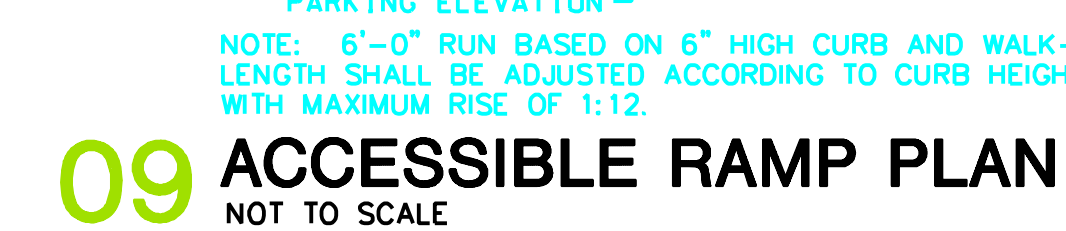
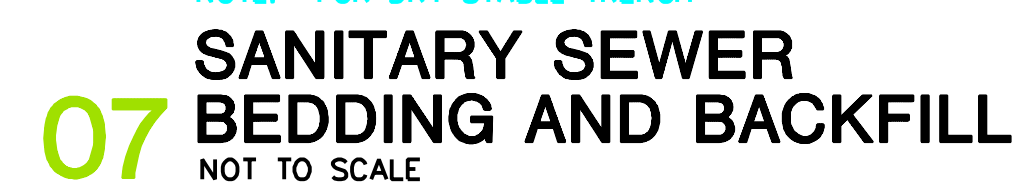
Drawing Title	SITE PIPING PLAN
Approved: Engineering Svc. Chief	
Approved: Medical Center Director	

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
	CHILLER PLANT EXPANSION
Building Number	
Location	VAMC HOUSTON, TX

Date	02/25/2003
Project No.	VA #560-58-110 SSA #1017-007-1
Drawing No.	C0.01
Dwg. of	

Veterans
Administration

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Date 02/25/2003

Project No.
VA #580-98-110
SSA #1017-007-1

Drawing No. 000

Dwg. of

	Veterans Administration
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**** GENERAL NOTES ****

1. THE BUILDING CODE USED IS SOUTHERN UNIFORM BUILDING CODE 1997 EDITION, AS MODIFIED BY THE CITY OF HOUSTON.
2. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS IN THE FILED PRIOR TO COMMENCING WORK. THE ENGINEER SHALL BE NOTIFIED OF ANY DISCREPANCIES WHICH MAY EXIST.
3. ALL FOOTINGS ARE TO BE PLACED ON FIRM AND CLEAN SOIL.
4. SEE ARCHITECTURAL DRAWINGS FOR FLOOR ELEVATIONS, SLOPE, AND THE LOCATION OF DEPRESSED FLOOR AREAS. THE CONTRACTOR SHALL COMPARE THE STRUCTURAL SECTIONS WITH THE ARCHITECTURAL SECTIONS AND REPORT ANY DISCREPANCY TO THE ARCHITECT PRIOR TO FABRICATING OR INSTALLING STRUCTURAL MEMBERS.
5. PRINCIPAL OPENINGS THROUGH THE FRAMING ARE SHOWN ON THESE DRAWINGS. THE GENERAL CONTRACTOR SHALL EXAMINE THE ARCHITECTURAL AND MECHANICAL DRAWINGS FOR THE REQUIRED OPENINGS AS HE SHALL PROVIDE FOR ALL OPENINGS WHETHER SHOWN ON THE DRAWINGS OR NOT AND SHALL VERIFY SIZE AND LOCATION OF ALL OPENINGS WITH THE MECHANICAL CONTRACTOR. ANY DEVIATION FROM THE OPENINGS SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE BROUGHT TO THE ENGINEER'S ATTENTION FOR APPROVAL.
6. SLABS ON GRADE SHALL BE PLACED ON FILL COMPACTED AT THE OPTIMUM MOISTURE CONTENT. +/- 3 PERCENT, TO 95 PERCENT STANDARD PROCTOR DENSITY.
7. DESIGN LIVE LOADS.
- | | POUNDS PER SQ. FT. |
|-----------|--------------------|
| ROOF..... | 20 |

8. LIVE LOAD REDUCTIONS

FOR ALL BEAMS, GIRDERS AND COLUMNS SUPPORTING MORE THAN 150 SQUARE FEET OF FLOOR AREA, REDUCED LIVE LOAD IS USED. THE LIVE LOAD IS REDUCED AT THE RATE OF 0.08 PERCENT PER SQUARE FOOT OF FLOOR AREA IN EXCESS OF 150 SQ. FT. SUPPORTED BY THE MEMBER OR AS DETERMINED BY THE FORMULA AT THE RATE OF $f = 23.10 (1 + \text{DEAD/LIVE LOAD})$ WHICHEVER IS MINIMUM. THE MAXIMUM LIMIT OF REDUCTION IS 60 PERCENT OF THE LIVE LOAD. FOR GARAGE AREA, THE MAXIMUM REDUCTION IS LIMITED TO 40 PERCENT ONLY.

9. FOUNDATION ALLOWABLE BEARING SOIL PRESSURE = 5460 PSF FOR TOTAL DEAD & LIVE @ LEAST 9'-0" BELOW EXISTING GRADE. SEE SOILS REPORT, BY PSI (# 286-15103, DATED JUNE 26, 2001) ALL ADDITIONAL SOIL TESTING SHALL BE THE RESPONSIBILITY OF THE CONSTRUCTION CONTRACTOR.

**** STEEL ****

1. STRUCTURAL STEEL ROLLED SHAPES SHALL CONFORM TO THE FOLLOWING ASTM DESIGNATIONS:

- ASTM A-36 ALL STEEL FRAMING INCLUDING COLUMNS, BASE PLATES, AND MISCELLANEOUS STEEL PLATES UNLESS NOTED OTHERWISE.
- ASTM A325 ALL BOLTS USED FOR CONNECTING STRUCTURAL MEMBERS.
- ASTM A307 ALL OTHER ANCHOR BOLTS UNLESS NOTED OTHERWISE.
- ASTM A501 ALL TUBULAR SECTIONS.

2. METAL DECK

FOR NON-COMPOSITE OVER BAR JOIST, GALVANIZED CORRUGATED METAL DECK WITH THE PROPERTIES GIVEN BELOW OR EQUIVALENT, SHALL BE USED FOR THE FOLLOWING AREAS:

	DEPTH	IN (4) /FT.	S IN (3) /FT.	Sn IN (3) FT	MAX. SPAN
ROOF	19"	0.169	0.186	0.192	6'-0"

ROOF DECK FASTENER

36/77 SUPPORT FASTENER LAYOUT WITH 5/8" PUDDLE WELDS, PROVIDE (4)#10 TEX. SCREWS PER SPAN FOR SIDE LAP FASTENERS.

3. ROOF DECK

ROOF DECK OVER STEEL BEAMS SHALL CONSIST OF RIGID INSULATION ATTACHED TO GALVANIZED CORRUGATED METAL DECK. LAP CORRUGATED METAL ENDS AND WELD ENDS AT EACH SIDE LAP OF CORRUGATED METAL DECK UNLESS DECK IS PRE-VENTED.

4. ALL WELDED WIRE FABRIC SHALL CONFORM TO ASTM A185. THE FOLLOWING WWF SHOULD BE USED FOR AREAS SPECIFIED BELOW:

FLOOR SLAB - 6x6 - W1.4x1.4
ROOF - SEE NOTE ABOVE.
FOR ADDITIONAL WWF, IF ANY, SEE FLOOR FRAMING PLAN.

5. GROUT USED UNDER COLUMN BASE PLATES SHOULD BE OF THE NON-SHRINKABLE VARIETY WITH MINIMUM COMPRESSIVE STRENGTH OF 6000 PSI IN 28 DAYS.
6. ALL STRUCTURAL STEEL DETAILS AND CONNECTIONS SHALL CONFORM TO THE AISC STANDARDS ASD 9th EDITION AND/OR LRFD 2nd EDITION.

7. NON-COMPOSITE BEAMS:

THE BEAM-TO-BEAM AND BEAM-TO COLUMN CONNECTION SHOULD DEVELOP THE END REACTION OF THE CONNECTED BEAM. THE END REACTION OF THE CONNECTED BEAM CAN BE OBTAINED FROM THE LOAD CARRYING CAPACITY OF THE BEAM ON THE BASIS OF EQUIVALENT UNIFORM LOADS GIVEN IN PART 2 (BEAMS & GIRDERS) OF AISC MANUAL 1990 EDITION. THE MINIMUM SHEAR CAPACITY OF 12 KIPS SHALL BE PROVIDED FOR ALL BEAMS 8 TO 10" DEEP WITH 8 KIPS FOR BEAMS OF DEPTH LESS THAN 8 INCHES. THE REACTIONS GIVEN ON THE DRAWING SUPERSEDE THIS NOTE.

8. FOR ALL HIGH STRENGTH BOLTS, HARDENED WASHERS SHALL BE PROVIDED ON NUT SIDE OF BOLT FOR TIGHTENING AS REQUIRED.

9. ALL WELDING SHALL CONFORM TO THE STANDARD OF THE AMERICAN WELDING SOCIETY, AWS D1.1 1994 STRUCTURAL WELDING CODE.

10. E70 ELECTRODES FOR ALL FIELD AND SHOP WELDING SHALL CONFORM TO AWS/AWS D1.1 1994 EDITION. ALL WELD SIZES SHALL BE PER AISC CODE.

11. ALL BEAMS AND GIRDERS SHALL BE CAMBERED EQUAL TO LENGTH /360 OR AS INDICATED ON STRUCTURAL DRAWINGS.

12. SPLICING STRUCTURAL MEMBERS WHERE NOT DETAILED ON THE DRAWING IS PROHIBITED WITHOUT PRIOR APPROVAL OF THE STRUCTURAL ENGINEER.

13. FIELD PENETRATION WELDS ALONG WITH COLUMN BASE PLATE DETAILS SHALL BE VISUALLY INSPECTED. A MINIMUM OF TWENTY-FIVE PERCENT OF ALL FIELD PENETRATION WELDS SHALL BE INSPECTED AT RANDOM.

14. ANY TEMPORARY SHORING OF BRACING DURING THE CONSTRUCTION PHASE BEFORE COMPLETION OF CONNECTION AND POURING OF FLOOR SLAB IS THE RESPONSIBILITY OF THE CONTRACTOR.

15. IF NOT SHOWN ON THE DRAWINGS, SUPPORT OF METAL DECK AROUND COLUMN CLOSURE, SCREED PLATES AROUND THE OPENINGS AND EDGE OF SLAB SHALL BE PROVIDED BY THE CONTRACTOR.

**** CONCRETE ****

1. ALL CONCRETE REINFORCING BARS SHALL CONFORM TO ASTM A615, GRADE 60 EXCEPT WHERE NOTED. NO. 3 BARS MAY CONFORM TO ASTM A615, GRADE 40.

2. CONCRETE IN ALL AREAS SHALL HAVE SAND AND GRAVEL AGGREGATE CEMENT CONFORMING TO ASTM C 150, TYPE I, IA, II, OR IIA, AND DESIGNATED COMPRESSIVE STRENGTH (FC) OF 3000 PSF IN 28 DAYS, UNLESS NOTED OTHERWISE.

3. ALL WELDED WIRE FABRIC SHALL CONFORM TO ASTM A185.

4. DETAILING OF CONCRETE REINFORCEMENT AND ACCESSORIES SHALL BE IN ACCORDANCE WITH ACI PUBLICATION 315, LATEST EDITION.

5. ALL MIXING, TRANSPORTATION, PLACING AND CURING OF CONCRETE SHALL BE DONE IN ACCORDANCE WITH THE RECOMMENDATIONS OF THE AMERICAN CONCRETE INSTITUTE.

6. CONCRETE PROTECTION FOR REINFORCEMENT OF POURED-IN-PLACE MEMBERS:

SEE SECTION 7.7 ACI 318-85 FOR CONDITIONS NOT NOTED:

FOOTINGS.....3 IN.

SLABS.....3/4 IN. EXCEPT WHERE NOTED.

COLUMNS.....1 1/2 IN.

SLAB ON GRADE.....1 IN. TOP

GRADE BEAMS.....1 1/2 IN. TOP, 3 IN. BOTTOM,
2 IN. SIDES

DRILLED PIER.....3 IN.

7. THERE SHALL BE NO HORIZONTAL CONSTRUCTION JOINTS IN CONCRETE POURS. ALL CONSTRUCTION JOINTS SHALL BE MADE IN THE CENTER OF SPANS WITH VERTICAL BULKHEADS. WHEN A BEAM INTERSECTS A GIRDER AT THIS POINT, THE JOINTS IN THE GIRDERS SHALL BE OFFSET A DISTANCE EQUAL TO TWICE THE WIDTH OF THE BEAM. THE LOCATION OF CONSTRUCTION JOINTS SHALL BE APPROVED BY THE STRUCTURAL ENGINEER. ADDITIONAL REINFORCING AT CONSTRUCTION JOINTS SHALL BE SPECIFIED BY THE ENGINEER.

8. ALL "CONTINUOUS" REINFORCEMENT SHALL HAVE MINIMUM LAP OF "C" TYPE (ACI 318-83, SECTION 12.5.1) AT SPLICES UNLESS NOTED OTHERWISE.

9. HORIZONTAL WALL REINFORCEMENT SHALL BE CONTINUOUS AND SHALL HAVE 90 DEGREE BENDS AND EXTENSIONS AT CORNERS AND INTERSECTIONS AS SHOWN ON TYPICAL BAR PLACING DETAILS.

10. ALL LAPS IN WWF SHOULD BE ONE MESH PLUS TWO INCHES AT SPLICES.

11. PROVIDE STANDARD BAR CHAIRS WITH PROTECTIVE TIPS AND SPACERS AT 5'-0" CENTERS FOR ALL SLABS AND BEAMS ABOVE GRADE. PROVIDE 3" X 6" X 20 GAGE SHEET METAL BAR CHAIRS AT 4'-0" MAXIMUM CENTERS EACH WAY FOR ALL TOP REINFORCING FOR SLABS ON GRADE.

MASONRY NOTES

1. HOLLOW LOAD BEARING MASONRY UNITS SHALL BE GRADE N UNITS CONFORMING TO THE ASTM DESIGNATION C90 AND IN ADDITION THE REQUIREMENTS OF THE QUALITY CONTROL STANDARDS OF THE CONCRETE MASONRY ASSOCIATION. MIN $f_m = 1500$ PSI
2. MASONRY UNITS SHALL HAVE BEEN CURED FOR NOT LESS THAN 28 DAYS WHEN PLACED IN THE STRUCTURE.

3. MORTAR SHALL BE FRESHLY PREPARED AND UNIFORMLY MIXED IN THE RATIO OF 1 PART PORTLAND CEMENT, 1/4 PART MINIMUM TO 1/2 PART MAXIMUM LIME PUTTY OR HYDRATED LIME, DAMP LOOSE SAND NOT LESS THAN 2 1/2 AND NOT MORE THAN 3 TIMES THE SUM OF THE VOLUMES OF THE CEMENT AND LIME USED, AND SHALL CONFORM TO ASTM DESIGNATION C270.

4. GROUT FOR POURING SHALL BE OF FLUID CONSISTENCY AND MIXED IN THE RATIO BY VOLUMES, 1 PART PORTLAND CEMENT, 2 1/4 PARTS MINIMUM TO 3 PARTS MAXIMUM DAMP LOOSE SAND WHERE THE GROUT SPACE IS LESS THAN 3 INCHES IN ITS LEAST DIMENSION.

5. FLUID CONSISTENCY SHALL MEAN THAT CONSISTENCY AS FLUID AS POSSIBLE FOR POURING WITHOUT SEGREGATION OF THE CONSTITUENT PARTS.

6. HYDRATED LIME SHALL CONFORM TO ASTM C207. QUICK LIME SHALL CONFORM TO ASTM C5.

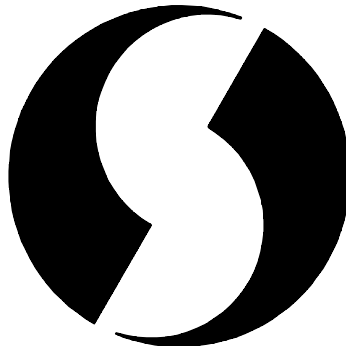
7. AGGREGATE FOR MORTAR SHALL CONFORM TO ASTM C144. AGGREGATE FOR GROUT SHALL CONFORM TO ASTM C404.

8. THE USE OF ADMIXTURES WILL NOT BE PERMITTED IN THE GROUT OR MORTAR UNLESS SUBSTANTIATING DATA IS SUBMITTED TO AND APPROVED BY THE STRUCTURAL ENGINEER OR THE ARCHITECT.

RECORD DRAWING

THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.

50% REVIEW	6-5-01
Revisions	Date



SHAH SMITH & ASSOCIATES, INC.

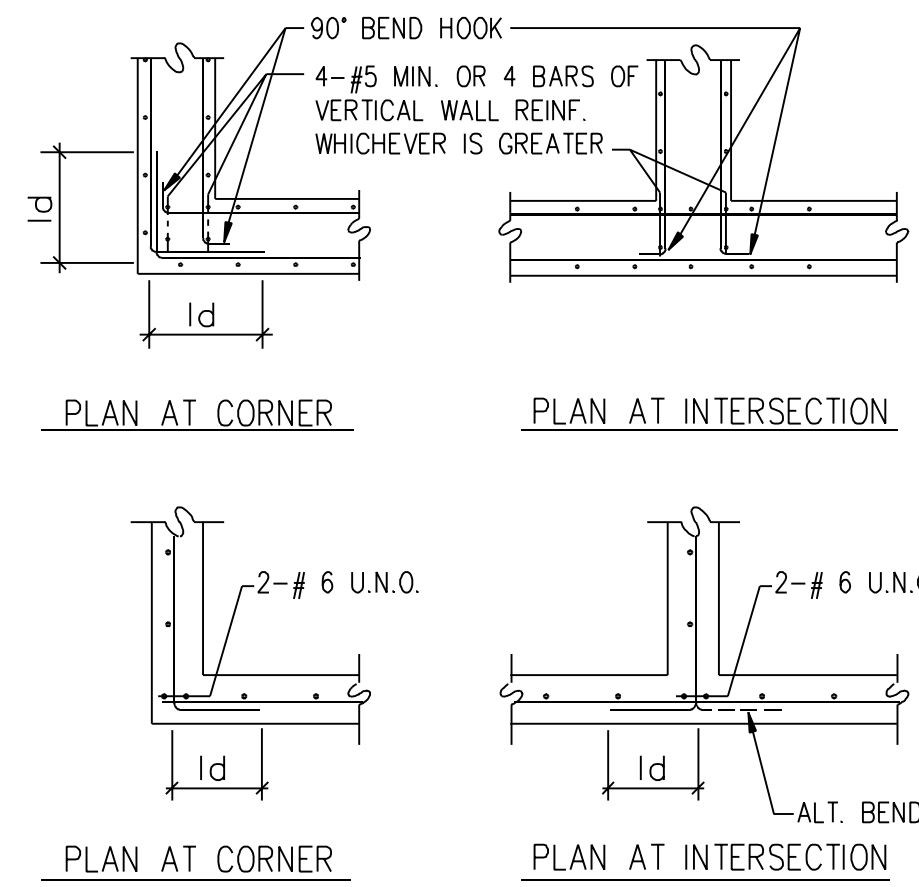
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563

Fax: 713-780-9209

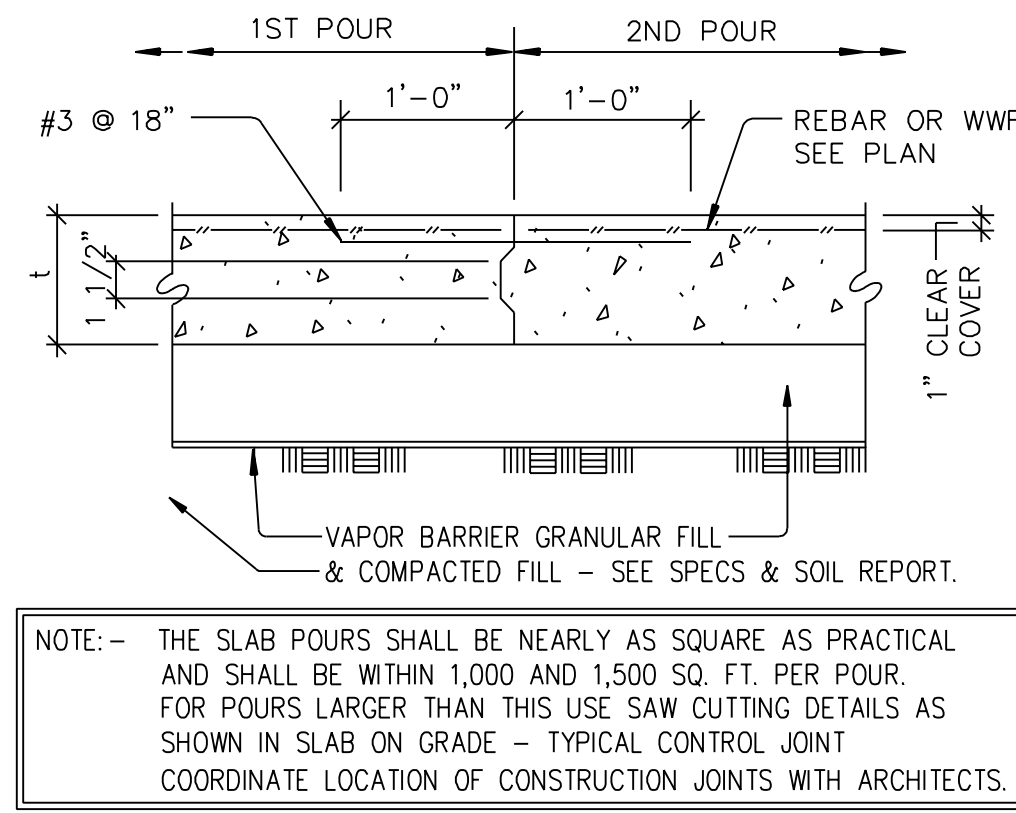
Drawing Title	GENERAL NOTES		
Approved: Engineering Svc Chief			
Approved: Medical Center Director			

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER		Date 7-31-01
CHILLER PLANT EXPANSION		Project No. VA #/ SSA#1017-007-1
Building Number 104/105	Checked YS	Drawn BW
Location VAMC HOUSTON, TX		Drawing No. S1.1
		Dwg. of

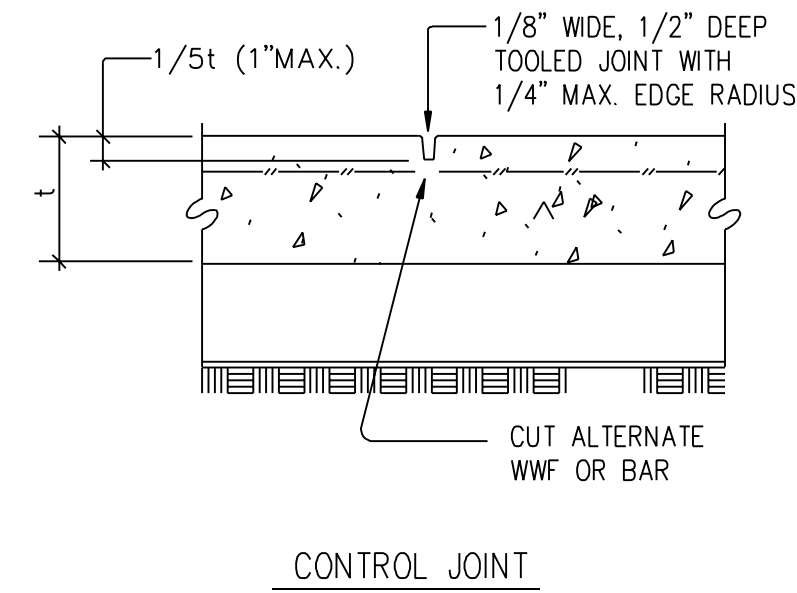
Veterans
Administration



1 TYPICAL BEAM OR WALL CORNER BAR PLACING DETAIL



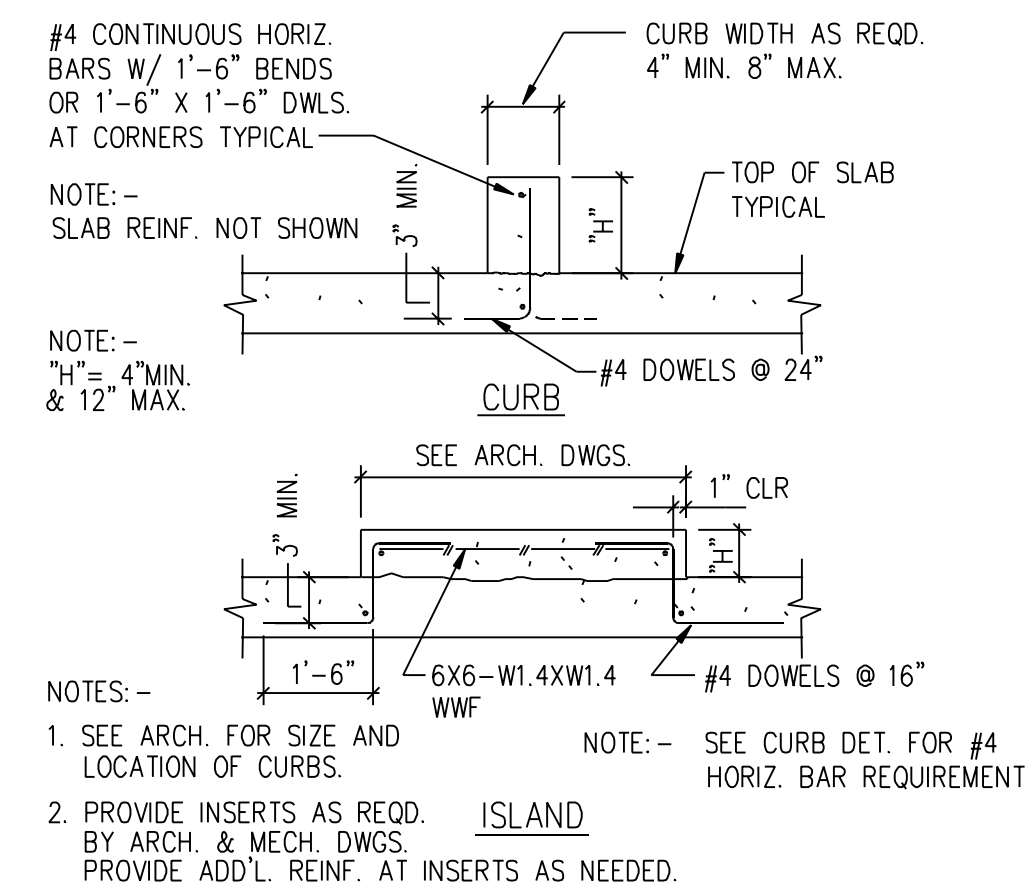
2 SLAB ON GRADE CONSTRUCTION JOINT



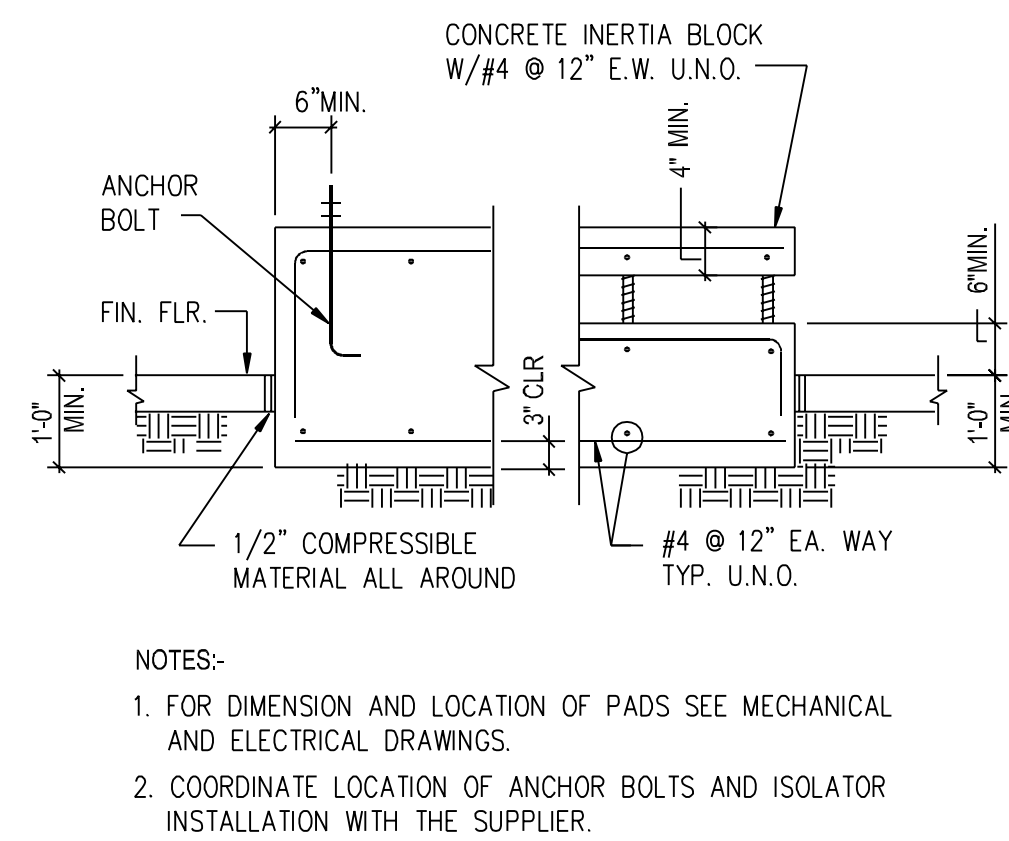
NOTES: -

1. LOCATE THESE JOINTS AT SUCH INTERVALS THAT MAXIMUM SLAB AREA BETWEEN THESE JOINTS DOES NOT EXCEED 1,000 SQ. FT. UNLESS LOCATIONS ARE SPECIFICALLY SHOWN ON STRUCTURAL PLANS.
2. SAWCUT MUST BE MADE WITHIN 12 HOURS OF PLACING OF CONCRETE.

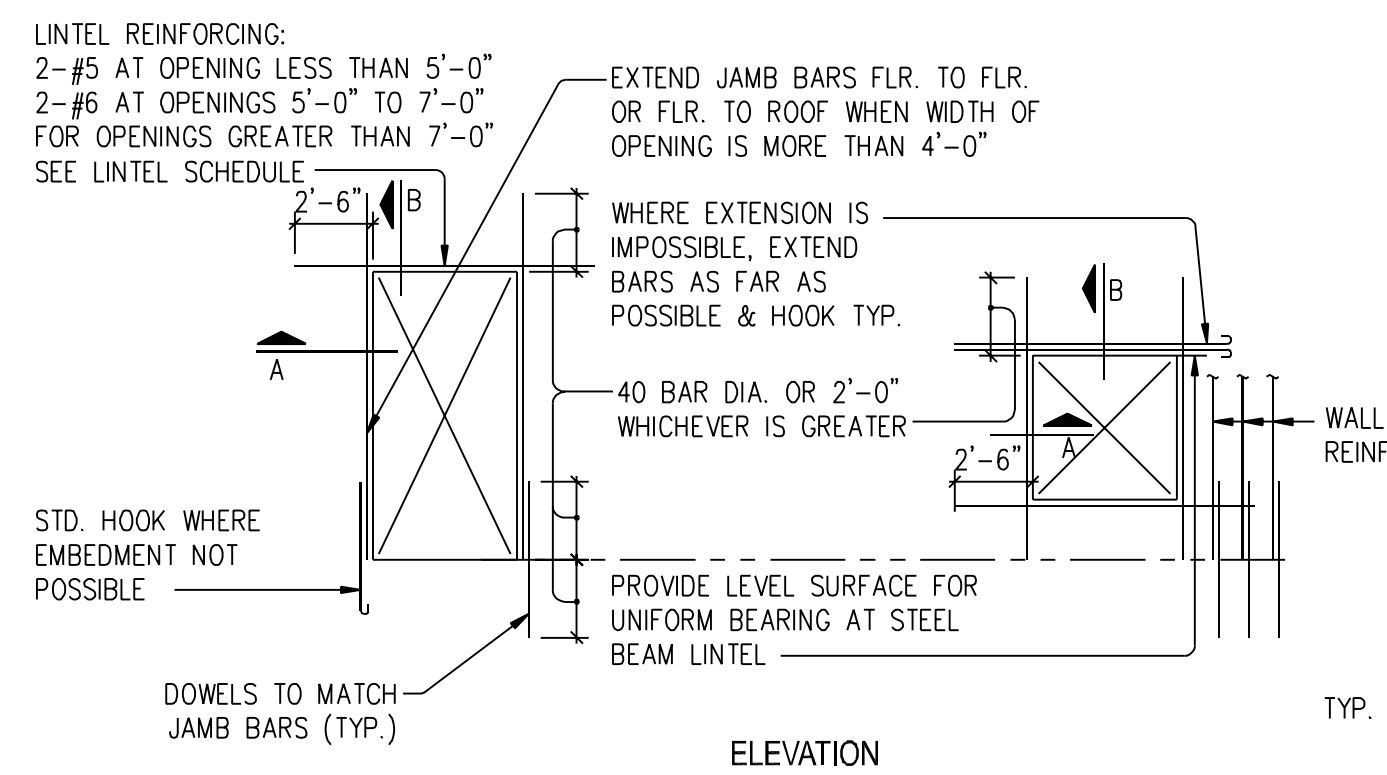
3 SLAB ON GRADE - TYP CONTROL JOINT



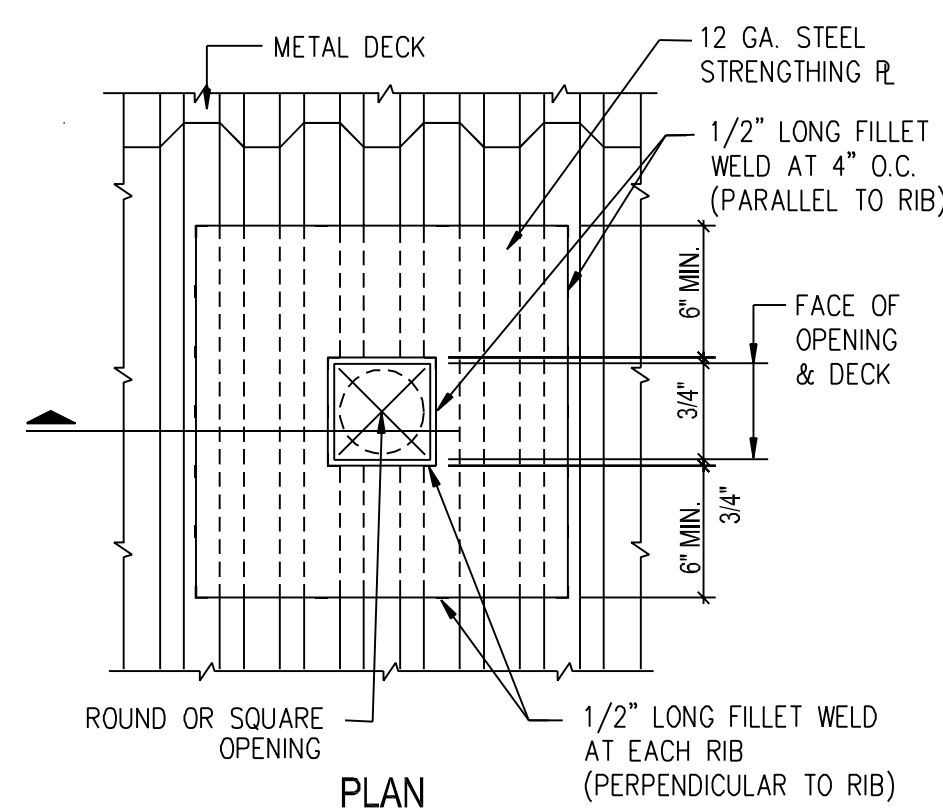
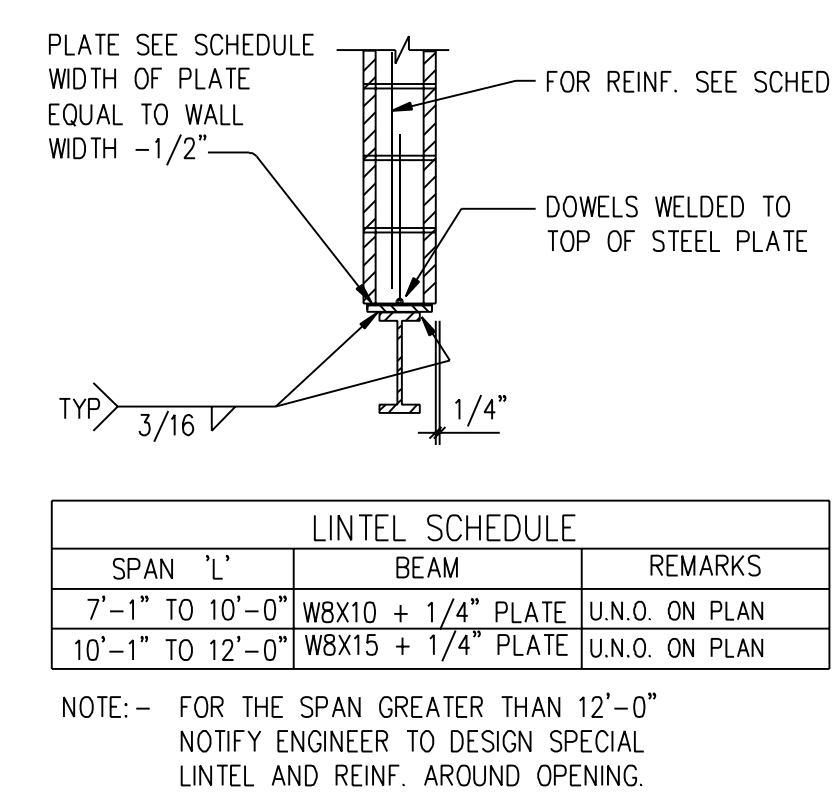
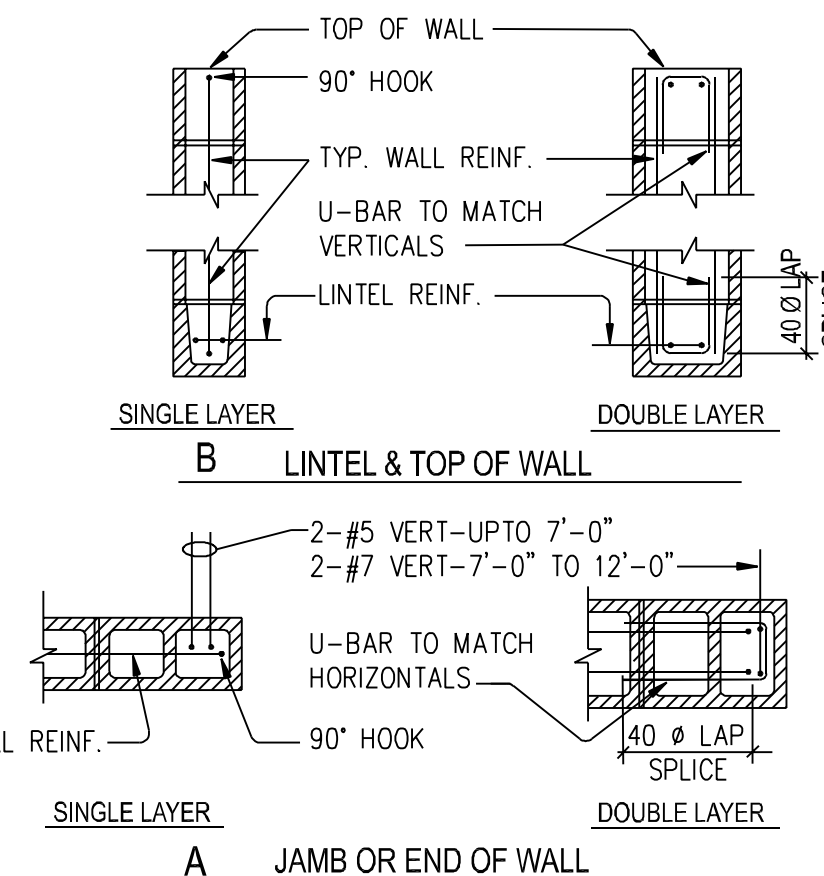
4 CONCRETE CURBS AND ISLANDS



5 CONCRETE BASE AND SLAB FOR EQUIPMENT

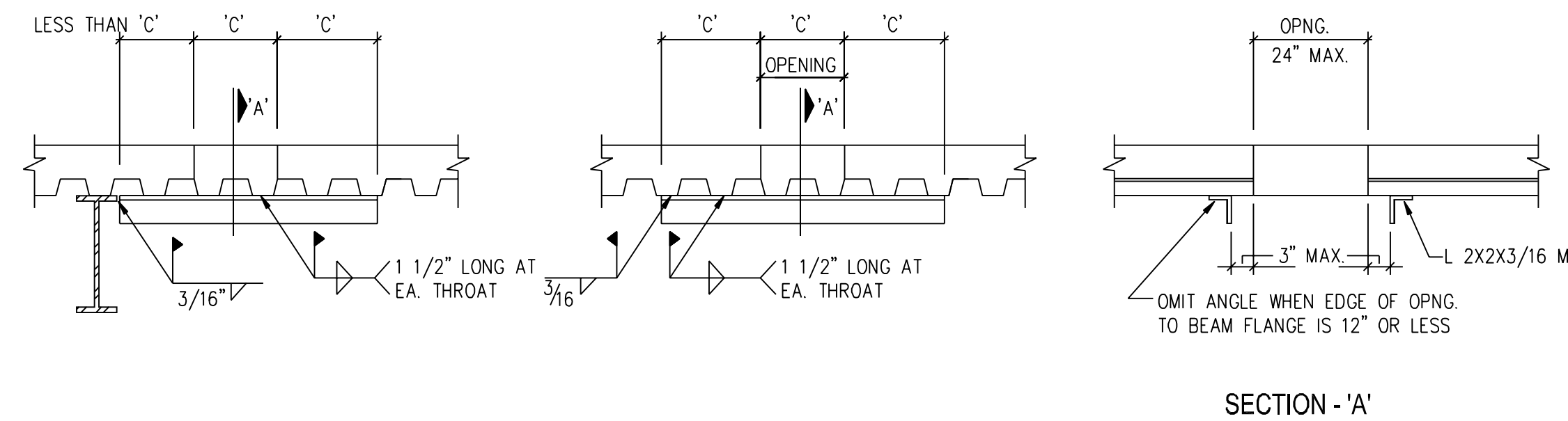
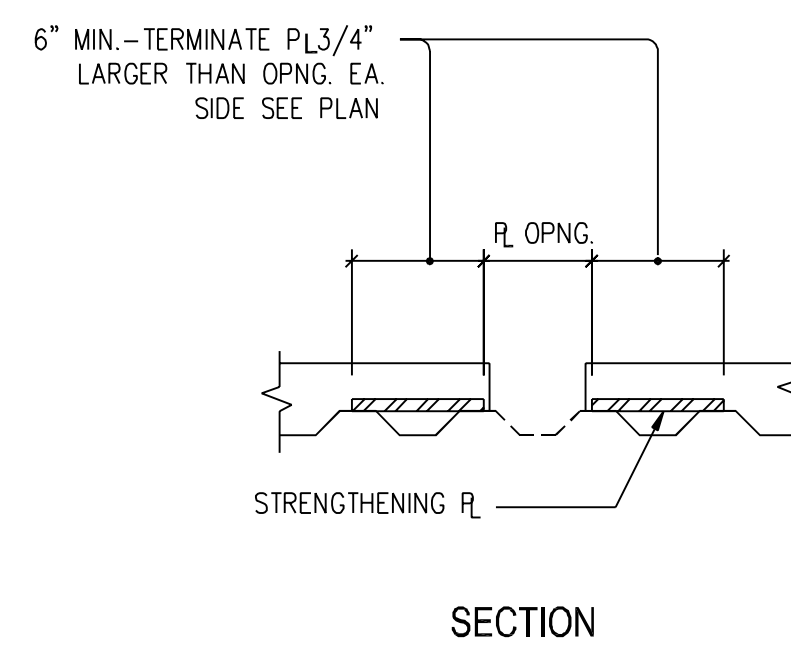


6 CMU MINIMUM REINF. AT WALL OPENING



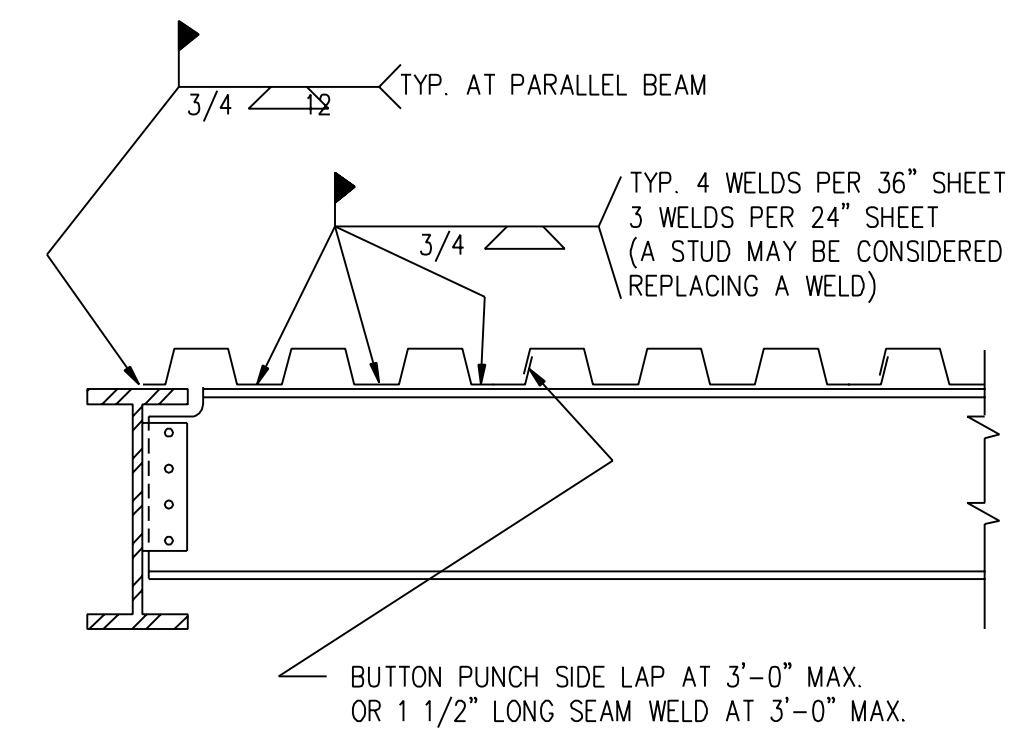
7 DECK OPENING (FOR OPENING GREATER THAN 4" & UPTO 9" DIA. OR SQ.)

NOTE: - FOR OPENINGS EQUAL TO OR SMALLER THAN 4" IN DIA. NO STRENGTHENING REINFORCEMENT IS REQUIRED



8 DECK OPENING (FOR OPENING LARGER THAN 9" & LESS THAN 24" DIA. OR SQ.)

NOTE: SEE PLAN FOR ANGLE SIZES FOR OPENINGS LARGER THAN 24"

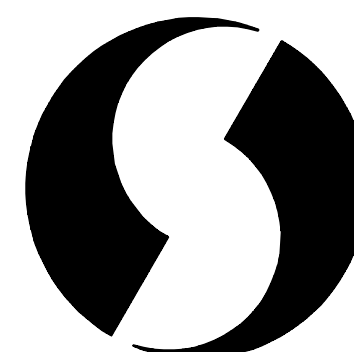


9 TYP WELDING DETAIL FOR METAL DECK ON STRUCT. STEEL

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

**CONCEPT
ENGINEERS, INC.**

50% REVIEW	6-5-01
Revisions	Date



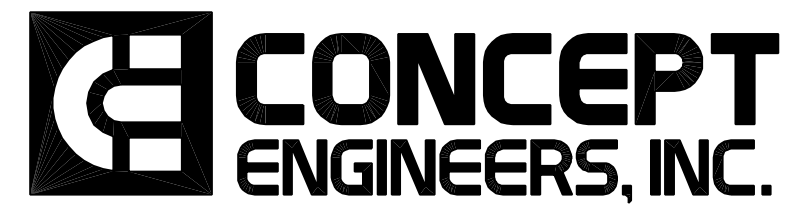
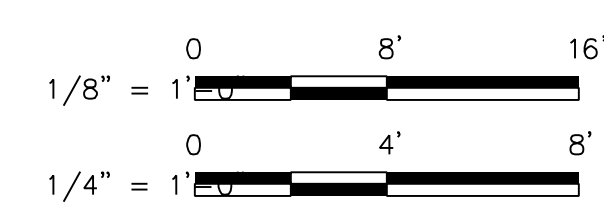
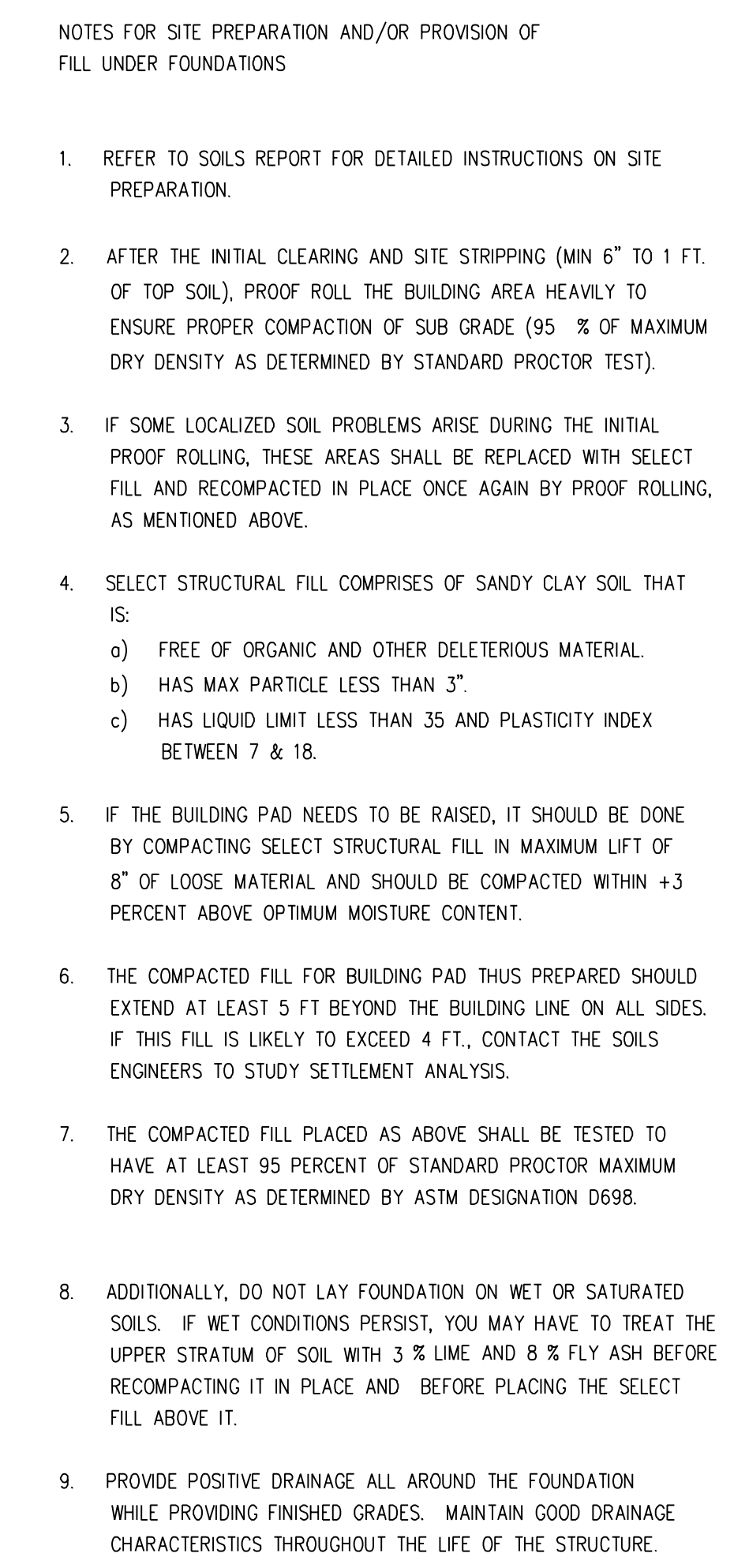
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	TYPICAL DETAILS
Approved: Engineering Svc Chief	
Approved: Medical Center Director	

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
Building Number	104/105
Location	VMC HOUSTON, TX

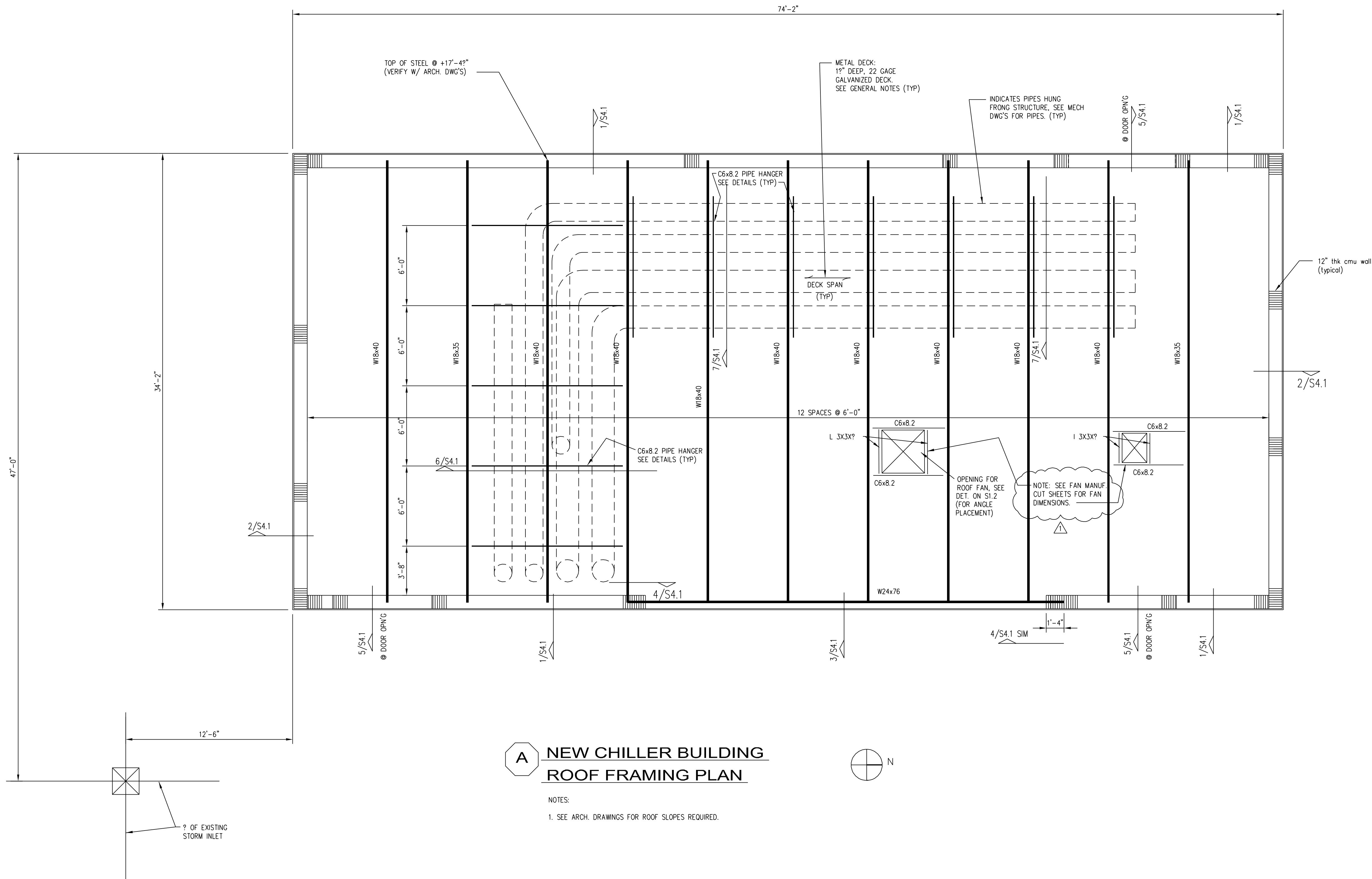
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Project No.	VA #7
Drawing No.	S1.2
Dwg.	of

Veterans
Administration



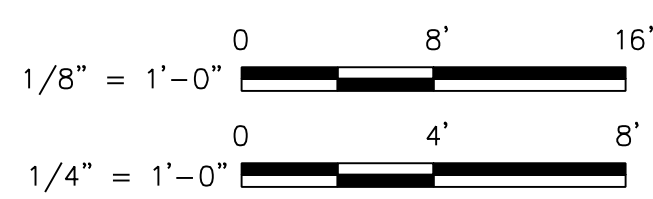
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION			Date 7-31-01	Veterans Administration
			Project No. VA #/ SSA #/ 17-007-1	
Building Number 104-105	Checked YS	Drawn BW	Drawing No. S2.1	
Location WAC HOUSTON, TX			Dwg. of	



A **NEW CHILLER BUILDING**
ROOF FRAMING PLAN

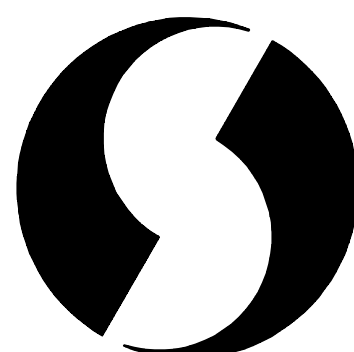
NOTES:
1. SEE ARCH. DRAWINGS FOR ROOF SLOPES REQUIRED.



RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

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REVISION #1	8-14-01
Revisions	Date



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Phone: 713-780-7563

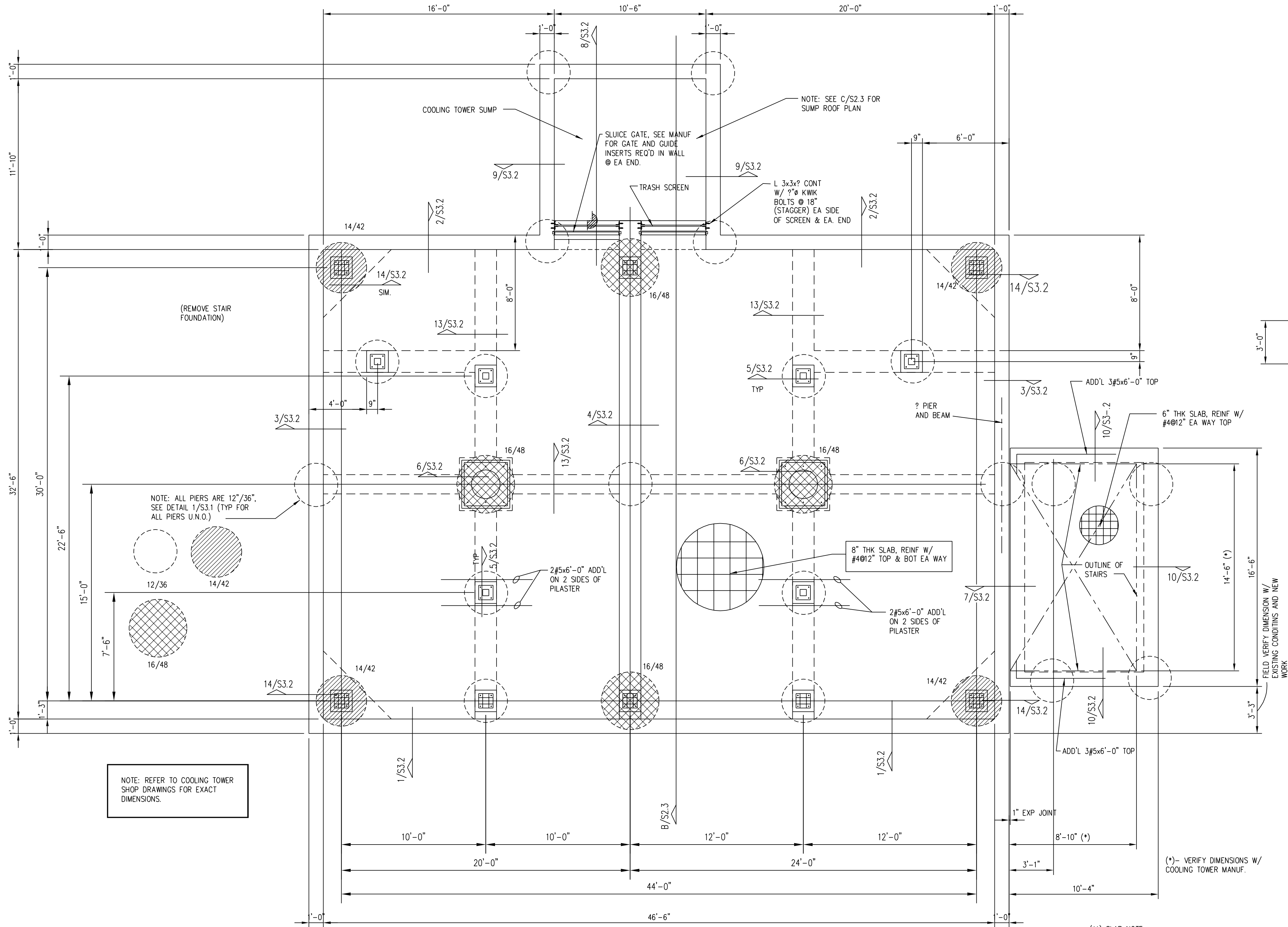
Fax: 713-780-9209

Drawing Title
NEW CHILLER BUILDING ROOF FRAMING PLAN
Approved: Engineering Svc Chief
Approved: Medical Center Director

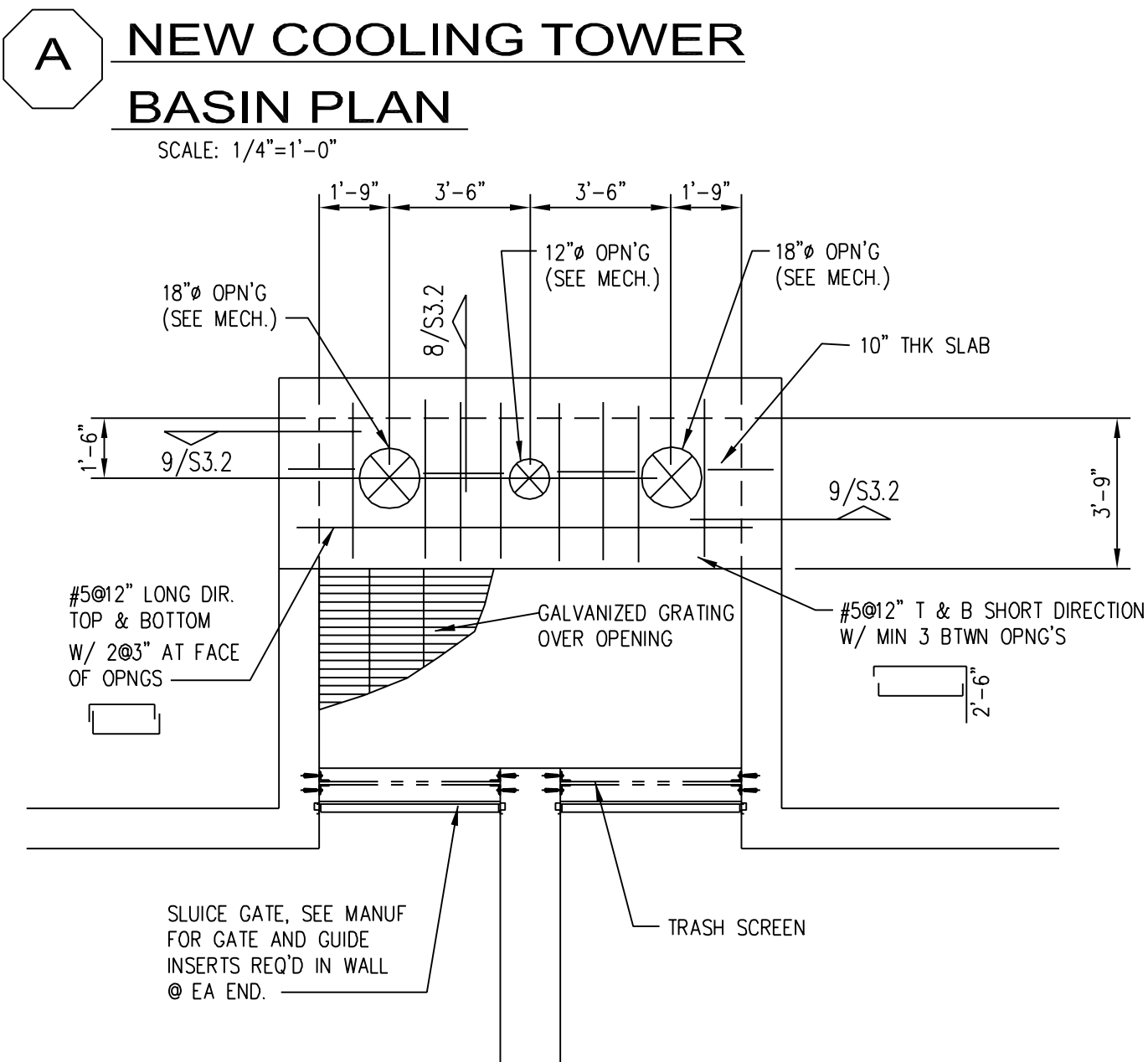
Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION
Building Number
104/105
Location
WAMC HOUSTON, TX

Date
7-31-01
Project No.
VA #17 SSA#1017-007-1
Drawing No.
S2.2
Dwg. of

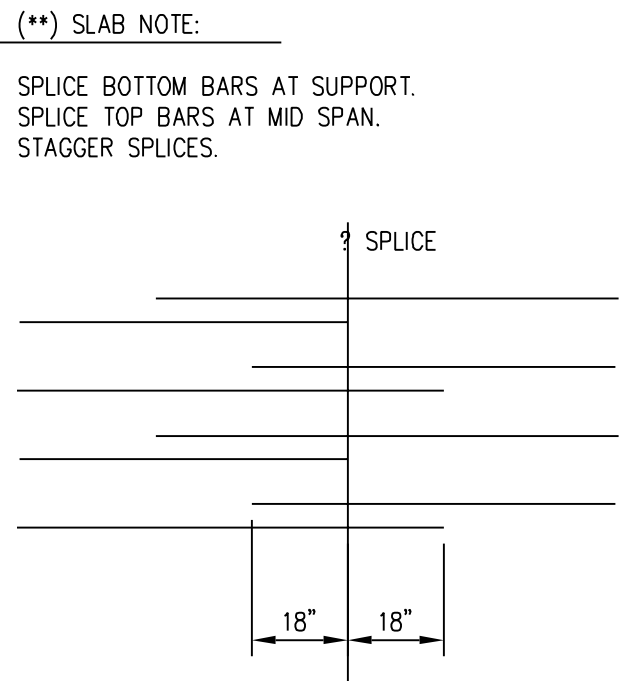
Veterans
Administration



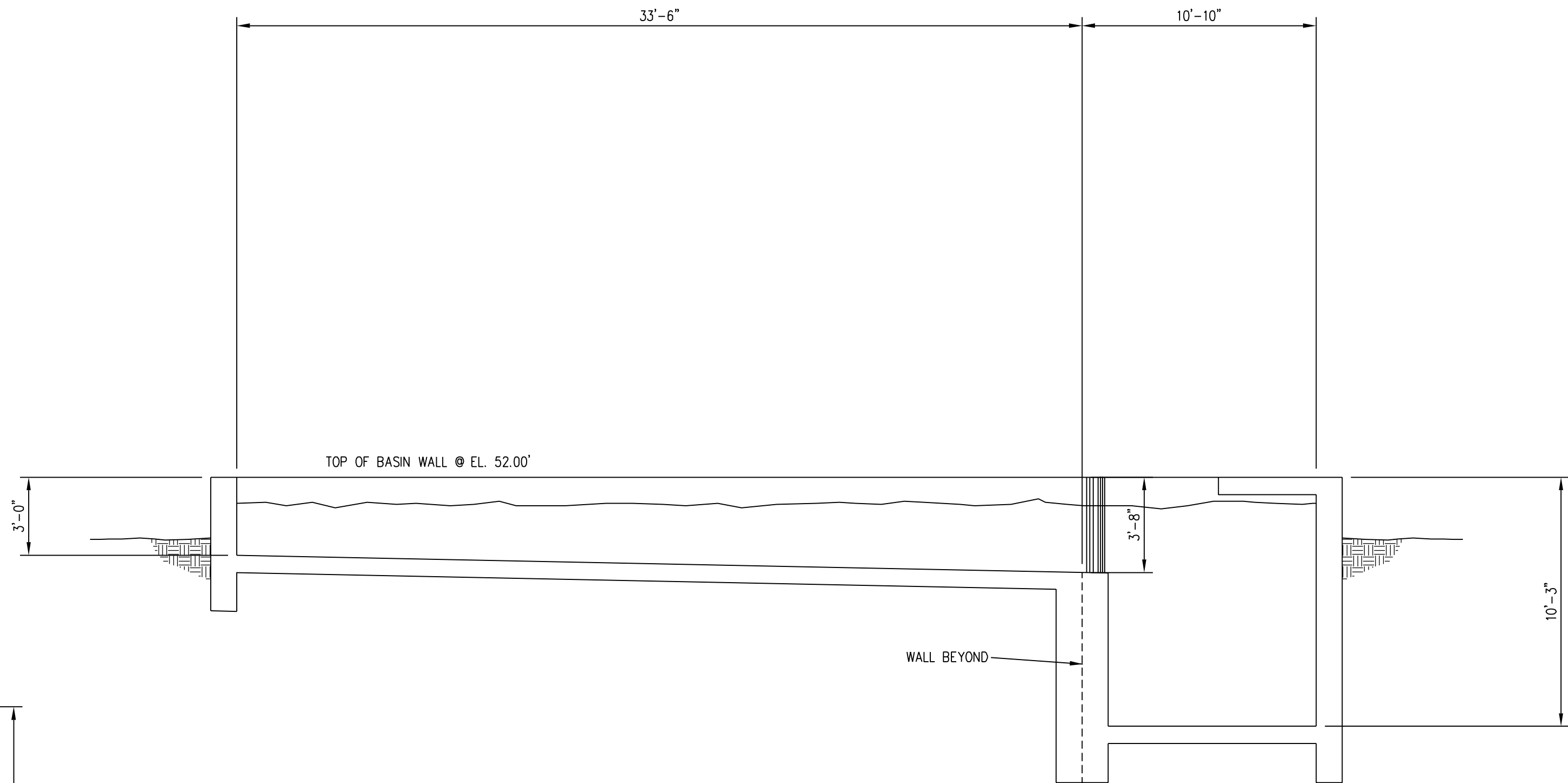
- NOTES:
- SEE ARCHITECTURAL DRAWINGS FOR SLAB EDGE DIMENSIONS AND WALL LOCATIONS. VERIFY ALL DIMENSIONS GIVEN ON THIS DRAWING WITH ARCHITECTURAL DRAWINGS BEFORE CONSTRUCTION.
 - VERIFY BRICK LEDGES, DROP LOCATIONS, SLAB SLOPES ETC. WITH THE ARCHITECTURAL DRAWINGS.
 - THIS DRAWING IS STRICTLY DRAWN TO SHOW STRUCTURAL DETAILS. IT IS NOT OUR INTENT TO INCLUDE AND SHOW ANY EMBEDMENTS OR OPENINGS REQUIRED FOR MECHANICAL/ELECTRICAL/PLUMBING SERVICES. SEE RELEVANT DRAWINGS FOR THOSE DETAILS.
 - SEE SHEET S3.1 FOR DETAILS, NOTES AND SPECS. ON THIS DESIGN.
 - PLEASE NOTE THAT PLACEMENT OF SELECT FILL UNDER THE FOUNDATION SLAB CAN SUBSTANTIALLY CHANGE THE SHRINK/SWELL POTENTIAL OF THE ON SITE SOILS. THE SOILS REPORT REQUIRES AT LEAST 3 FEET OF PROPERLY COMPACTED STRUCTURAL FILL BETWEEN THE SLAB AND NATURAL SOIL. SEE SITE PREPARATION NOTES ON S2.1 AND IN THE SOILS REPORT.
 - DUE TO HIGH PLASTICITY OF SOILS AT THIS SITE, 6" DEEP VOID BOXES MAY BE PROVIDED UNDER GRADE BEAMS IF SO DESIRED. THESE VOID BOXES WORK EXTREMELY WELL IF POSITIVE DRAINAGE HAS BEEN PROVIDED ALL AROUND THE BUILDING. IF NOT, THESE VOID BOXES UNDER THE BEAMS HAVE BEEN FOUND TO HAVE ADVERSE EFFECT.



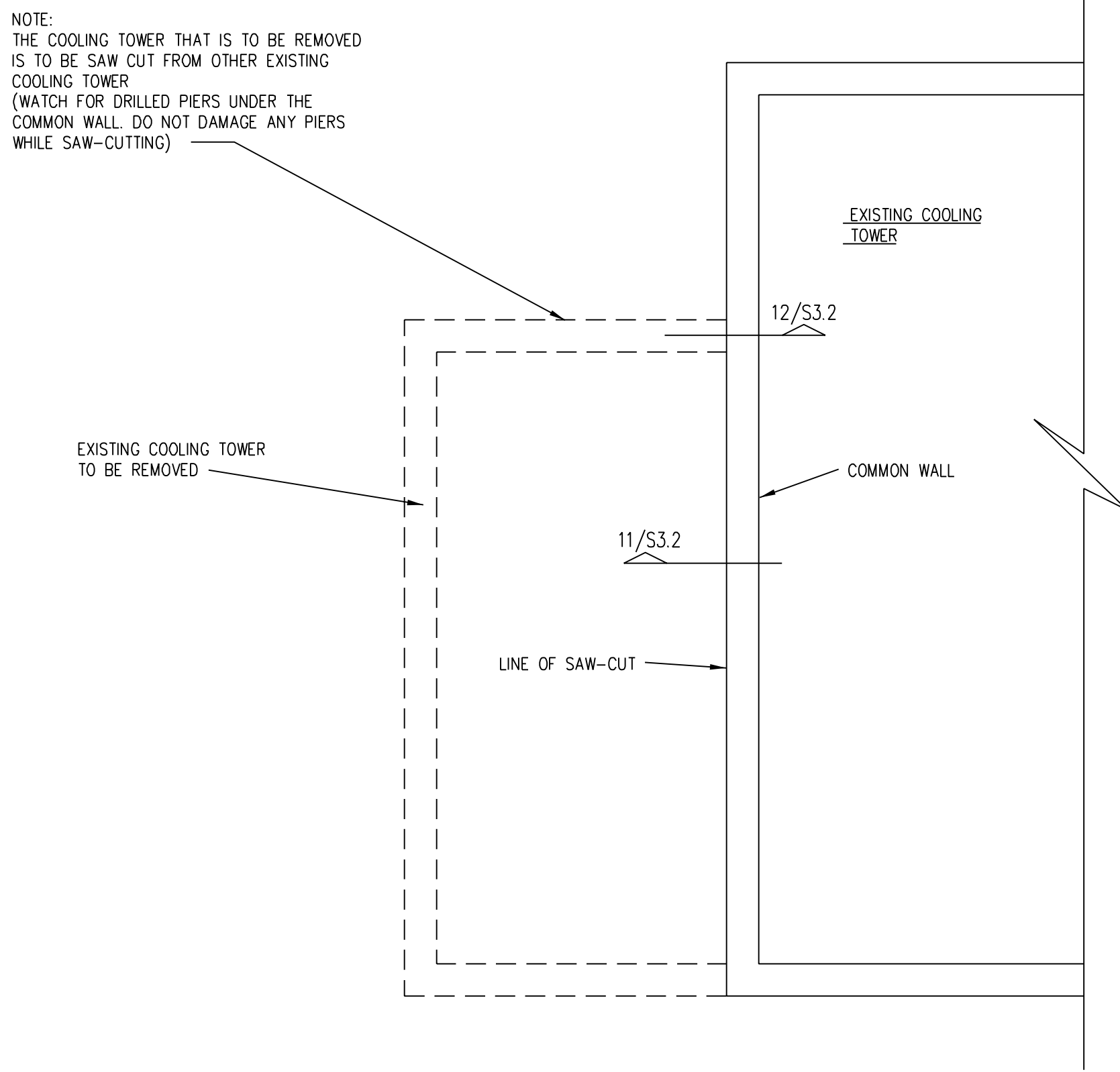
C NEW COOLING TOWER SUMP ROOF PLAN



(**) SLAB NOTE:
SPlice BOTTOM BARS AT SUPPORT.
SPlice TOP BARS AT MID SPAN.
STAGGER SPICES.

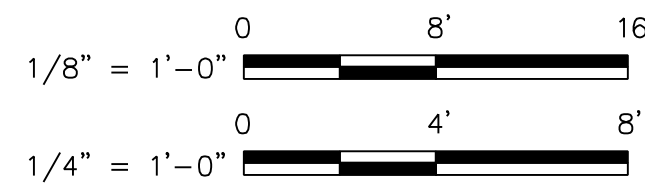


B NEW COOLING TOWER BASIN SECTION



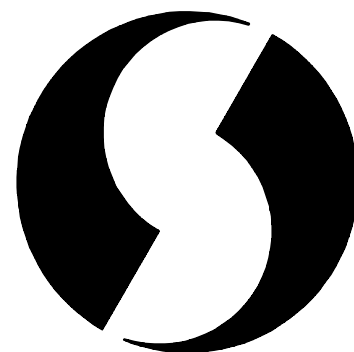
D EXISTING COOLING TOWER PARTIAL PLAN

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.



CONCEPT ENGINEERS, INC.

REVISION # 1		8-14-01
REVISION # 2		10-29-01
REVISION # 3	COOLING TOWER COORDINATION	11-27-01
Revisions		Date



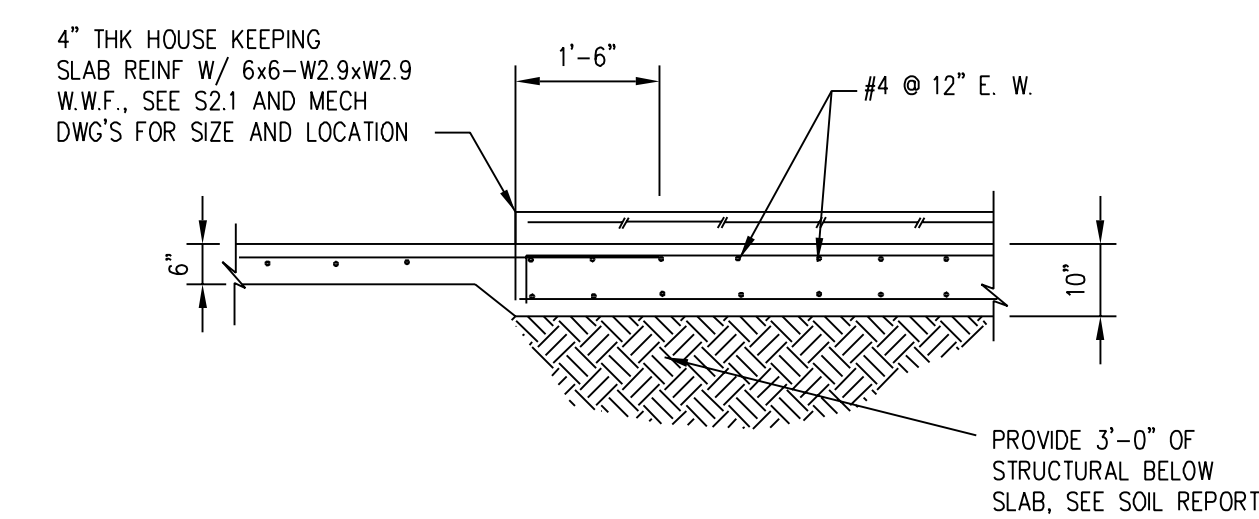
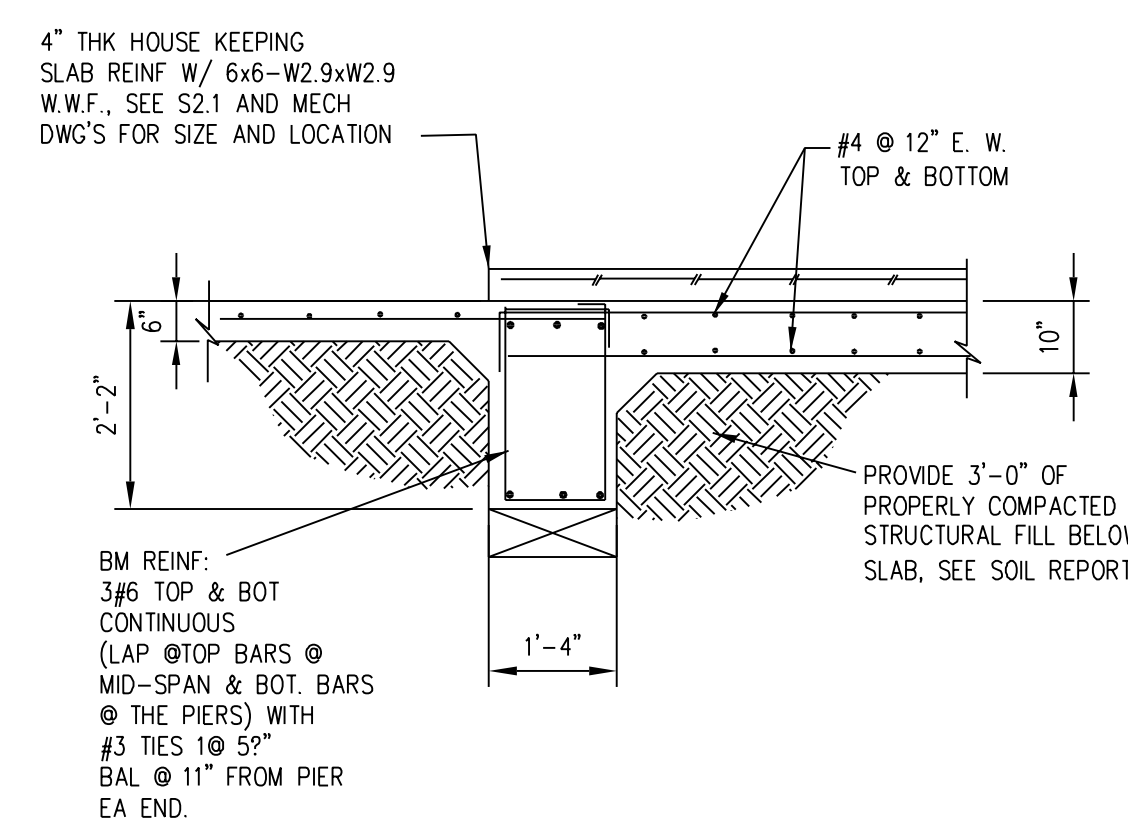
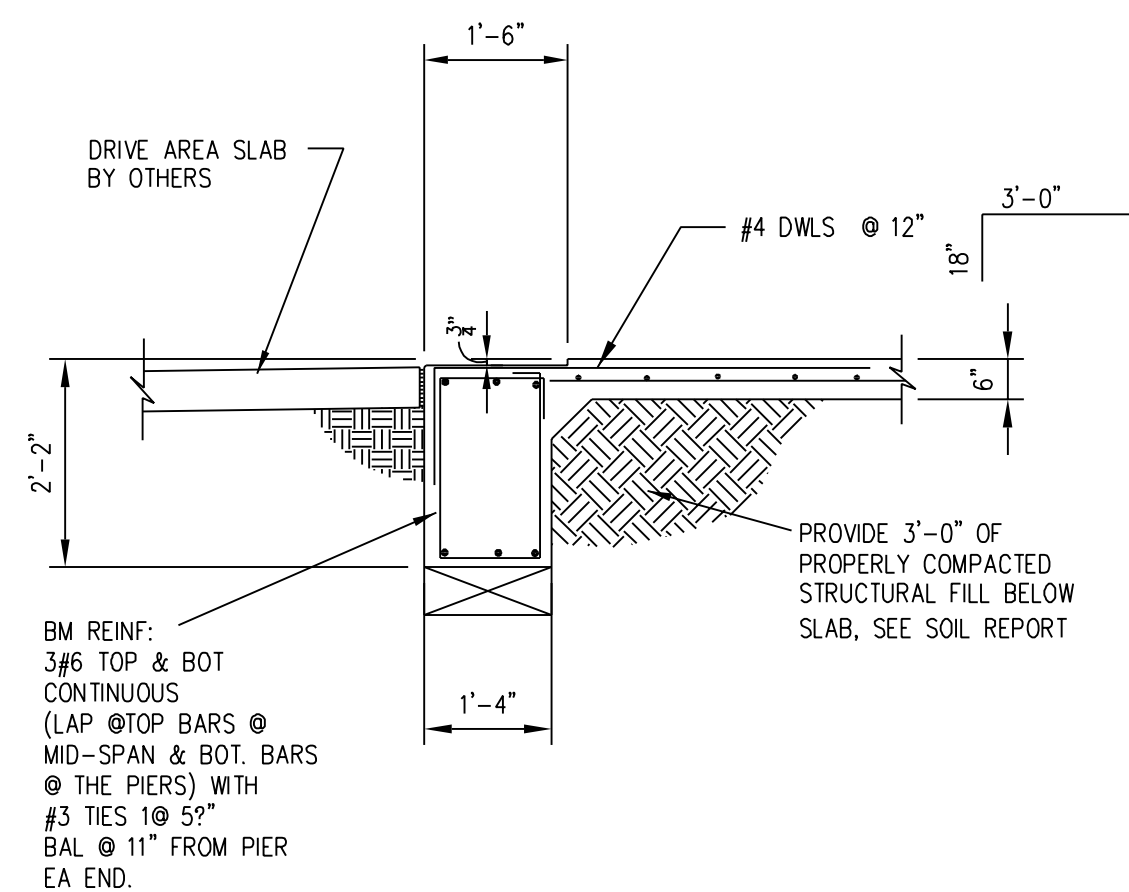
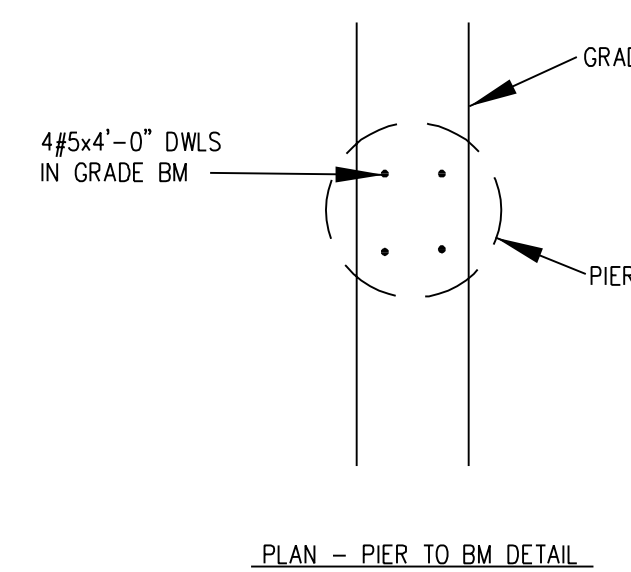
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	NEW COOLING TOWER FOUNDATION PLAN
Approved: Engineering Svc Chief	
Approved: Medical Center Director	

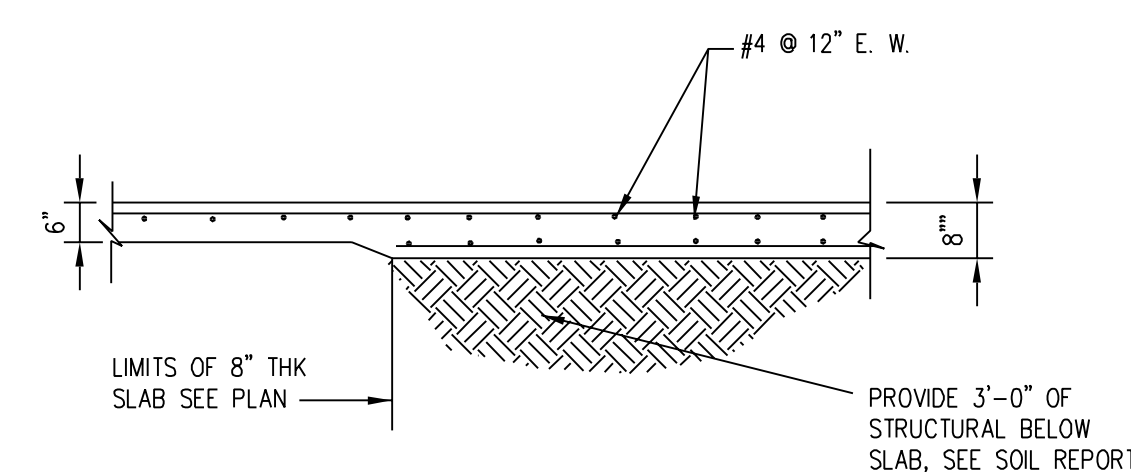
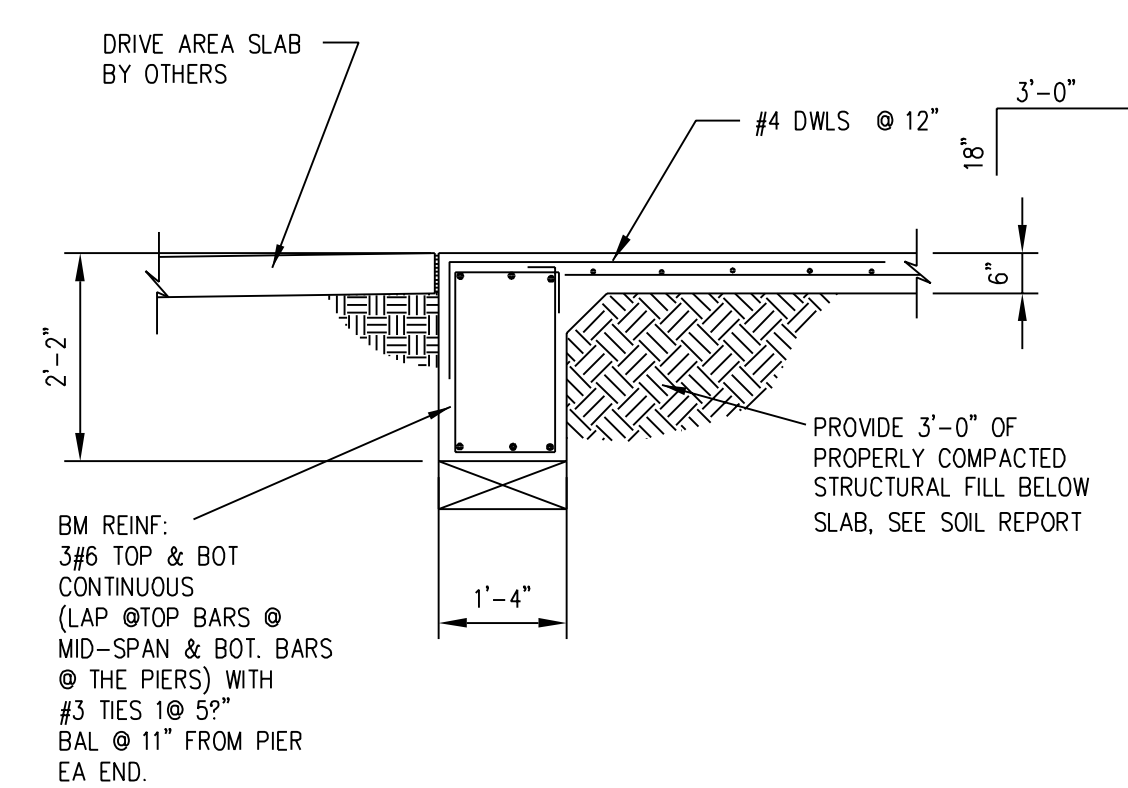
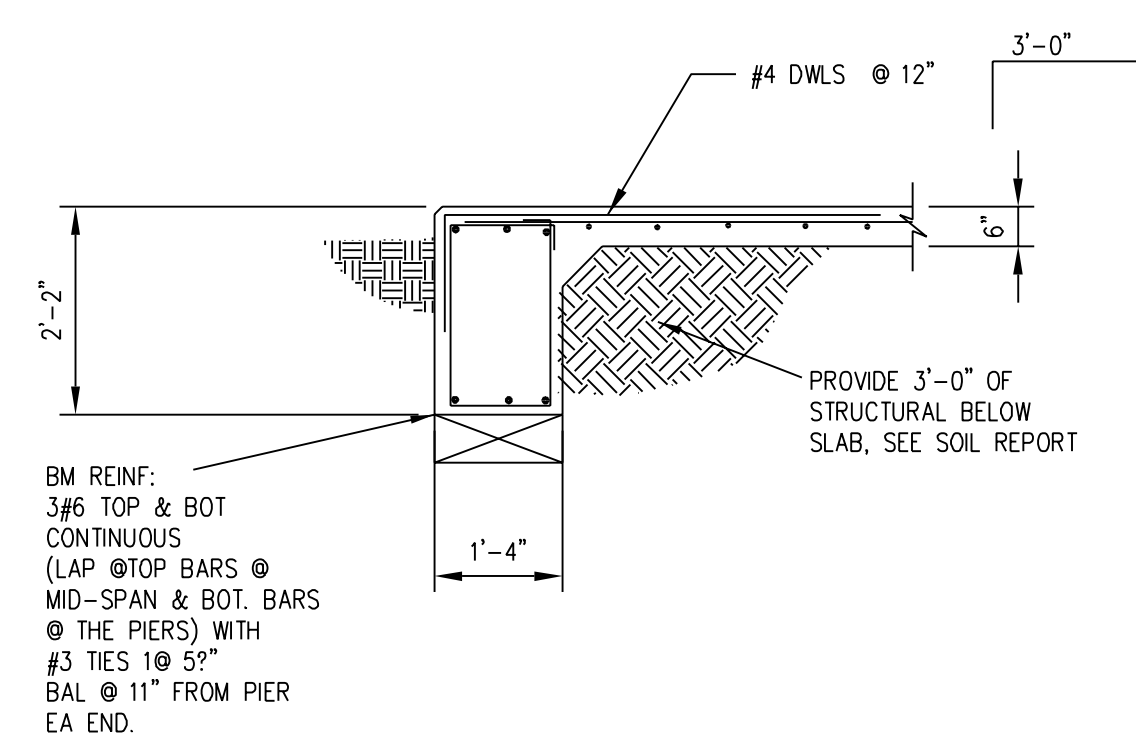
Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
Building Number	104/105
Location	WAMC HOUSTON, TX
Checked	YS
Drawn	BW

Date	7-31-01
Project No.	VA #17
SSA #1017-007-1	
Drawing No.	S2.3
Dwg.	of

Veterans Administration

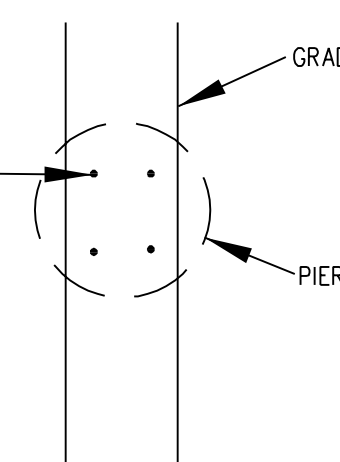


5 SECTION

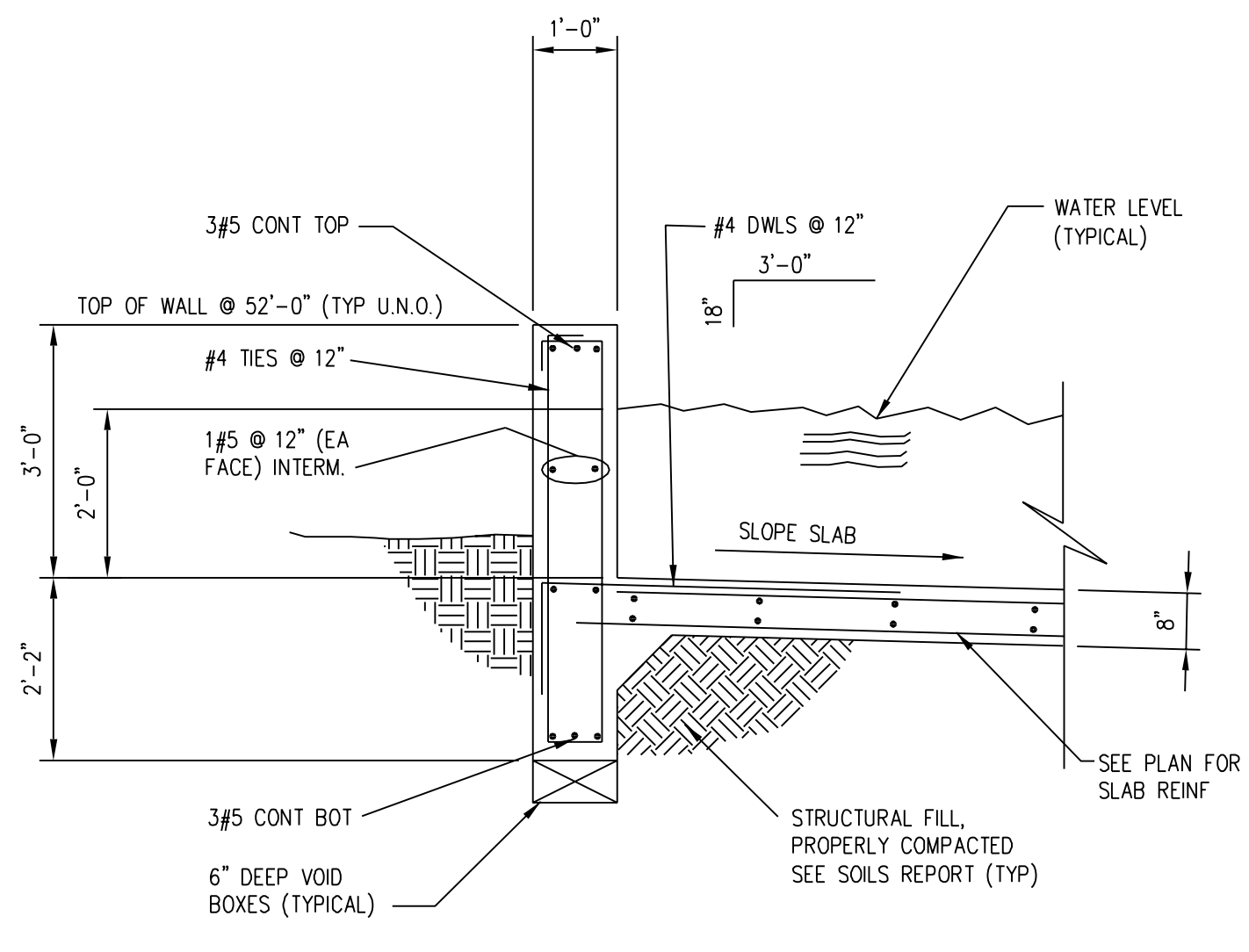


NOTE: ADD'L REBAR AROUND SLAB OPENINGS
NOT SHOWN FOR CLARITY. (SEE PLAN)

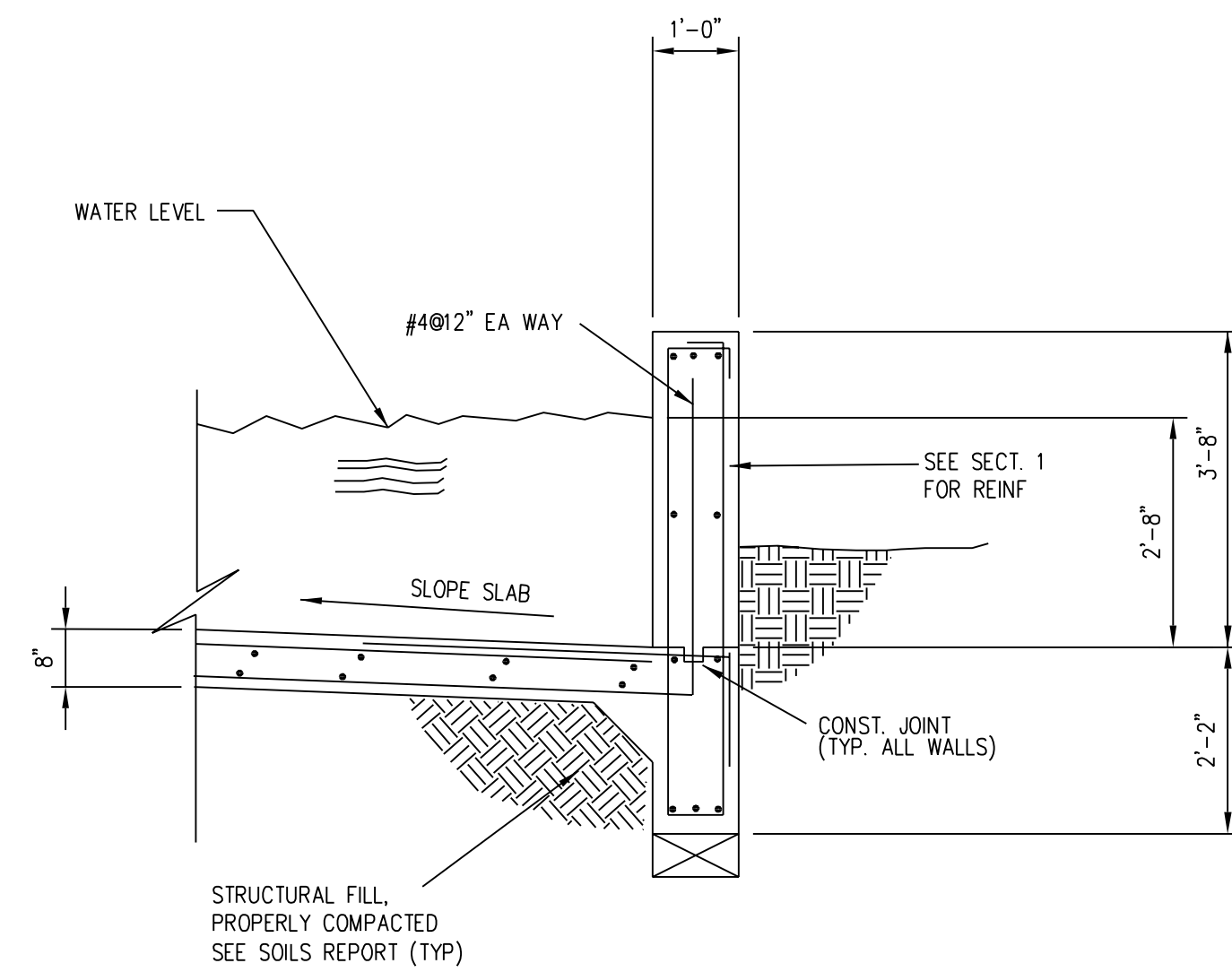
8 SECTION



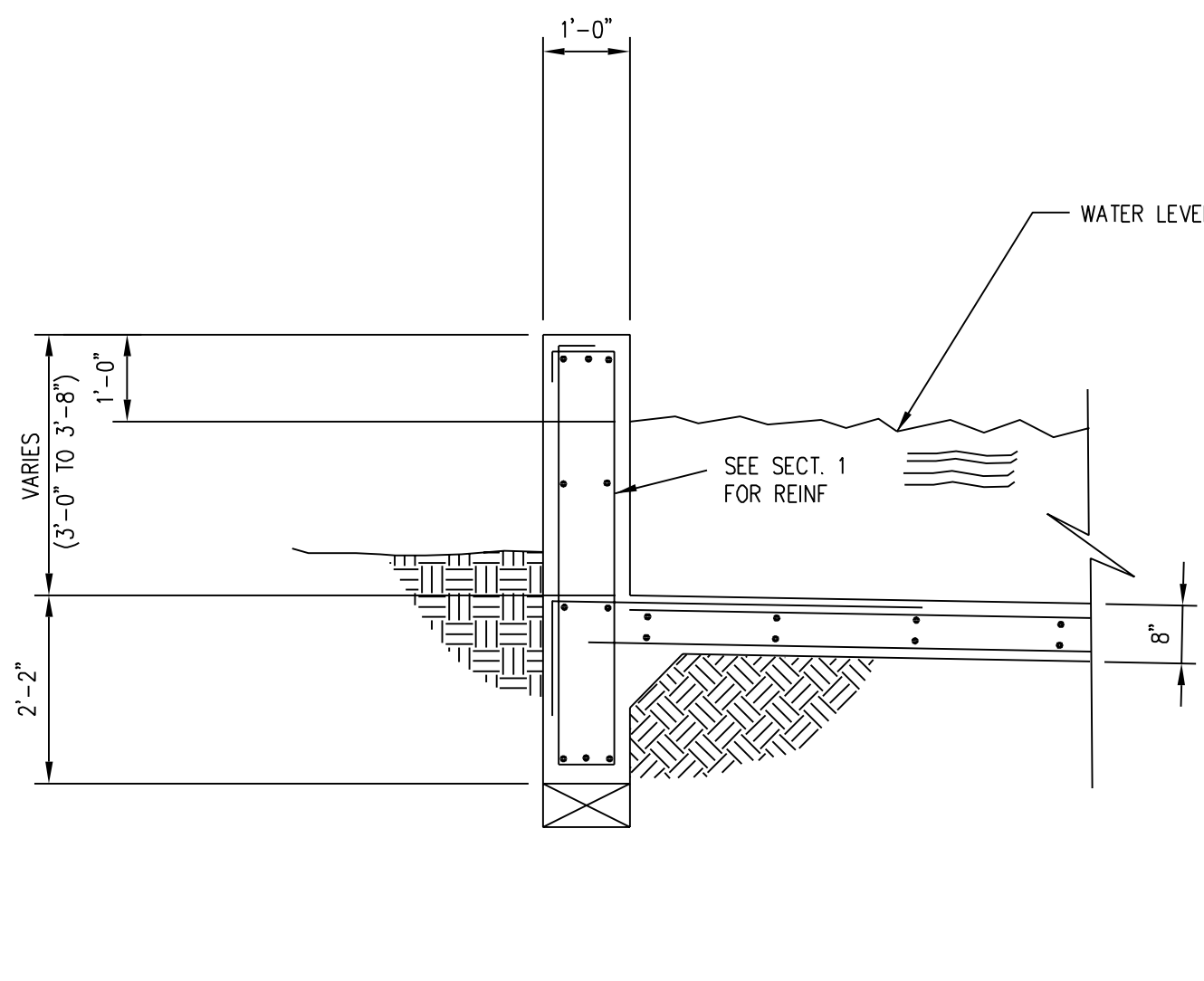
Veterans Administration



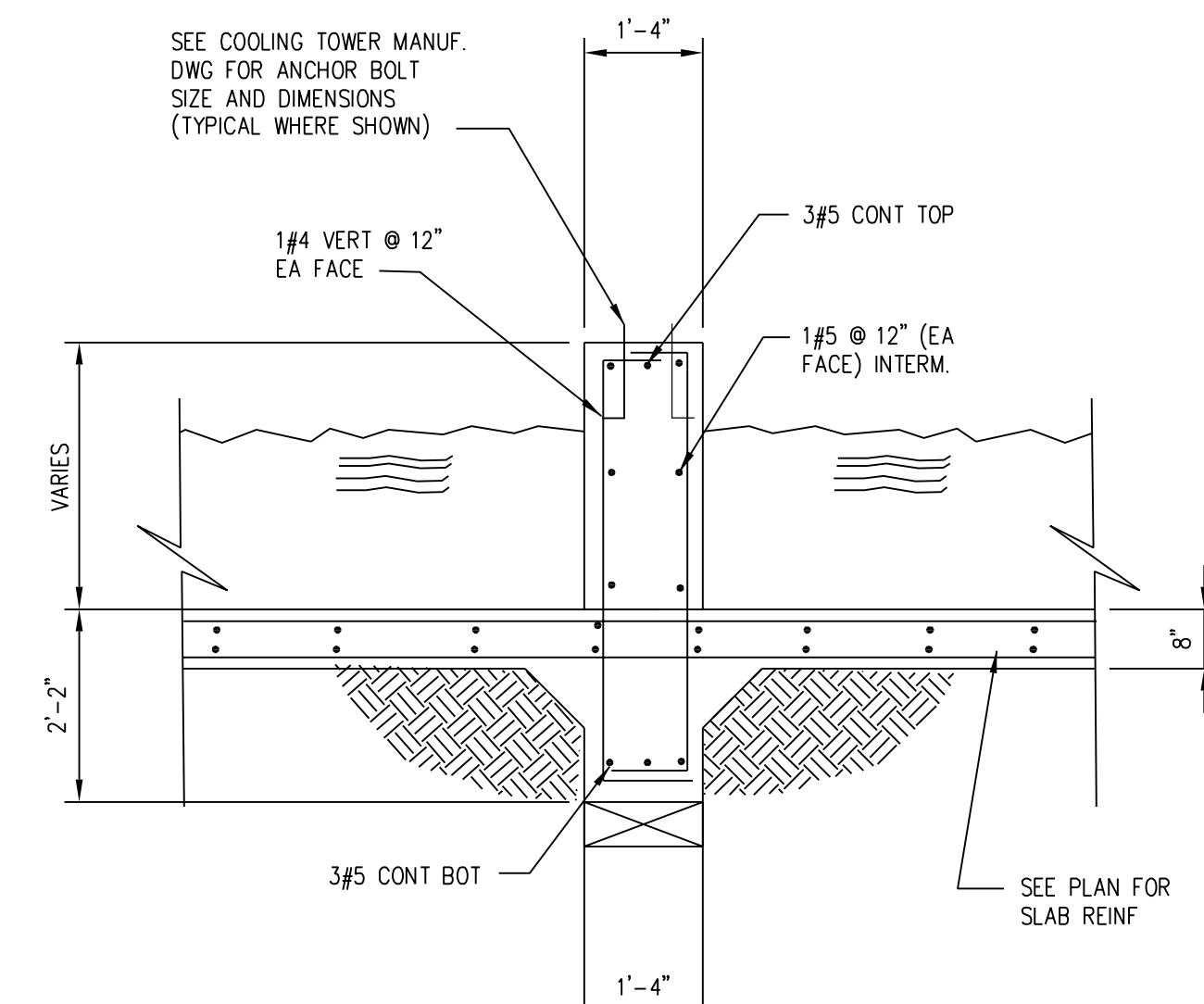
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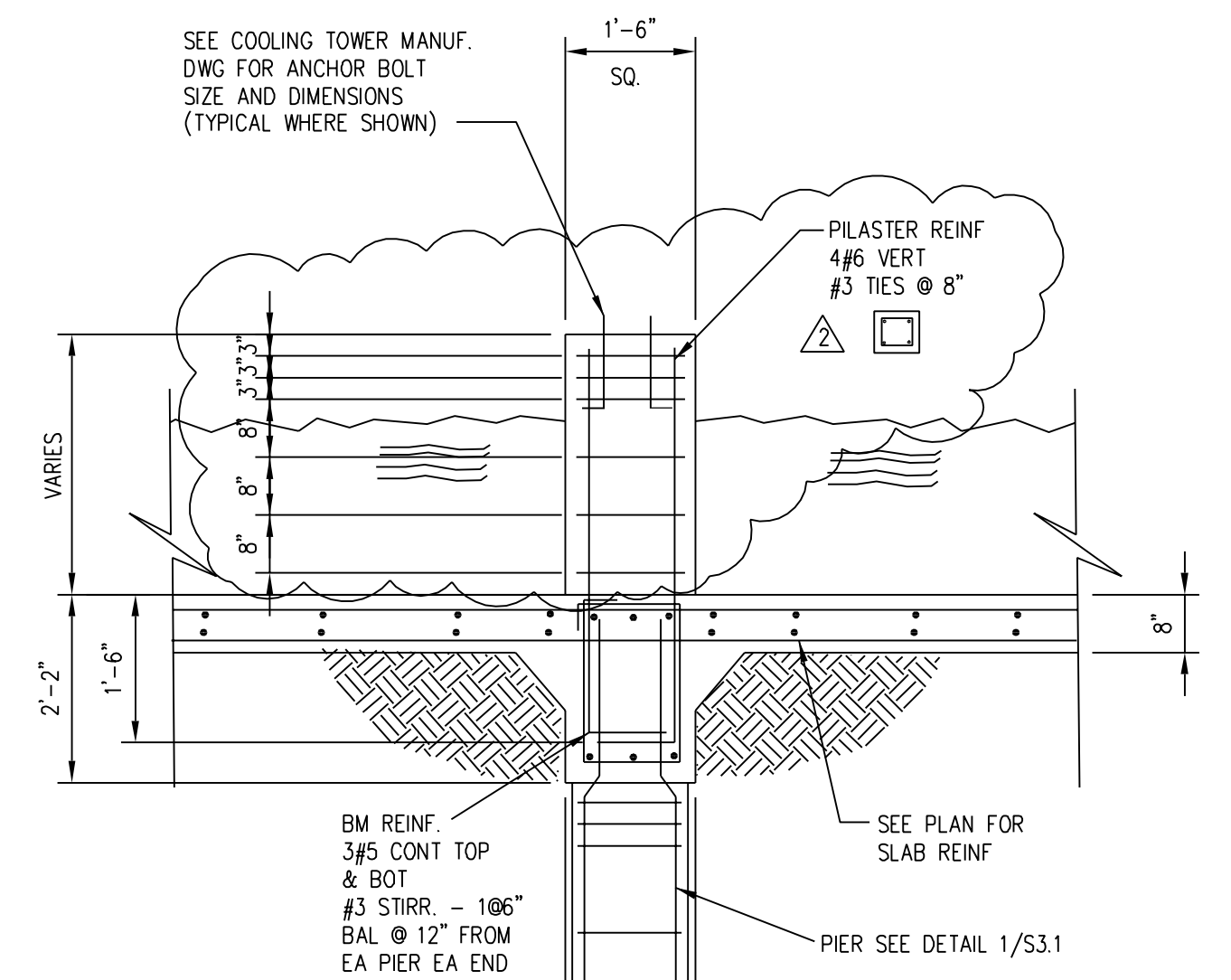
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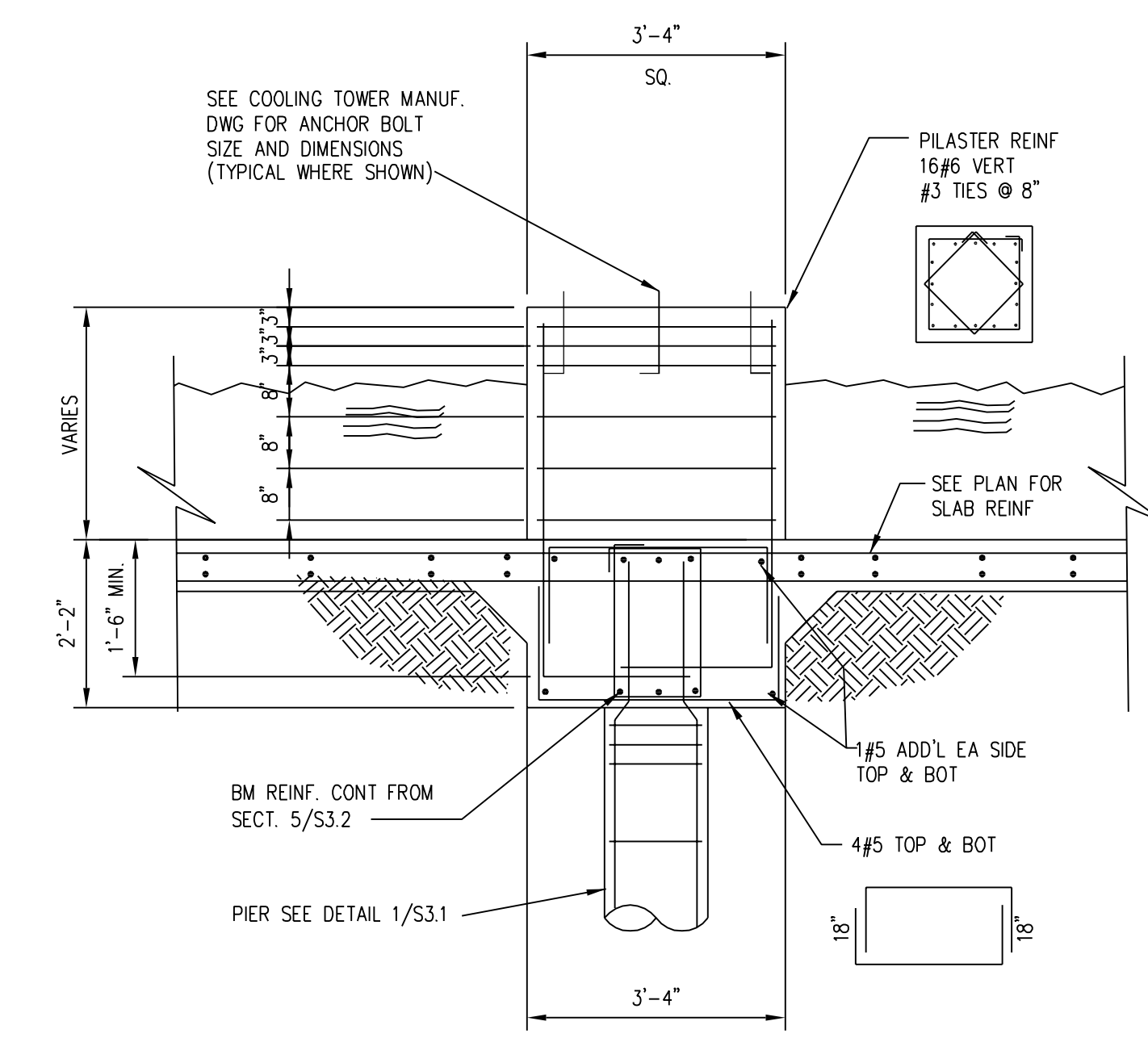
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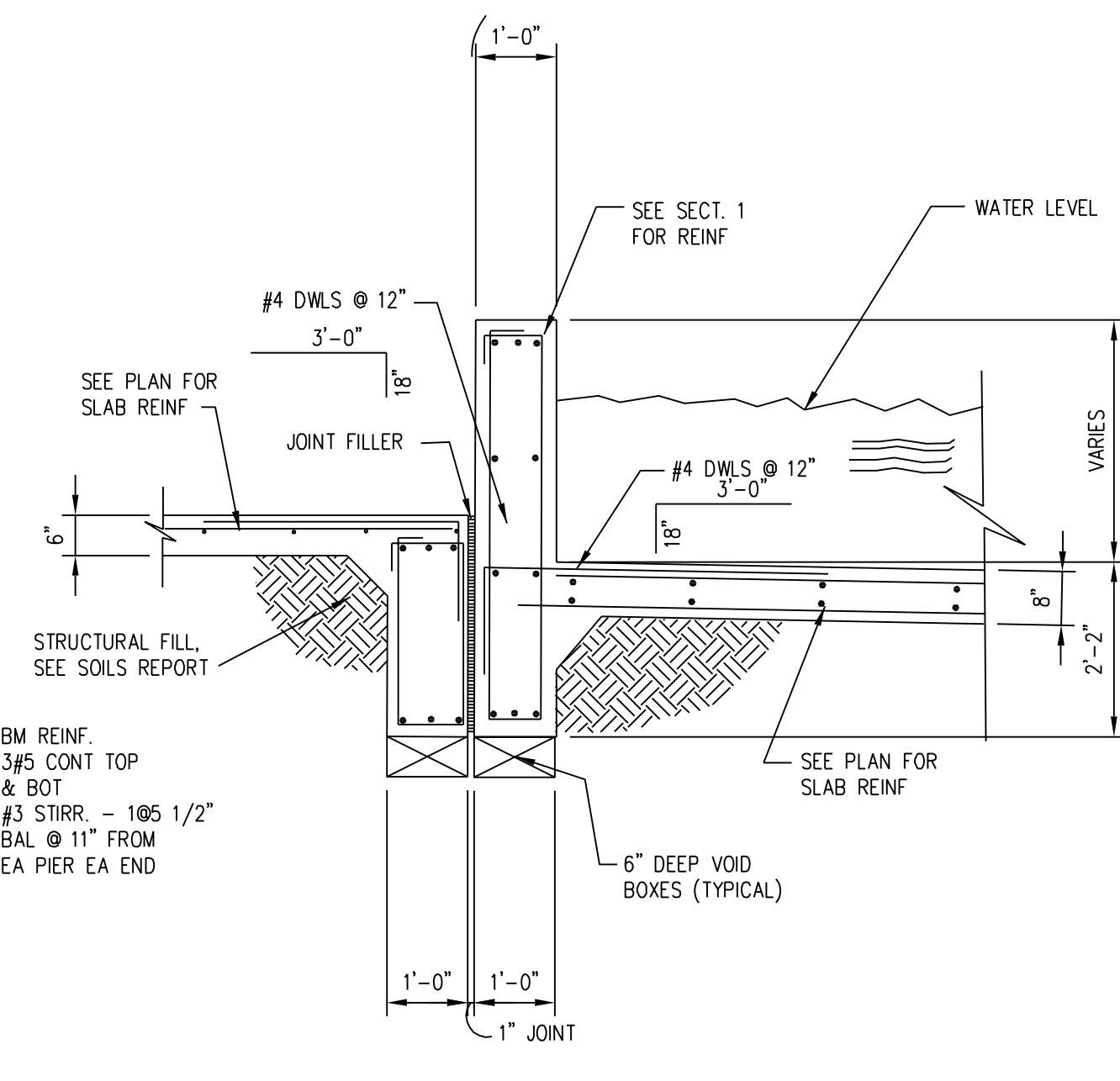
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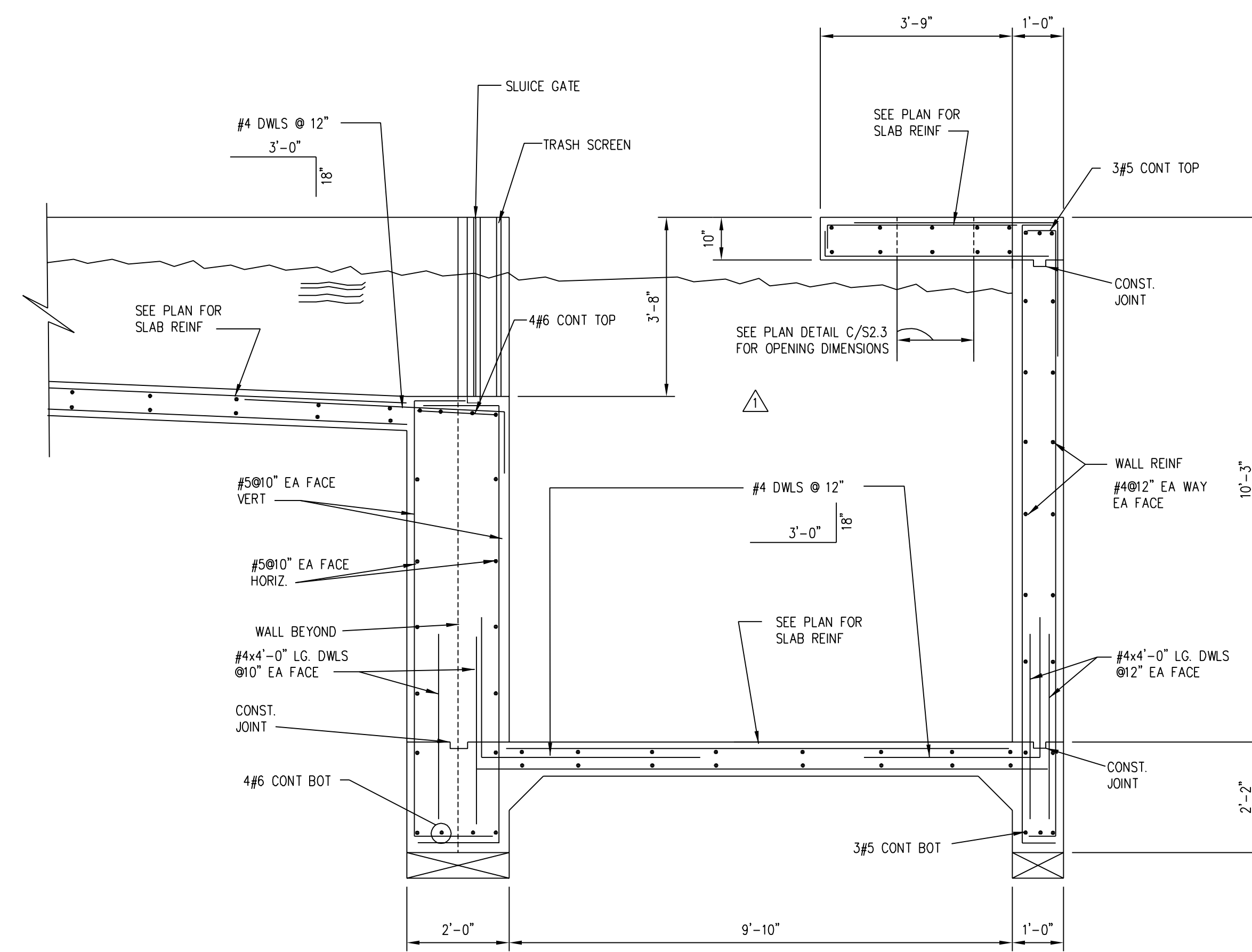
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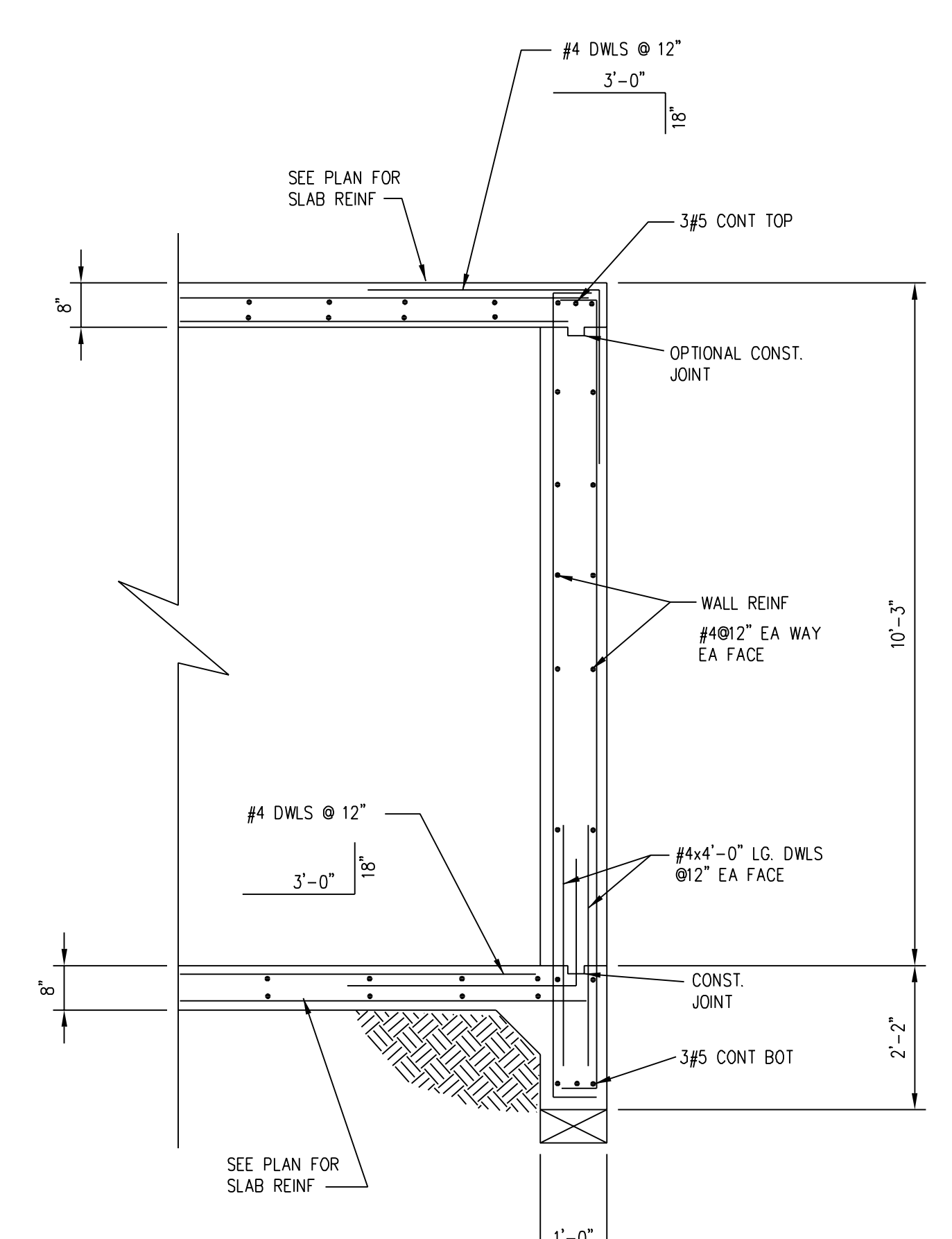
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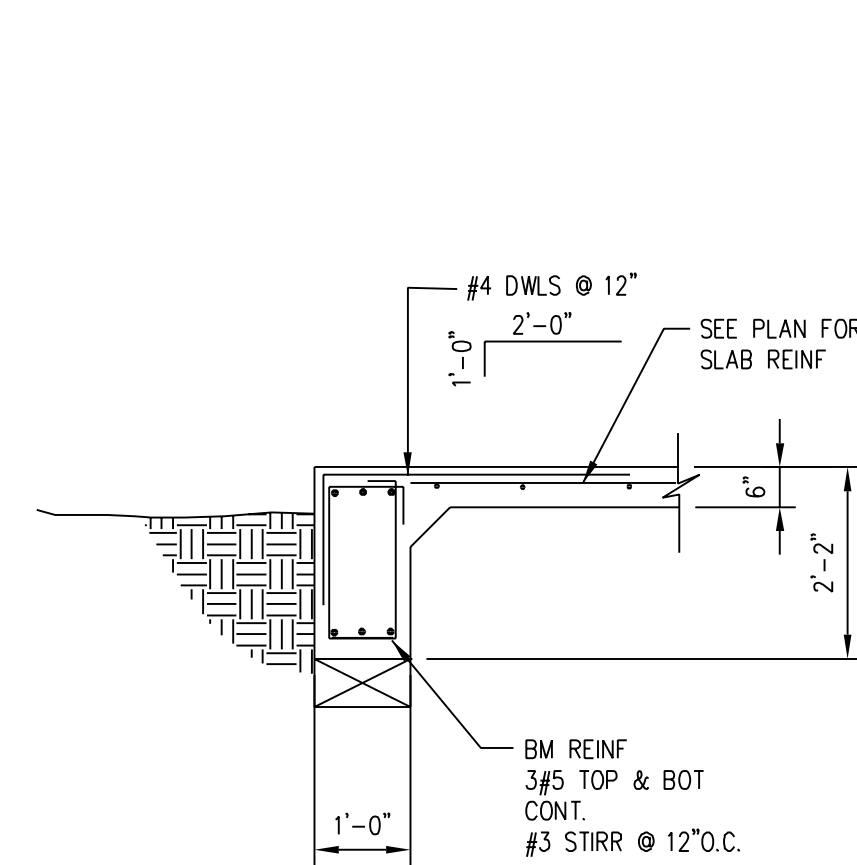
7 SECTION



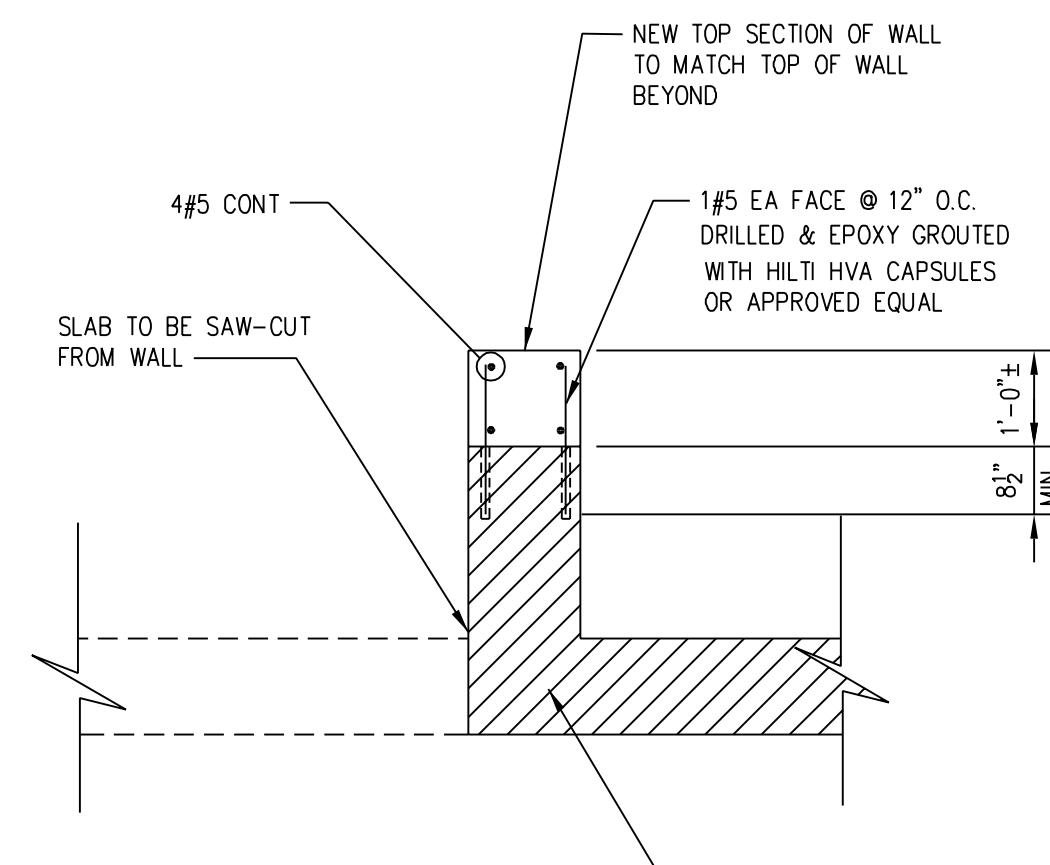
8 SECTION



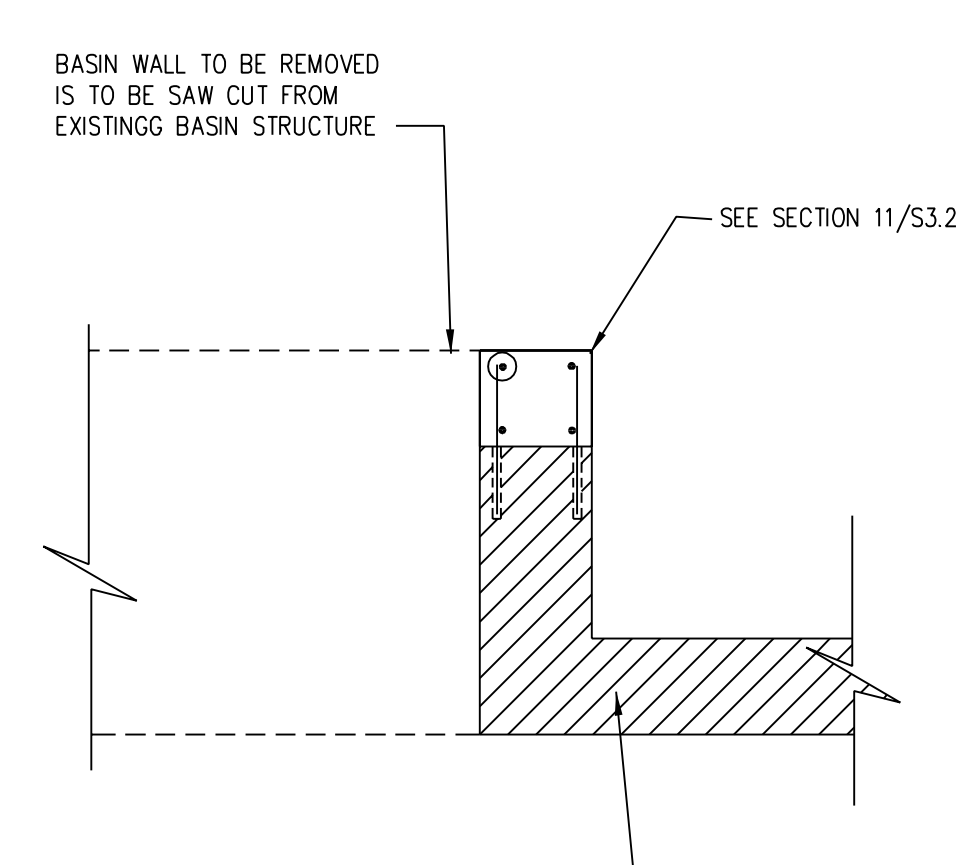
9 SECTION



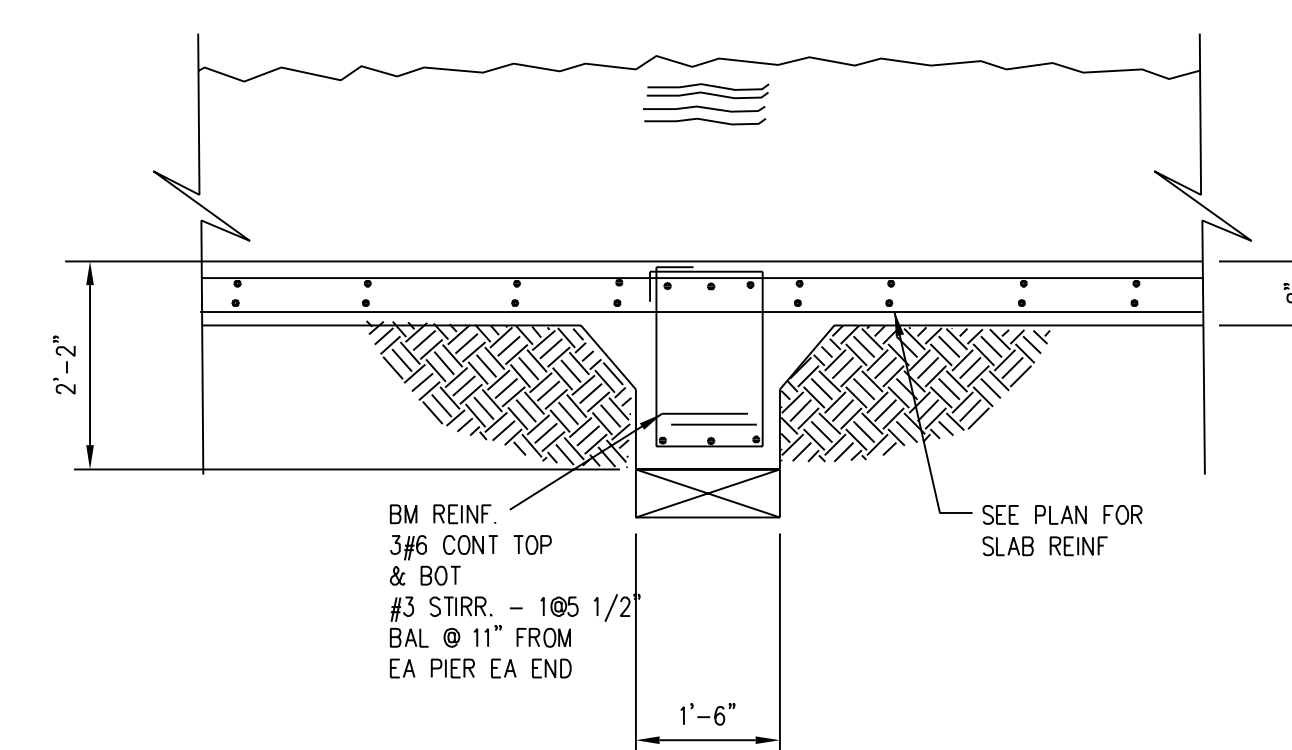
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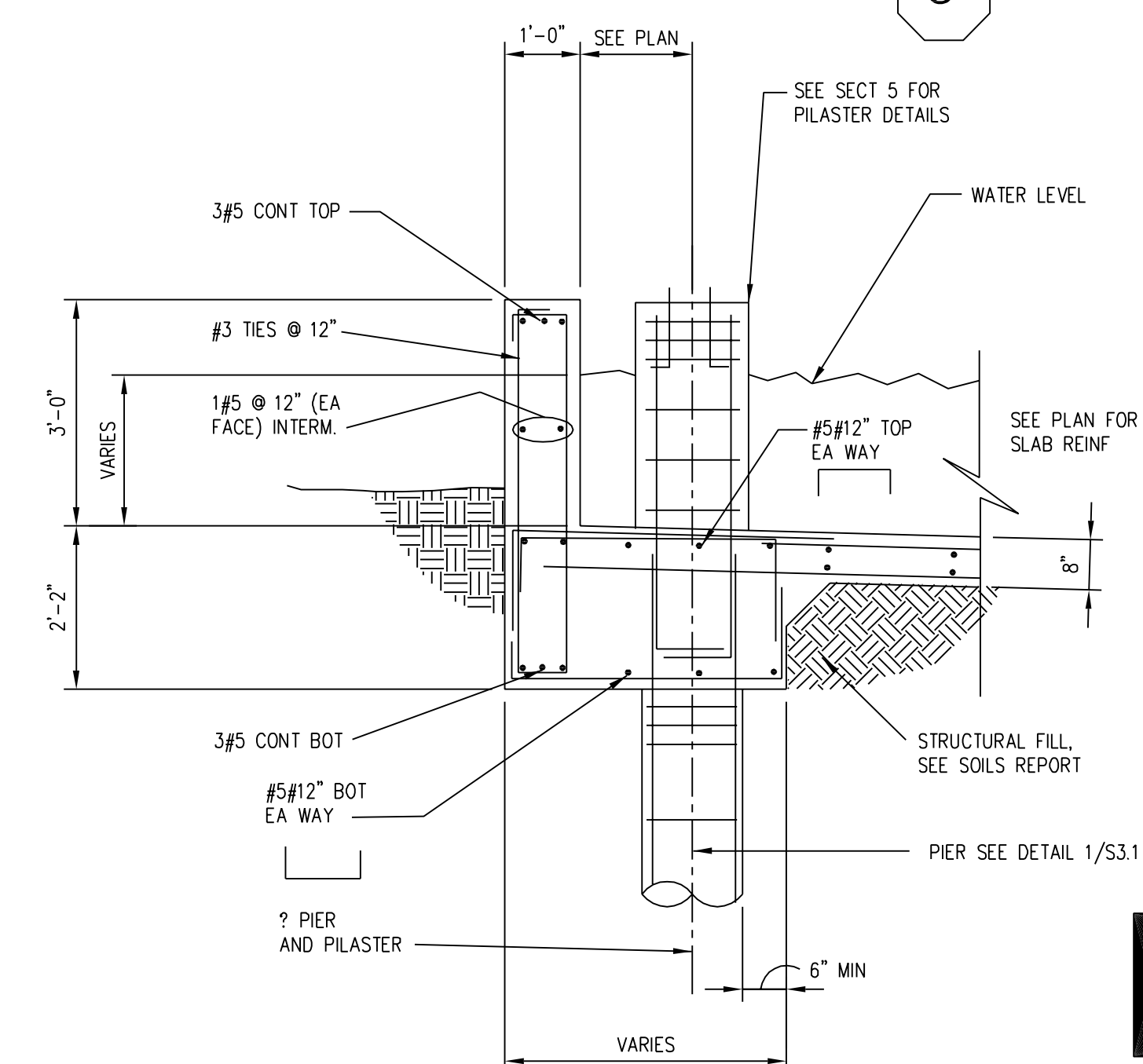
11 SECTION



12 SECTION



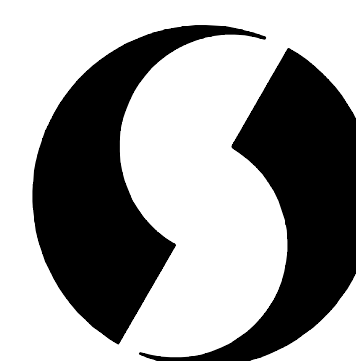
13 SECTION



RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.

CONCEPT ENGINEERS, INC.

REVISION # 1	8-14-01
REVISION # 2	10-29-01
Revisions	Date



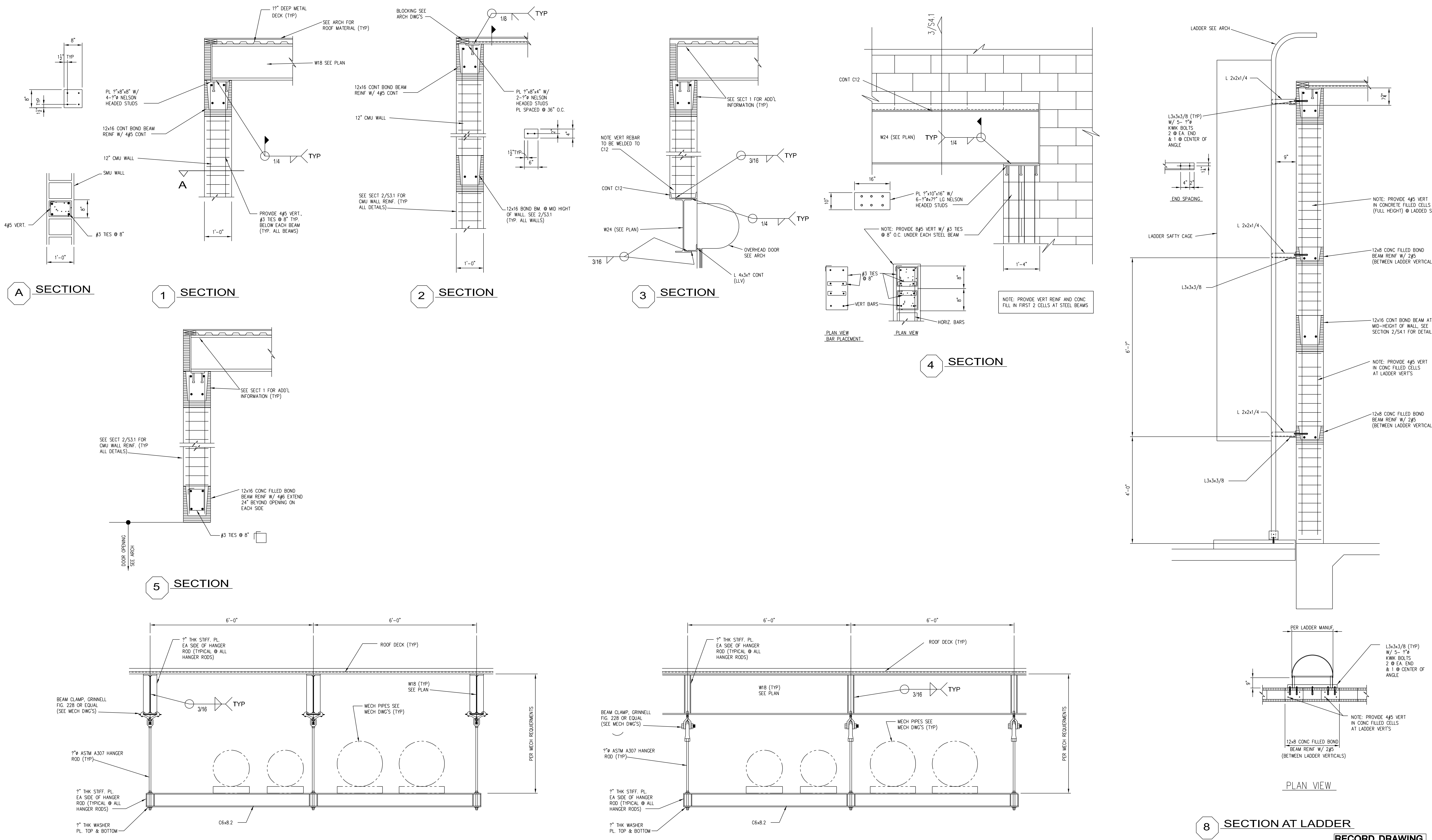
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	COOLING TOWER BASIN FOUNDATION DETAILS	
Approved: Engineering Svc Chief		
Approved: Medical Center Director		

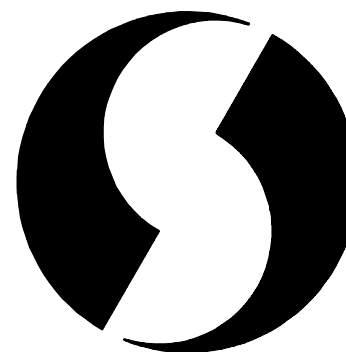
Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER	
Building Number	104/105	Checked
Location	WAMC HOUSTON, TX	Drawn

Date	7-31-01
Project No.	VA #17
Drawing No.	S3.2
Dwg. of	

Veterans Administration



REVISION # 1	8-14-01
Revisions	Date



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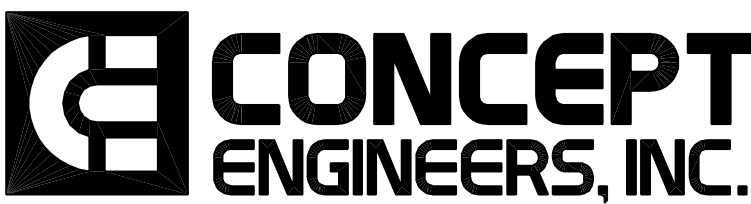
Drawing Title	SECTION AND DETAILS	
Approved: Engineering Svc Chief		
Approved: Medical Center Director		

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION	
Building Number	104/105	Checked YS
Location	VAMC HOUSTON, TX	

Date	07-31-01
Project No.	VA #/ SSA #017-007-1
Drawing No.	S4.1
Dwg.	of

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Administration

RECORD DRAWING
THIS DRAWING HAS BEEN REVISIED TO SHOW
SIGNIFICANT CHANGES IN THE WORK PAGE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, CHANGES AND OTHER DATA.

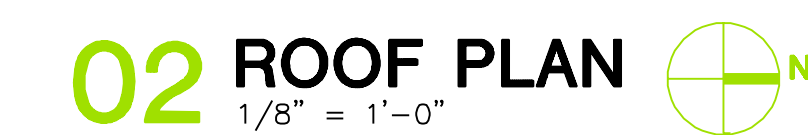




DOOR SCHEDULE					
MARK	SIZE	DESCRIPTION	FRAME	HARDWARE	REMARKS
D1	6'-0"x7'-0"x 3/4"	PAIR HOLLOW METAL FLAT SURFACE ASTRAGAL ON ACTIVE DOOR RIGHT HAND DOOR ACTIVE HALF GLASS DOOR LITES	HOLLOW METAL 5 3/4"x2"	6 BUTTS McKINNEY TA 2314 4 1/2"x4" ANRP x260 1 EXIT DEVICE 8813 x 713 ETJ x 644 SURFACE STRIKE 1 FLOOR BOLT IVES 458 x 260 1 HEAD BOLT IVES 458 x 260 1 CLOSER 351 PH9 ON ACTIVE LEAF 1 STOP/HOLDER GJ 704X x S3 ON INACTIVE LEAF	SARGENT HARDWARE, NUMBERS USED SCHLAGE AND VON DUPRIN ACCEPTED. HARDWARE FINISH 260. WEATHERSTRIP EXTERIOR DOORS AT HEAD AND JAMBS WITH MAY 794-A. SEE SILL DETAIL FOR THRESHOLD AND BOTTOM SEAL. MAY 856-A RAIN DRIP CAP AT EXTERIOR HOLLOW METAL FRAME HEADS. DOOR LITES: ②
D2	6'-0"x7'-0"x 1 3/4"	SAME AS D1			
D3	30'-0 3/8"x 11'-11 7/8"	STEEL VERTICAL ROLLING TYPE A	NONE	NONE	25 PSF DADE COUNTY WIND LOAD. COMPLETE WEATHERSTRIP PACKAGE.
D4	8'-0 3/8"x 7'-10 3/4"	STEEL VERTICAL ROLLING TYPE B	SAME AS D3		

KEYED NOTES

- 1 REMOVE CONCRETE CURING COMPOUND AND STAINS AND APPLY 2 COATS SONNEBORN LAPIDOLUX PER MANUFACTURER'S INSTRUCTIONS.
- 2 DOOR LITES: FRAME TYPE 1, STANDARD FOR 1/4" GLASS, STANDARD SIZE $\pm 24 \times 32$. 1/4" POLISHED SAFETY WIRE GLASS, FORM 1 SQUARE MESH. DOOR LITES TO BE SILICONE SHOP GLAZED.
- 3 MEGAFLOX COATINGS TO BE APPLIED BY PPG APPROVED CONTRACTOR.

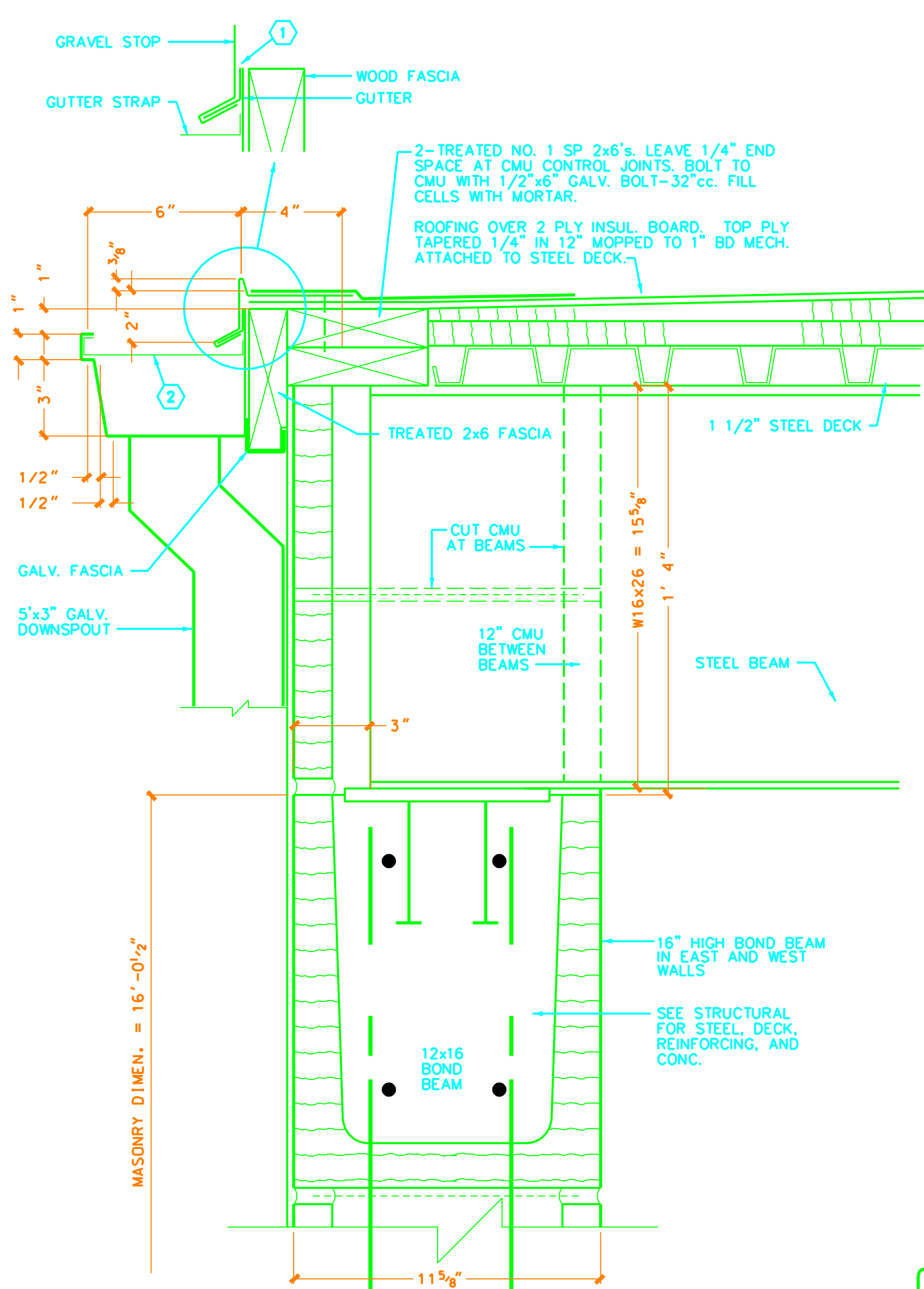


RECORD DRAWING
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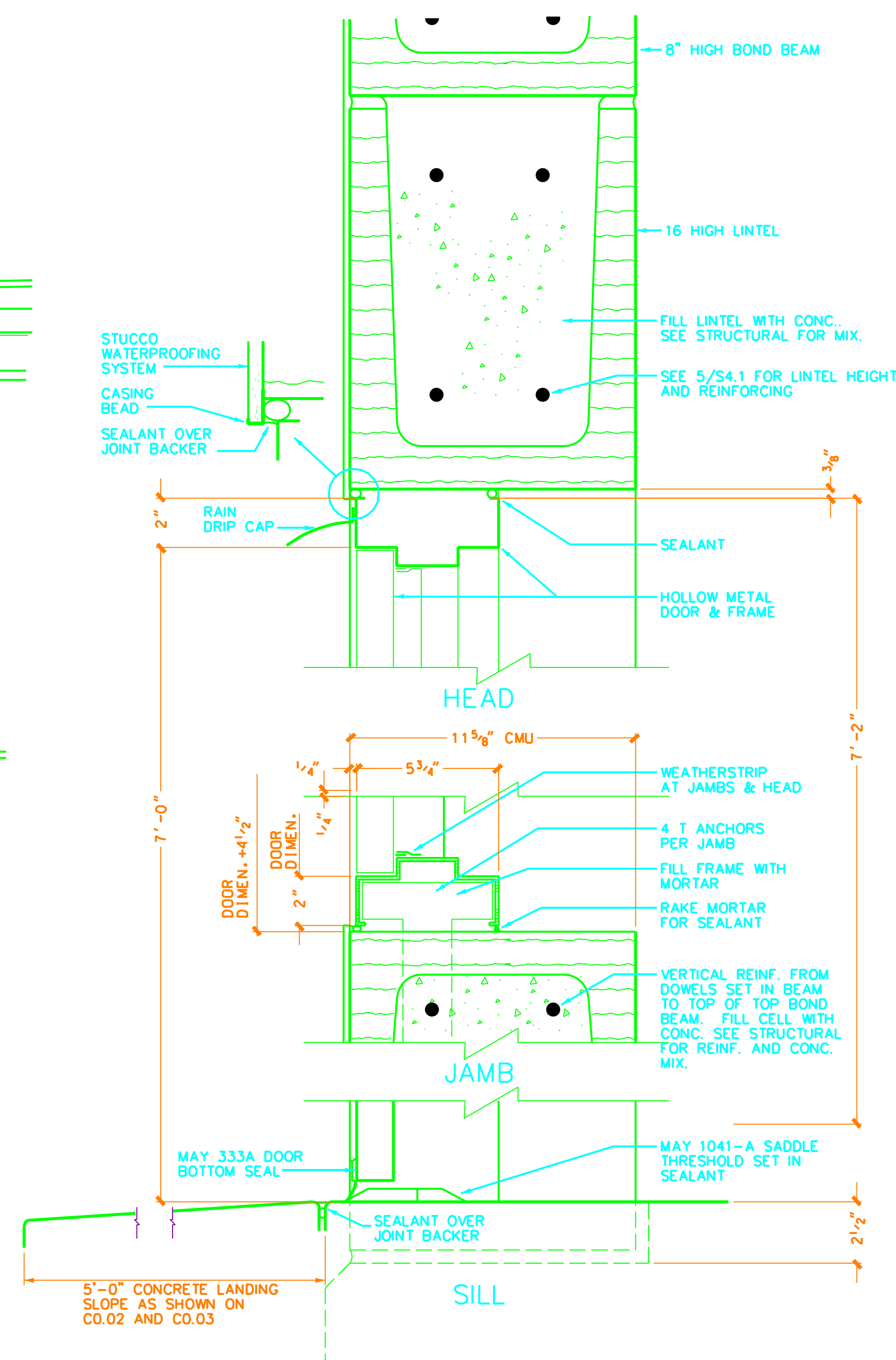
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Phone: 713-780-7563 Fax: 713-780-9209

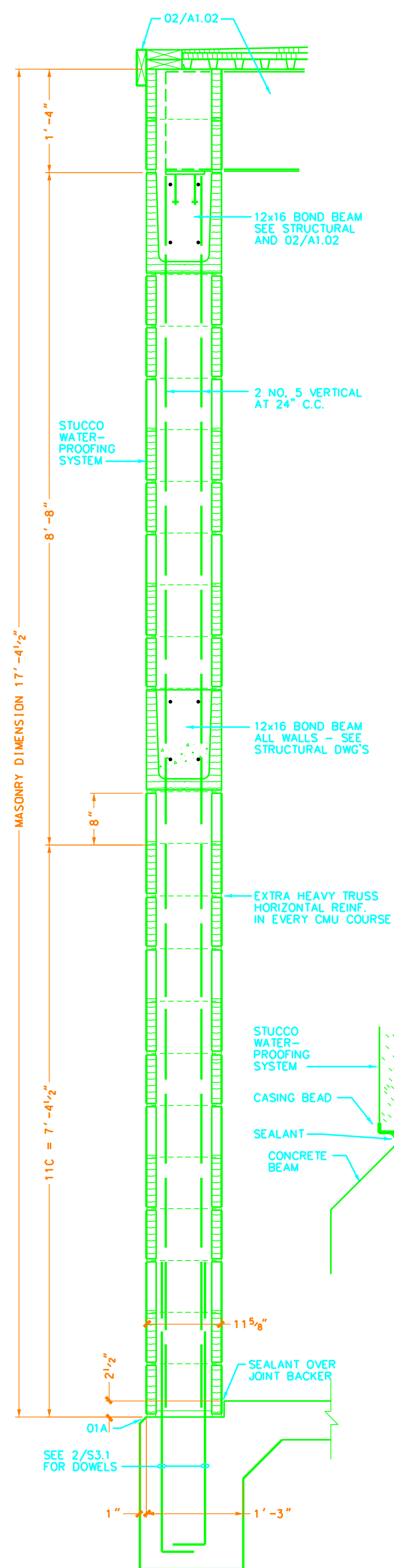
Drawing Title ARCHITECTURAL PLANS AND ELEVATIONS	Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION			Date 02/25/2003	Veterans Administration
	Approved: Engineering Svc Chief			Project No. VA #580-68-110 SSA #1017-007-1	
	Building Number 104/105	Checked JBS	Drawn MDT	Drawing No. A1.01	
	Approved: Medical Center Director			Location VAMC HOUSTON, TX	



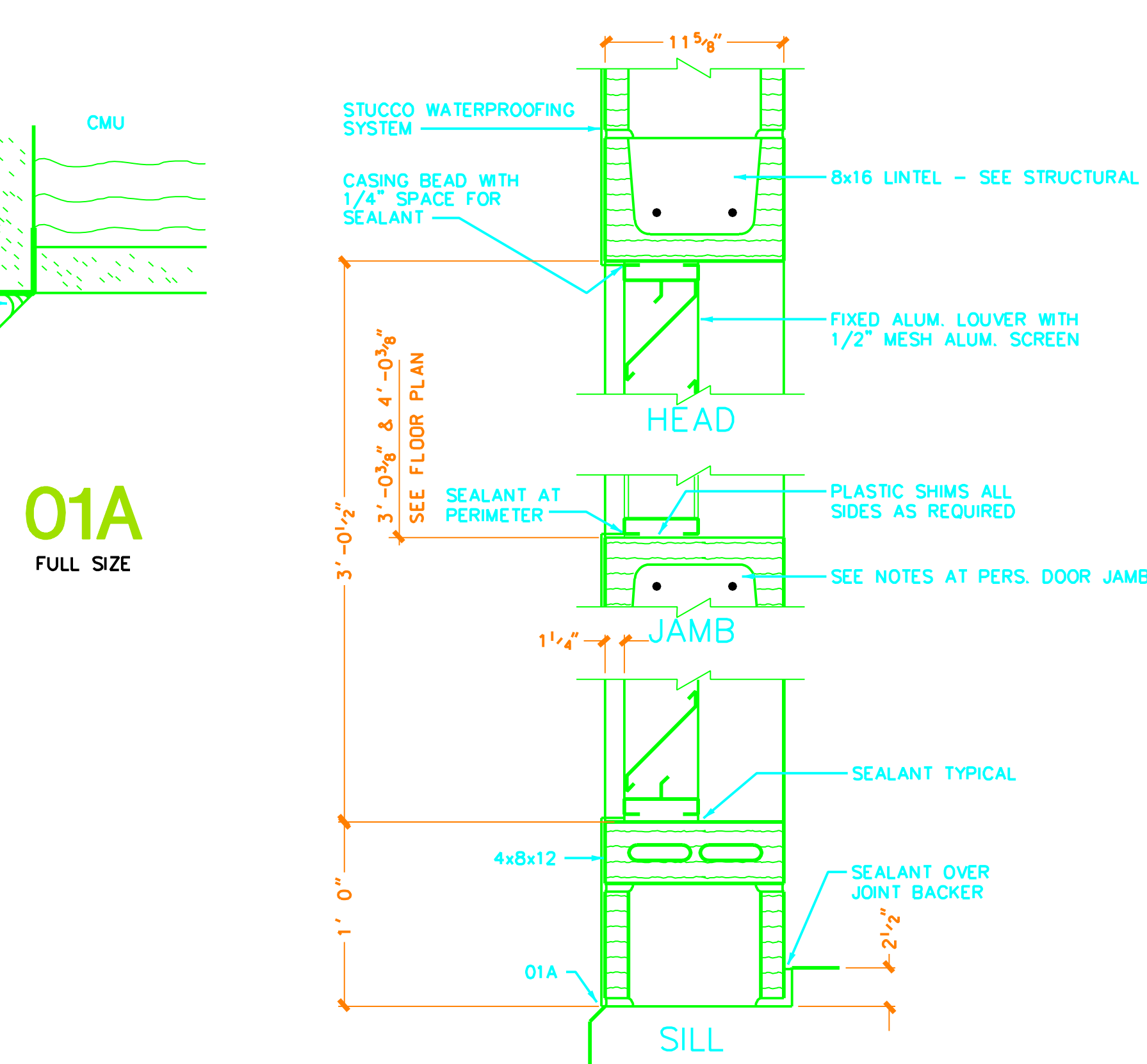
02 ROOF EDGE EAST AND WEST WALLS



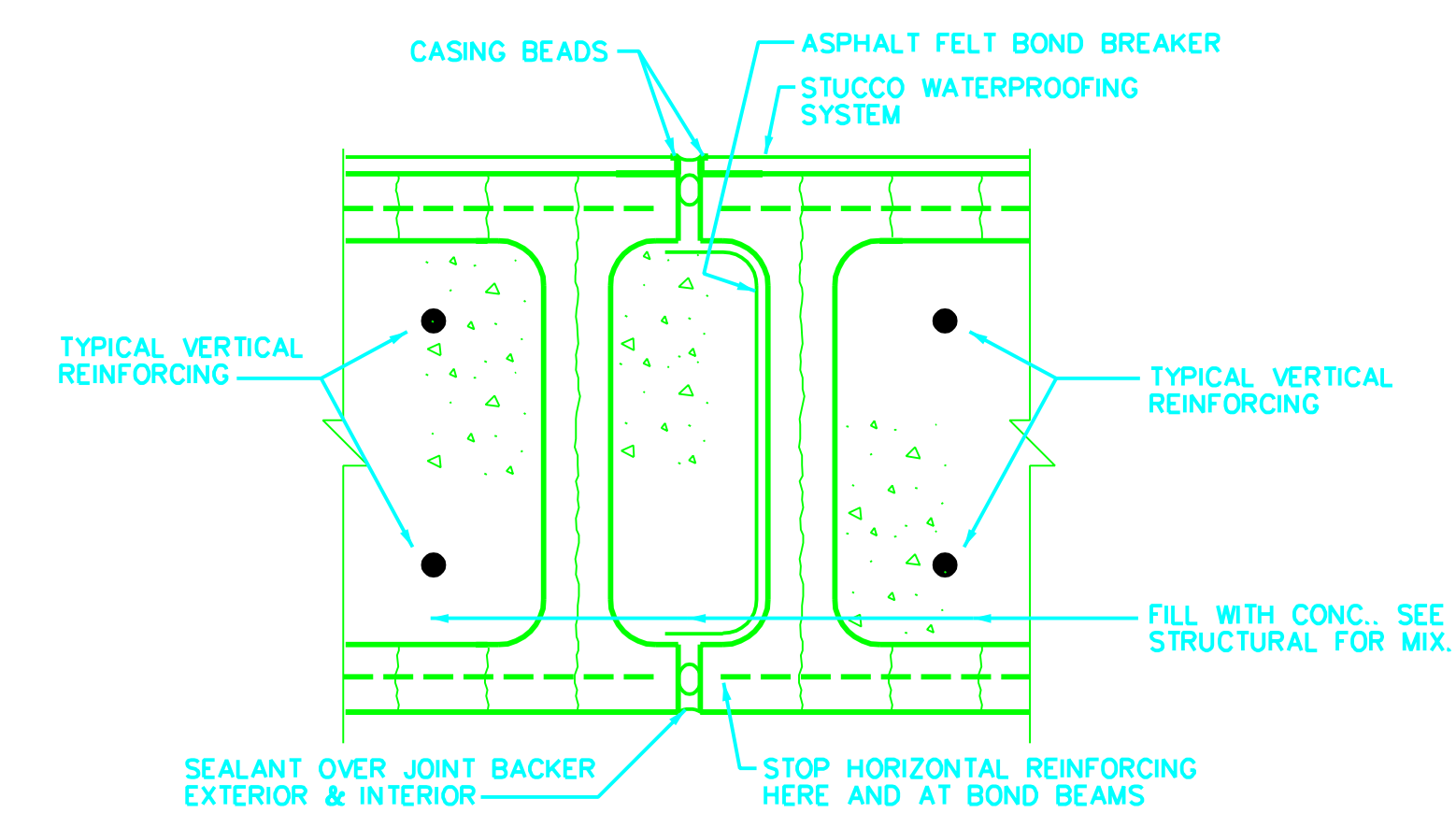
04 PERSONNEL DOOR DETAILS



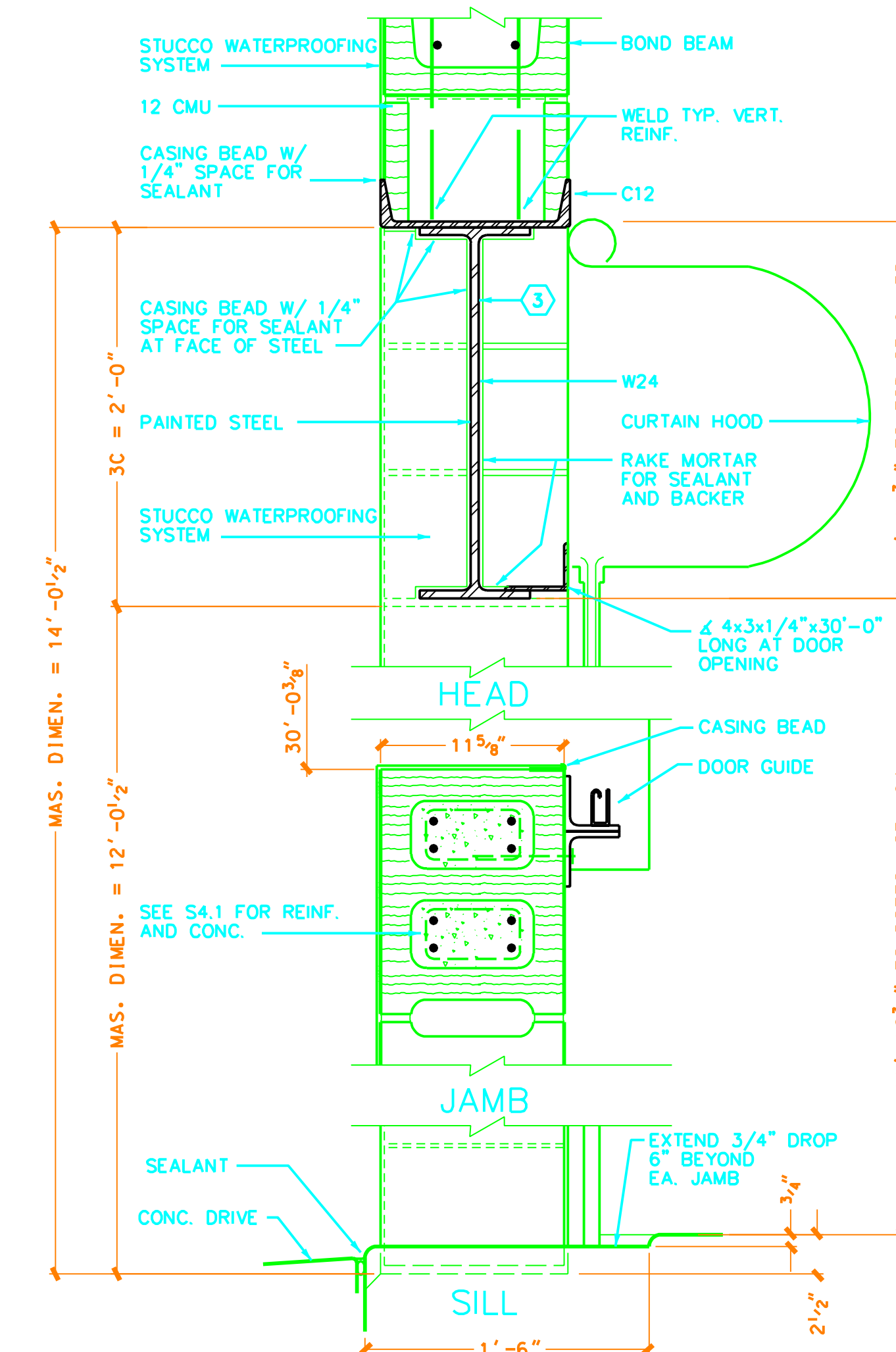
01 TYPICAL WALL SECTION



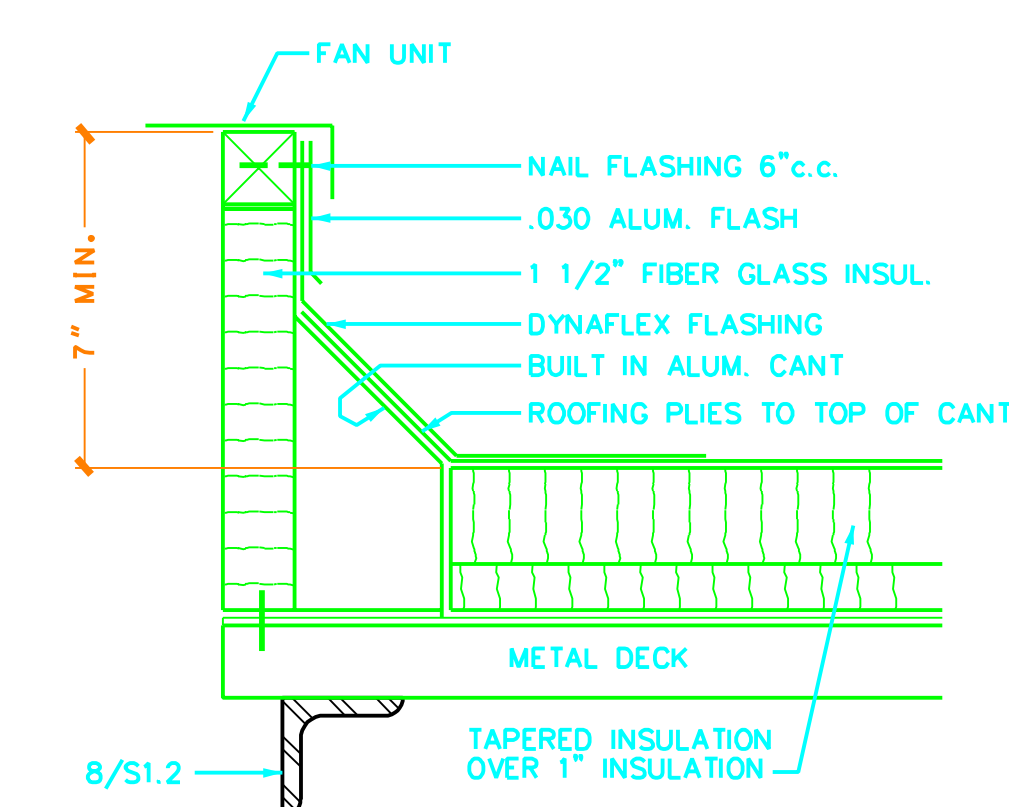
03 LOUVER DETAILS



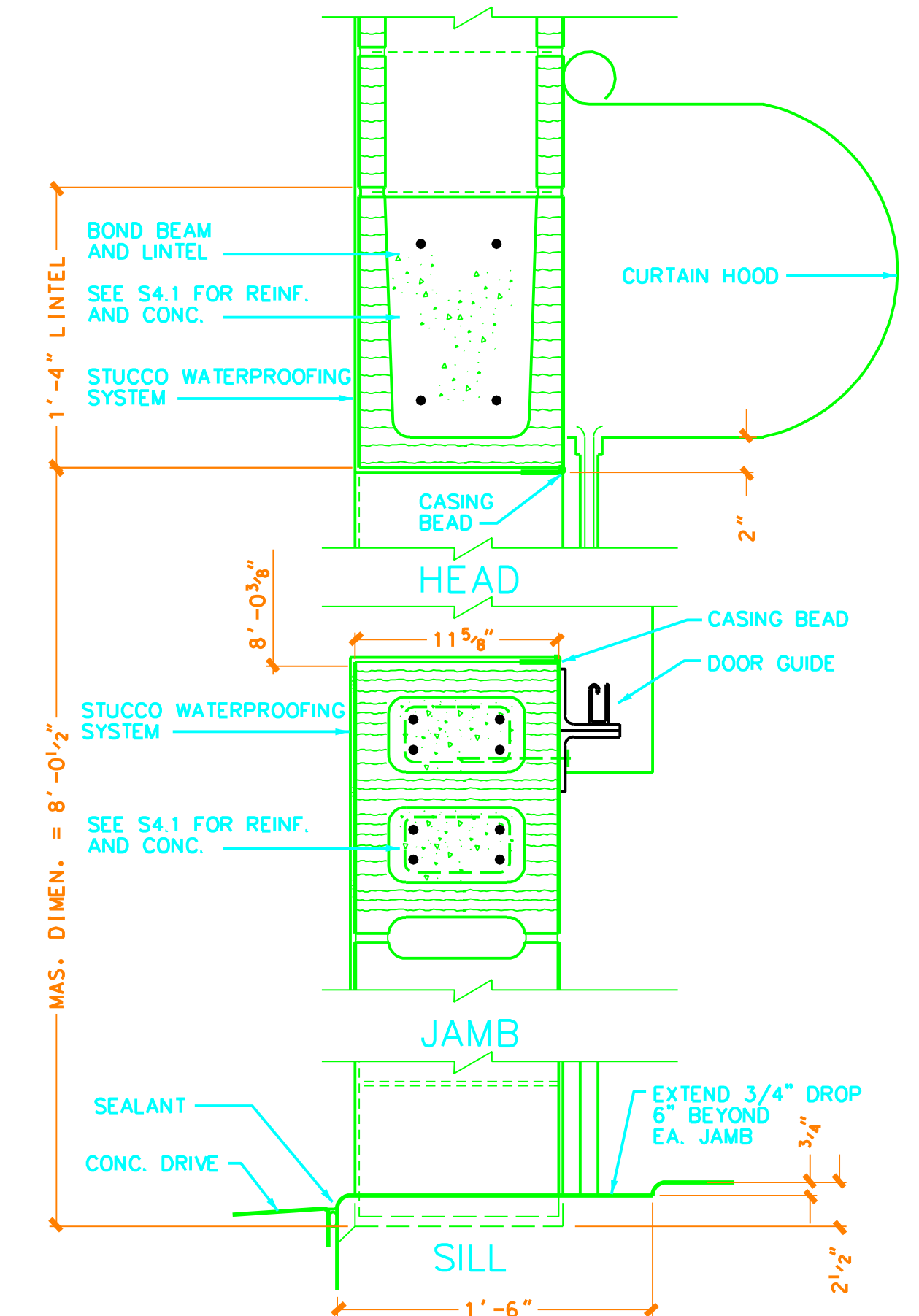
06 MASONRY CONTROL JOINT



07 VERT. ROLLING DOOR D3
1 1/2" = 1'-0"



08 ALUMINUM CANT AT FAN CURB

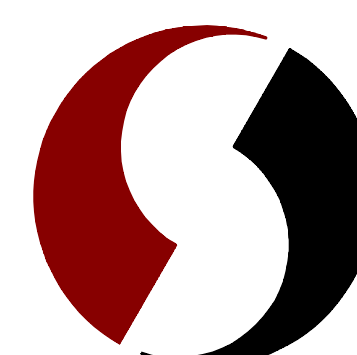


09 VERT. ROLLING DOOR D4

KEYED NOTES

- ① CONTINUOUS GALVANIZED CLEAT.
- ② GUTTER STRAP
- ③ PRIOR TO DOOR INSTALLATION FINISH INTERIOR SURFACES OF BEAM.

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

[illegible]**SHAH SMITH & ASSOCIATES, INC.**

2825 WILCREST, SUITE 350
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Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title ARCHITECTURAL DETAILS	Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION			Date 02/25/2003	Veterans Administration
				Project No. VA #580-68-110 SSA #1017-0007-1	
	Approved: Engineering Svc Chief	Building Number 104105	Checked JDS	Brown HOT	
	Approved: Medical Center Director	Location VAMC HOUSTON, TX			
			Drawing No. A1.02	Drawn of	

Veterans Administration

LEGEND									
SYMBOL	ABBREV.	DESCRIPTION	SYMBOL	ABBREV.	DESCRIPTION	SYMBOL	ABBREV.	DESCRIPTION	
		SUPPLY DUCT		CHS	CHILLED WATER SUPPLY			SOLENOID VALVE	
		O.A. RETURN OR EXH.DUCT		CHR	CHILLED WATER RETURN			BALL VALVE	
	VD	VOLUME DAMPER		CWS	CONDENSING WATER SUPPLY			FIRESTAT	
	FD OR SD	FIRE OR SMOKE DAMPER		CWR	CONDENSING WATER RETURN		HG	HOT GAS	
		THERMOSTAT		HWS	HOT WATER SUPPLY		L	REFRIGERANT LIQUID	
		HUMIDISTAT		HWR	HOT WATER RETURN		S	REFRIGERANT SUCTION	
		SAN. WASTE OR VENT STACK		#5	# OF STEAM SUPPLY			GATE VALVE	
		WASTE OR VENT NO.		#R	# OF STEAM RETURN			GLOBE VALVE	
	VTR	VENT THRU ROOF		A	COMPRESSED AIR			CHECK VALVE	
	FCO	FLOOR CLEANOUT			THERMOMETER WELL			VALVE BOX	
	CW	DOM. COLD WATER			THERMOMETER			GAGE COCK	
	HW	DOM. HOT WATER SUPPLY			UNION			BUTTERFLY VALVE	
		DOM. HOT WATER RETURN			STRAINER			PLUG VALVE	
		VENT			REDUCER			TWO-WAY CONTROL VALVE	
	SD	STORM DRAIN			GAGE			THREE-WAY CONTROL VALVE	
	G	GAS			FLEXIBLE JOINT		PRV	PRESSURE REDUCING VALVE	
	F	FIRE			ANCHOR		RV	PRESSURE RELIEF VALVE	
	FS	FIRE SPRINKLER			VENTURI FLOW ELEMENT			EXISTING ITEMS TO REMAIN	
	ACD	AUXILIARY CONDENSATE DRAIN		MVD	MOTORIZED VOLUME DAMPER			NEW ITEMS	
		DIFF. GR. OR REG. MARK			TEMPERATURE SENSOR			EXISTING ITEMS TO BE REMOVED	
		CFM			ANNUBAR				

PUMP SCHEDULE									
MARK	SERVICE	TYPE	SIZE		FLOW RATE GPM	TOTAL HEAD FT H ₂ O	HP/VOLTS/PH/HZ	RPM	REMARKS
			SUCT.	DISCH.					
104-CHP-8, 104-CHP-9	CHILLED WATER	HORIZ. SPLIT	10	8	2143	53	40/460/3/60	1780	ITT A-C SERIES 8100 10x8x12L OR EQUAL
104-CWP-8, 104-CWP-9	CONDENSER WATER	VERTICAL TURBINE	18	12	3750	60	100/460/3/60	1770	LAYNE BOWLER - 1800 - SINGLE STAGE
104-CWP1	CONDENSER WATER	VERTICAL TURBINE	12	8	1050	60	30/460/3/60	1770	LAYNE BOWLER - 1800 - 11EH - 2 STAGE

NOTES

- VERTICAL TURBINE PUMP HEAD IS IN ADDITION TO COLUMN LOSSES AND STATIC HEAD BETWEEN SUCTION AND DISCHARGE OF PUMP.
- NPSHR FOR CONDENSER WATER PUMPS IS 28.1 FT.

CHILLER SCHEDULE															
MARK	CAP. TONS	MAX. KW	VOLTS/ PH/HZ	CHILLER						CONDENSER					REMARKS
				GPM	EWT ° F	LWT ° F	NO PASSES	FOULING FACTOR	MAXHD LOSS FT.	GPM	EWT ° F	NO PASSES	FOULING FACTOR	MAXHD LOSS FT.	
104-CCU-8	1250	917	4160/3/60	2143	52	38	2	.00025	15.2	3750	86	2	.0005	19.5	YORK YKHHHBJ2 - DCE FURNISHED BY OWNER, CONTRACTOR INSTALLED
104-CCU-9	1250	917	4160/3/60	2143	52	38	2	.00025	15.2	3750	86	2	.0005	19.5	YORK YKHHHBJ2 - DCE FURNISHED BY OWNER, CONTRACTOR INSTALLED

NOTES

- OWNER HAS PREPURCHASED 104-CCU-8 AND 104-CCU-9. CONTRACTOR IS RESPONSIBLE FOR UNLOADING, INSPECTING AND INSTALLING CHILLERS.
- CHILLER PERFORMANCE DATA PROVIDED FOR REFERENCE ONLY.

COOLING TOWER SCHEDULE							
MARK	EWT ° F	LWT ° F	WBT ° F	CAPACITY GPM	MOTOR HP/ VOLTS/ PH/HZ	MOTOR RPM	REMARKS
104-CT-8	95°F	85°F	80°F	3750	60/460/3/60	1800	CERAMIC UL-2030-60-10P4
104-CT-9	95°F	85°F	80°F	4800/1050	75/460/3/60	1800	CERAMIC UL-2430-75-10P4

NOTES

- FURNISHED AND INSTALLED BY OWNER ON CONTRACTOR PROVIDED BASIN. ALL CONNECTIONS TO FIELD INSTALLED PIPING BY CONTRACTOR.
- COOLING TOWER DATA PROVIDED FOR REFERENCE ONLY.

LOUVER SCHEDULE					
MARK	TYPE	DIMENSIONS	% FREE AREA	MAX FACE VELOCITY (FPM)	REMARKS
LVR-1, 2	FIXED DRAINABLE BLADE, ALUMINUM	4'x4'	55	700	RUSKIN ELF 375X
LVR-3, 4	FIXED DRAINABLE BLADE, ALUMINUM	3'x3'	55	700	RUSKIN ELF 375X

NOTES

- MILL FINISH SUITABLE FOR FIELD APPLICATION OF PAINT.

INSTRUMENT IDENTIFICATION

EXAMPLE SYMBOLS

- FIRST LETTER (S)
SUCCEEDING LETTERS
- INSTRUMENT TAG NUMBER
- FIELD MOUNTED INSTRUMENT
- DISTRIBUTED CONTROL SYSTEM DISPLAY POINT
(DISPLAYED)
- INDICATING LIGHT, WHERE XXX DESIGNATES FUNCTION
- HARDWIRE
- CONTROL AIR
- SOFTWARE LINK OR ELECTRICAL SIGNAL

VIBRATION ISOLATOR SCHEDULE

EQUIPMENT	ON GRADE		
	BASE TYPE	ISOL TYPE	MIN. DEFL.
COMPRESSORS			
1.5 kW (2 HP) AND OVER:			
8-13 REV/S (500-750 RPM)	-	D	-
13.1 REV/S (751 RPM) AND OVER	-	D	-
PUMPS			
BASE MOUNTED:			
11 kW THRU 30 kW (15 HP THRU 40 HP)	I	S	25 (1.0)

NOTES

- SEE SPECIFICATIONS FOR ISOLATOR SYMBOL DESIGNATION.

EXHAUST FAN SCHEDULE

MARK	SERVICE	TYPE	CFM	SP 'H O	HP	VOLTS	PH/HZ	FAN RPM	INTERLOCK WITH	REMARKS
104-EF-1	PURGE/ EXHAUST	UP BLAST	16600	.4	3	480	3/60	381	DDC CONTROLS	COOK 445R-10B
104-EF-2	EXHAUST	DOWN BLAST	1200	.125	.167	120	1/60	1075	N/A	COOK 120 ACE-B

NOTES

- COORDINATE ROOF PENETRATION SIZE AND CURB DIMENSIONS WITH INTEGRAL FAN CURB CAP DIMENSIONS.

CONTROL AIR COMPRESSOR

MARK	SERVICE	TYPE	V/PH/HZ	REMARKS
104-CAC-1	CONTROL AIR	DUPLEX	480/3/60	SIZED AND PROVIDED BY DDC CONTROLS CONTRACTOR

NOTES

- DDC CONTRACTOR TO PROVIDE ELECTRICAL SERVICE REQUIREMENTS FOR COMPRESSOR TO ELECTRICAL CONTRACTOR.

CONTROL AIR DRYER

MARK	SERVICE	V/PH/HZ	REMARKS
104-CAD-1	CONTROL AIR	120/1/60	SIZED AND PROVIDED BY DDC CONTROLS CONTRACTOR

NOTES

- DDC CONTRACTOR TO PROVIDE ELECTRICAL SERVICE REQUIREMENTS FOR DRYER TO ELECTRICAL CONTRACTOR.

ELECTRIC UNIT HEATER SCHEDULE

MARK	LOCATION	MOUNTING HEIGHT	CFM	MAX RPM	MOTOR HP	OUTPUT BTUH 70 F EAT	CAP KW	VOLTS	PH/HZ	REMARKS
104-EUH-1, 2	BLD 104	AT ROOF	1040	1550	1/10	51180	15	480	3/60	REZNOR EGE-15 KW

NOTES

- PROVIDE UNIT WITH INTEGRAL THERMOSTAT AND MOUNTING BRACKET.

INSTRUMENT IDENTIFICATION LETTERS

FIRST LETTERS		SUCCEEDING LETTERS		
MEASURING OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A ANALYSIS (+)		ALARM		
B BLOCK (ON/OFF)				
C			CONTROL	
D	DIFFERENTIAL			
E VOLTAGE		SENSOR (PRIMARY ELEMENT)		
F FLOW RATE	RATIO (FRACTION)			
G	GAUGE	GLASS, VIEWING DEVICE		
H HAND				HIGH
I CURRENT (ELECTRICAL)		INDICATE		
J POWER	SCAN			
K TIME, SCHEDULE	TIME RATE OF CHANGE		CONTROL STATION	
L LEVEL		LIGHT		LOW
M				MIDDLE, INTERMEDIATE
N SURGE				
O		ORIFICE, RESTRICTION		
P PRESSURE, VACUUM		POINT (TEST) CONNECTION		
Q QUANTITY, EVENT	INTEGRATE, TOTALIZE			
R RADIATION		RECORD		
S SPEED, SOLENOID, FREQUENCY	SAFETY		SWITCH	
T TEMPERATURE			TRANSMIT	
U MULTIVARIABLE		MULTI- FUNCTION	MULTI- FUNCTION	MULTI- FUNCTION
V VIBRATION, MECHANICAL ANALYSIS			VALVE, DAMPER, LOUVER	
W WEIGHT, FORCE		WELL		
X TORQUE	X AXIS			
Y EVENT, STATE OR PRESENCE	Y AXIS		RELAY, COMPUTE, CONVERT	
Z POSITION, DIMENSION	Z AXIS		DRIVER, ACTUATOR, UNCLASSIFIED FINAL CONTROL ELEMENT	

- (+)-WHEN USED, EXPLANATION IS SHOWN ADJACENT TO INSTRUMENT SYMBOL. SEE ABBREVIATIONS AND LETTER SYMBOL.

GENERAL NOTES

THESE GENERAL NOTES APPLY TO ALL MECHANICAL DRAWINGS.

IN ANY CASE WHERE A PIPE SHOWN ON A PLAN SHEET DIFFERS FROM THAT SHOWN IN A SCHEMATIC OR DETAIL, USE THE LARGER OF THE TWO SIZES SHOWN.

ALL ELEVATIONS INDICATED IN THIS WAY (8'-0") ARE THE ELEVATIONS FROM THE FINISHED FLOOR DIRECTLY BELOW TO THE BOTTOM OF THE BARE PIPE.

SEAL OPENINGS AROUND PIPE PENETRATIONS IN EXTERIOR WALLS WITH WEATHER PROOFING MATERIAL.

PIPE HANGER LOCATIONS MAY BE SHOWN ON CERTAIN DRAWINGS, BUT NOT ALL THE HANGER LOCATIONS ARE SHOWN. PROVIDE HANGERS AT MAXIMUM SPANS AS REQUIRED BY SPECIFICATIONS, WHERE NOT INDICATED ON DRAWINGS.

LEGENDS ON THIS SHEET ARE STANDARD LEGENDS AND NOT ALL THE INFORMATION APPLIES TO THIS PROJECT.

RECORD DRAWING

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Revisions	Date



SHAH SMITH & ASSOCIATES, INC.

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Phone: 713-780-7563

Fax: 713-780-9209

Drawing Title

**MECHANICAL SCHEDULES
AND LEGENDS**

Approved: Engineering Svc. Chief

Approved: Medical Center Director

Project Title

HOUSTON VETERANS AFFAIRS MEDICAL CENTER

CHILLER PLANT EXPANSION

Building Number

Checked

Drawn

Location

VAMC HOUSTON, TX

Date

02/25/2003

Project No.

VA #560-58-110
SSA #1017-007-1

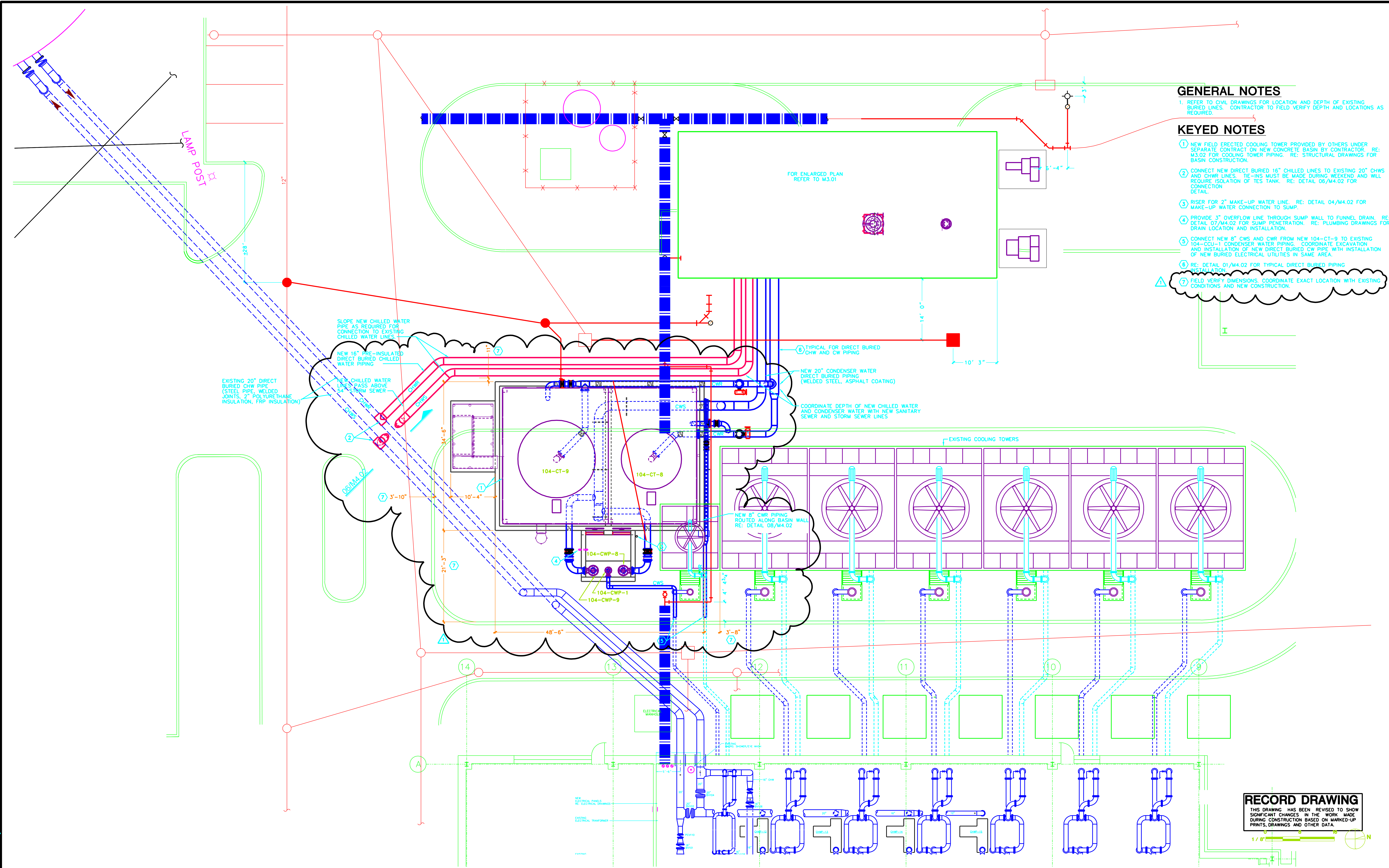
Drawing No.

M0.01

Dwg.

of

Veterans
Administration



- ### GENERAL NOTES
1. REFER TO CIVIL DRAWINGS FOR LOCATION AND DEPTH OF EXISTING BURIED LINES. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATIONS AS REQUIRED.
- ### KEYED NOTES
- ① NEW FIELD ERECTED COOLING TOWER PROVIDED BY OTHERS UNDER SEPARATE CONTRACT ON NEW CONCRETE BASIN BY CONTRACTOR. RE: M3.02 FOR COOLING TOWER PIPING. RE: STRUCTURAL DRAWINGS FOR BASIN CONSTRUCTION.
 - ② CONNECT NEW DIRECT BURIED 16" CHILLED LINES TO EXISTING 20" CHWS AND CHWR LINES. TE-INS MUST BE MADE DURING WEEKEND AND WILL REQUIRE ISOLATION OF YES TANK. RE: DETAIL 06/M4.02 FOR CONNECTION DETAIL.
 - ③ RISER FOR 2" MAKE-UP WATER LINE. RE: DETAIL 04/M4.02 FOR MAKE-UP WATER CONNECTION TO SUMP.
 - ④ PROVIDE 3" OVERFLOW LINE THROUGH SUMP WALL TO FUNNEL DRAIN. RE: DETAIL 07/M4.02 FOR SUMP PENETRATION. RE: PLUMBING DRAWINGS FOR DRAIN LOCATION AND INSTALLATION.
 - ⑤ CONNECT NEW 8" CWS AND CWR FROM NEW 104-CT-9 TO EXISTING 104-CGU-1 CONDENSER WATER PIPING. COORDINATE EXCAVATION AND INSTALLATION OF NEW DIRECT BURIED CW PIPE WITH INSTALLATION OF NEW BURIED ELECTRICAL UTILITIES IN SAME AREA.
 - ⑥ RE: DETAIL 01/M4.02 FOR TYPICAL DIRECT BURIED PIPING INSTALLATION.
 - ⑦ FIELD VERIFY DIMENSIONS. COORDINATE EXACT LOCATION WITH EXISTING CONDITIONS AND NEW CONSTRUCTION.

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.

COOLING TOWER COORDINATION	
Revisions	Date
	29NOV01



SHAH SMITH & ASSOCIATES, INC.
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Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	MECHANICAL PIPING PLAN
Approved: Engineering Svc. Chief	
Approved: Medical Center Director	

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
	CHILLER PLANT EXPANSION
Building Number	
Location	VAMC HOUSTON, TX

Date	02/25/2003
Project No.	VA #560-58-110 SSA #1017-007-1
Drawing No.	M2.01
Dwg. of	

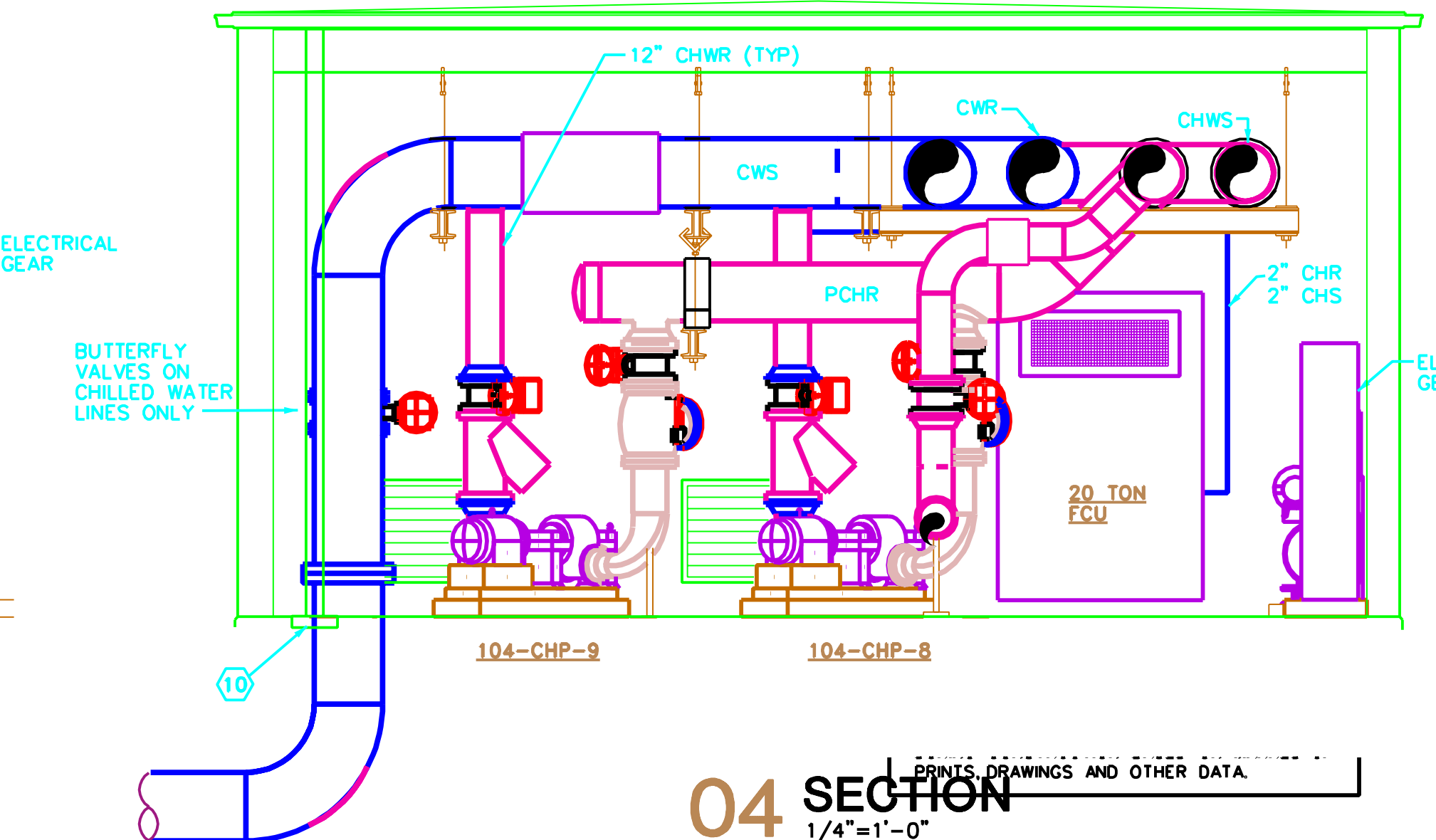
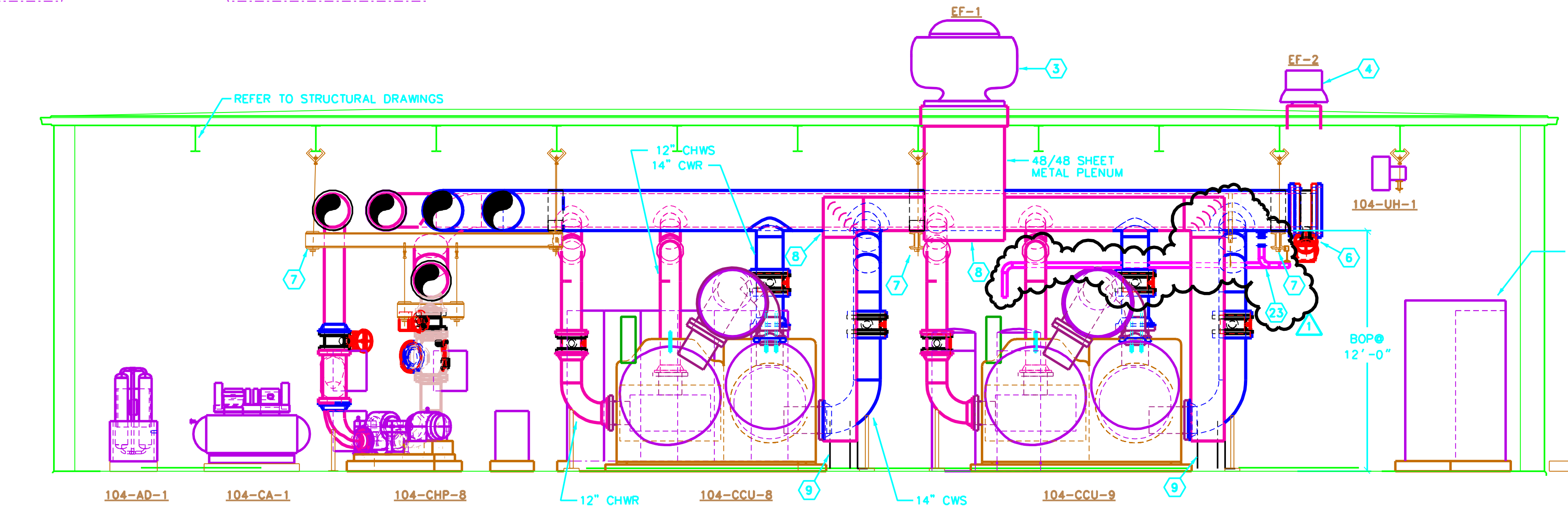
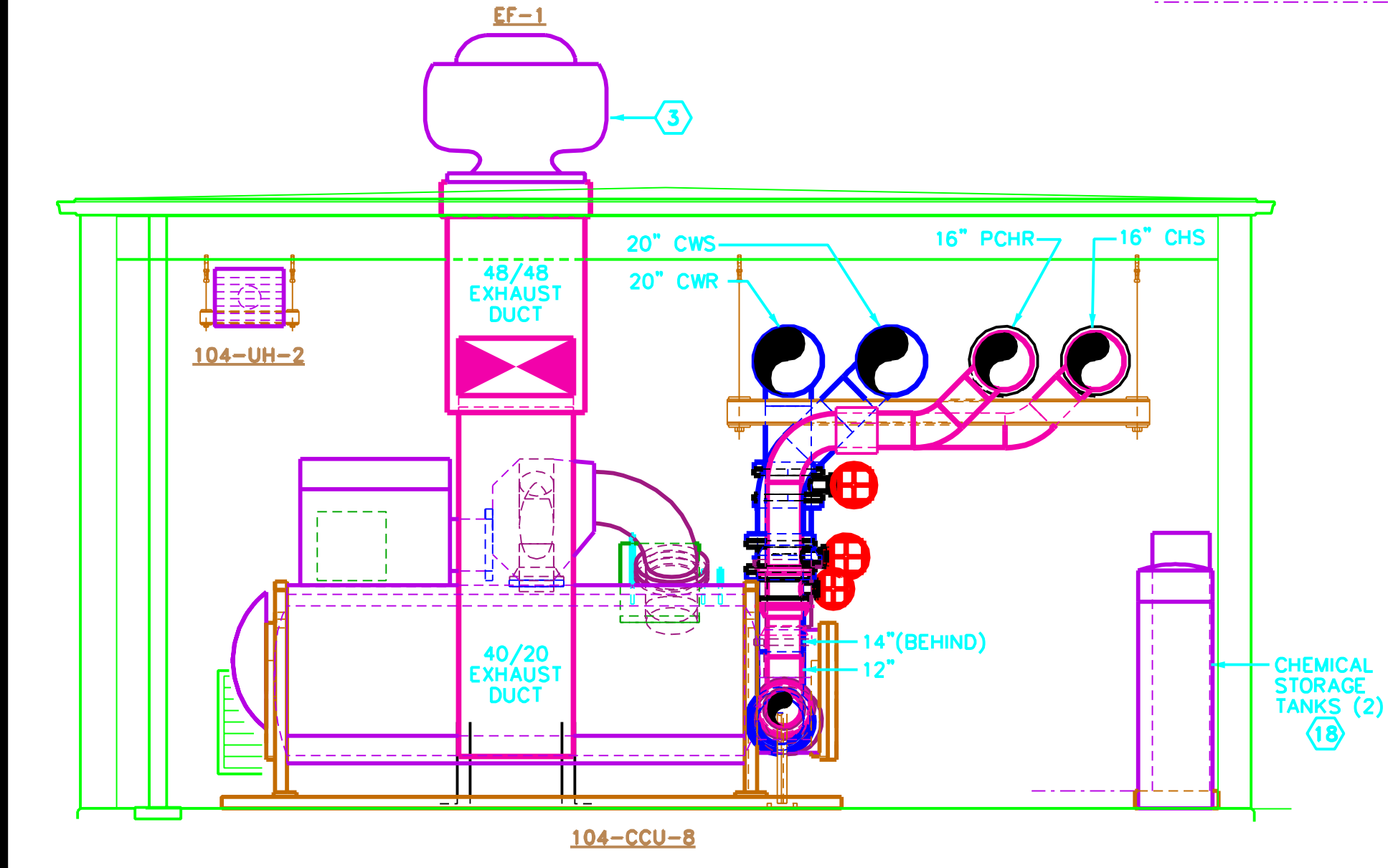
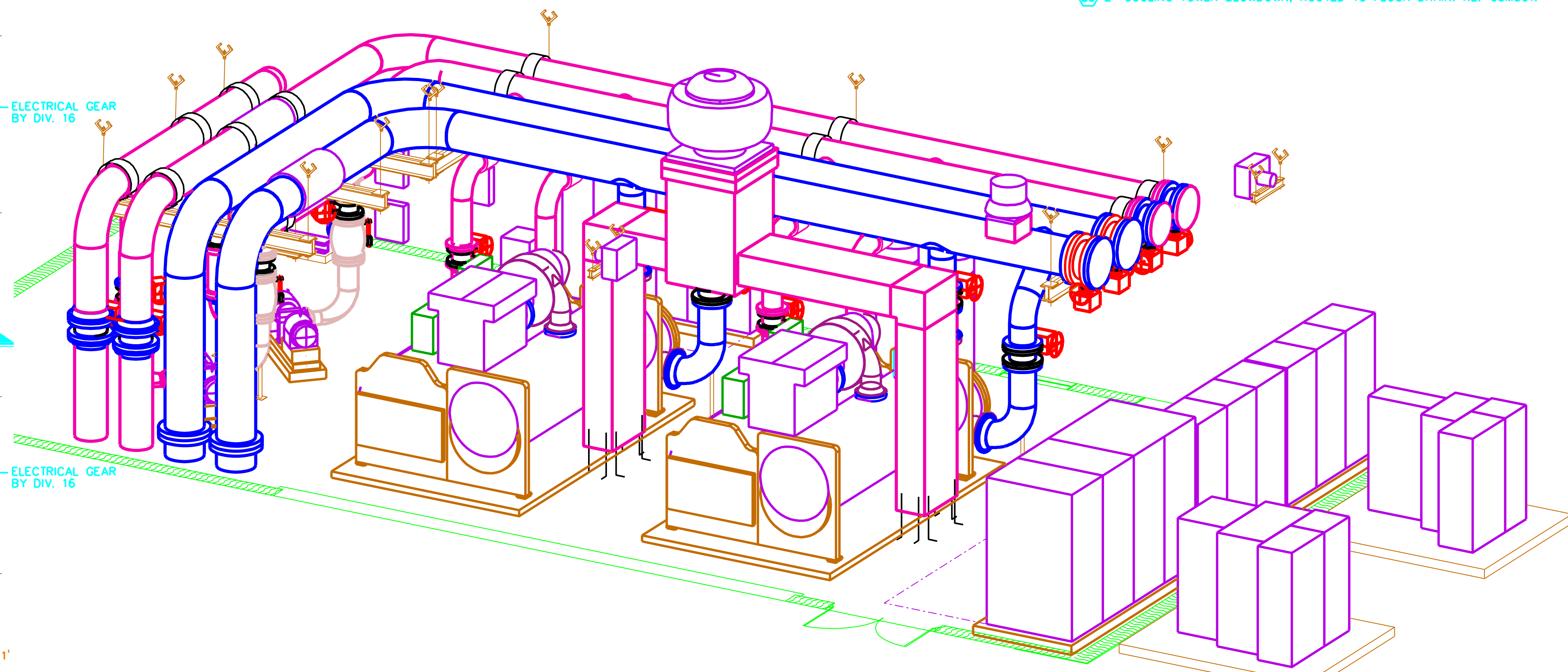
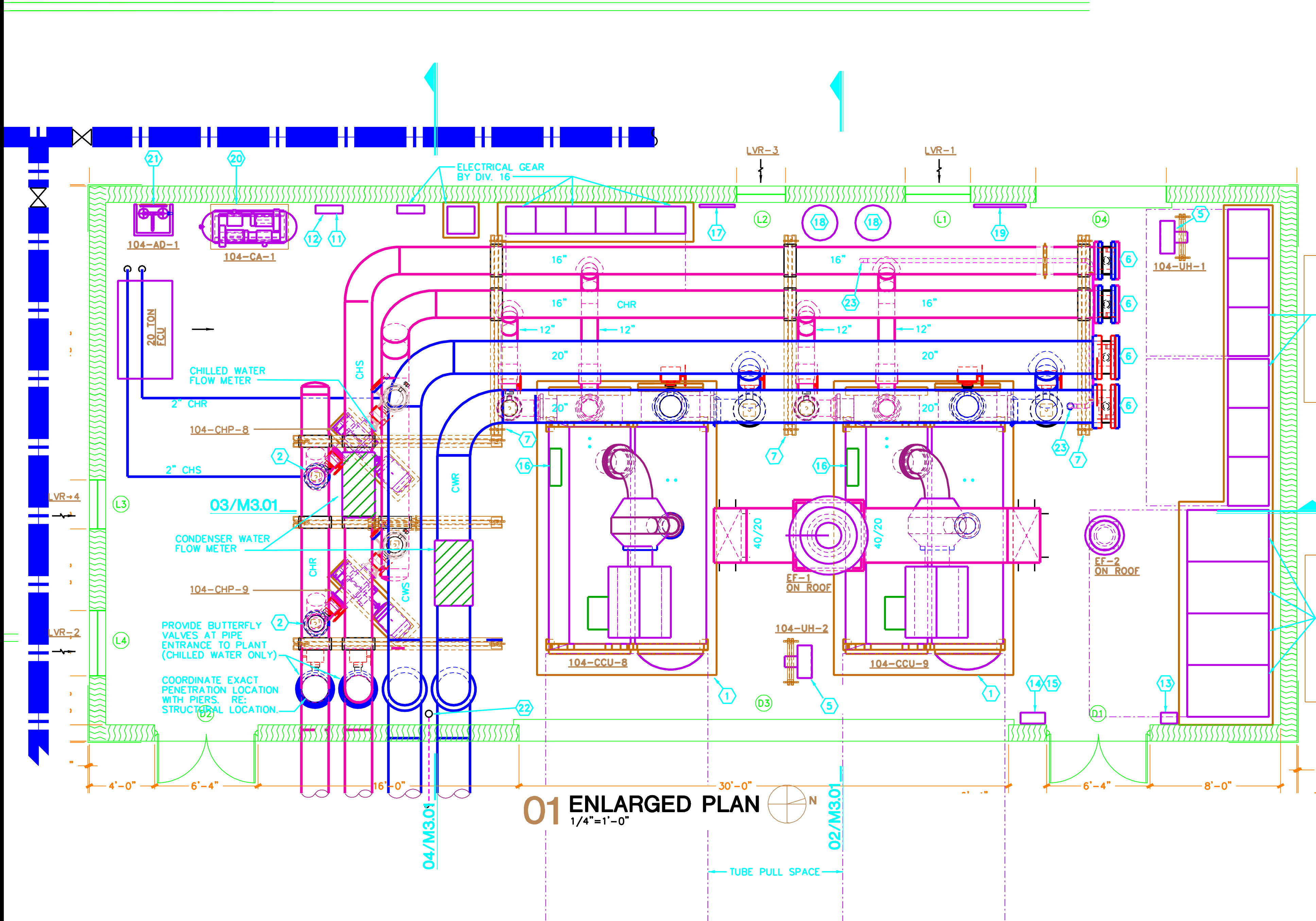
Veterans
Administration

GENERAL NOTES

1. ITEMS SHOWN BOLD ARE NEW.
2. NOT ALL PIPE SUPPORTS SHOWN FOR CLARITY. SUPPORT PIPES PER CRITERIA IN SPECIFICATION.

KEYED NOTES

1. INSTALL OWNER FURNISHED CENTRIFUGAL CHILLER ON 4" HOUSEKEEPING PAD (RE: STRUCTURAL DWGS FOR PADS). PROVIDE PIPING CONNECTIONS PER DETAIL 03/M4.01.
2. PROVIDE HORIZONTAL SPUT CASE PUMP ON 4" HOUSEKEEPING PAD. (RE: STRUCTURAL DWG FOR PADS). PROVIDE PIPING CONNECTIONS PER DETAIL 01/M4.01.
3. PROVIDE UP BLAST EXHAUST FAN ON CURB (RE: ARCHITECTURAL DWG FOR CURB). RE: DETAIL 05/M4.01 FOR INSTALLATION OF FAN.
4. PROVIDE DOWN DISCHARGE EXHAUST FAN ON CURB (RE: ARCHITECTURAL DWG FOR CURB). RE: DETAIL 5/M4.01 FOR INSTALLATION OF FAN.
5. PROVIDE ELECTRIC UNIT HEATERS WITH INTEGRAL THERMOSTAT HUNG FROM STRUCTURE WITH BEAM CLAMP AND 1/2" ALL THREAD ROD.
6. PROVIDE BUTTERFLY VALVE RATED FOR END OF SERVICE AND BLIND FLANGE AS WHERE INDICATED.
7. PROVIDE PIPE SUPPORT PER DETAIL 06/M4.01. COORDINATE EXACT LOCATIONS WITH ROOF STRUCTURE.
8. PROVIDE FLANGED CONNECTION TO ALLOW VERTICAL PORTION OF DUCT TO BE REMOVED FOR CHILLER ACCESS.
9. PROVIDE ANGLE IRON FEET TO SUPPORT VERTICAL PORTION OF DUCTWORK. BOLT FEET TO SLAB. PAINT PER DIVISION 9.
10. PROVIDE PIPE PENETRATION THROUGH SLAB PER DETAIL 08A & 08B/M4.01.
11. PROVIDE REFRIGERANT MONITOR, SAMPLING TUBES, ALARMS, INTERLOCKS, AND INTER CONNECTIONS TO BUILDING AUTOMATION SYSTEM. RE: DETAIL 11/M4.01 FOR INSTALLATION SCHEMATIC. (STACKED ABOVE)
12. BUILDING AUTOMATION SYSTEM PANEL AND ENCLOSURE. (STACKED BELOW)
13. PROVIDE COMBINATION BELL AND STROBES IN WEATHERPROOF ENCLOSURES INITIATED BY REFRIGERANT MONITOR WHEN REFRIGERANT LEVELS EXCEED SPECIFIED AMOUNT.
14. PROVIDE BREAK GLASS SWITCH THAT TURNS ON EF-1 REGARDLESS OF TEMPERATURE IN MACHINERY ROOM BY CONTACT CLOSURE SIGNAL TO BUILDING AUTOMATION SYSTEM.
15. PROVIDE SIGNAGE OUTSIDE DOOR FOR EMERGENCY SHUTDOWN PROCEDURES PER SPECIFICATIONS.
16. PROVIDE OPERATIONAL SCHEMATIC AND INFORMATION PLAQUE ON CHILLER CONTROL PANEL PER SPECIFICATIONS.
17. BROMINATOR BY WATER TREATMENT CONTRACTOR. RE: SCHEMATIC 03/M5.01. (ALTERNATE BID #1)
18. CHEMICAL STORAGE TANKS BY WATER TREATMENT CONTRACTOR. (ALTERNATE BID #1)
19. CONTROLLER, METERING PUMPS AND APPURTENANCES BY WATER TREATMENT CONTRACTOR. (ALTERNATE BID #1)
20. CONTROL AIR COMPRESSOR BY CONTROLS CONTRACTOR. RE: SCHEMATIC 03/M4.01.
21. CONTROL AIR DRYER BY CONTROLS CONTRACTOR. RE: 03/M4.02.
22. PNEUMATIC CONTROL LINE TO COOLING TOWER BURIED WITH CONDENSER WATER PIPE. SIZED BY BUILDING AUTOMATION CONTRACTOR. MECHANICAL CONTRACTOR TO INSTALL FROM STUB-UP IN MACHINERY ROOM TO STUB-UP AT COOLING TOWER. UNDER GROUND TUBING TO BE TYPE "K" COPPER.
23. 2" COOLING TOWER BLOWDOWN, ROUTED TO FLOOR DRAIN. RE: 03/M5.01.



RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.

0 4 8
1 / 4" = 1'-0"

Revisions	Date
	29NOV01

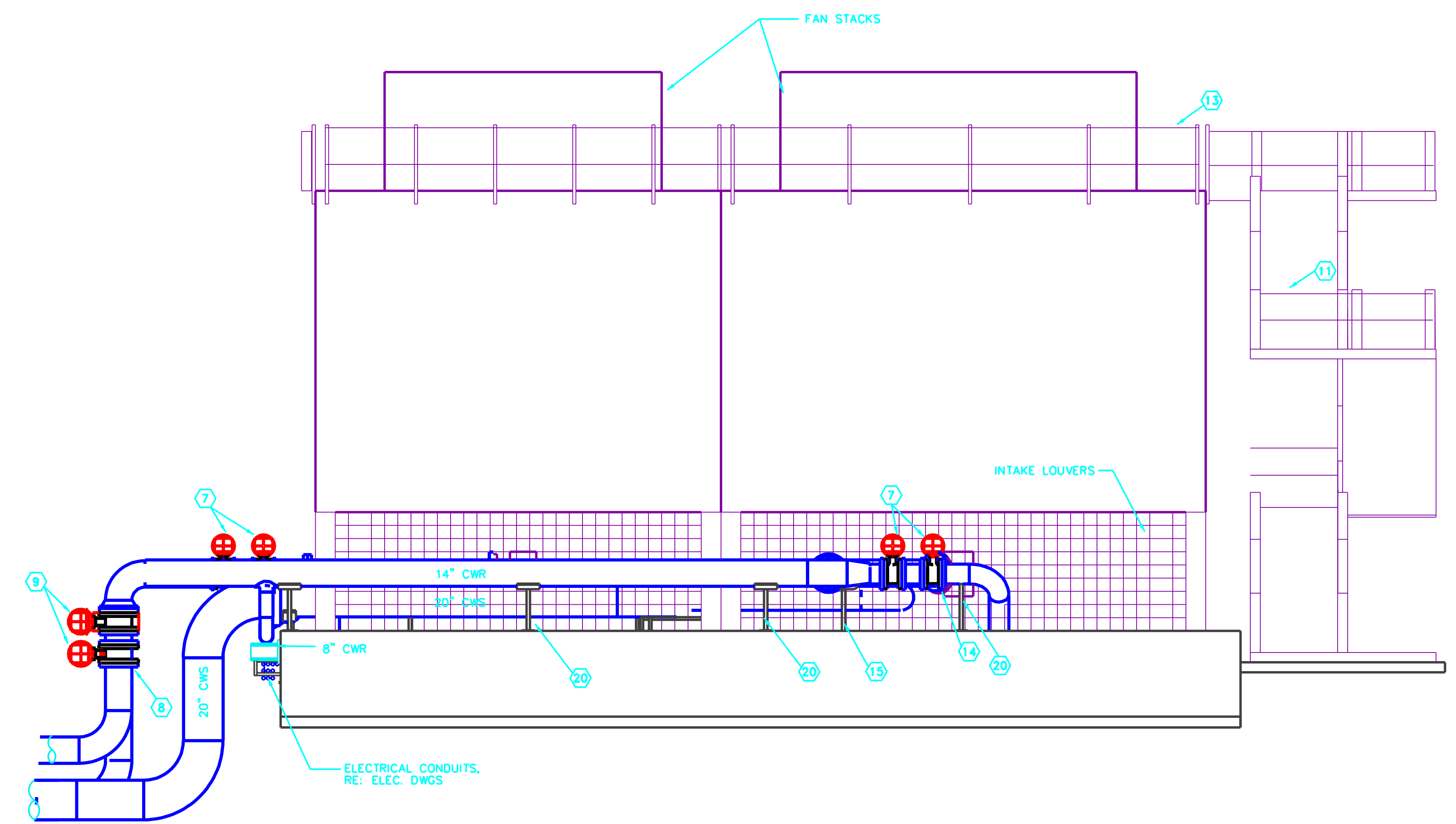
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	MECHANICAL PIPING PLAN AND SECTIONS
Approved: Engineering Svc. Chief	
Approved: Medical Center Director	

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
	CHILLER PLANT EXPANSION
Building Number	
Location	VAMC HOUSTON, TX

Date	02/25/2003
Project No.	VA #580-58-110 SSA #1017-007-1
Drawing No.	M3.01
Dwg. of	

Veterans Administration



104-CWP-8

BUTTERFLY VALVE

4

20" CWS

14" CWR

6

17

8" CWR

19

20

20

20" CWS

CHECK VALVE

8" CWS

18

9'-10"

RE: 08/M4.02

ELECTRICAL CONDUITS, RE: ELEC. DWGS

03 SECTION

1/4"=1'-0"

20" CWS

14" CWR

20

19

GRADE

16" CWS

16" CWR

04 SECTION

1/4"=1'-0"

RECORD DRAWING

- 17 ROUTE 8" CONDENSER WATER PIPE ALONG BASIN WALL. RE-DETAIL OR MAKE FOR SUPPORT AND PIPE INSTALLATION.
- 18 CONNECT 8" CONDENSER WATER PIPING TO EXISTING PIPING.
- 19 UNISTRUT F1000 FIBERGLASS CHANNELS WITH F1036 GENERAL MOUNTING FITTINGS AND F561 V-GE-PIV BASE OR EQUAL, SECURED TO SUMP WALL WITH STAINLESS STEEL HARDWARE.
- 20 HOT DIPPED GALVANIZED PIPE STAND SECURED TO BASIN WALL WITH STAINLESS STEEL HARDWARE.

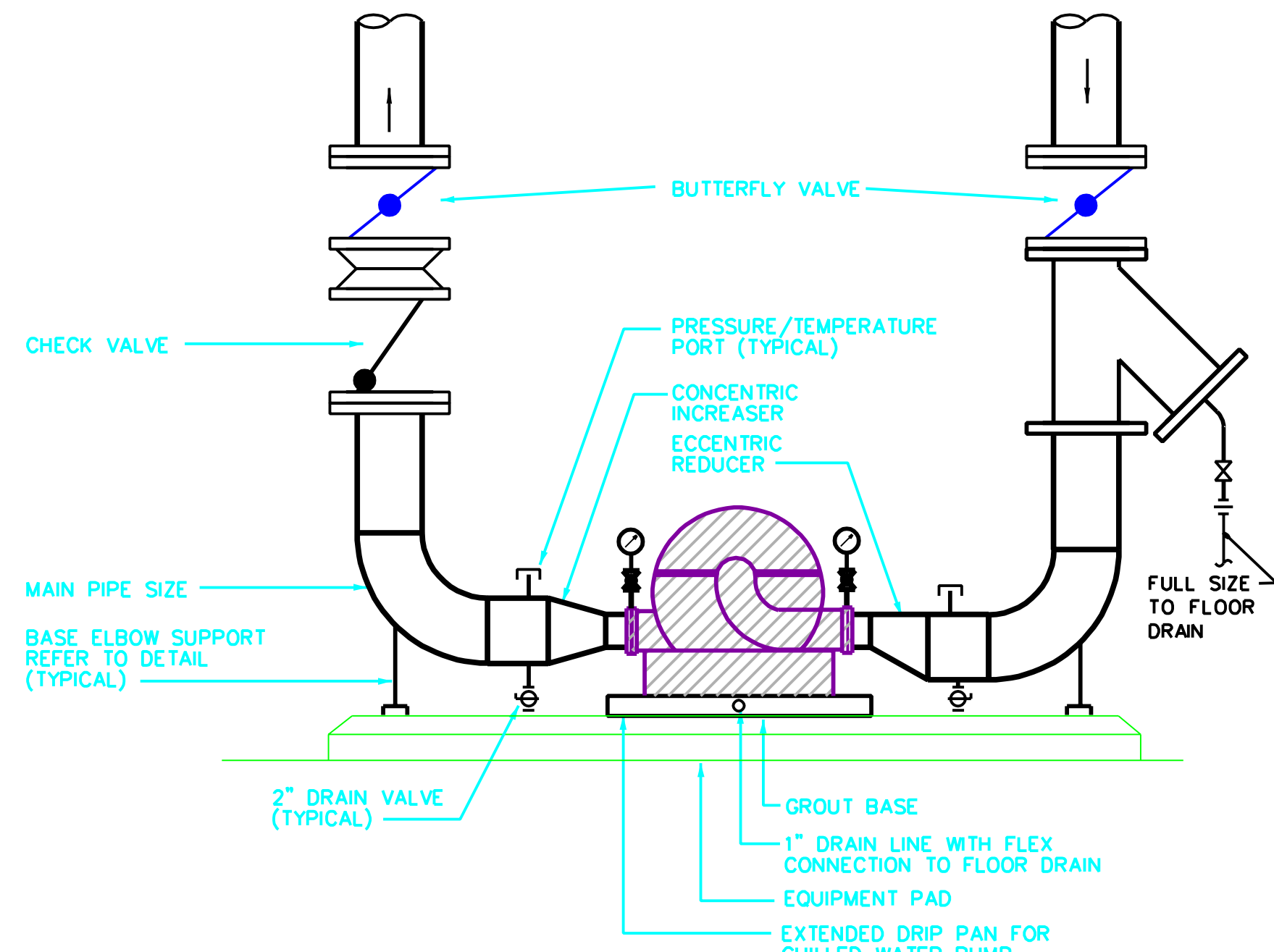


Fax: 713-780-9209

Date	02/25/2003
Project No.	VA #580-98-110 SSA #1017-007-1
Drawing No.	M3.02
Dwn	of

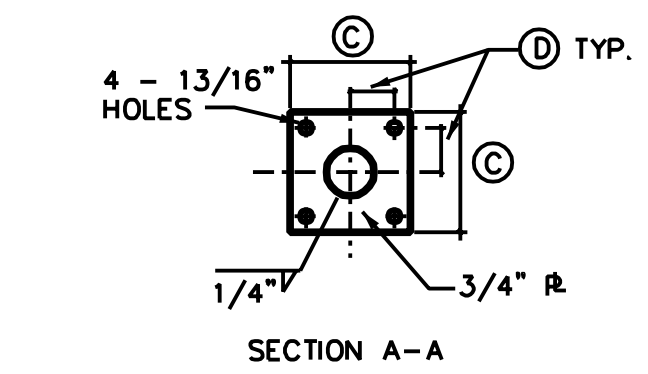
Veterans Administration

RECORD DRAWING
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SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.



01 DOUBLE SUCTION PUMP DETAIL
NOT TO SCALE

SIZE SCHEDULE			
(A)	(B)	(C)	(D)
12" UP	6"	12"	4"

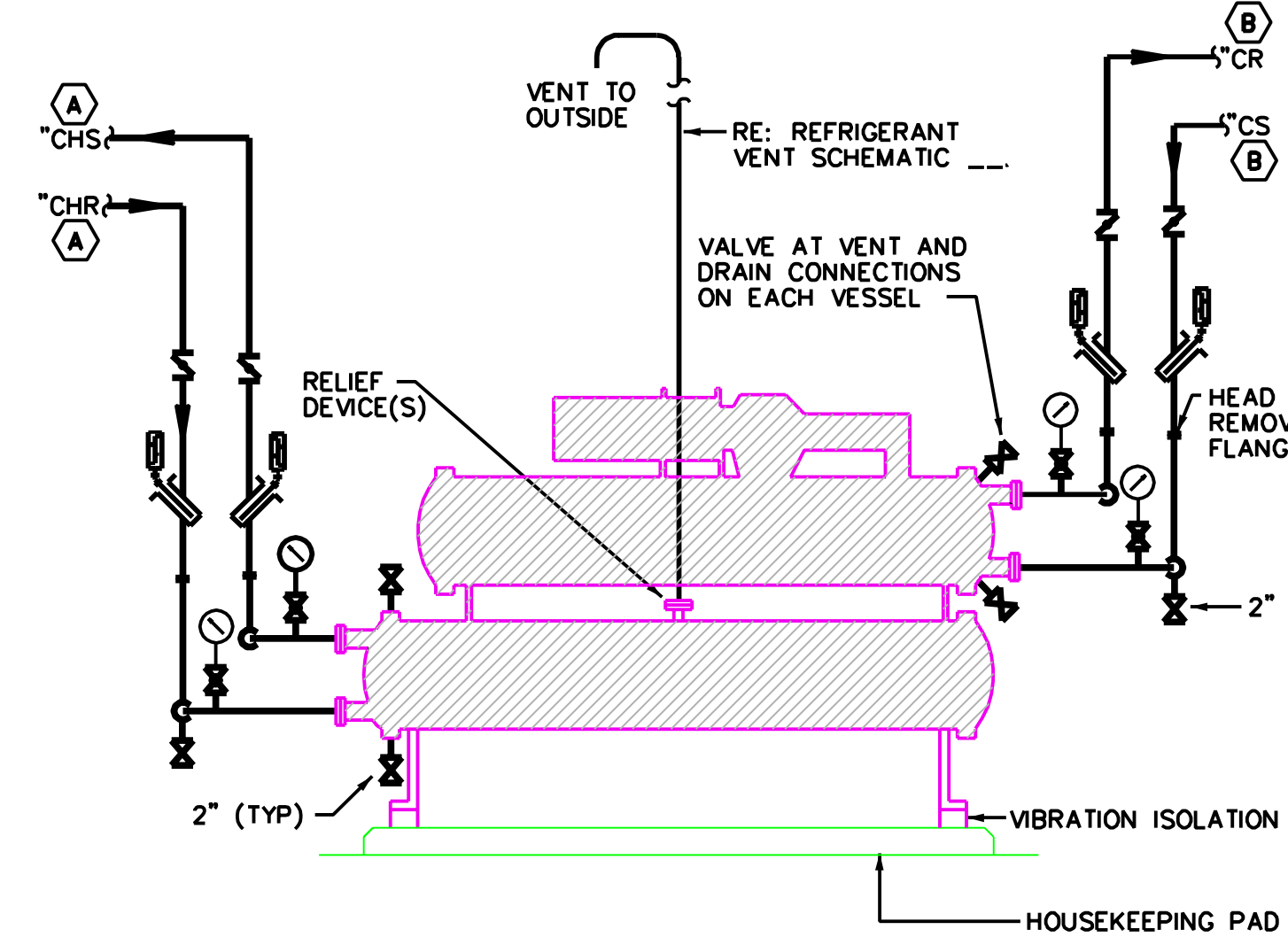


NOTE:
1. INSULATE BASE ELBOW SUPPORTS ON ALL LOW TEMPERATURE SERVICE.

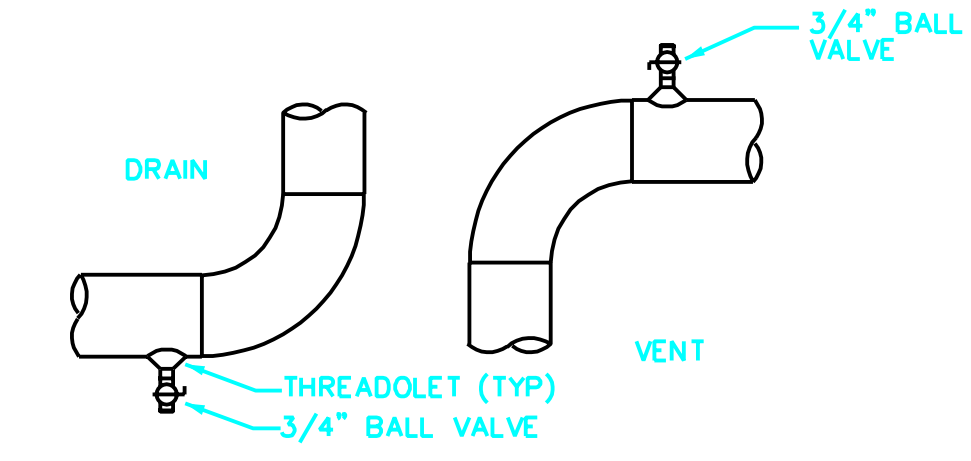
02 BASE ELBOW SUPPORT DETAIL
NO SCALE

SIZE SCHEDULE		
UNITS	(A)	(B)
104-CCU-8	12"	14"
104-CCU-9	12"	14"

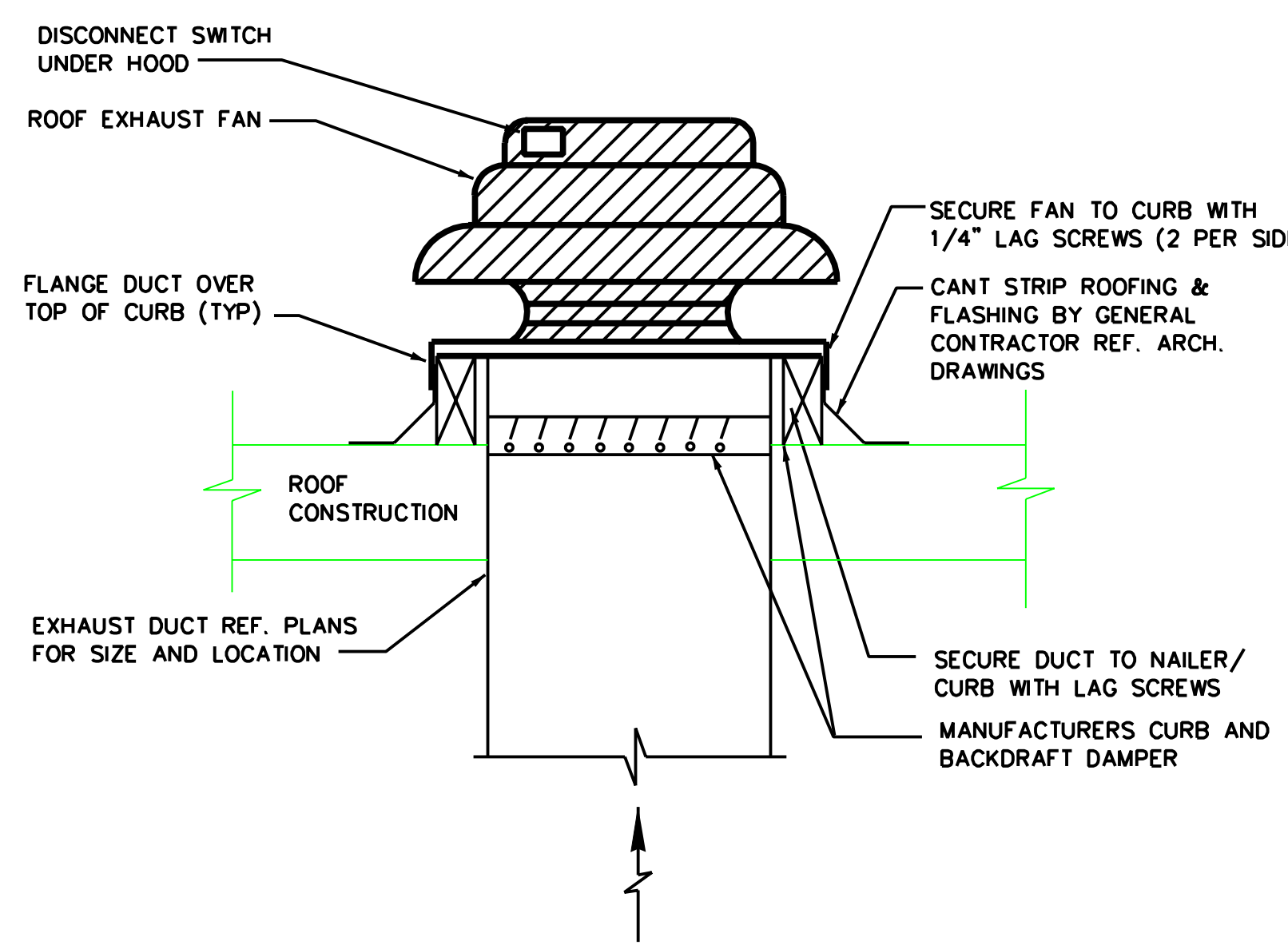
- NOTES:
1. SUBMIT SHOP DRAWINGS OF PIPING IF NOZZLE ARRANGEMENTS VARY FROM ARRANGEMENT SHOWN ON PLAN VIEW.
2. LOCATE PIPING TO FACILITATE WATER BOX COVER REMOVAL.



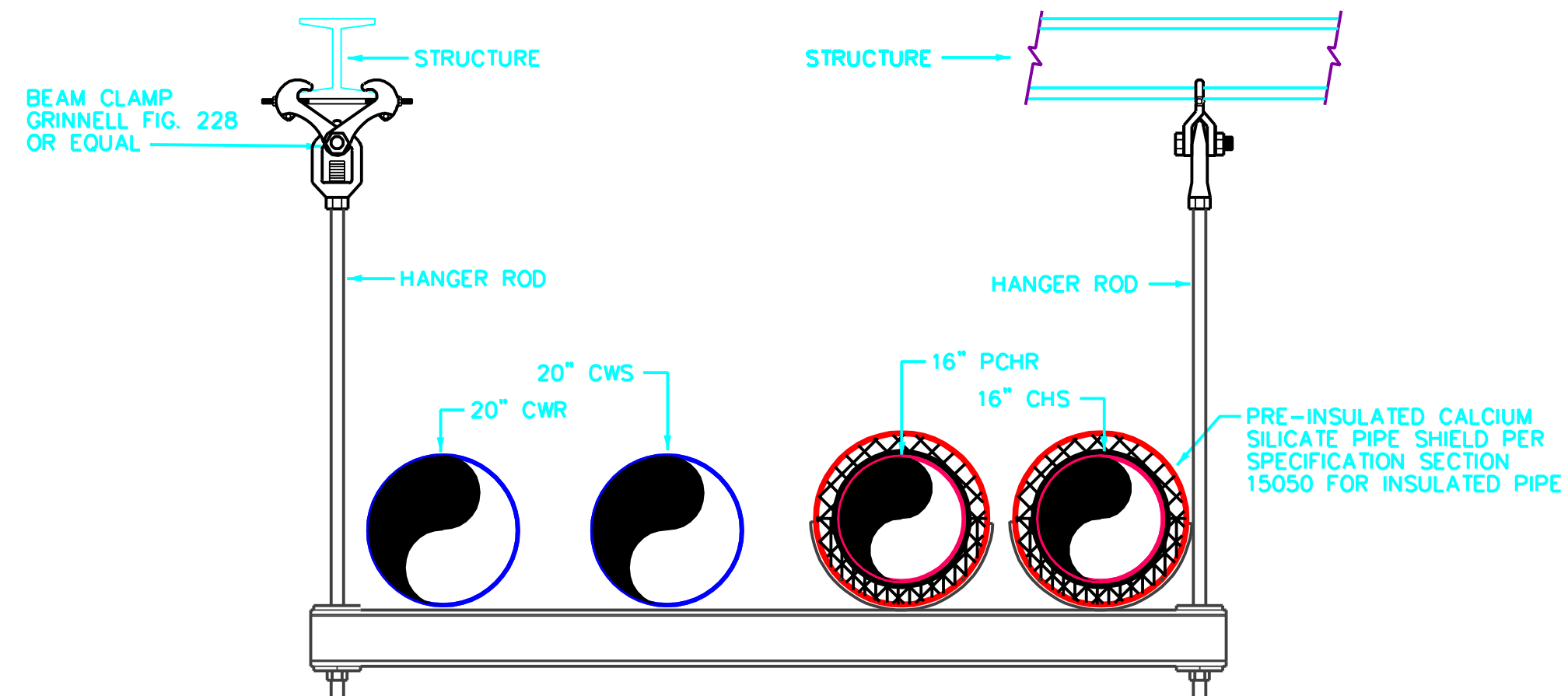
03 CENTRIFUGAL CHILLER PIPING DETAIL
NO SCALE



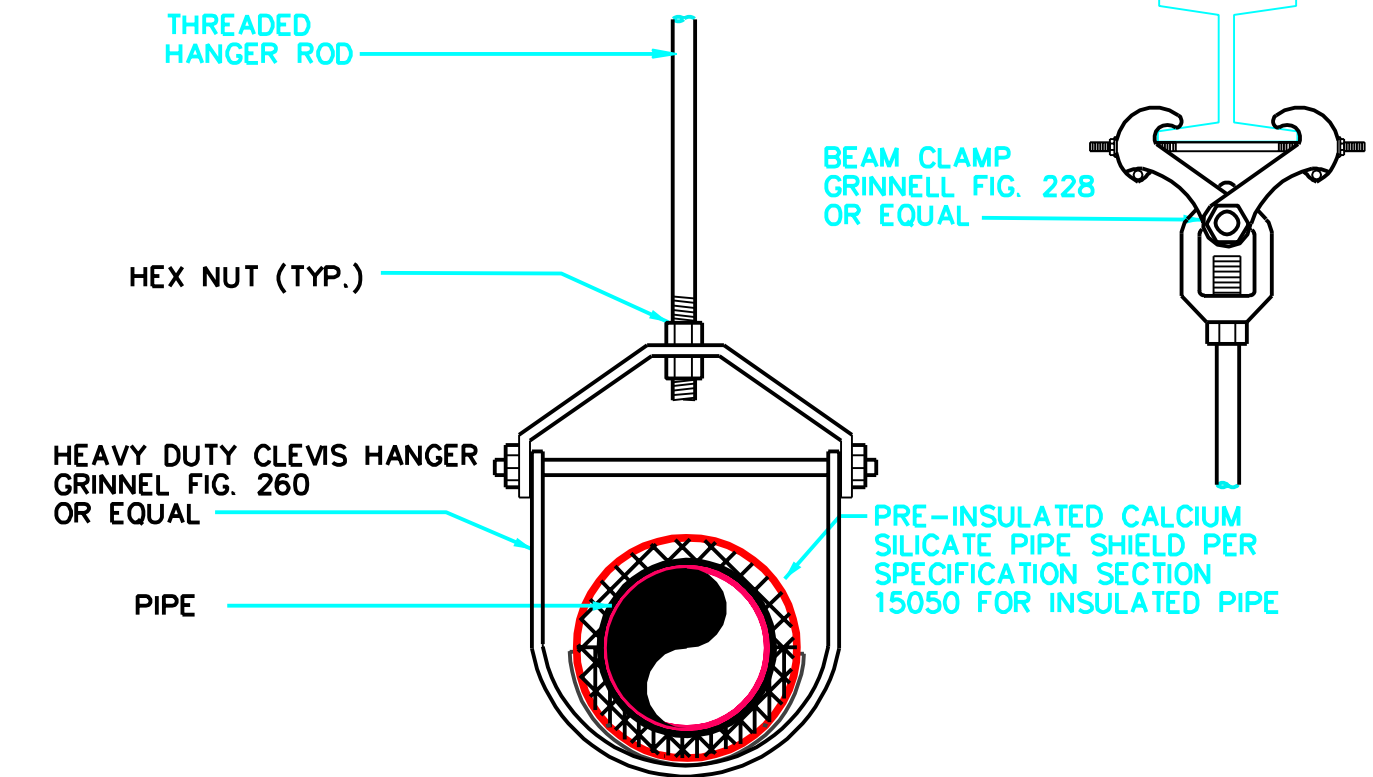
04 DRAIN AND VENT DETAIL
NO SCALE



05 ROOF MOUNTED EXHAUST FAN
NOT TO SCALE

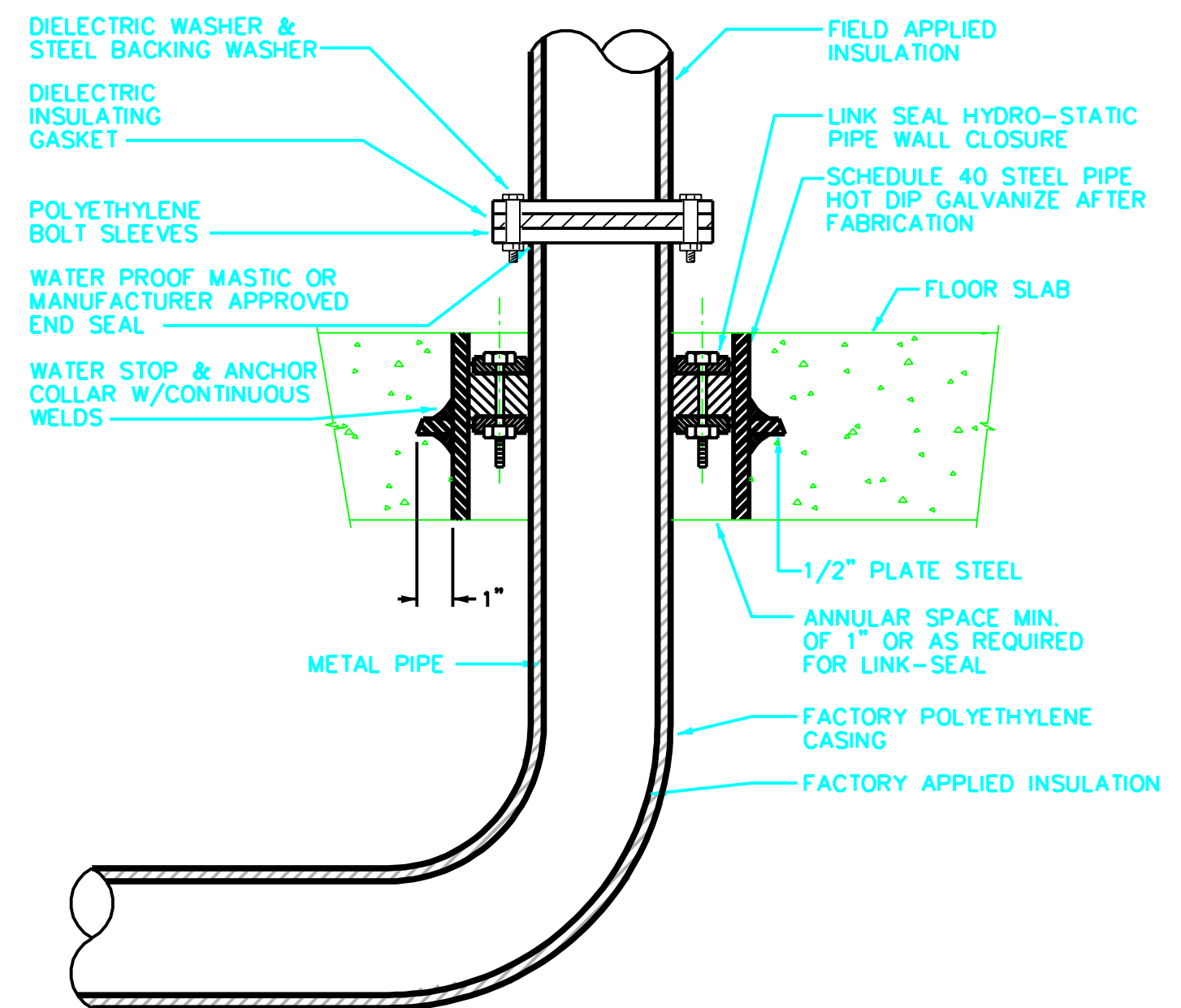


06 PIPE SUPPORT DETAIL
NO SCALE

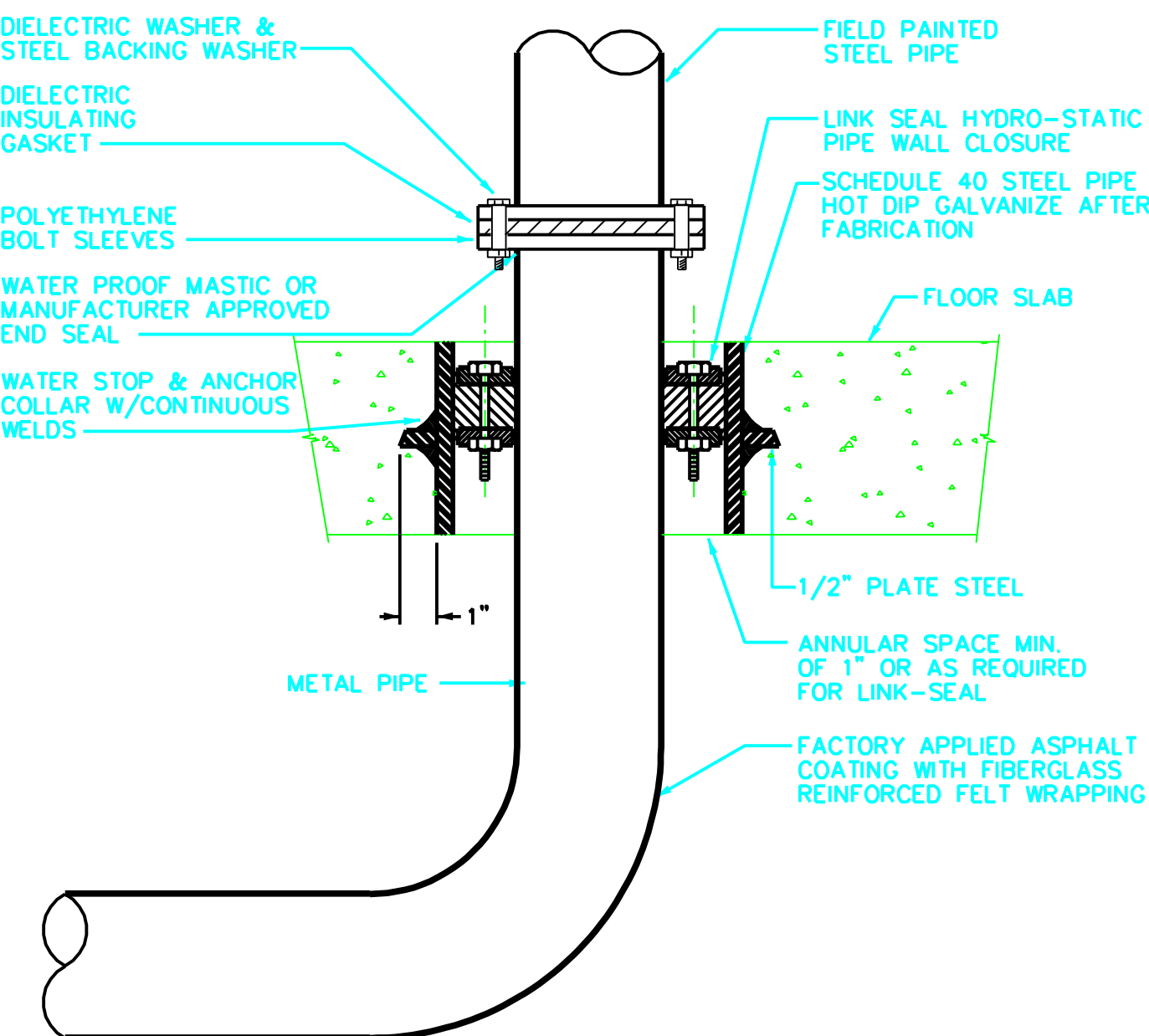


LOAD CARRYING CAPACITIES OF THREADED HANGER ROD MATERIALS - ASTM A36, A575 GR. 1020 OR A576 GR1020		
NOMINAL DIAMETER, IN.	ROOT AREA OF COARSE THREAD SQUARE IN.	MAXIMUM SAFE LOAD POUNDS, ROD TEMPERATURE, 650 DEG. F
5/8 (MIN.)	0.202	1810
3/4	0.302	2710
7/8	0.419	3770
1	0.552	4960
1 1/4	0.889	8000
1 1/2	1.293	11630
1 3/4	1.744	15690
2	2.232	20680
2 1/4	3.021	27200
2 1/2	3.716	33500
2 3/4	4.619	41600
3	5.621	50600

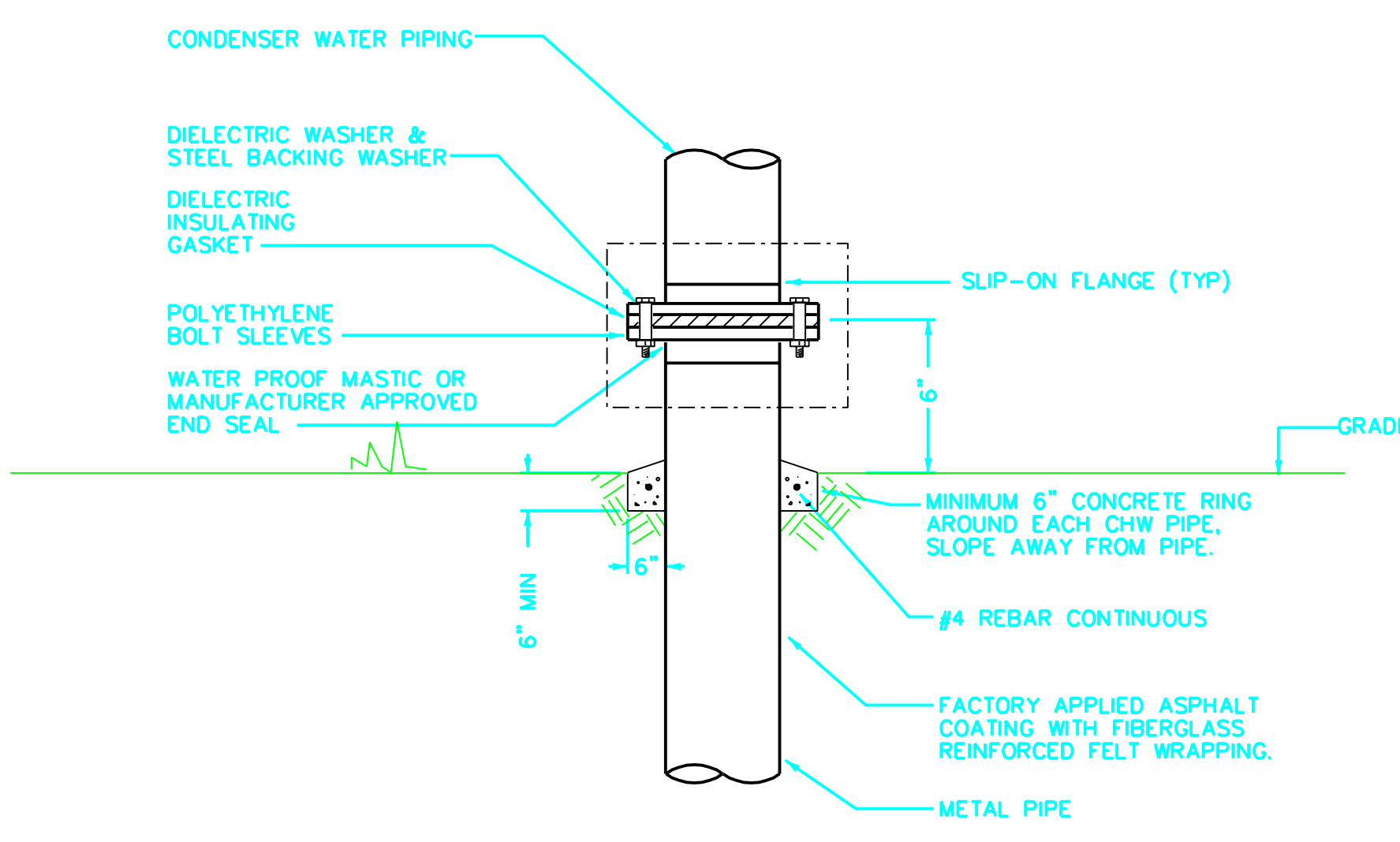
07 PIPE SUPPORT DETAIL PS-C
NO SCALE



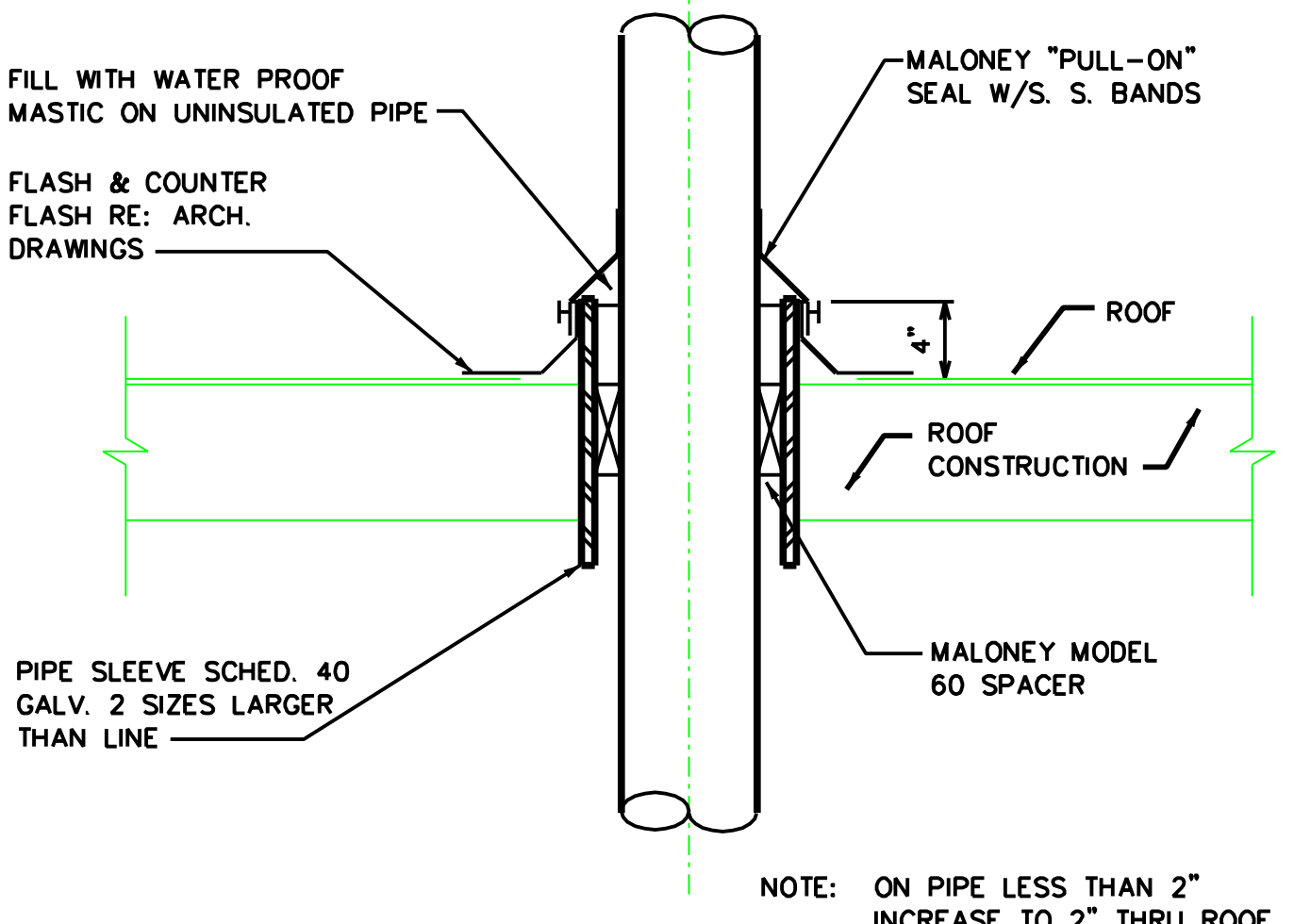
08A CHILLED WATER PIPE THROUGH FLOOR DETAIL
NO SCALE



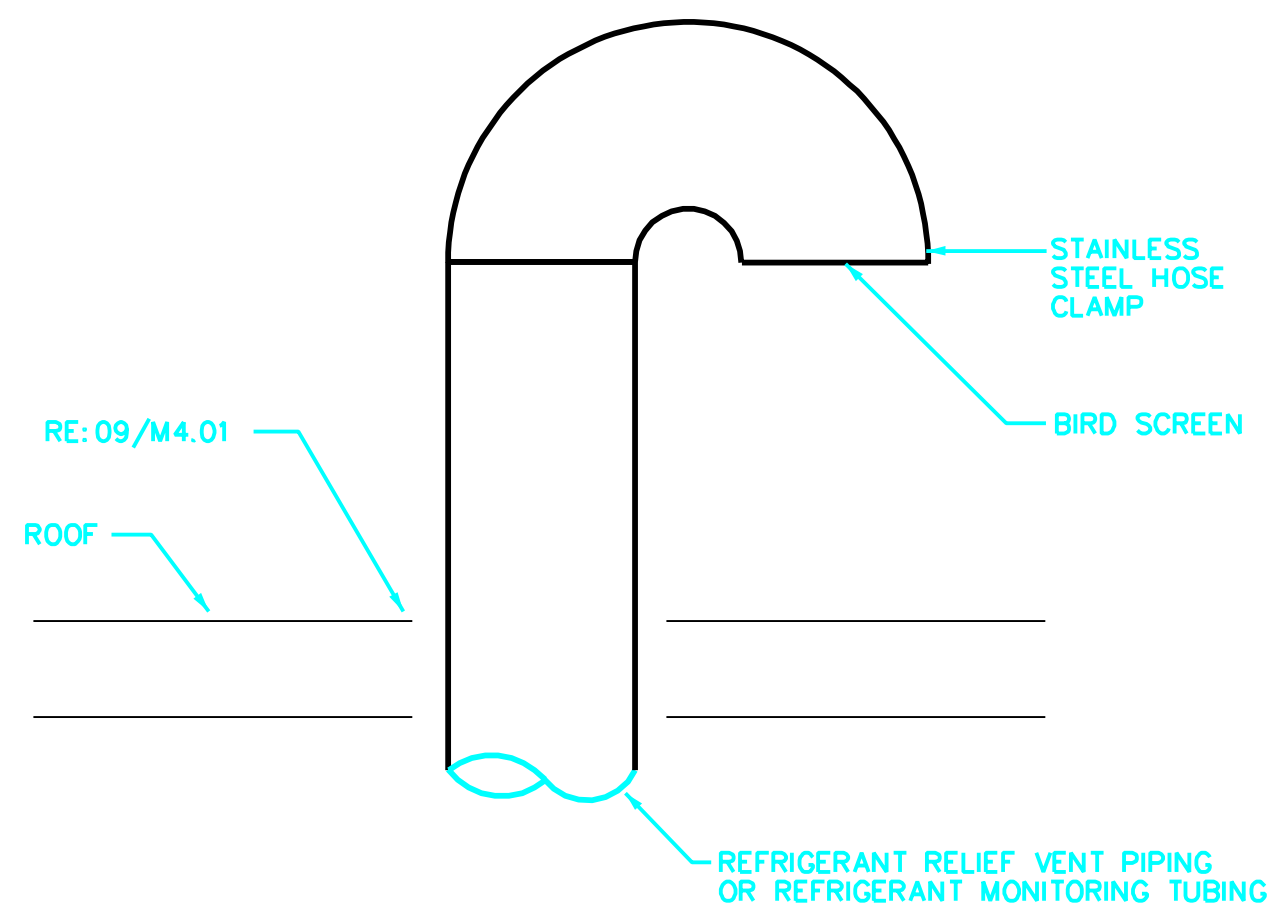
08B CONDENSER WATER PIPE THROUGH FLOOR DETAIL
NO SCALE



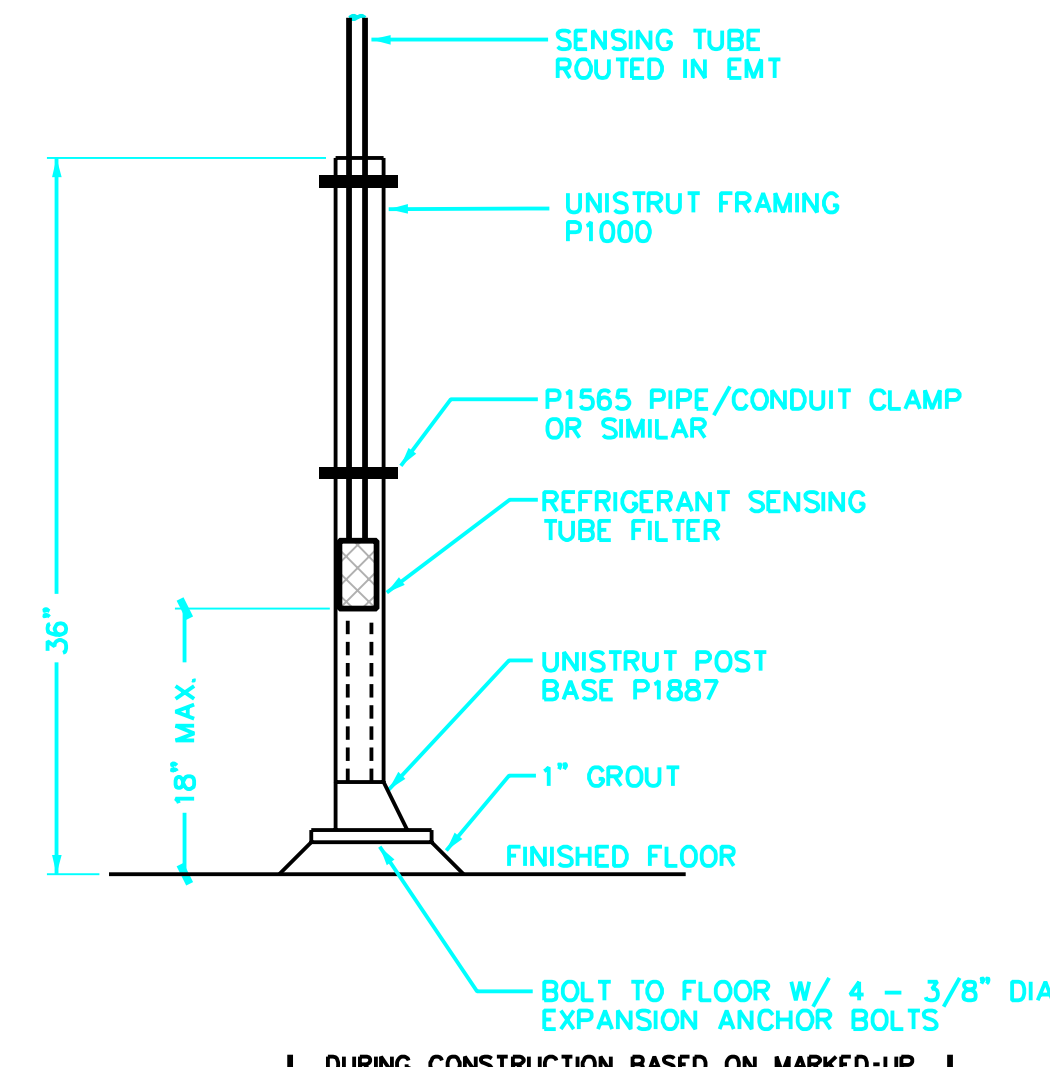
08C CONDENSER WATER TRANSITION FOR DIRECT BURIED TO ABOVE GRADE
NO SCALE



09 PIPE THRU ROOF DETAIL
NO SCALE



10 REFRIGERANT RELIEF PIPE TERMINATION DETAIL
NO SCALE



11 REFRIGERANT MONITORING REFERENCE TUBING DETAIL
NO SCALE

RECORD DRAWING
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Revisions	Date



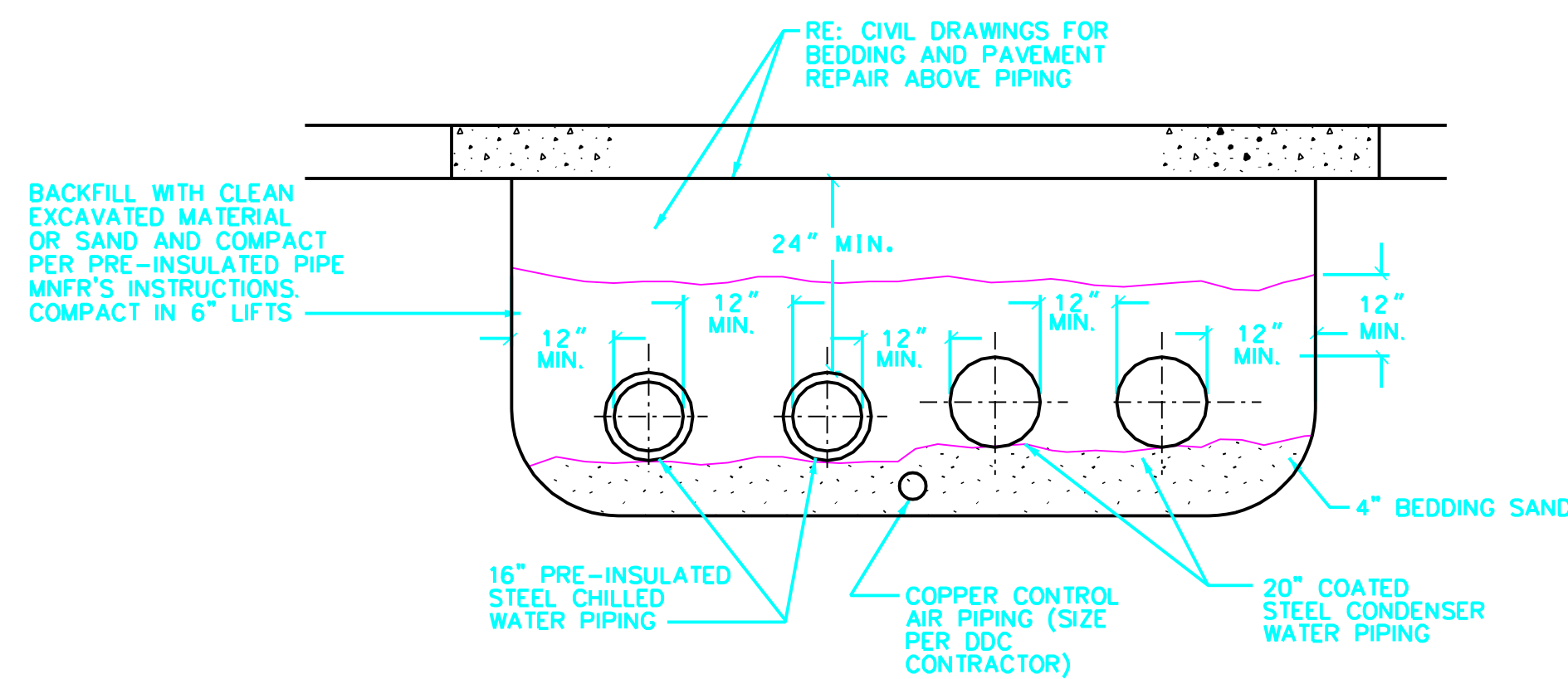
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title MECHANICAL DETAILS
Approved: Engineering Svc. Chief
Approved: Medical Center Director

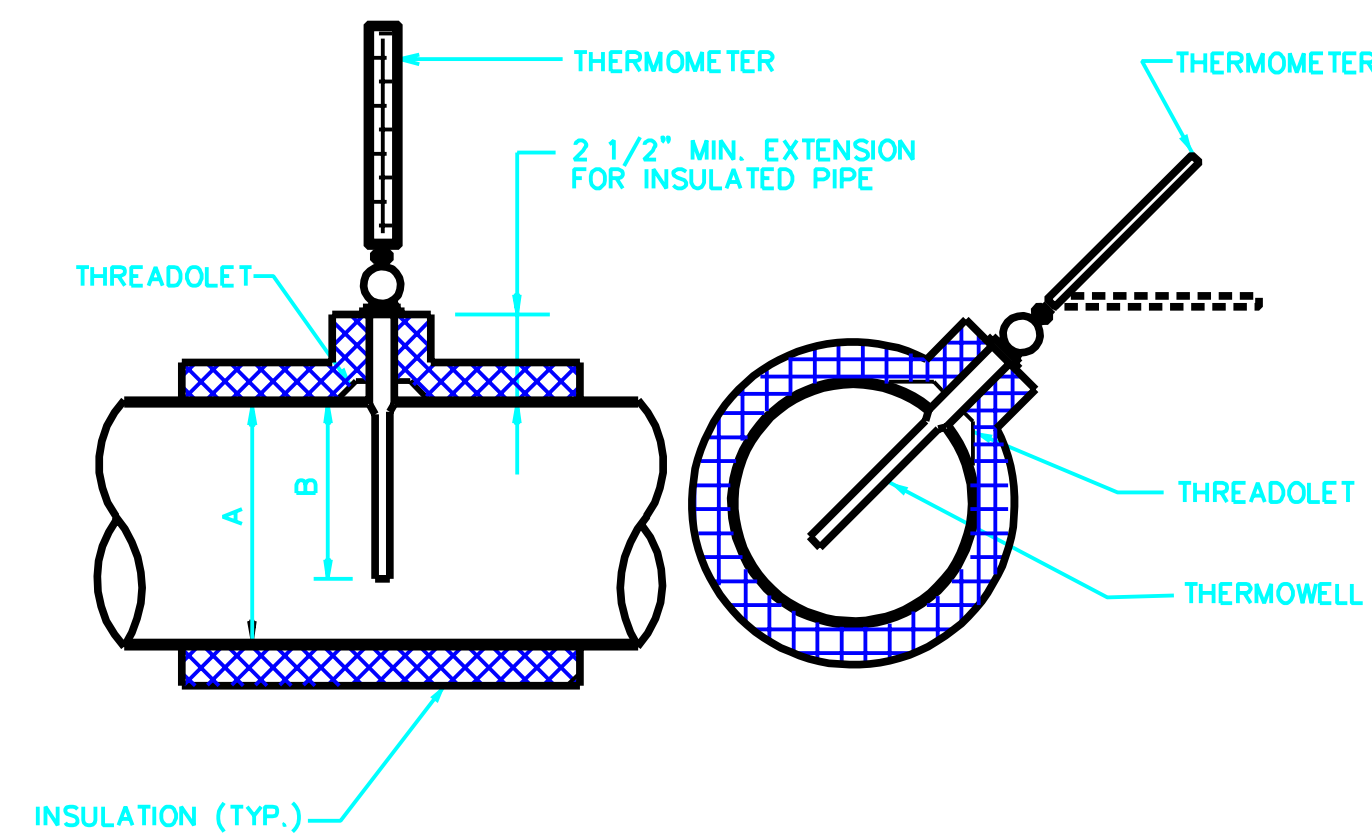
Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION
Building Number
Location VAMC HOUSTON, TX

Date 02/25/2003
Project No. VA #560-58-110 SSA #1017-007-1
Drawing No. M4.01
Dwg. of

Veterans Administration

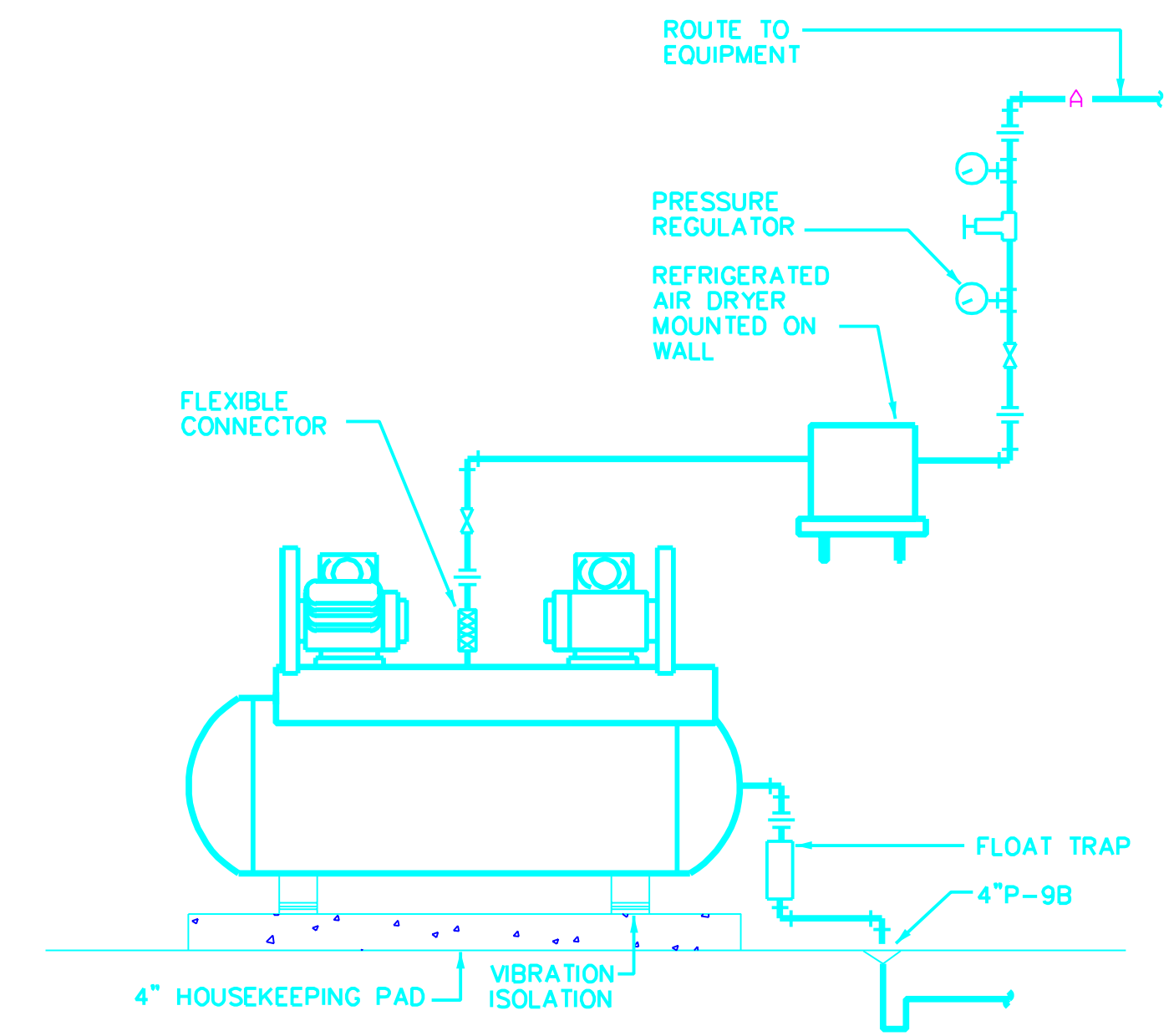


01 TRENCH DETAIL
NOT TO SCALE

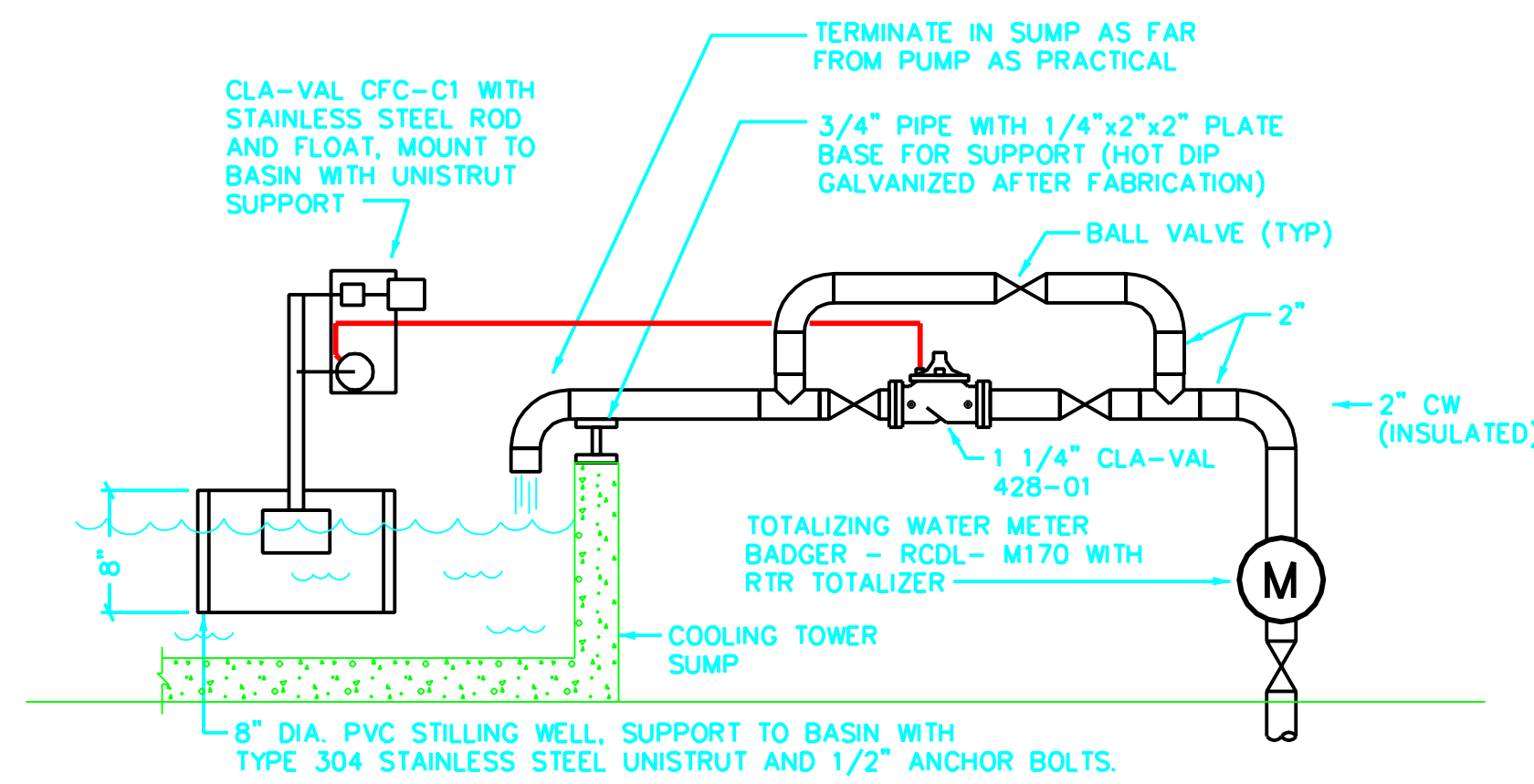


02 THERMOMETER WELL DETAIL
NOT TO SCALE

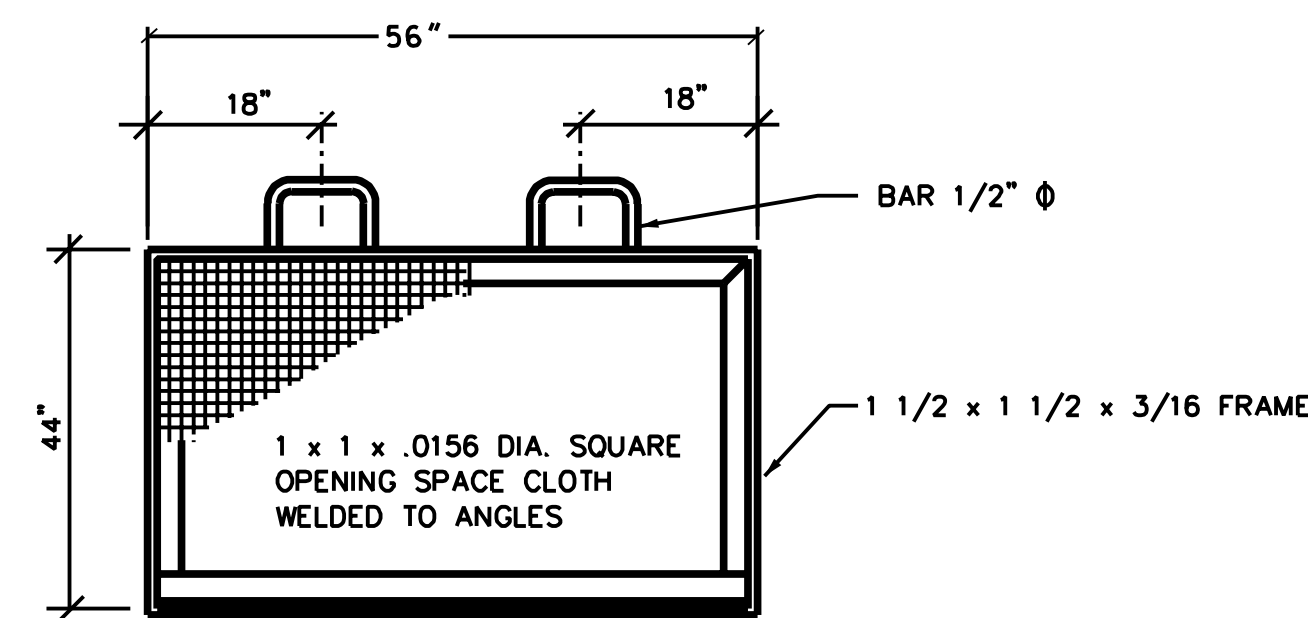
INSERTION LENGTH	
A	B
3"	2"
4"	3"
5"	4"
6"	5"
8" AND 10"	7 1/2"
12" & LARGER	11 1/2"



03 DUPLEX AIR COMPRESSOR SCHEMATIC
NO SCALE

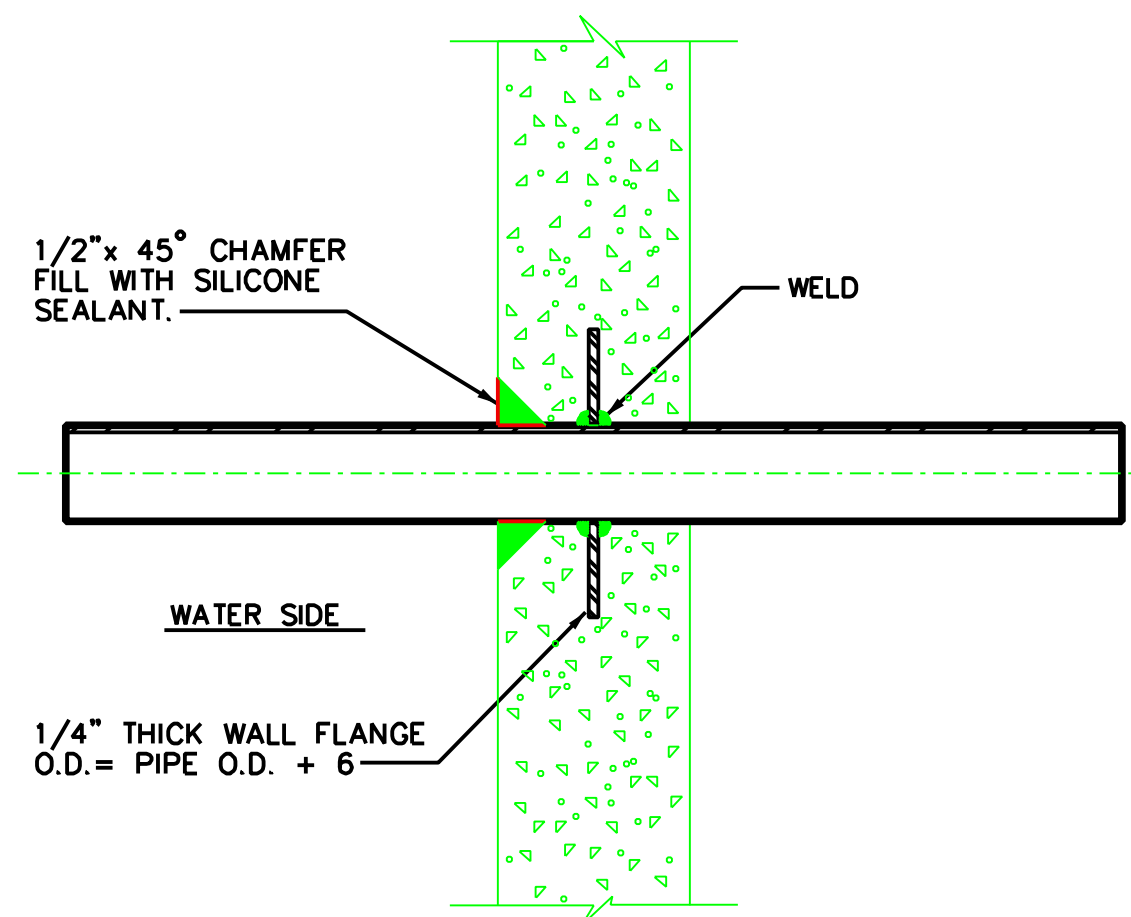


04 FLOAT VALVE DETAIL
NOT TO SCALE

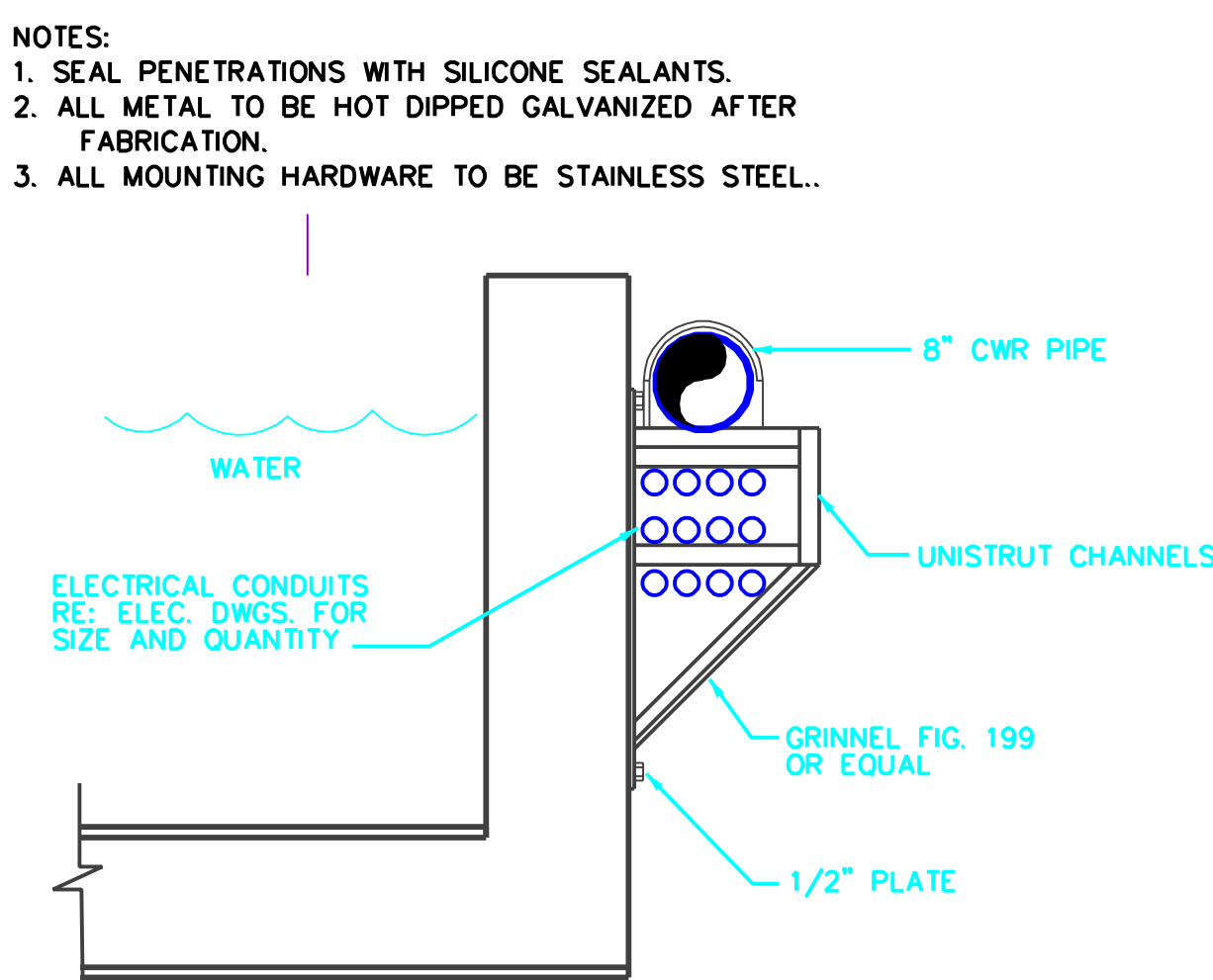


- NOTES:
1. ALL WELDED CONSTRUCTION.
 2. ALL MATERIAL IS HOT DIPPED GALVANIZED AFTER CONSTRUCTION.
 3. PROVIDE ONE SCREEN AT EACH SUMP INLET. RE: STRUCTURAL DWGS FOR BRACKETS.
 4. COORDINATE OVERALL DIMENSIONS WITH SUMP DIMENSIONS PRIOR TO FABRICATION. (RE: STRUCTURAL DWGS)

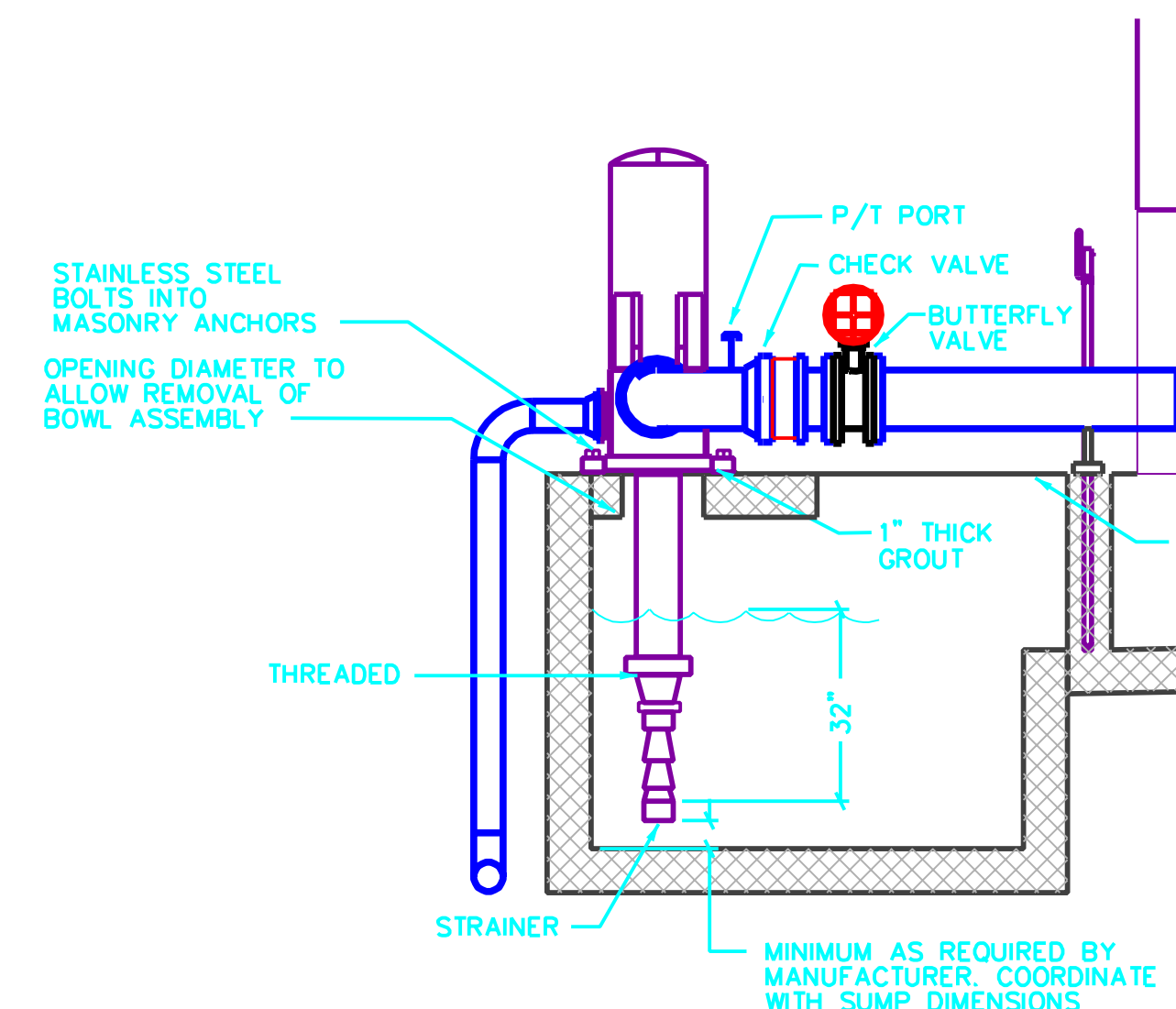
05 CONDENSER WATER SUMP TRASH SCREEN DETAIL
NO SCALE



07 PIPE PENETRATION DETAIL FOR COOLING TOWER OVER FLOW
NOT TO SCALE

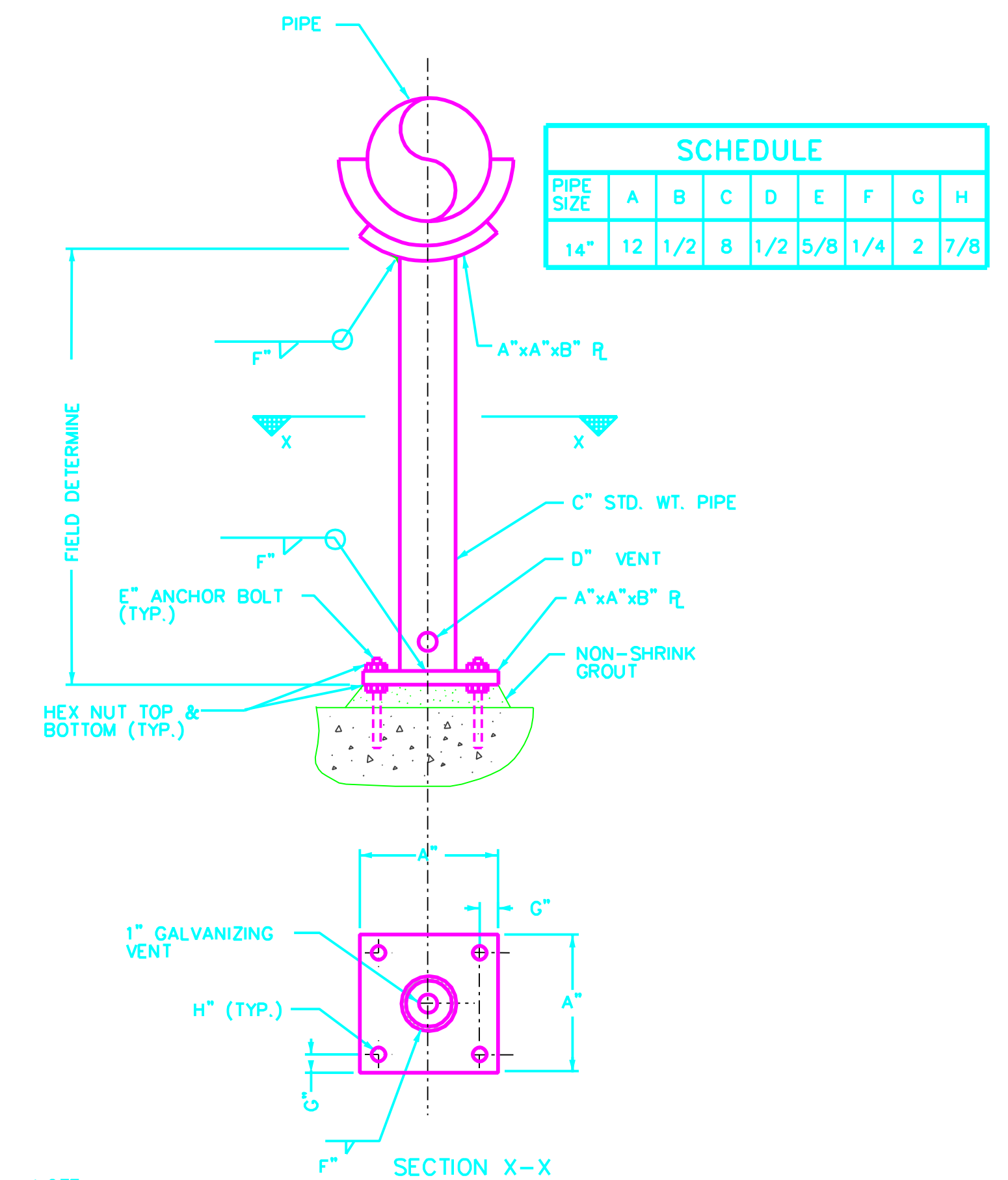


08 SECTION DETAIL
NOT TO SCALE



- NOTE:
1. EXISTING SUMP DIMENSION BASED ON SCHEDULED PUMPS. IF PROVIDING PUMPS BY OTHER MANUFACTURER, PROVIDE MANUFACTURER'S SUMP LAY-OUT AND REVISED SUMP AS NECESSARY.

09 SECTION DETAIL
NOT TO SCALE

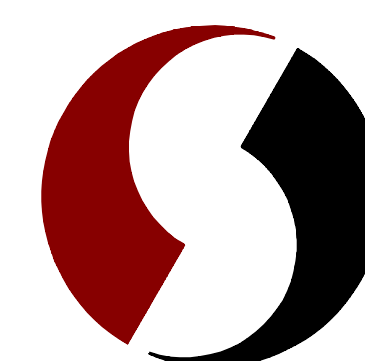


- NOTE:
1. PIPE SUPPORTS OUTDOORS SHALL BE HOT DIP GALVANIZED AFTER FABRICATION.
 2. ALL OTHER PIPE SUPPORTS SHALL BE PRIMED AND PAINTED PER SPECIFICATIONS.

10 PIPE SUPPORT DETAIL PS-H
NOT TO SCALE

RECORD DRAWING
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Revisions	Date
29NOV01	



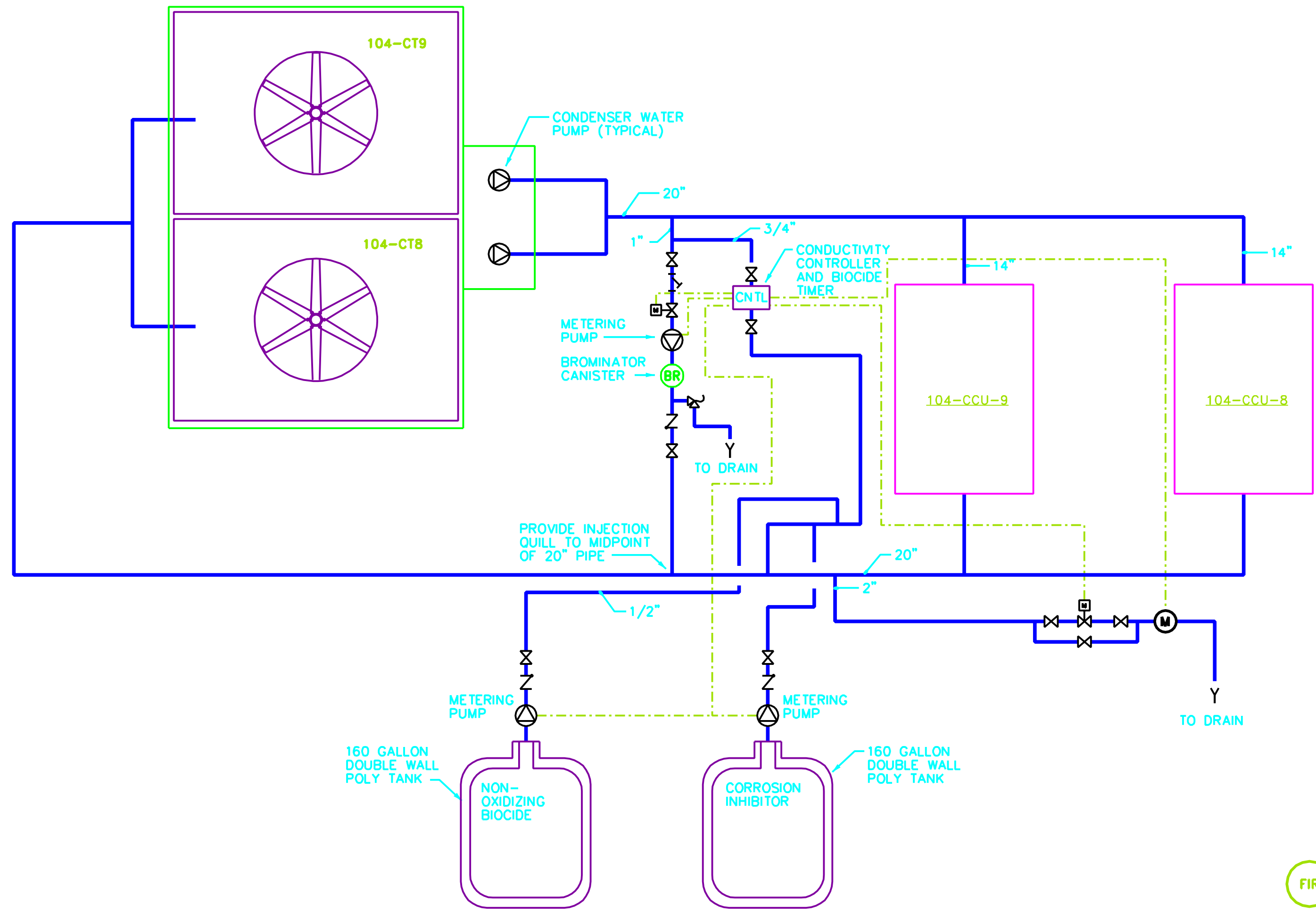
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	MECHANICAL DETAILS
Approved: Engineering Svc. Chief	
Approved: Medical Center Director	

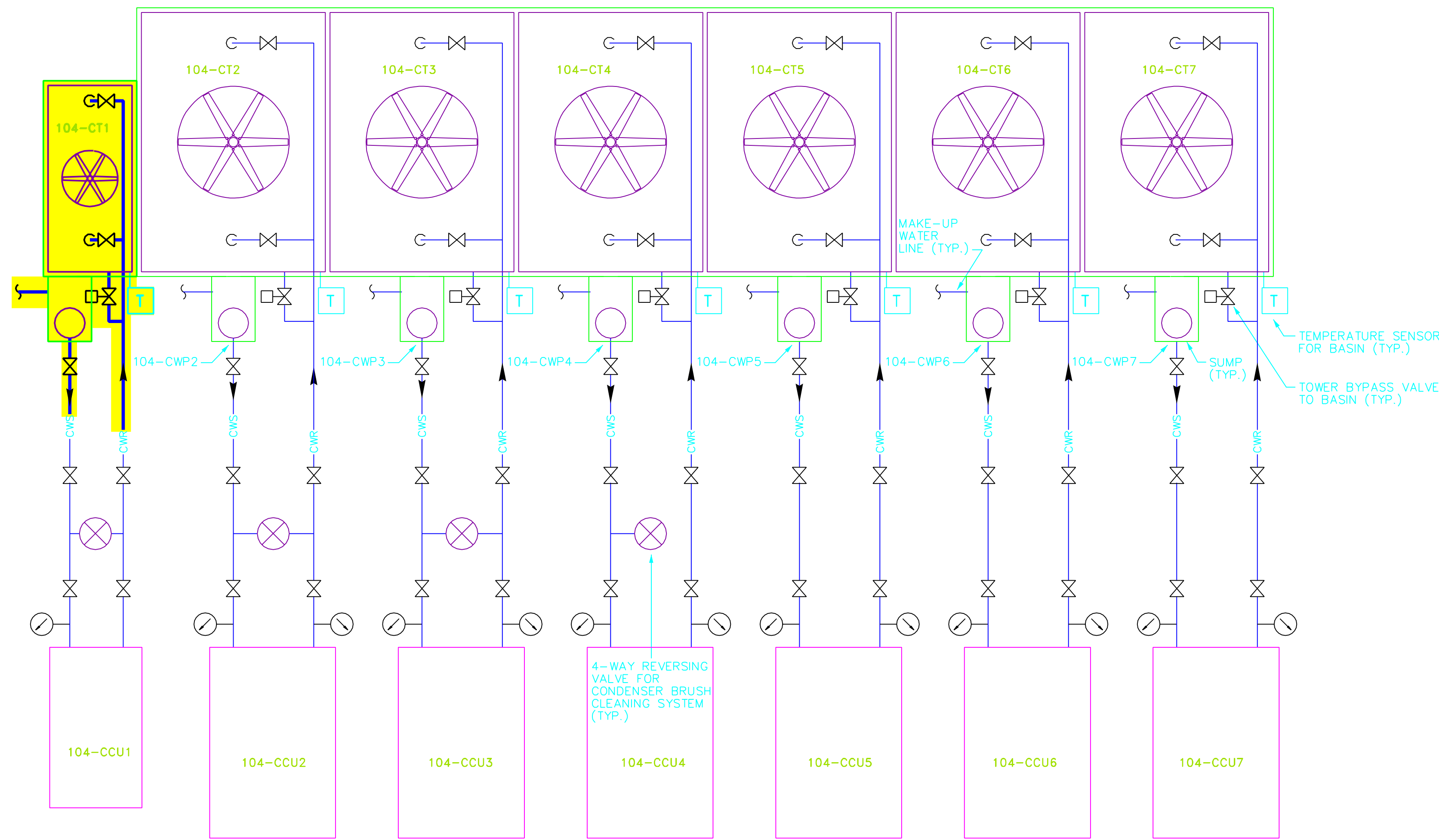
Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
Building Number	104/106
Location	VAMC HOUSTON, TX

Date	02/25/2003
Project No.	VA #560-58-110 SSA #1017-007-1
Drawing No.	M4.02
Dwg. of	

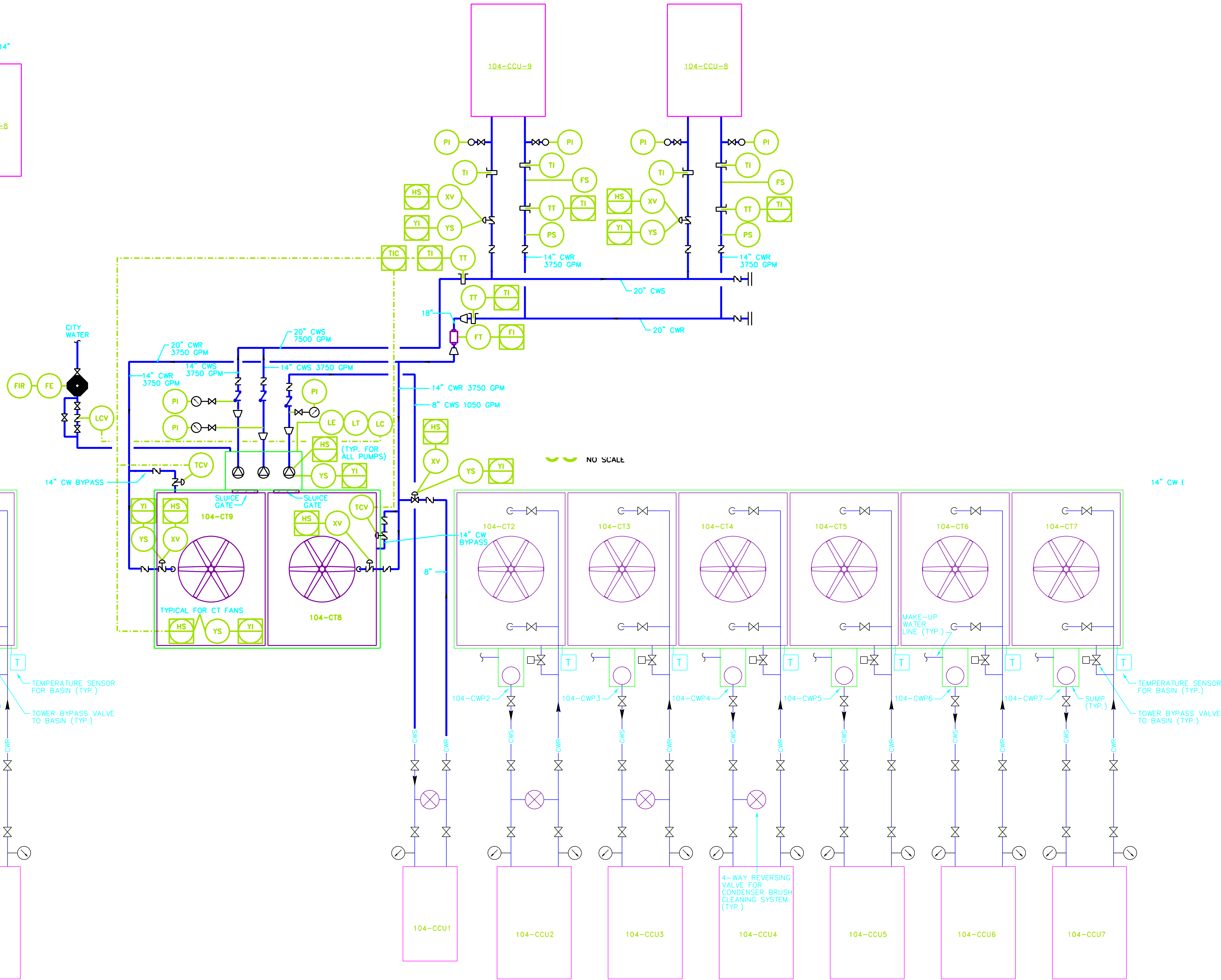
Veterans Administration



03 CONDENSER WATER TREATMENT SCHEMATIC
(ALTERNATE BID #1)
NO SCALE



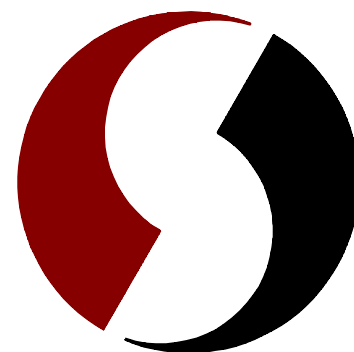
01 CONDENSER WATER FLOW DIAGRAM - DEMOLITION
NO SCALE



02 CONDENSER WATER FLOW DIAGRAM - RENOVATION
NO SCALE

RECORD DRAWING
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SIGNIFICANT CHANGES IN THE WORK MADE
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Revisions	Date



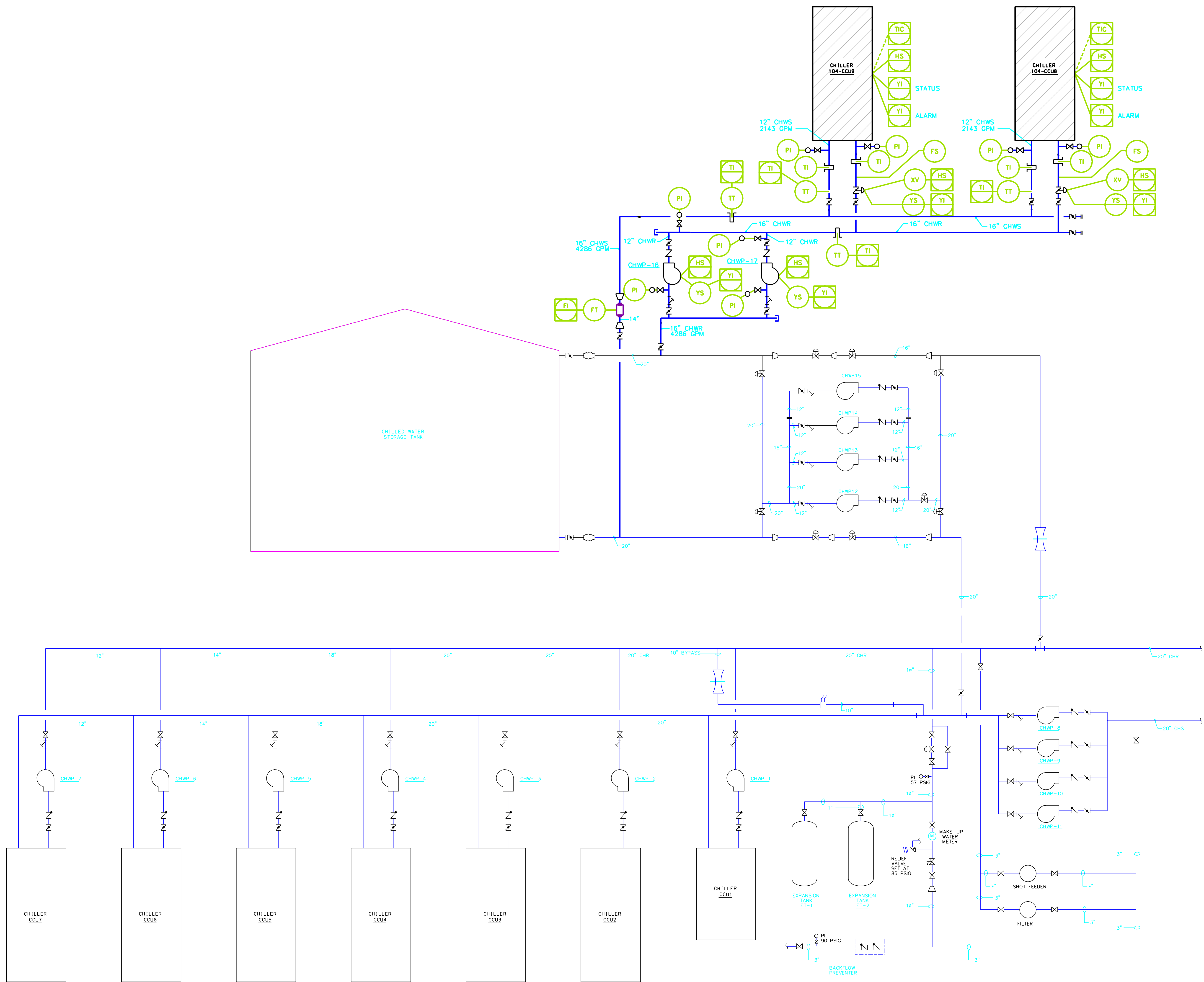
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title	CONDENSER WATER FLOW DIAGRAM		
Approved: Engineering Svc. Chief			
Approved: Medical Center Director			

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION		
Building Number	104/106	Checked	ASS
Location	VAMC HOUSTON, TX		

Date	02/25/2003
Project No.	VA #560-58-110 SSA #1017-007-1
Drawing No.	104-01
Dwg. of	

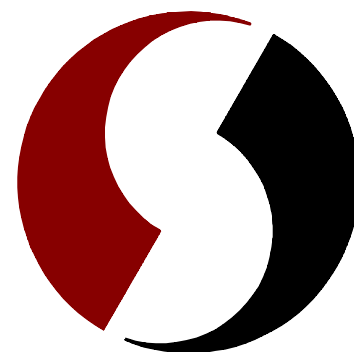
Veterans
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01 CHILLED WATER FLOW DIAGRAM
NOT TO SCALE

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Revisions	Date



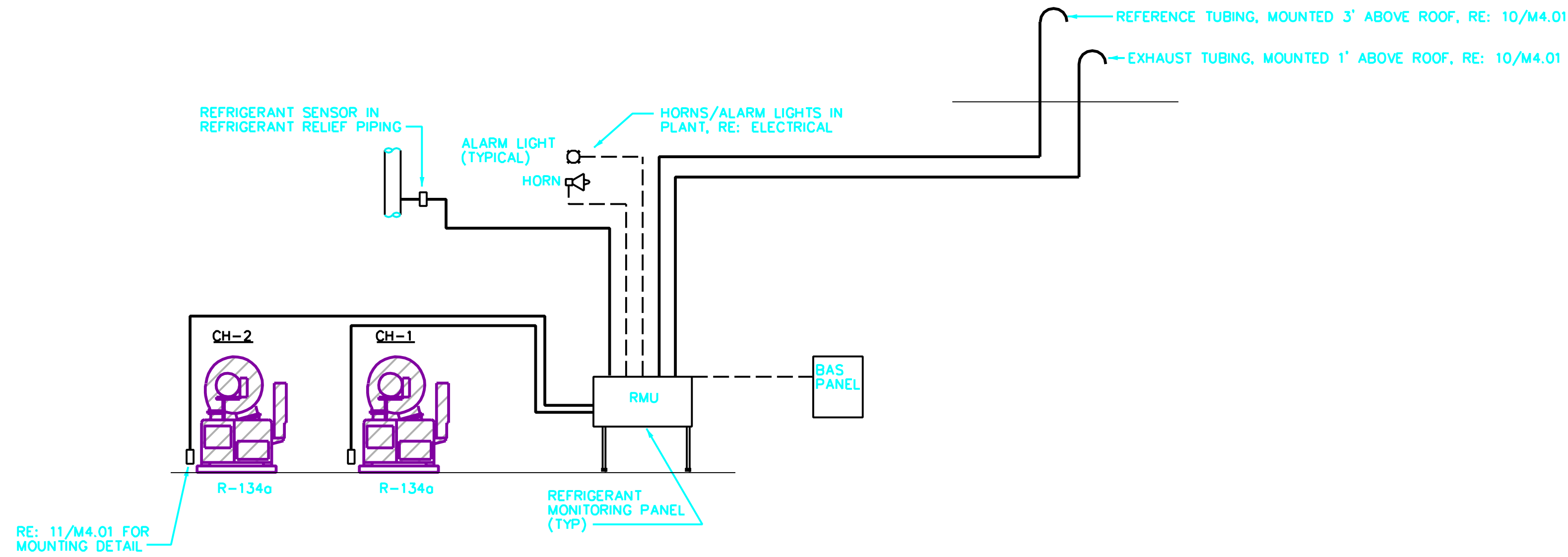
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
CHILLED WATER FLOW DIAGRAM
Approved: Engineering Svc. Chief
Approved: Medical Center Director

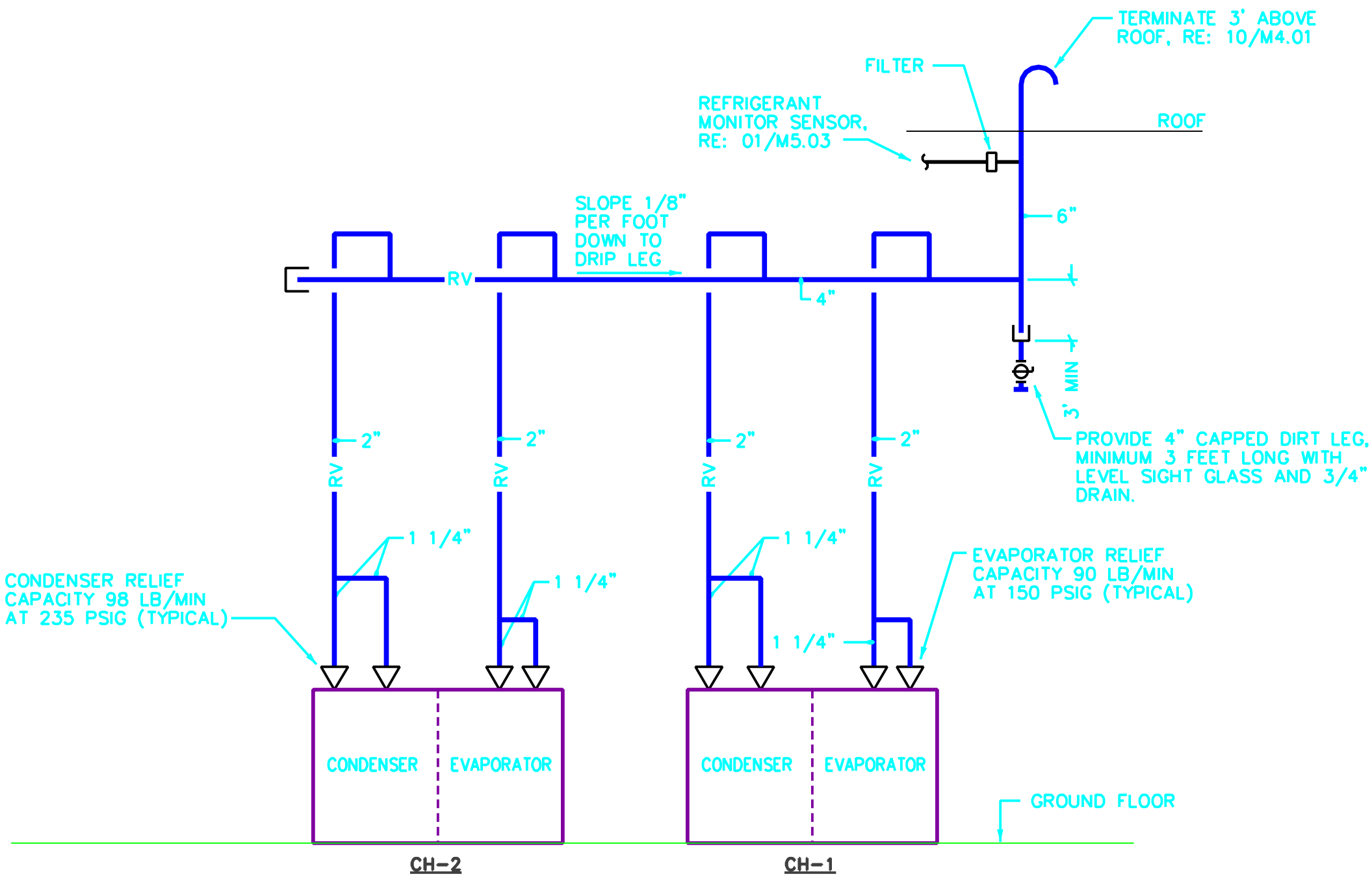
Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

Date
02/25/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
NX02
Dwg. of

Veterans
Administration



01 REFRIGERANT MONITORING SCHEMATIC
NO SCALE



02 REFRIGERANT RELIEF VENT
PIPING SCHEMATIC
NO SCALE

RECORD DRAWING
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Revisions	Date



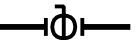
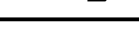
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

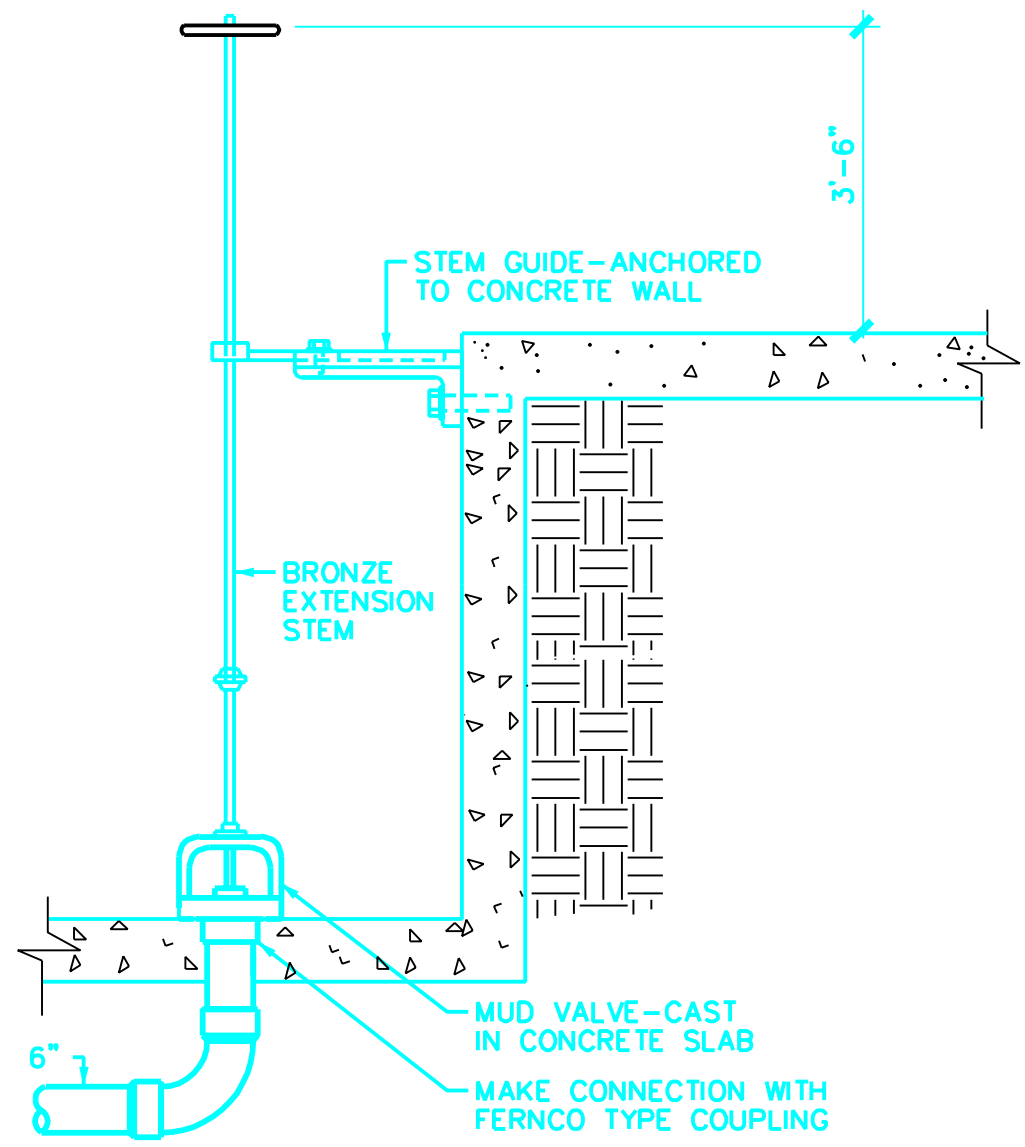
Drawing Title
MECHANICAL PIPING SCHEMATICS
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

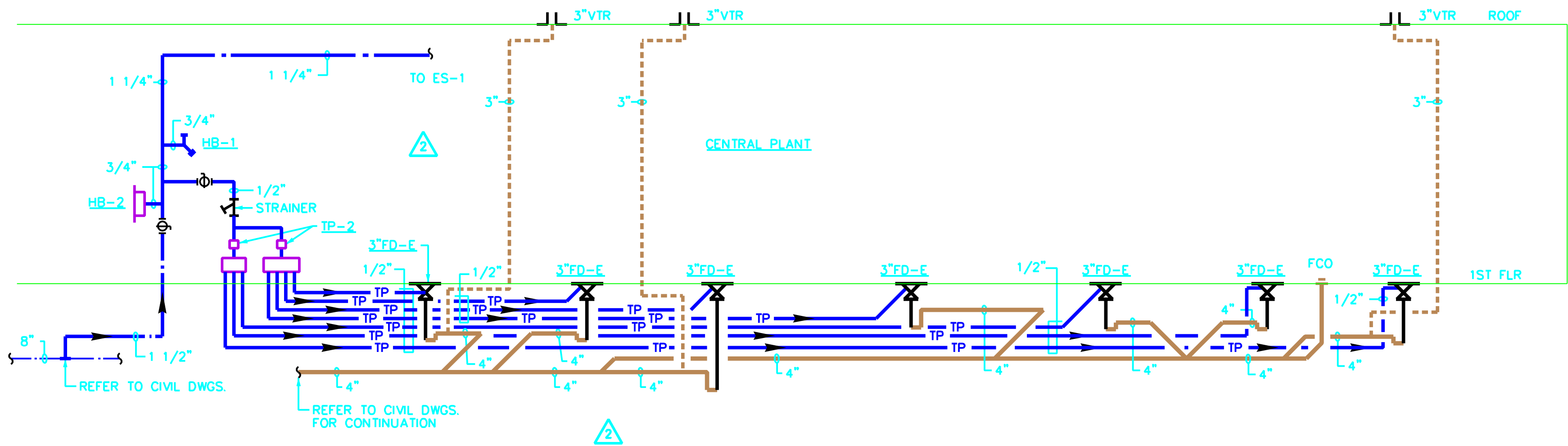
Date
02/25/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
105X03
Dwg. of

Veterans
Administration

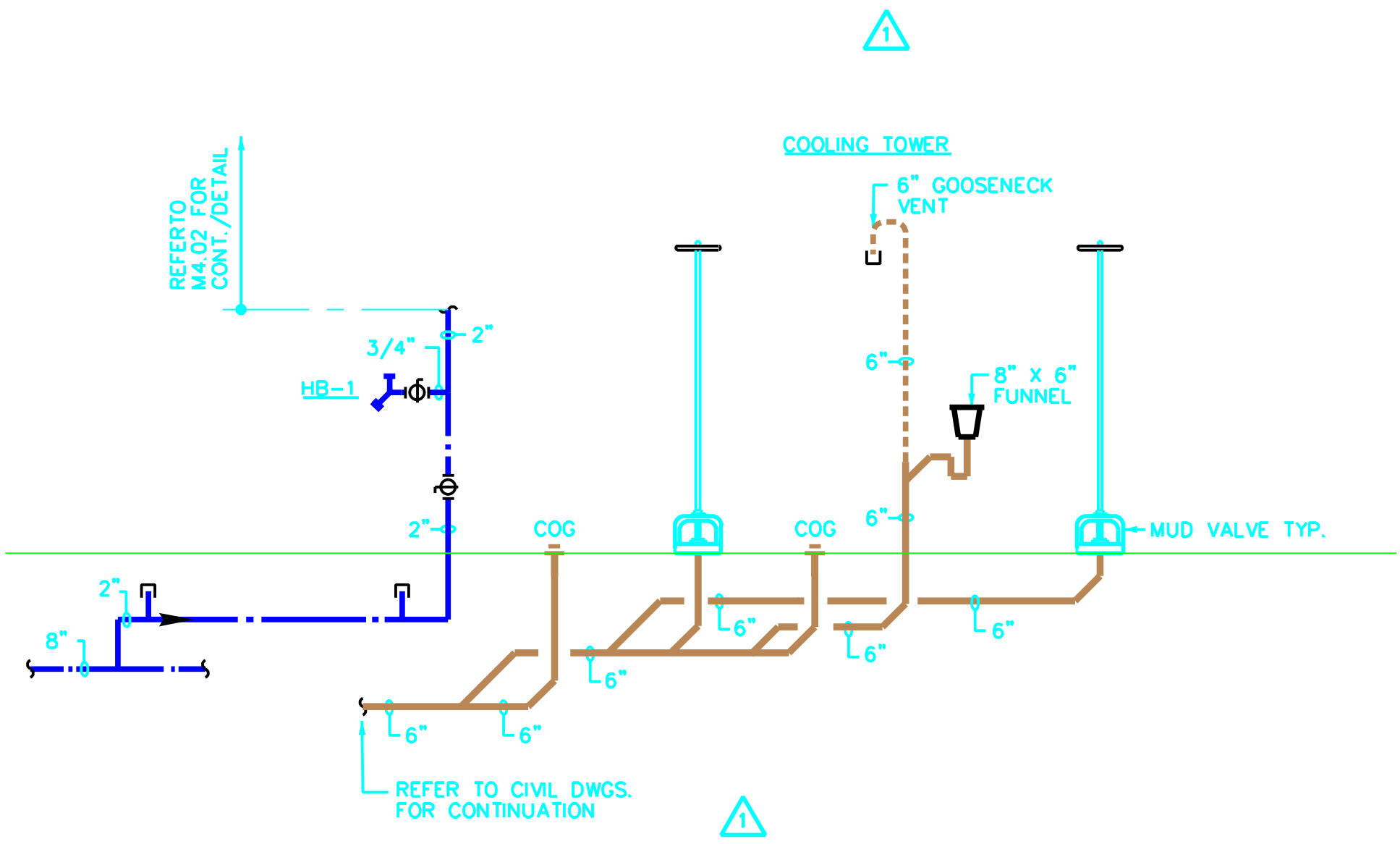
LEGEND					
SYMBOL	ABBREV	DESCRIPTION	SYMBOL	ABBREV	DESCRIPTION
-----		VENT			BALL VALVE
-----		SANITARY SEWER UNDERGROUND			GATE VALVE
--- --	CW	DOMESTIC COLD WATER		FCO	FLOOR CLEANOUT
		EXISTING PIPING--TO REMAIN			THREADED PLUG
		EXISTING TO BE REMOVED		COG	CLEANOUT AT GRADE
				I.E.	INVERT ELEVATION
				AFF	ABOVE FINISHED FLOOR



1 MUD VALVE DETAIL
NOT TO SCALE



2 PLUMBING RISER DIAGRAM
NOT TO SCALE



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Revisions	Date



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2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
PLUMBING LEGEND AND DETAILS
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

Date
02/25/2003
Project No.
VA #580-58-110 SSA #1017-007-1
Drawing No.
P0.01
Dwg. of

Veterans
Administration



1. INSULATE OUTDOOR DOMESTIC COLD WATER PIPING WITH ALUMINUM JACKET. PROVIDE FREEZE PROTECTION HEAT TRACING SYSTEM ON ALL OUTDOOR DOMESTIC WATER PIPING.

- ① 4" SANITARY SEWER - I.E. = -45.10', 33DFU+22GPM
REFER TO CIVIL DWG C0.01 FOR CONT.
- ② 6" SANITARY - SEWER-I.E. = 45.10'
REFER TO CIVIL DWG C0.01 FOR CONT.
- ③ 3" VENT FROM BELOW & UP VENT THRU ROOF
- ④ CONNECT 6" SANITARY TO BASIN WITH MUD VALVE. REFER TO 1/PD.01
FOR DETAIL
- ⑤ 8" DOMESTIC WATER MAIN. REFER TO CIVIL DWGS.
- ⑥ REMOVE EXISTING PIPING
- ⑦ CONNECT 2" DOMESTIC WATER TO SERVICE MAIN. SEE C0.01 FOR CONTINUATION
- ⑧ UNDERGROUND DOMESTIC WATER LINE
- ⑨ RACK PIPING TO COOLING TOWER WALL & ROUTE OVER
GRATING, SECURED TO WALL
- ⑩ 2" DOMESTIC COLD WATER W/BALL VALVE
AND THREADED PLUG FOR CONNECTION TO COOLING
TOWER FLOAT VALVES.
- ⑪ ROUTE 1 1/2" DOMESTIC COLD WATER THRU WALL-TOIGHT
TO UNDERSIDE OF GRATING.
- ⑫ INSULATE & HEAT TRACE HOSE BIBB, AND ASSOCIATED ABOVE
GROUND PIPING & VALVES.
- ⑬ PROVIDE 6" SANITARY W/8" X 6" FUNNEL DRAIN WITH P-TRAP RACKED ALONG
W/ RIM OF FLOOR DRAIN. 2" 24" ABOVE GRADE. 6" VENT FROM DWG WITH
GOOSENECK OPEN END WITH 100 MESH STAINLESS STEEL BUG SCREEN SECURED
WITH STAINLESS STEEL ADJUSTABLE HOSE CLAMPS.
- ⑭ NEW 1 1/2" DOMESTIC WATER. REFER TO C0.01 FOR CONTINUATION.
- ⑮ REMOVE EXISTING UNDERGROUND IRRIGATION SPRINKLER PIPING
- ⑯ REMOVE IRRIGATION SPRINKLER HEAD
- ⑰ RELOCATE EXISTING IRRIGATION SPRINKLER HEAD-TO BE REUSED
- ⑱ CONNECT NEW 2" IRRIGATION SPRINKLER LINE TO EXISTING
- ⑲ CONNECT NEW 1" IRRIGATION SPRINKLER LINE TO EXISTING
- ⑳ CONNECT NEW 1 1/2" IRRIGATION LINE TO EXISTING SPRINKLER LINE.
- ㉑ EXISTING-TO REMAIN
- ㉒ NEW PIPING
- ㉓ EXISTING-TO BE REMOVED

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

01 PLUMBING PIPING PLAN

SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Date	02/25/2003
Project No.	VA #580-98-110 SSA #1017-007-1
Drawing No.	P2.01
Dwn of	

Veterans Administration

SYMBOLS

	1'x4' FLUORESCENT LIGHT FIXTURE
	1'x4' FLUORESCENT LIGHT FIXTURE WITH EMERGENCY BATTERY PACK
	HID LIGHT FIXTURE
	PANELBOARD
	JUNCTION OR PULL BOX
	MOTOR, 3PHASE HORSEPOWER AS NOTED
	MOTOR, SINGLE PHASE HORSEPOWER AS NOTED
	GROUND ROD
	GROUND ROD IN TEST WELL
	CONNECTION POINT
	GROUND WIRE - #4/0 BARE STRANDED COPPER UNLESS NOTED OTHERWISE
	HOME RUN TO SWGR, MCC OR PANELBOARD, CIRCUIT DESIGNATION AS NOTED
	FEEDER DESIGNATION - SEE ONE LINE DIAGRAM FOR CONDUIT AND WIRE SIZES
	KEYED NOTE
	CONDUIT TURNED UP
	CONDUIT TURNED DOWN
	CAPPED CONDUIT
	SINGLE POLE TOGGLE SWITCH - 48\"/>
	3 - THREE WAY SWITCH M - MANUAL MOTOR STARTER
	20A, 125V DUPLEX RECEPTACLE MOUNT 54\"/>
	PHOTOCELL
	NON-FUSED DISCONNECT SWITCH
	FUSED DISCONNECT SWITCH
	COMBINATION STARTER FVNR
	FVNR STARTER
	MEDIUM VOLTAGE CIRCUIT BREAKER CONTROL SWITCH
	SQUARE D POWERLOGIC METER
	MOTOR PROTECTION RELAY
	INDICATING LIGHT - LETTER DENOTES COLOR G-GREEN R-RED C-CLEAR B-BLUE W-WHITE A-AMBER
	MAIN FIRE ALARM PANEL - 6' AFF TO TOP OF PANEL
	FIRE ALARM PULL STATION 48\"/>
	FIRE ALARM COMBINATION AUDIO/VISUAL DEVICE 80\"/>
	FIRE ALARM VISUAL DEVICE - 15/75 CANDELA 80\"/>
	FIRE ALARM HEAT DETECTOR
	4\"/>
	PUSHBUTTON - MOUNT 48\"/>
	BREAK GLASS STATION FOR REFRIGERANT MONITOR SYSTEM - 48\"/>
	COMBINATION HORN/SSTROBE FOR REFRIGERANT MONITOR SYSTEM - 80\"/>

	MEDIUM VOLTAGE DRAWOUT POWER CIRCUIT BREAKER 3P SIZE AS NOTED
	MOTOR CIRCUIT PROTECTOR - SIZE AS NOTED. THREE POLES UNLESS OTHERWISE NOTED.
	MOLDED CASE CIRCUIT BREAKER - SIZE AS NOTED. THREE POLES UNLESS OTHERWISE NOTED.
	DISCONNECT SWITCH - 3P UNLESS NOTED OTHERWISE
	FUSE - QUANTITY AND RATING AS NOTED
	FUSE-PULLOUT TYPE
	TRANSFORMER - DESCRIPTION, CONNECTION AND RATING AS NOTED
	TRANSFORMER - OPEN DELTA CONNECTION
	TRANSFORMER - DELTA CONNECTION
	TRANSFORMER - WYE CONNECTION
	TRANSFORMER - GROUNDED WYE CONNECTION
	ZERO SEQUENCE CT - RATIO AS NOTED
	CURRENT TRANSFORMER (CT) - QUANTITY AND RATIO AS NOTED
	VOLTAGE TRANSFORMERS (VT) - QUANTITY AND RATIO AS NOTED
	SURGE ARRESTER, METAL-OXIDE, QUANTITY, RATING AND MCOV AS NOTED
	MOTOR STARTER - NEMA SIZE AS INDICATED
	SPACE HEATER
	GROUND CONNECTION
	15KV ELBOW SEPARABLE CONNECTOR (DEAD FRONT TRANSFORMER)
	MEDIUM VOLTAGE CABLE TERMINATION (STRESS CONE)
	PROTECTIVE RELAY FUNCTION - ANSI DESIGNATION AS NOTED
	INDICATING LIGHT - LETTER DENOTES COLOR G-GREEN R-RED C-CLEAR B-BLUE W-WHITE A-AMBER
	COOLING TOWER VIBRATION SWITCH TRANSDUCER PROVIDED UNDER DIV 15
	COOLING TOWER OIL SWITCH PROVIDED UNDER DIV 15
	COOLING TOWER VIBRATION SWITCH MONITOR PROVIDED UNDER DIV 15
	COOLING TOWER VIBRATION SWITCH RESET PUSHBUTTON
	POWER FACTOR CORRECTION CAPACITOR

NOTE:
NOT ALL SYMBOLS AND ABBREVIATIONS SHOWN ARE USED ON THE PLANS.

ABBREVIATIONS

A	AMPERE (S)	MAX	MAXIMUM
AIC	AMPERES INTERRUPTING CAPACITY	MCC	MOTOR CONTROL CENTER
AFF	ABOVE FINISHED FLOOR	MCP	MOTOR CIRCUIT PROTECTOR
ATS	AUTOMATIC TRANSFER SWITCH	MCOV	MAX. CONTINUOUS OPERATING VOLTAGE
BRK	BREAKER	MFAP	MASTER FIRE ALARM PANEL
BLDG	BUILDING	MIN	MINIMUM
C	CONDUIT	MLO	MAIN LUGS ONLY
CAOP	CHILLER AUXILIARY OIL PUMP	MOT	MOTOR
CB	CIRCUIT BREAKER	MPR	MOTOR PROTECTION RELAY
CCP	CHILLER CONTROL PANEL	MR	MULTI-RATIO
CU	COPPER	MTD	MOUNTED
CKT	CIRCUIT	NC	NORMALLY CLOSED
CLF	CURRENT LIMITING FUSE	NIC	NOT IN CONTRACT
DEG	DEGREE	NO	NORMALLY OPEN
DIV	DIVISION	NTS	NOT TO SCALE
DCS	DISTRIBUTED CONTROL SYSTEM	No	NUMBER
DWG	DRAWING	OFCI	OWNER FURNISHED - CONTRACTOR INSTALLED
EA	EACH	O.C.	OVERCURRENT
F	FREQUENCY	PFCC	POWER FACTOR CORRECTION CAPACITOR
FLA	FULL LOAD AMPERES	PF	POWER FACTOR
FVNR	FULL VOLTAGE NON-REVERSING	RCP	REMOTE CONTROL PANEL
GAOP	GEAR AUXILIARY OIL PUMP	RGS	RIGID GALVANIZED STEEL
GFCI	GROUND FAULT CIRCUIT INTERRUPTER	RMS	ROOT MEAN SQUARE
GND	GROUND	SYM	SYMMETRICAL
HAS	HOUSTON AIRPORT SYSTEM	SYN	SYNCHRONOUS
HOA	HAND-OFF-AUTOMATIC	SWGR	SWITCHGEAR
HP	HORSEPOWER	UON	UNLESS OTHERWISE NOTED
HPF	HIGH POWER FACTOR	V	VOLT(S)
HPS	HIGH PRESSURE SODIUM	VIB	VIBRATION
HTG	HEATING	WP	WEATHERPROOF
JB	JUNCTION BOX		
KA	KILOAMPERES		
KV	KILOVOLT		
KVA	KILOVOLT AMPERES		
KWHD	KILOWATT-HOUR DEMAND		
LR	LOCAL-REMOTE		
LRA	LOCKED ROTOR AMPERES		
LS	LIMIT SWITCH		

LOAD ANALYSIS

4.16KV SERVICE	
DESCRIPTION	KVA
LARGEST MOTOR(1200HP)	970
25% LARGEST MOTOR	243
OTHER MOTOR LOADS	1379
LIGHTING (125%)	2.0
RECEPTACLES	1.5
TOTAL	2595.5
THE TOTAL LOAD IS 360A AT 4.16KV, 3 PHASE. AMPACITY OF 3-500 KCMIL 5KV INSULATED CONDUCTORS IS 470A PER NEC TABLE 310-77. THEREFORE, INCOMING FEEDERS WILL CARRY THE LOAD.	

LIGHTING FIXTURE SCHEDULE						
TYPE	MANUFACTURER AND CATALOG NUMBER	DESCRIPTION	VOLTAGE	LAMPS	MOUNTING	REMARKS
FA	LITHONIA *AF10332120GEB20 WGAFPV	4' INDUSTRIAL FLUORESCENT, 10% UPLIGHT, WIRE GUARD,	120V	3-F03218 85 CRI	CHAIN 10' AFF TO BOTTOM, UON ADJUST LOCATIONS SHOWN ON PLANS AS REQUIRED TO SUIT PIPING LAYOUT	3 LAMP ELECTRONIC BALLAST
FB	LITHONIA *AF10332120GEB20 PS1400WGAFPV	4' INDUSTRIAL FLUORESCENT, 10% UPLIGHT, WIRE GUARD, EMERGENCY BATTERY PACK (1)	120V	3-F03218 85 CRI	CHAIN 10' AFF TO BOTTOM UON	3 LAMP ELECTRONIC BALLAST UNSWITCHED
HA	LITHONIA *TWH150M120	WALL MOUNTED AREA LIGHT FIXTURE, GLASS REFRACTOR, CORROSION RESISTANT	120V	1-150W MH	WALL MOUNT BOTTOM OF FIXTURE 12" ABOVE GRADE	
XA	LITHONIA *LESWIR120/277ELN	SINGLE FACE EXIT SIGN WITH EMERGENCY BATTERY PACK	120V	LED	WALL 92" AFF	UNSWITCHED

(1) EMERGENCY BATTERY PACK TO OPERATE ONE LAMP AT 1400 LUMENS FOR 90 MINUTES, PROVIDE A TEST SWITCH AND MOUNT SO SWITCH EXTENDS THROUGH BALLAST COVER ADJACENT TO LAMPS, ROUTE ONE UNSWITCHED LEG TO FIXTURE. ON LOSS OF NORMAL POWER BATTERY PACK WILL OPERATE ONE LAMP FOR A MINIMUM OF 90 MINUTES. REFER TO MANUFACTURER WIRING DIAGRAM.

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW SIGNIFICANT CHANGES IN THE WORK MADE DURING CONSTRUCTION BASED ON MARKED-UP PRINTS, DRAWINGS AND OTHER DATA.

Revisions	Date



SHAH SMITH & ASSOCIATES, INC.
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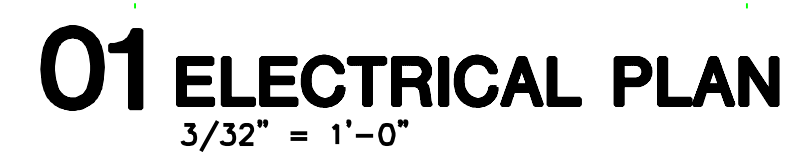
Drawing Title
ELECTRICAL LEGEND, NOTES AND SCHEDULE
Approved: Engineering Svc. Chief
Approved: Medical Center Director

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER		
CHILLER PLANT EXPANSION		
Building Number	Checked	Drawn
104/106	AEB	JN
Location		
VAMC HOUSTON, TX		

Date
02/25/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
E0.01
Dwg. of

Veterans
Administration

- 1 REMOVE DISCONNECT SWITCH. REMOVE WIRE BACK TO SOURCE. REMOVE CONDUIT TO 2' BELOW GRADE.
- 2 REMOVE DISCONNECT SWITCH. REMOVE WIRE BACK TO SOURCE. REMOVE CONDUIT TO 2' BELOW GRADE OUTSIDE AREA OF NEW WORK.
- 3 REMOVE DISCONNECT SWITCH. REMOVE WIRE BACK TO SOURCE. REMOVE CONDUIT TO 2' BELOW GRADE. THERE IS NO LOAD CONNECTED TO THE DISCONNECT SWITCH. PANEL NLC IS LOCATED INSIDE MCC2.
- 4 REMOVE WIRING FROM COOLING TOWER DISCONNECT SWITCHES TO HIGH RISE BUILDING. REMOVE CONDUIT THAT IS UNDER THE LOCATION OF THE NEW COOLING TOWERS. SEE NEW WORK ON SHEET E2.01, SEE ONE LINE DIAGRAM ON SHEET E2.01 FOR THE EXACT LOCATION OF THE UNDERGROUND CONDUITS.
- 5 RELOCATE LIGHT FIXTURES AND DIRECT BURIED POLE. FIXTURES ARE 250W HPIS, 480V MOUNTED ON A 30" HIGH ALUMINUM POLE. REMOVE WIRE BACK TO FIXTURE LOCATED SOUTH OF THIS POLE. REMOVE CONDUIT AS REQUIRED TO SERVE POLE IN NEW LOCATION. SEE SHEET E2.01 FOR NEW WORK.



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DURING CONSTRUCTION BASED ON MARKED-UP
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[illegible]

Approved: Medical Center Director

E1.01

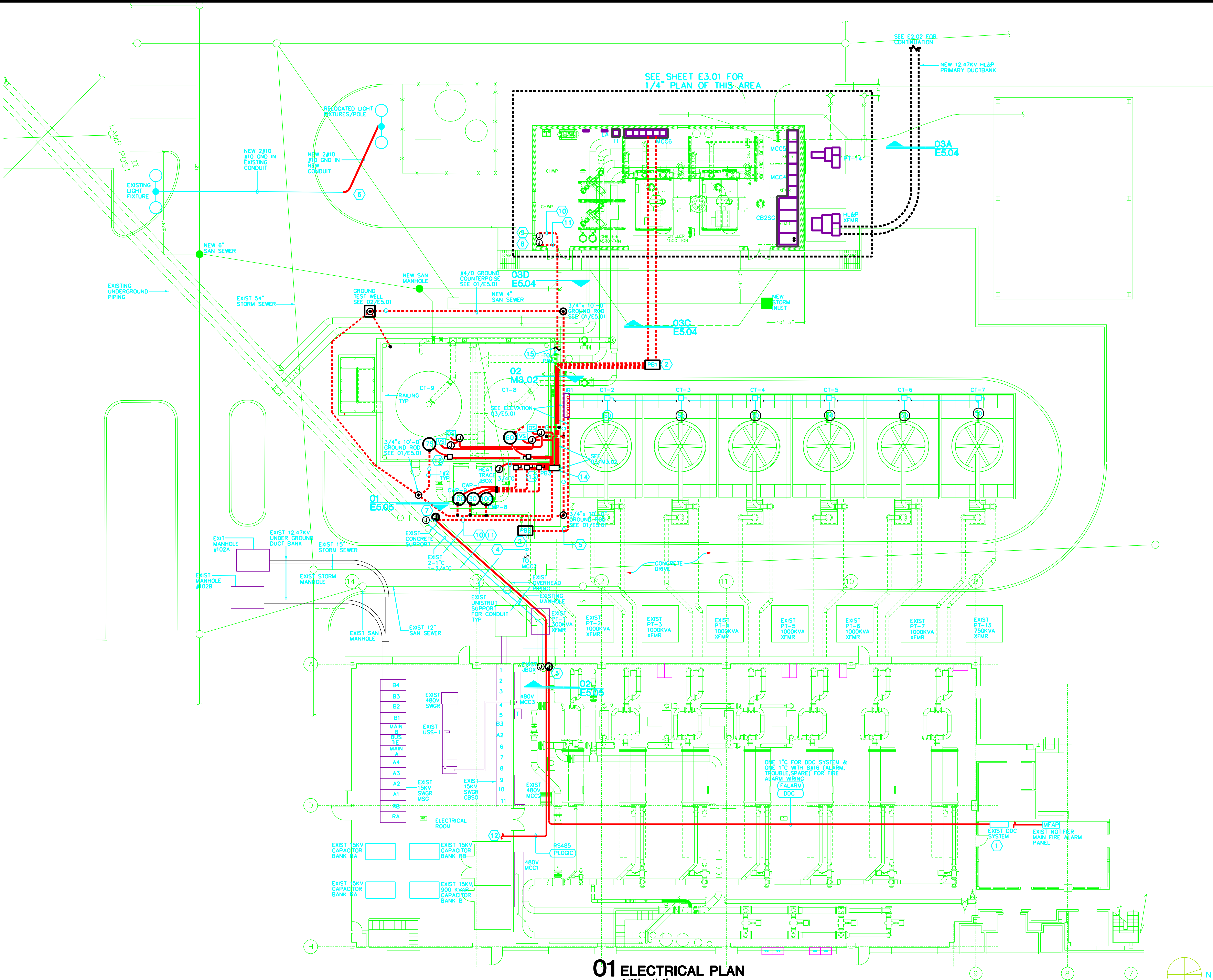
Veterans Administration

GENERAL NOTES

1. ALL BELOW GRADE CONDUITS SHALL BE ENCASED IN CONCRETE. SEE DETAIL 03/E5.04.
2. ALL ABOVE GROUND CONDUIT AT COOLING TOWERS SHALL BE RGS.

KEYED NOTES

1. COORDINATE EXACT TERMINATION OF CONDUIT INTO DDC PANEL WITH OWNER.
2. NEW 30" X 48" X 36"D PULL BOX WITH BASE, QUAZITE #PG3048DA36, 30" X 48" X 18" PULL BOX WITH NO BASE STACKED ON TOP OF 36"D BOX, QUAZITE #PG3048BA18 WITH BOLTED DOWN COVER #PG3048HA00, PROVIDE COVER LOGO MARKED "ELECTRICAL".
3. NEW WALL MOUNTED JUNCTION BOX. SEE DETAIL 02/E5.05.
4. NEW 6#4 #10G IN EXISTING 1 1/2"C. TYPICAL FOR COOLING TOWERS CT-2 THROUGH CT-7. SEE E4.03 FOR ONE LINE DIAGRAM.
5. NEW 6#4 #10G IN NEW 1 1/2"C. TYPICAL FOR COOLING TOWERS CT-2 THROUGH CT-7. SEE E4.03 FOR ONE LINE DIAGRAM.
6. EXTEND EXISTING CONDUIT INTO RELOCATED POLE BASE. PROVIDE NEW WIRE FROM RELOCATED FIXTURE TO EXISTING FIXTURE.
7. 12" X 12" X 12" NEMA 3R DIVIDED PULL BOX WITH HINGED FRONT COVER. MOUNT PULL BOX ON CONCRETE FOUNDATION 24" ABOVE GRADE. ONE HALF OF PULL BOX IS FOR FIRE ALARM WIRING AND THE OTHER HALF IS FOR DDC AND COMMUNICATION WIRING. COMMUNICATION WIRING CONSISTS OF TELEPHONE AND POWERLOGIC. SEE DETAIL 01/E5.05.
8. JUNCTION BOX FOR FIRE ALARM WIRING. SEE E3.01 FOR 1/4" PLAN.
9. JUNCTION BOX FOR CONTROL/COMMUNICATION WIRING. SEE E3.01 FOR 1/4" PLAN.
10. ONE 1 1/2" CONDUIT FOR COMMUNICATION WIRING AND ONE 1 1/2" SPARE CONDUIT WITH PULL STRING. INSTALL DDC WIRING, TELEPHONE WIRING AND POWERLOGIC WIRING IN ONE CONDUIT. DDC AND TELEPHONE WIRING PROVIDED BY OTHERS. POWERLOGIC WIRING FOR SKV METERING SHALL BE AS INDICATED ON SHEET E4.02.
11. 4#16, 4#16 SPARE IN 1 1/2"C FOR FIRE ALARM WIRING.
12. COORDINATE FINAL CONNECTION INTO EXISTING POWERLOGIC COMMUNICATION LOOP IN ROOM WITH OWNER.
13. MOUNT DISCONNECT SWITCH ON A HOT DIPPED GALVANIZED UNISTRUT RACK. TYPICAL FOR DISCONNECT SWITCHES AT COOLING TOWER.
14. NEW 36"L X 18"D X 24"H NEMA 4X PULLBOX WITH BOLTED REMOVABLE FRONT COVER.
15. RACK CONDUITS BELOW POWER CONDUITS SHOWN. SEE MECHANICAL SHEET M3.02 FOR DETAILS.



RECORD DRAWING
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Revisions	Date



SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
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Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
ELECTRICAL SITE PLAN RENOVATION
Approved: Engineering Svc. Chief
Approved: Medical Center Director

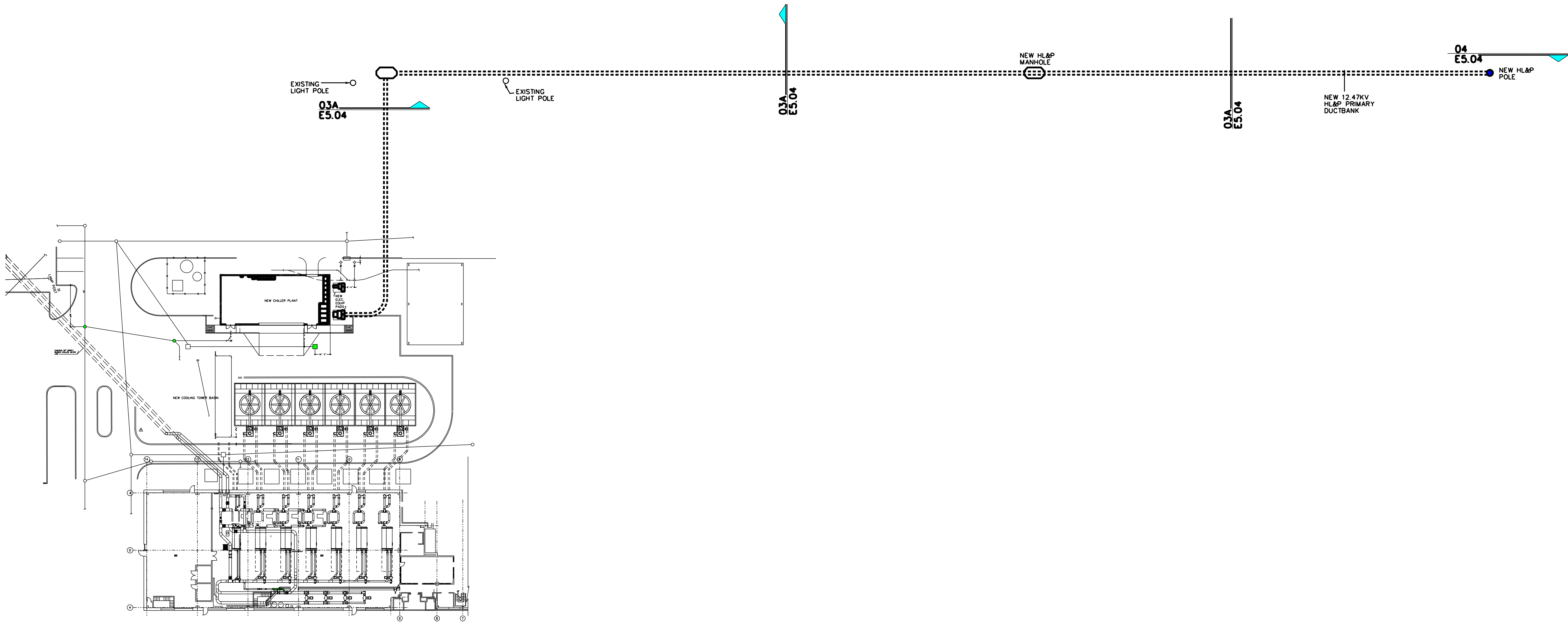
Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

Date
02/25/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
E2.01
Dwg. of

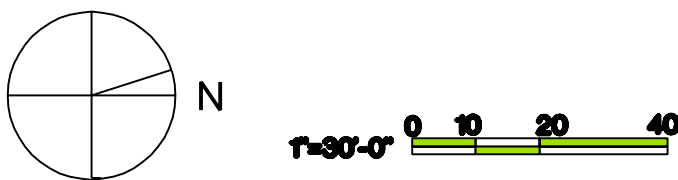
Veterans Administration

GENERAL NOTES

- 1. NEW WORK IS SHOWN BOLD.
- 2. A 1" CLEARANCE SHALL BE MAINTAINED BETWEEN HL&P DUCTBANK AND ALL NON-HL&P FACILITIES CROSSING THE DUCTBANK.
- 3. A FIVE FOOT HORIZONTAL CLEARANCE SHALL BE MAINTAINED BETWEEN HL&P DUCTBANK AND ALL NON-HL&P FACILITIES RUNNING PARALLEL TO THE DUCTBANK.
- 4. COORDINATE 12.47KV DUCTBANK, MANHOLE AND TRANSFORMER PAD CONSTRUCTION REQUIREMENTS WITH HL&P.



01 12.47 KV DISTRIBUTION PLAN
1" = 30'-0"



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DURING CONSTRUCTION BASED ON MARKED-UP
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Revisions	Date



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Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
DRAWING TITLE
Approved Engineering Svc Chief
Approved Medical Center Director

Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/105
Location
VAMC HOUSTON, TX

Date
02/26/2003
Project No.
VA #580-98-110 SSA #1017-007-1
Drawing No.
E2.02
Drawn of

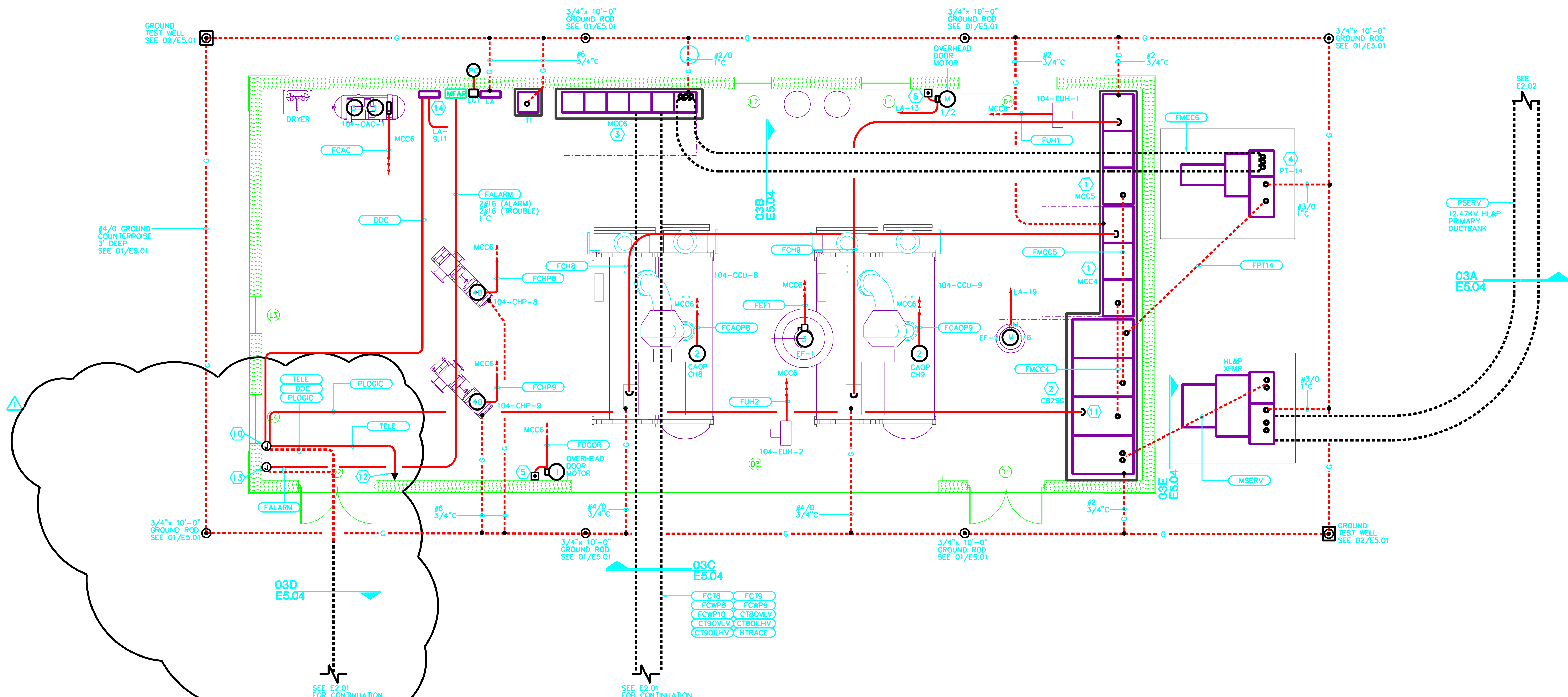
Veterans
Administration

GENERAL NOTES

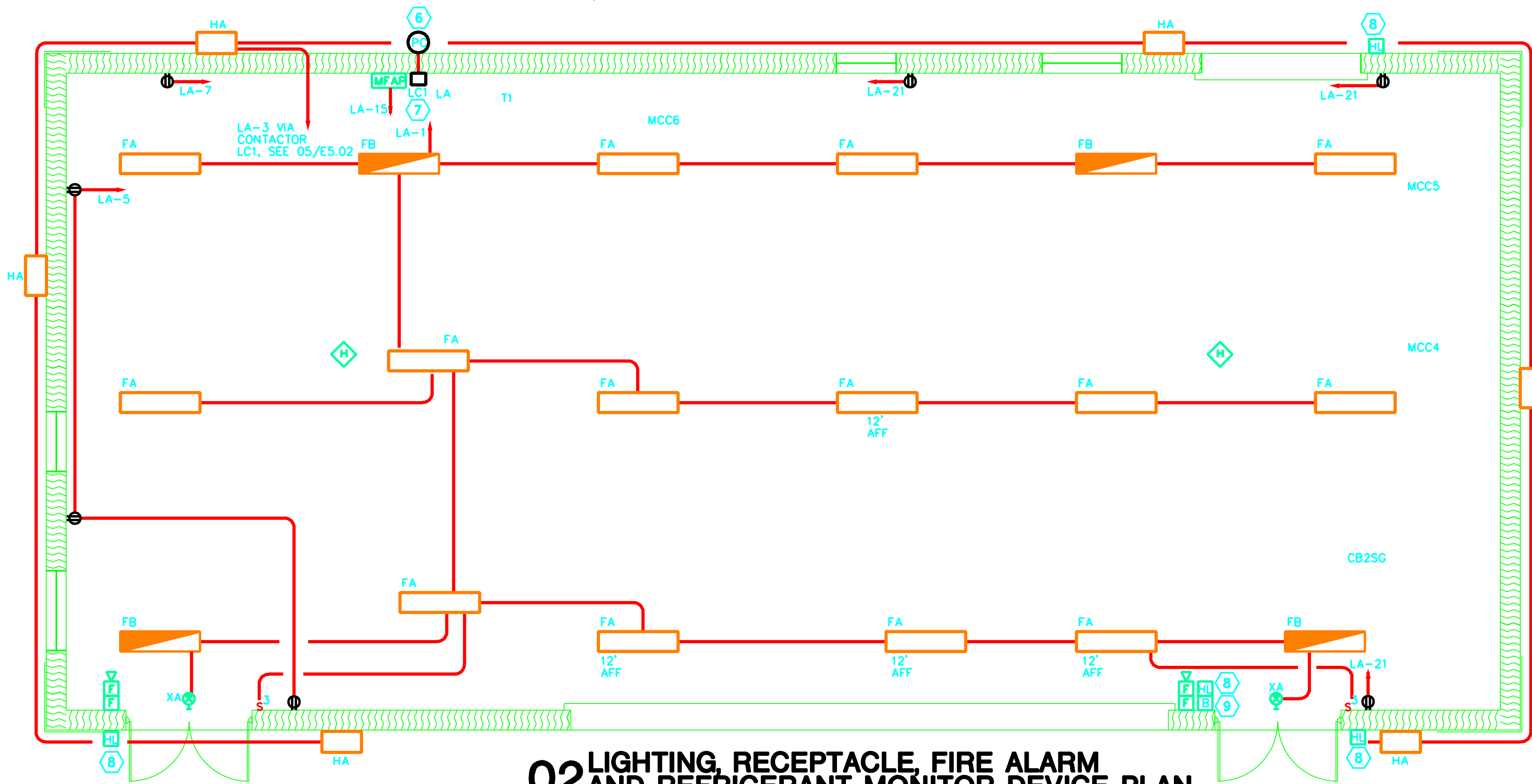
1. NEW WORK IS SHOWN BOLD.
2. ALL BELOW GRADE CONDUITS SHALL BE ENCASED IN CONCRETE.

KEYED NOTES

- ① SEE DETAIL 01/E5.04 FOR ELEVATION.
② SEE DETAIL 02/E5.04 FOR ELEVATION.
③ SEE DETAIL 02/E4.04 FOR ELEVATION.
④ SEE DETAIL 04/E5.01.
⑤ PUSHBUTTON CONTROL STATION PROVIDED UNDER DIVISION 08330 AND INSTALLED UNDER DIVISION 16. PROVIDE 5/8" IN 3/4" C FROM PUSHBUTTON STATION TO DOOR MOTOR CONTROLLER ABOVE.
⑥ MOUNT PHOTOCELL 12' ABOVE GRADE.
⑦ SEE LIGHTING CONTACTOR CONTROL DIAGRAM 05/E5.02.
⑧ SEE DETAIL 06/E5.02 FOR REFRIGERANT MONITOR CONTROL DIAGRAM.
⑨ EMERGENCY SHUTDOWN BREAK GLASS STATION. SEE DETAIL 07/E5.02.
⑩ 6" X 6" X 4" DEEP JUNCTION BOX MOUNTED 24" AFF FOR DDC, TELEPHONE, POWERLOGIC AND FUTURE WIRING.
⑪ CONNECT TO POWERLOGIC METERS. SEE ONE LINE DIAGRAM ON SHEET E4.02.
⑫ 4" SQUARE JUNCTION BOX MOUNTED 48" AFF FOR TELEPHONE OUTLET. OUTLET PROVIDED BY OWNER. ROUTE ONE 3/4" C WITH PULL STRING FROM OUTLET BOX TO JUNCTION BOX ON SOUTH WALL.
⑬ 4" SQUARE JUNCTION BOX MOUNTED 24" AFF FOR FIRE ALARM WIRING.
⑭ DDC PANEL AND REFRIGERANT MONITOR PANEL, ONE ABOVE THE OTHER. PROVIDED UNDER DIVISION 15. CONNECT CIRCUIT 9 TO DDC PANEL AND CIRCUIT 11 TO REFRIGERANT MONITOR PANEL.

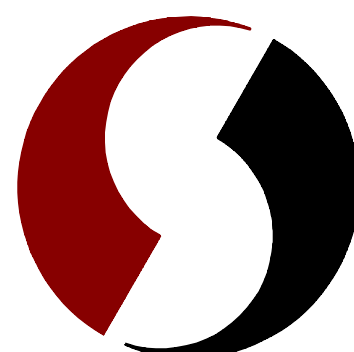


01 ELECTRICAL POWER PLAN
1/4" = 1'-0"

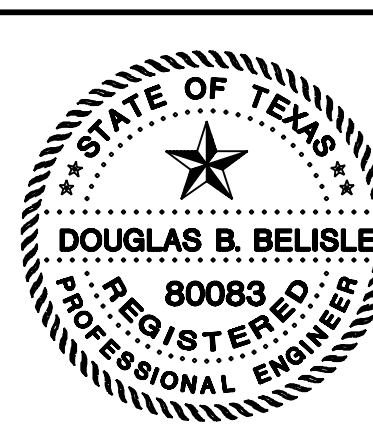


02 LIGHTING, RECEPTACLE, FIRE ALARM AND REFRIGERANT MONITOR DEVICE PLAN
1/4" = 1'-0"

Revisions	Date
COOLING TOWER COORDINATION	29NOV01



SHAH SMITH & ASSOCIATES, INC.
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Drawing Title
ENLARGED ELECTRICAL POWER, LIGHTING & COMMUNICATION PLANS
Approved: Engineering Svc. Chief
Approved: Medical Center Director

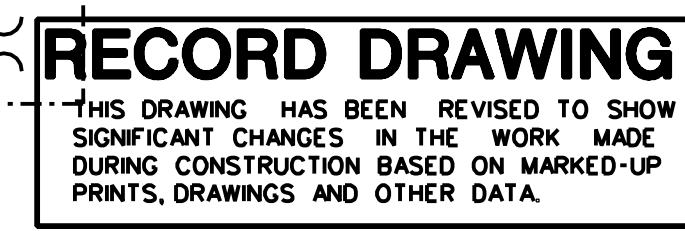
Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

Date
02/25/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
E3.01
Dwg. of

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A. ALL EQUIPMENT SHOWN IS EXISTING

1 EXISTING SQUARE D POWER METER COMMUNICATION LOOP SHALL BE CONNECTED TO THE NEW SQUARE D POWER METERS. COORDINATE EXACT CONNECTION POINT IN EXISTING CENTRAL PLANT SWITCHGEAR ROOM WITH OWNER. SEE SHEET E4.02 FOR NEW POWER METERS.



Fax: 713-780-9209

Approved: Engineering Svc Chief

Approved: Medical Center Director

Project Title	HOUSTON VETERANS AFFAIRS MEDICAL CENTER
---------------	---

CHILLER PLANT EXPANSION

Building Number	104/105
-----------------	---------

Location	VAMC HOUSTON, TX
----------	------------------

Date 02/25/2003

Project No.
VA #580-98-110
SSA #1017-007-1

Drawing No.

E4.01

Dwg. of

Veterans Administration

- 1 CHILLER MOTOR SIZES ARE SHOWN IN HORSEPOWER.
- 2 SQUARE D POWER METER 3020PM650 WITH DISPLAY.
- 3 CONNECT TO EXISTING POWERLOGIC METER COMMUNICATION LOOP IN CENTRAL PLANT SWITCHGEAR ROOM.



RECORD DRAWING
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SIGNIFICANT CHANGES IN THE WORK MADE
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[illegible]

2825 WILCREST, SUITE 350
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Drawing Title COMPOSITE ONE LINE-DIAGRAM
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER			
CHILLER PLANT EXPANSION			
Building Number 104105	Checked NER	Drawn JN	
Location VAMC HOUSTON, TX			

Date	02/25/2003	Veterans Administration
Project No.	VA #580-68-110 SSA #1017-007-1	
Drawing No.	E4.02	
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Veterans Administration

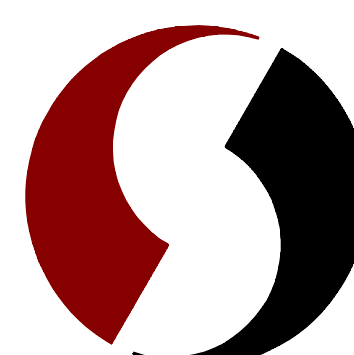
A. NEW WORK IS SHOWN BOLD.

B. DEMOLITION WORK IS SHOWN SHADED.

- ① DISCONNECT MOTOR. REMOVE WIRE BACK TO MOTOR CONTROL CENTER. REMOVE EXTERIOR CONDUIT IN AREA OF NEW WORK. ABANDON REMAINING BELOW GRADE CONDUIT AND CAP. LABEL STARTER AS 6-APARE.
- ② REMOVE WIRE BACK TO MOTOR CONTROL CENTER. REMOVE EXTERIOR CONDUIT IN AREA OF NEW WORK. ABANDON REMAINING BELOW GRADE CONDUIT AND CAP. LABEL BREAKER IN PANELBOARD AS 6-APARE.
- ③ REMOVE WIRE BACK TO MOTOR CONTROL CENTER. REMOVE CONDUIT IN AREA WHERE NEW COOLING TOWERS WILL BE INSTALLED.
- ④ NEW CONDUCTORS IN EXISTING CONDUIT.
- ⑤ NEW CONDUCTORS IN NEW CONDUIT.
- ⑥ NEW PULLBOX/JUNCTION BOX. SEE SHEET E2.01.



RECORD DRAWING

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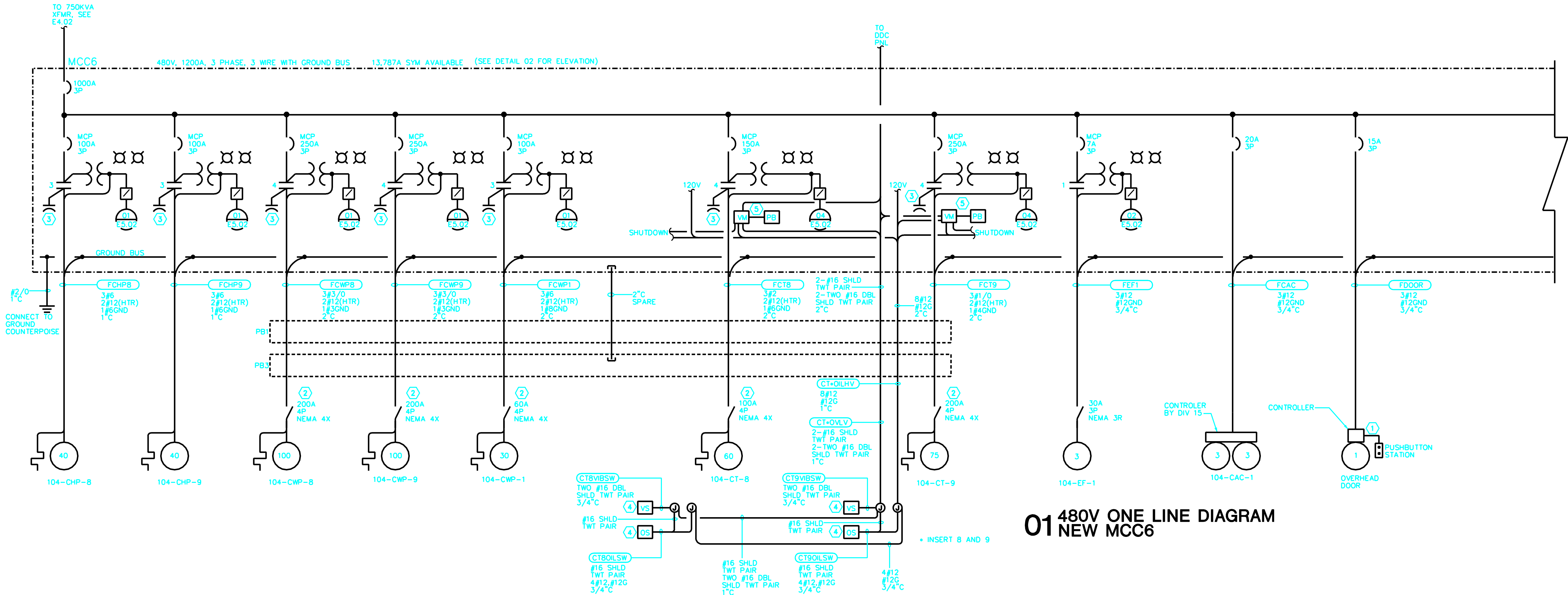
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HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title EXISTING MCC2 DEMOLITION & RENOVATION	Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION			Date 02/25/2003	Veterans Administration
	Approved: Engineering Svc. Chief 			Project No. VA #580-88-110 SSA #1017-007-1	
Approved: Medical Center Director 	Building Number 104/126	Checked AEB	Drawn JN	Drawing No. E4.03	Veterans Administration
	Location VAMC HOUSTON, TX			Dwa. of	

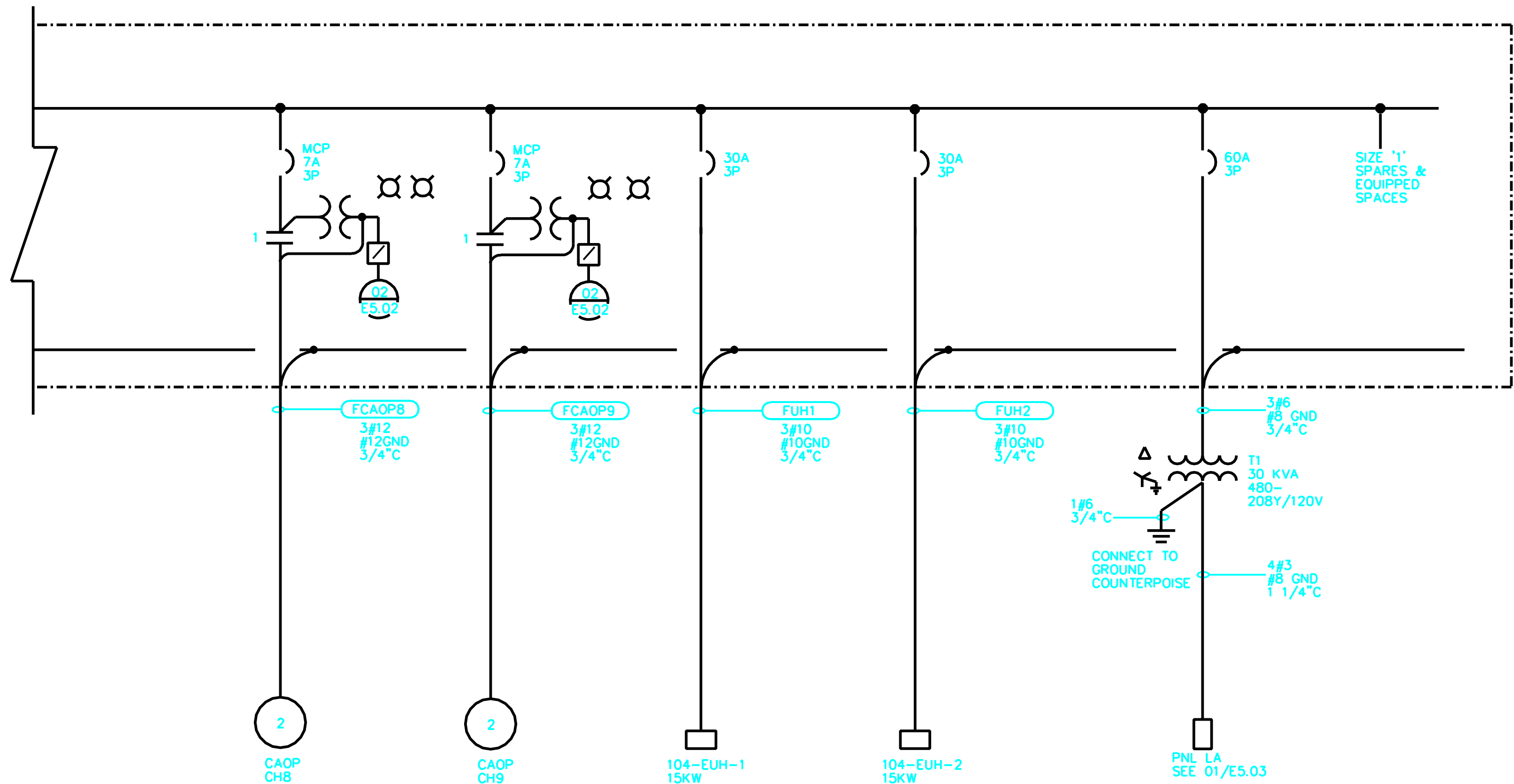
Veterans Administration

KEYED NOTES

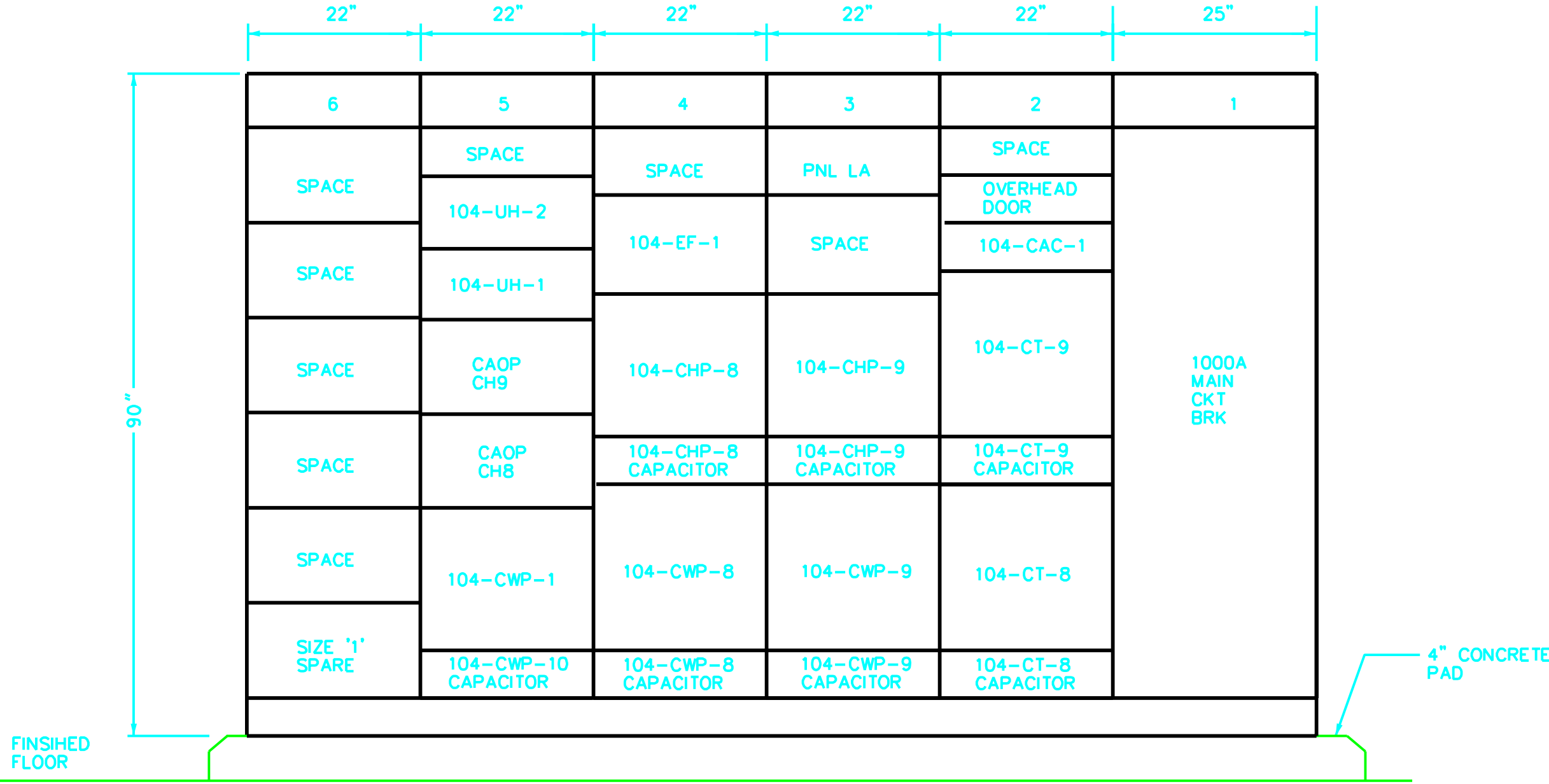
- 1 CONTROLLER AND PUSHBUTTON STATION PROVIDED WITH DOOR. WIRING BETWEEN CONTROLLER PUSHBUTTON STATION SHALL BE 5#14, 3/4" C.
- 2 FOURTH POLE IS FOR SPACE HEATER CIRCUIT.
- 3 SIZE MOTOR POWER FACTOR CORRECTION CAPACITOR TO IMPROVE POWER FACTOR TO 0.95. VERIFY WITH MOTOR MANUFACTURER.
- 4 SEE DETAIL 03/E5.02.
- 5 VIBRATION SWITCH PUSHBUTTON AND MONITOR PROVIDED UNDER DIVISION 15 AND INSTALLED UNDER DIVISION 16.



01 480V ONE LINE DIAGRAM
NEW MCC6



02 MCC6 ELEVATION
NTS



03 MCC6 ELEVATION
NTS

NOTE:
MCC6 IS OWNER FURNISHED AND CONTRACTOR INSTALLED.

RECORD DRAWING
THIS DRAWING HAS BEEN REVISED TO SHOW
SIGNIFICANT CHANGES IN THE WORK MADE
DURING CONSTRUCTION BASED ON MARKED-UP
PRINTS, DRAWINGS AND OTHER DATA.

SHAH SMITH & ASSOCIATES, INC.
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Drawing Title 480V ONE LINE-DIAGRAM AND MCC ELEVATION	Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER CHILLER PLANT EXPANSION	Date 02/25/2003
Approved: Engineering Svc. Chief	Building Number 104/106	Project No. VA #560-58-110 SSA #1017-007-1
Approved: Medical Center Director	Location VAMC HOUSTON, TX	Drawing No. E4.04
	Checked AEB	Dwg. of
	Drawn JN	

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Administration

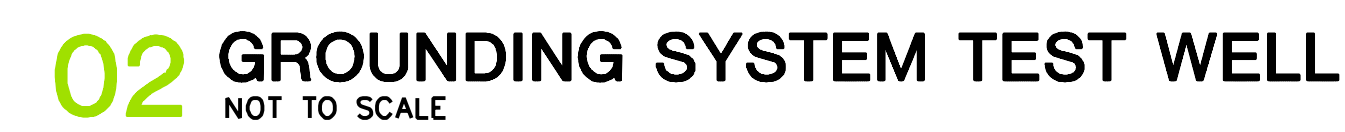


KEYED NOTES

- 1 R-RATED FUSES SIZED BY MANUFACTURER.
- 2 SOLID STATE MULTI-FUNCTION MPR, MULTILIN 1469
- 3 RELAY SETTINGS WILL BE SET BY ENGINEER AFTER SUBMITTAL OF ALL MOTOR DATA.
- 4 MOTOR PROTECTION RELAY TRIP INFORMATION TO CCP.
- 5 RS-485 SERIAL COMMUNICATION..
- 6 CONNECT TO CONTROL COUNTERPOISE, SEE E3.01.
- 7 MOTOR SURGE CAPACITORS AND SURGE ARRESTERS.
- 8 PROVIDE CIRCUIT BREAKERS OR PULLOUT FUSES FOR CONTROL POWER CIRCUITS.
- 9 6 STATOR MOTOR BEARING RTDS.
- 10 CURRENT SIGNAL FOR PERCENT LOADING INDICATION.
- 11 FUSES SIZED BY MOTOR CONTROLLER MANUFACTURER.
- 12 PROVIDE CAPACITORS TO IMPROVE THE POWER FACTOR TO THE MINIMUM OF 0.95. THE MAXIMUM SIZE RECOMMENDED BY THE MOTOR MANUFACTURER.

5KV MOTOR CONTROLLERS ARE OWNER FURNISHED
AND CONTRACTOR INSTALLED.

RECORD DRAWING
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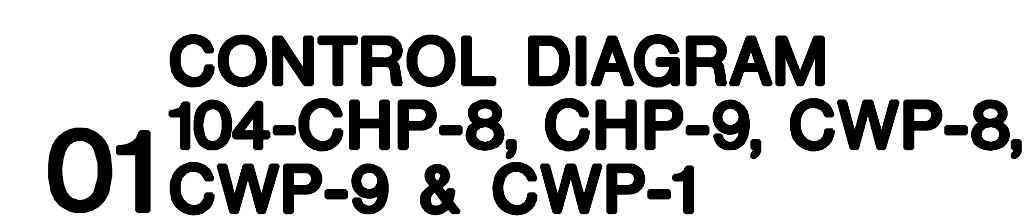


Drawing Title
ELECTRICAL DETAILS
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER			
CHILLER PLANT EXPANSION			
Building Number 104105		Checked AEB	Drawn JN
Location 144MC-HOUSTON, TX			

Date	02/25/2003	Veterans Administration
Project No.	VA #560-68-110 SSA #1017-007-1	
Drawing No.	E5.01	
Scale	3/4" = 1'-0"	

Veterans Administration



① SIZE CPT TO CARRY MOTOR SPACE HEATER

NOTE:
ALL CIRCUITS MUST
HAVE AN EQUIPMENT
GROUNDING CONDUCTOR,
NOT SHOWN FOR CLARITY

RECORD DRAWING
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DURING CONSTRUCTION BASED ON MARKED-UP
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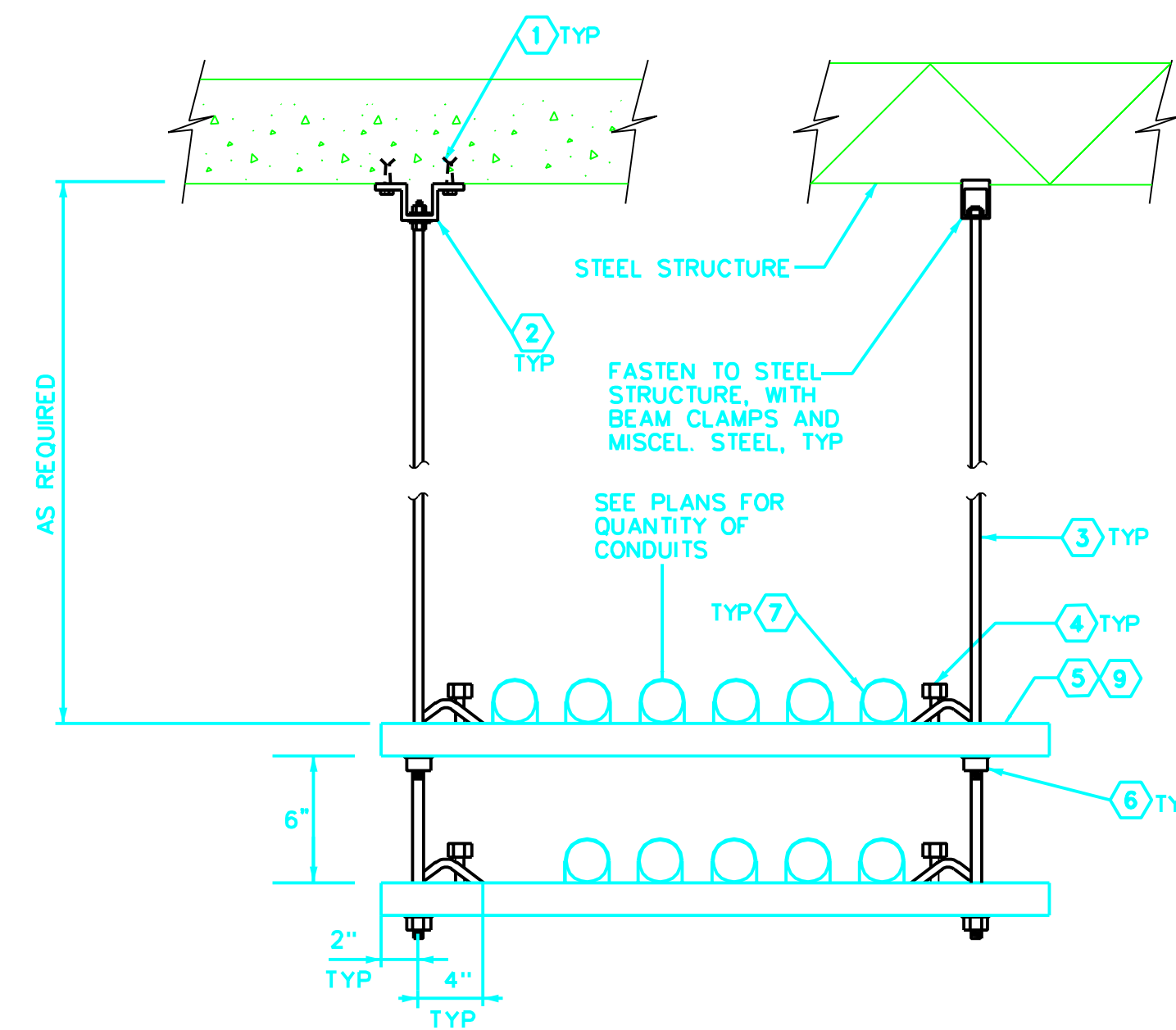
Veterans Administration

100A AMP BUS 3PHASE 4 WIRE 120/208VOLT 22,000A RMS SYM
ALL BREAKERS 20A UNLESS OTHERWISE NOTED

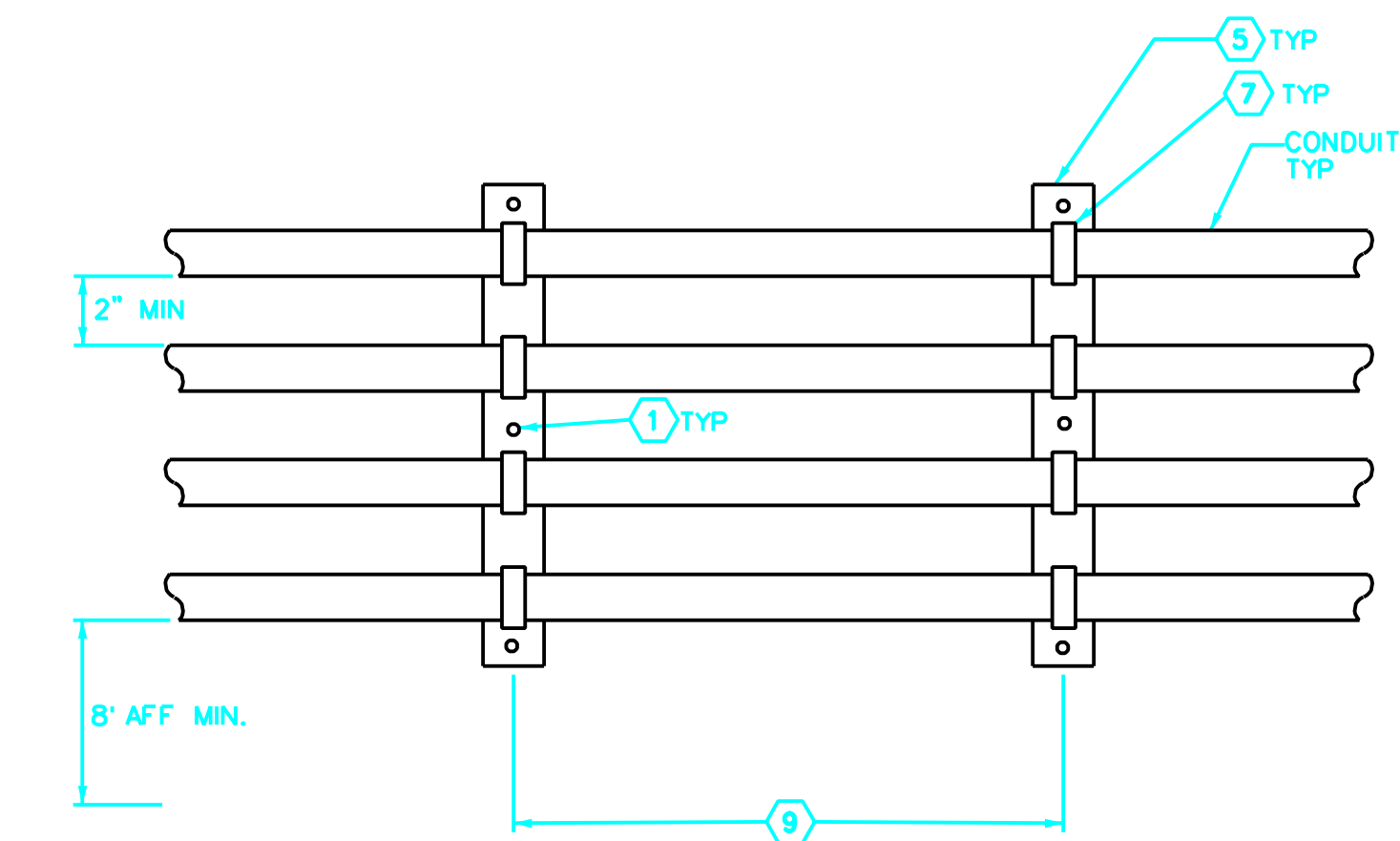
WIRE SIZE	VA			100A SP		VA	WIRE SIZE
2#12, #12G, 3/4" C	1170	LIGHTS-INTERIOR	1	2	SPARE		
2#12, #12G, 3/4" C	1110	LIGHTS-EXTERIOR	3	4	SPARE		
2#12, #12G, 3/4" C	540	RECEPTACLES	5	6	SPARE		
2#12, #12G, 3/4" C	600	AIR DRYER	7	8	SPARE		
2#12, #12G, 3/4" C	200	DDC PNL	9	10	SPACE		
2#12, #12G, 3/4" C	200	REFR MON PNL	11	12	SPACE		
2#12, #12G, 3/4" C	600	OVERHEAD DOOR	13	14	SPACE		
2#12, #12G, 3/4" C	600	MAIN FIRE ALARM PANEL	15	16	SPACE		
2#10, #10G, 3/4" C	800	COOLING TOWER HEAT TRACE	17	18	SPACE		
2#12, #12G, 3/4" C	528	EF-2	19	20	SPACE		
2#12, #12G, 3/4" C	540	RECEPTACLES	21	22	SPACE		
2#12, #12G, 3/4" C	100	LCI CONTROL POWER	23	24	SPACE		
		SPARE	25	26	SPACE		
		SPARE	27	28	SPACE		
		SPARE	29	30	SPACE		

6988 VA CONNECTED 6988 VA DEMAND

01 PANELBOARD SCHEDULE

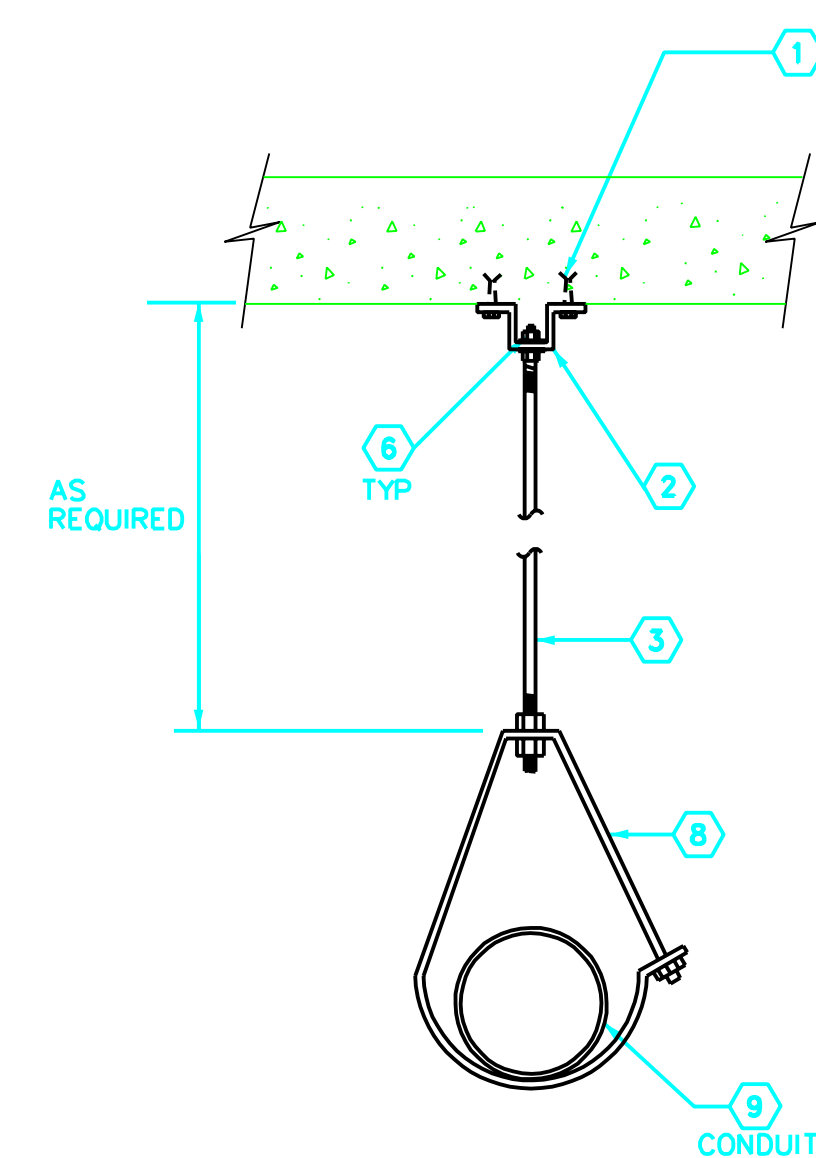


02 TYPICAL CONDUIT TRAPEZE SUPPORT

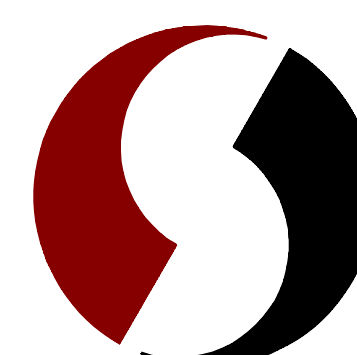


03 TYPICAL WALL MOUNTED CONDUIT RACK

- 1 GALVANIZED 3/8" X 2 1/2" LONG EXPANSION ANCHOR.
- 2 GALVANIZED P2682 UNISTRUT BEAM CLAMP W/HANGER CLEVIS OR EQUIV.
- 3 GALVANIZED 3/8" DIAMETER ALL THREAD ROD.
- 4 GALVANIZED P1386 UNISTRUT BEAM CLAMP OR EQUIV.
- 5 GALVANIZED P1000 UNISTRUT CHANNEL OR EQUIV.
- 6 GALVANIZED 3/8" HEX NUT AND LOCKWASHER.
- 7 CONDUIT CLAMP.
- 8 CABLE CLAMP.
- 9 SUPPORTS AT 8' MAX SPACING.
- 10 CIRCUIT ROUTED TO MCC6S FOR UNDERGROUND ROUTING IN DUCTBANK.



04 CONDUIT SUPPORT DETAIL

[illegible]

SHAH SMITH & ASSOCIATES, INC.
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Drawing Title
ELECTRICAL DETAILS
Approved: Engineering Svc Chief
Approved: Medical Center Director

Project Title HOUSTON VETERANS AFFAIRS MEDICAL CENTER			
CHILLER PLANT EXPANSION			
Building Number 104105	Checked JEB	Drawn JN	
Location VAMC HOUSTON, TX			

Date	02/25/2003
Project No.	VA #560-98-110 SSA #1017-007-1
Drawing No.	E5.03

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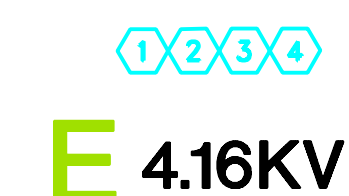
RECORD DRAWING
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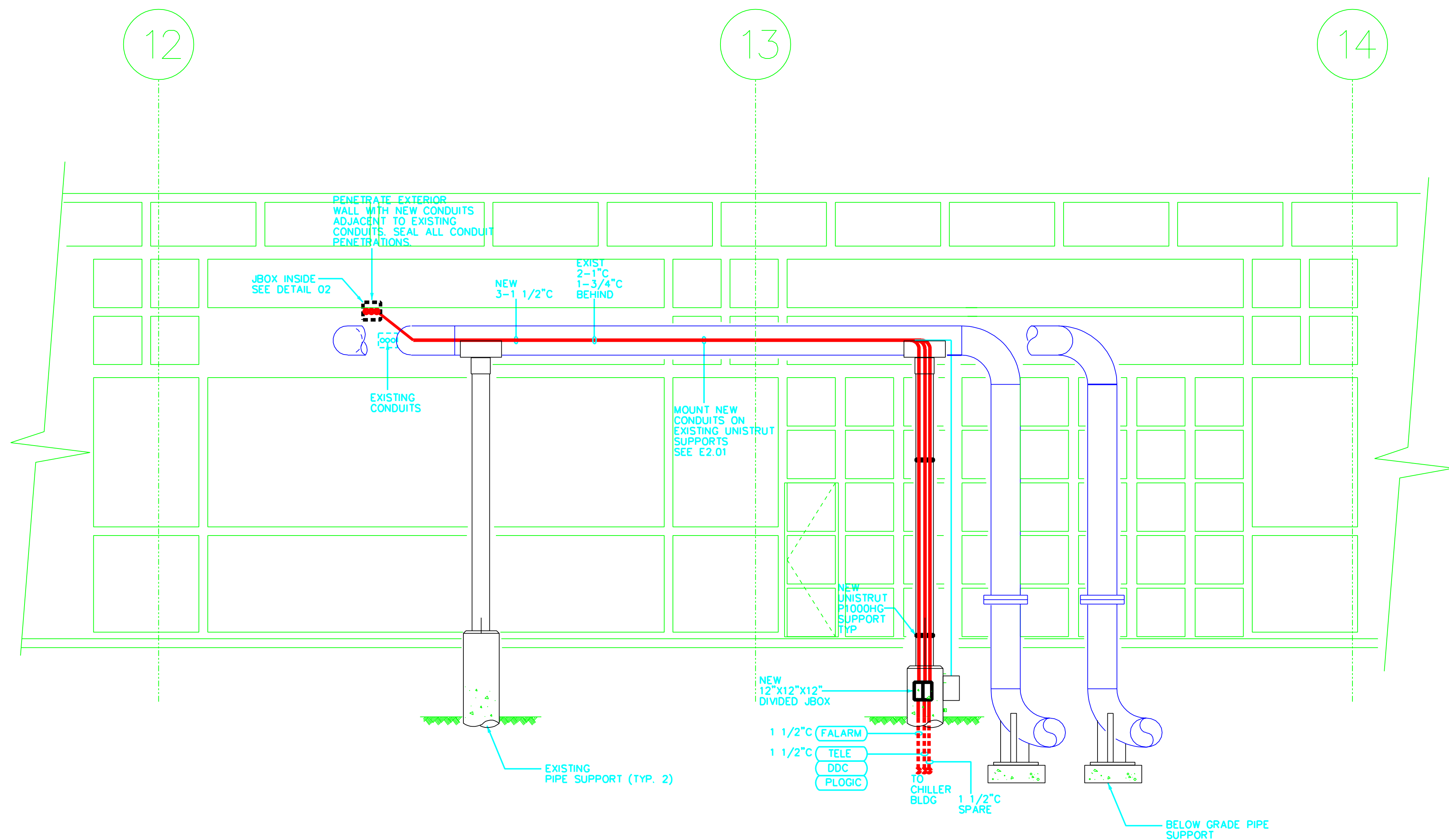
- A12.47KV**



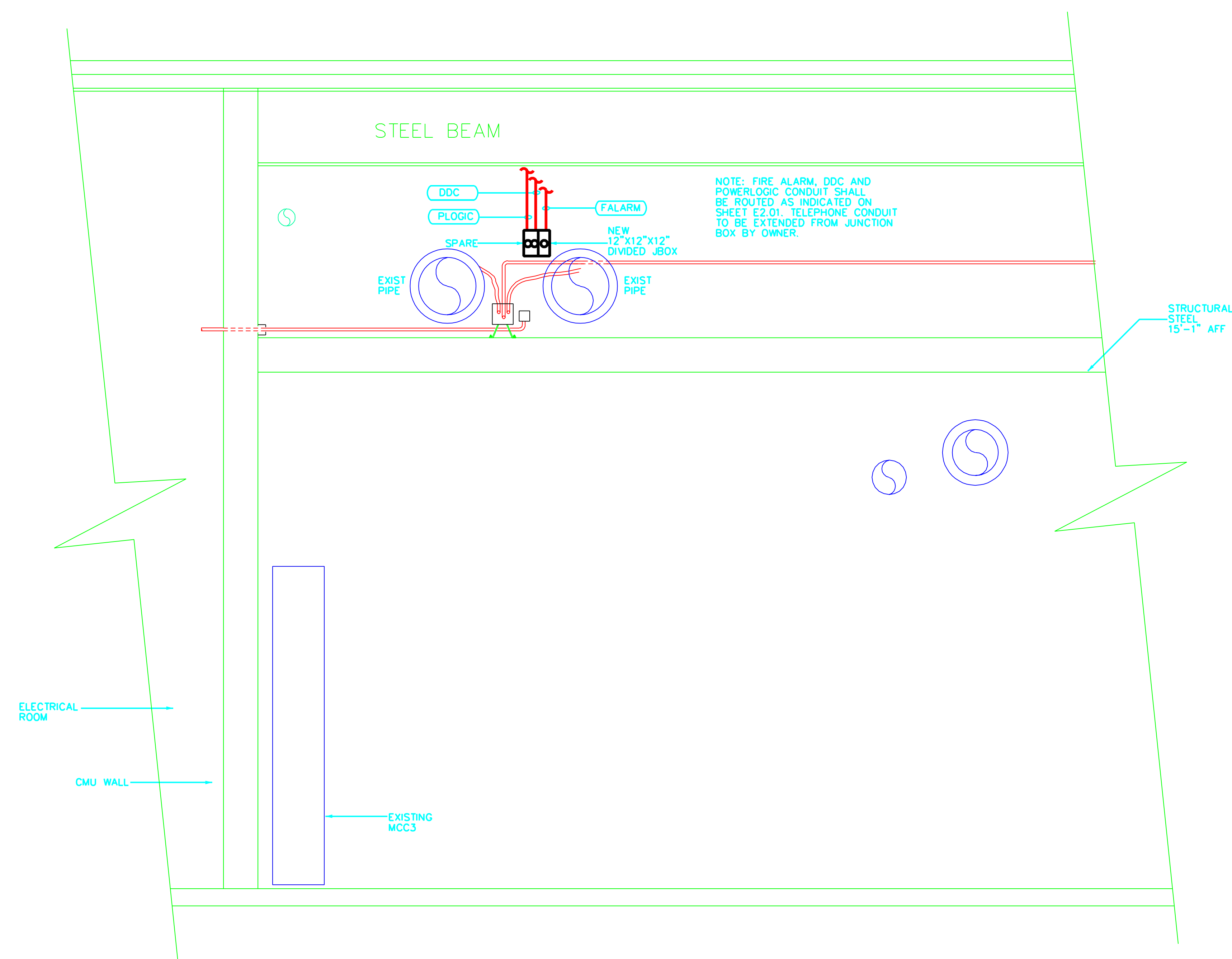
03 DUCT BANK DETAILS



1017
11:16:52 AM 02/25/2003 ADAMS
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**01 SECTION AT BUILDING 104
LOOKING EAST**
1/4"=1'-0"



**02 SECTION AT EXTERIOR WALL
LOOKING WEST**
1/4"=1'-0"

RECORD DRAWING
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PRINTS, DRAWINGS AND OTHER DATA.

0 4 8
1/4"

Revisions	Date



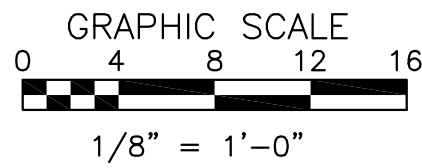
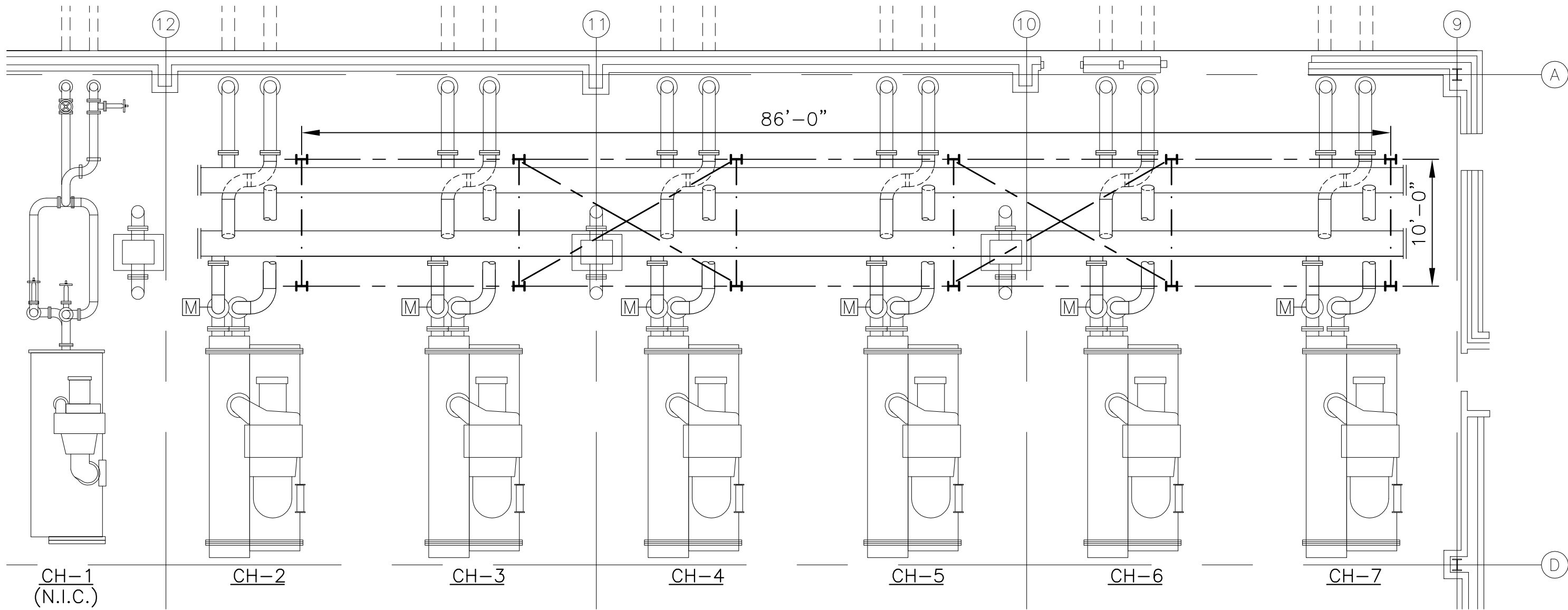
SHAH SMITH & ASSOCIATES, INC.
2825 WILCREST, SUITE 350
HOUSTON, TEXAS 77042
Phone: 713-780-7563 Fax: 713-780-9209

Drawing Title
ELECTRICAL DETAILS
Approved: Engineering Svc Chief
Approved: Medical Center Director

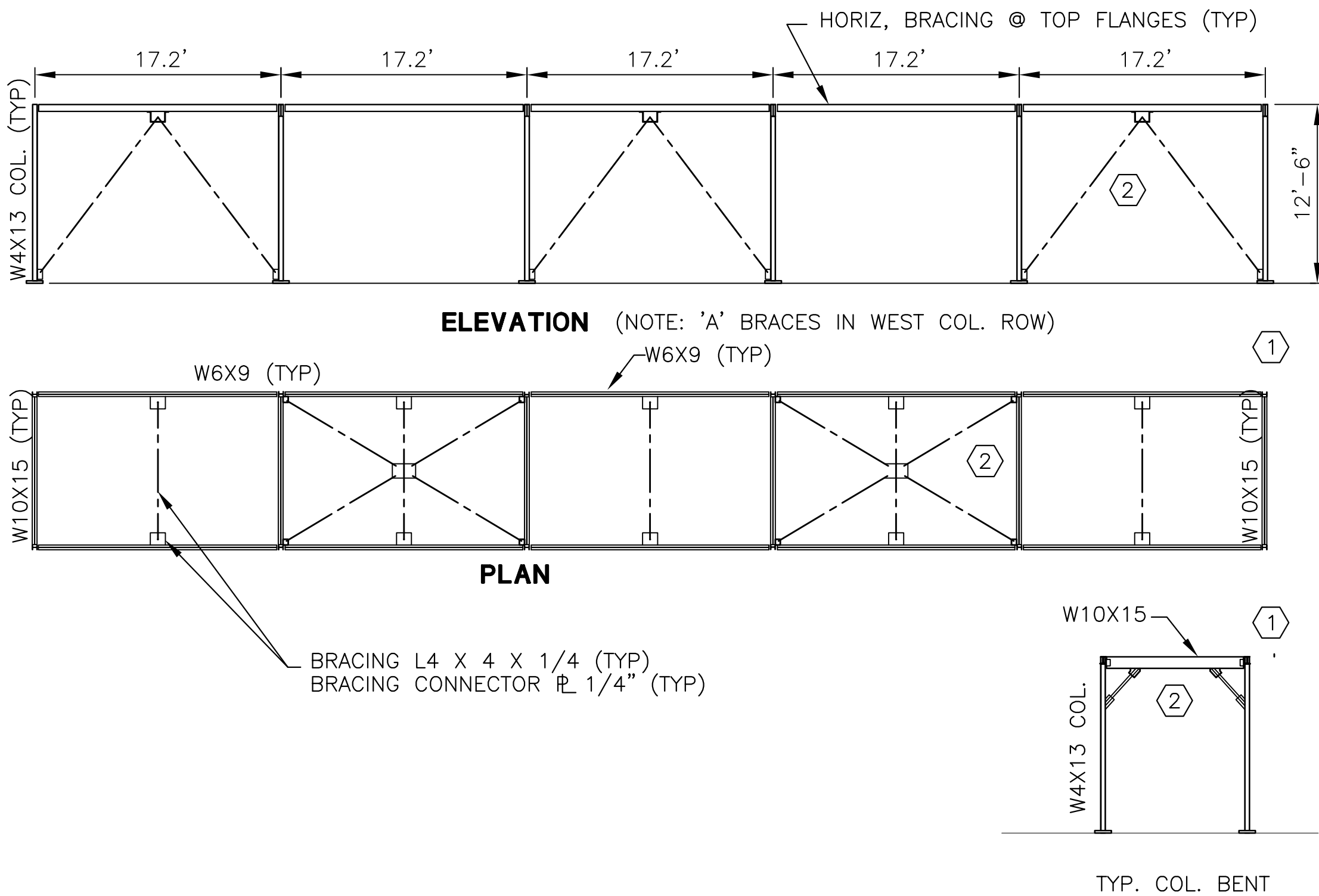
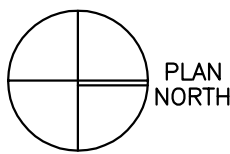
Project Title
HOUSTON VETERANS AFFAIRS MEDICAL CENTER
CHILLER PLANT EXPANSION
Building Number
104/106
Location
VAMC HOUSTON, TX

Date
02/26/2003
Project No.
VA #560-58-110 SSA #1017-007-1
Drawing No.
E5.05
Dwg. of

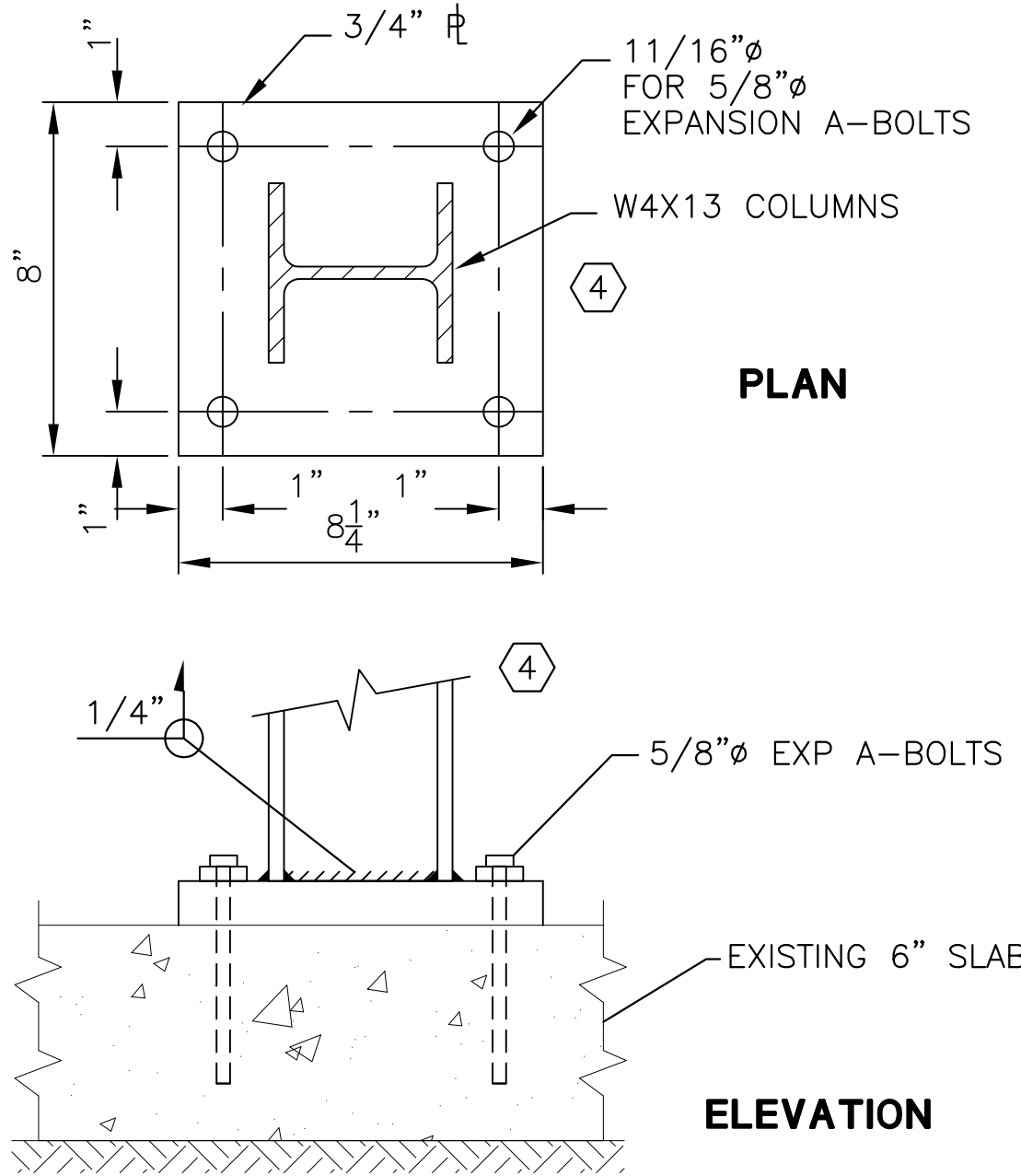
**Veterans
Administration**



1 PARTIAL PLAN - BLDG. 104-STRUCTURAL
 SCALE: 1/8"= 1'-0"



A DETAIL - CONDENSER WATER HEADER SUPPORT
 SCALE: 1/8"= 1'-0"



B DETAIL - COLUMN BASE PLATE
 SCALE: 1"= 3"

GENERAL NOTES:

- STEEL SHAPES AND PLATES SHALL BE ASTM A-36.
- FABRICATION AND ERECTION SHALL BE IN COMPLIANCE WITH AISC CODE OF STANDARD PRACTICE.
- FIELD WELDING SHALL BE IN COMPLIANCE WITH STANDARD PRACTICES OF THE AWS.

KEYED NOTES:

B/S1.0

- CONNECTION PLATES FOR BEAM SHALL BE FOR BOLTED WEB CONNECTIONS AND SHOP WELDED TO COLUMNS, IN COMPLIANCE WITH AISC SPECIFICATIONS.
- CONNECTION PLATES FOR BRACING SHALL BE 1/4" PLATE, SHOP WELDED TO COLUMNS AND BEAMS.
- HORIZONTAL BRACING SHALL BE INSTALLED AT THE TOP FLANGE OF BEAMS WITH VERTICAL LEGS EXTENDING DOWN. 'A' BRACES SHALL BE INSTALLED AT THE CENTERLINES OF BEAMS AND COLUMNS WITH OUTSTANDING LEGS EXTENDING IN.
- COLUMN BASE PLATES SHALL BE LOCATED AND SET IN THE SLAB PRIOR TO WELDING COLUMNS TO BASE PLATES. COLUMNS SHALL BE WELDED ALL AROUND WITH 1/4" FILLET.

THE SEAL APPEARING ON THIS DOCUMENT WAS
 AUTHORIZED BY WILLIAM J. CORRETT, P.E. 17225
 ON 11/29/2007.



DATE:	11/29/2007
SCALE:	AS INDICATED
PROJECT MANAGER:	PAT CORRETT, P.E.
DESIGN ENGINEER:	PAT CORRETT, P.E.
CLIENT:	

VA MEDICAL CENTER
 VAMC-104 COOLING TOWER REPLACEMENT
 PARTIAL PLAN-STRUCTURAL

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 PERFORMANCE ENGINEERING
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 Tel (512)-444-4565
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NO	DATE	REVISIONS	BY
3	11/29/07	RELEASED FOR CONSTRUCTION	CPC
2	11/20/07	RELEASED FOR 50 % COMPLETION	CPC
1	11/12/07	RELEASED FOR REVIEW	CPC

S1.0